Packings, Lattices and Configurations

Lecture notes for MAT670, TU Graz, Summer Term 2016
First typeset draft from the handwritten notes

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About this draft

These notes are a first typeset pass at the handwritten notes for the course *Packings, Lattices and Configurations* (MAT670), taught in eleven two-hour lectures at TU Graz in the summer term of 2016. The course description read:

> This course will introduce a variety of different tools and models to explore some of the theory of geometric configuration spaces that underlie physical systems. ... By the end of the course, you should have some familiarity with the geometry and topology of configuration spaces, Morse theory, the geometry of lattices, maximal density arrangements, random packings and statistical mechanics.

The structure, order, examples and choice of proofs follow the handwritten notes. Telegraphic notes have been written out as prose, sketches have been redrawn, and gaps have been filled. The colour code is:

- Black text is a cleaned-up rendering of what is on the page.
- Blue text was *not* in the handwritten notes. It covers completed definitions and proofs, missing hypotheses, references, and remarks on results that appeared after 2016. Longer insertions are marked [Added].
- Orange margin notes **Fix:** mark places where the notes contained an error that has been corrected, saying what changed. **?** marks a doubtful reading of the handwriting, or a step that is still open.
- Grey margin notes describe the original page (crossed-out text, omitted doodles, and so on).

The interludes. Several chapters are entirely new and are set in black for readability, but all of them are added material:

A From Dynkin diagrams to Lie algebras and Lie groups (after Lecture 6);

B Lattices, quadratic forms and the geometry of numbers (after A);

C The torus, Diophantine approximation and aperiodic order (after B);

D Min-max, configuration spaces and braids (after Lecture 11);

E Random packings, entropy and information (after D);

F Computation and certified proof (after E);

G Packings in the laboratory (after F);

H Dynkin diagrams everywhere (at the end).

Extensions. Sections titled "Extension" inside the lectures are also new material set in black: Rogers' simplex bound (end of Lecture 2) and why root lattices pack well (end of Lecture 6).

Sources. The scans used are `MAT670_1`, `_2`, `_3`, `_46`, `_78`, `_9X` and `_XI` on the course web page. `MAT670_emsemble` is a bundled copy of Lectures 1–3. `MAT670_classification` is an earlier scan of Lectures 4–5, the first 31 pages of `MAT670_46` minus one recap page. Both were transcribed independently and used only as a cross-check.

Known open items in this draft.

- Lecture 3: the "improved" radial Blichfeldt gauge is stated in the notes without proof. No proof has been supplied here; a numerical search in low dimensions found no counterexample.
- Lecture 10: the deformation theorem for $0 < \theta < \pi/N$ had no proof in the notes. The proof given in blue is the editor's and should be checked. Pages 24–26 of the scan are mostly headings. The notes themselves mark this lecture "needs to be rewritten, out of order / missing pieces".
- Lecture 11: the count of locally jammed configurations in the open-ended channel may be off at the ends, giving $\binom{\ell-k-1}{k}$ rather than $\binom{\ell-k+1}{k}$. It is flagged but not changed.

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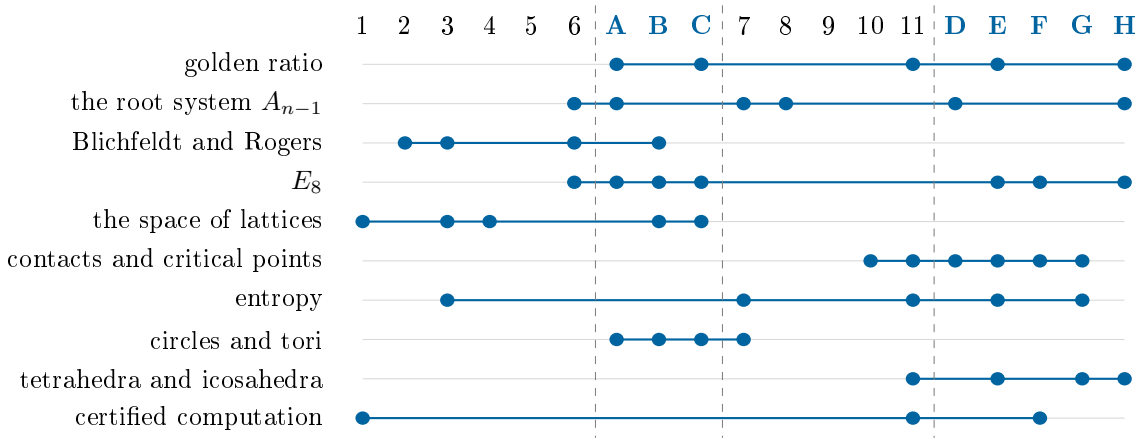
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Threads through the notes

These notes are not meant to be read in a straight line. The same few objects keep turning up in different clothes, and following them is the most fun way through. Below are some of the threads, each with its stops. Lectures are numbered 1–11 and the new interludes are lettered A–H.



Chapters are listed in book order (lectures in black, interludes in blue); a dot marks a stop on the thread.

The golden ratio. Lecture 11 counts the jammed states of a narrow channel with binary words having no two adjacent 1s, and Fibonacci numbers appear. The most likely fraction of 1s is $(5-\sqrt{5})/10$, and the entropy per site is $\log\varphi$. Interlude E recognizes this as the Shannon capacity of the "golden mean" channel. Interlude C meets the Fibonacci word again: it is a Sturmian word of slope φ^{-2}, the cutting sequence of an irrational line through the square lattice, its gaps obey the three-gap theorem, and the golden ratio is the worst approximable number (Hurwitz). Folding the E_8 diagram with $\tau=\varphi$ gives the non-crystallographic group H_4 and icosahedral quasicrystals (Interlude C). Interlude A sends the binary icosahedral group to the extended E_8 diagram (McKay). The icosahedron itself is not made of regular tetrahedra (Interlude E).

The root system A_{n-1}. It is built in Lecture 6 as the vectors e_i-e_j in the hyperplane $\sum x_i=0$. In Lecture 8 the same hyperplanes $x_i=x_j$ cut $\mathrm{Conf}(N,\mathbb{R})$ into Weyl chambers. In Interlude A the roots are the eigenvalues of conjugation on the matrix units E_{ij}; the maximal torus of $\mathrm{SU}(n)$ is $\mathbb{R}^{n-1}$ modulo the A_{n-1} lattice; and its Weyl alcove is the triangle of gaps between points on a circle from Lecture 7. Complexify the hyperplanes and divide by Σ_N, and you get the braid group (Interlude D).

Blichfeldt. His 1929 gauge trick bounds the density in Lectures 2 and 3. Rogers' simplex bound (end of Lecture 2) replaces his spreading of mass by a cutting of Voronoi cells, and is sharp in the plane, where it is Thue's theorem. His 1914 lemma proves Minkowski's convex body theorem (Interlude B). The Cohn–Elkies bound is a "two-sided" Blichfeldt gauge, with positivity demanded on both sides of the Fourier transform, and with Viazovska's magic function it settles dimension 8 (Interlude B). The gauges for long bodies in Lecture 3 (capsules, cylinders) are the subject of G. Fejes Tóth and W. Kuperberg, *Blichfeldt's density bound revisited* (Math. Ann. 1993), which leads on to the cylinder bounds of A. Bezdek and W. Kuperberg (1990) and W. Kusner (2014).

E_8. It is constructed in Lecture 6 as $D_8 \cup (D_8 + h)$ and from the Hamming code. In Interlude A it is $248 = 8 + 240$: Cartan subalgebra plus root spaces, and its density $\pi^4/384$ can be read off from $\det C = 1$. In Interlude B its theta series is the Eisenstein series E_4, so it has $240\,\sigma_3(m)$ vectors of norm $2m$, and Viazovska's theorem makes it the densest packing in dimension 8. In Interlude C it folds to H_4, and in Interlude E the Golay code builds its big sibling, the Leech lattice.

Why root lattices? They are within a few percent of Rogers' bound in dimensions 2–8 (end of Lecture 6). Their density can be read off from the determinant of the Cartan matrix (Interlude A). The Weyl group's irreducibility makes them eutactic and the abundance of roots makes them perfect, so Voronoi's theorem makes each one a local optimum. Their $O(n^2)$ roots are too few in high dimension, where Interlude B's random lattices take over.

The space of lattices. It appears in Lecture 4 with Mahler's criterion, and in Interlude B with Hermite's constant, reduction of quadratic forms and Siegel's mean value theorem. From the last of these comes the Minkowski–Hlawka lower bound $\delta \geq 2^{-d} \cdot 2\zeta(d)$, which was the lower bound quoted in Lectures 1 and 3. In Interlude C dynamics on the space takes over: bounded orbits correspond to badly approximable numbers (Dani), and Margulis's proof of the Oppenheim conjecture.

Contacts, forces and critical points. The contact graphs and force balance of Lectures 10–11 make the max-min distance function a "min-type" Morse function. Interlude D turns this into min-max: the Tammes problem as the top critical value, Betti numbers of configuration spaces as lower bounds on the number of critical configurations, and kissing numbers. Interlude E reads the same force-balance condition as jamming, with Maxwell counting and isostaticity. The slack in the icosahedral kissing configuration (Interlude E) is what lets the jitterbug of Lecture 11 move.

Entropy. Lecture 7 derives the Boltzmann distribution and the Gaussian fluctuations of a gas in a box. Lecture 11 introduces Edwards' entropy of jammed states and solves hard rods. In Interlude E the partition function of hard spheres becomes the volume of a configuration space; the parking problem has an exact answer; and maximum entropy, Shannon's channel capacity and sphere packing in high dimensions turn out to be one circle of ideas, with Kabatiansky–Levenshtein (Lecture 3) as a coding theorem.

Circles and tori. Points on a circle (Lecture 7) have a triangle of gaps. Interlude A finds that triangle as the space of conjugacy classes of SU(3). Interlude C cuts the torus along an irrational line ("shredding the torus"), proves equidistribution, and finds at most three gaps. Interlude B counts lattice points in a circle and asks how wrong πr^2 can be (the Gauss circle problem).

Tetrahedra and icosahedra. Twelve balls around one fit at the vertices of an icosahedron with room to spare. Lecture 11's jitterbug uses that room, and Interlude E proves it with rotating tropical rings. Interlude H sees the same fact in a Coxeter diagram: $\{3, 3, 5\}$ is spherical, not flat, so five regular tetrahedra close up around an edge only on $\mathbb{S}^3$, in the 600-cell. In the FCC lattice, octahedra fill exactly the gap the tetrahedra leave. In the laboratory, tetrahedral dice and colloidal clusters show what happens in between (Interlude G).

Certified computation. Thue's triangle lemma (Lecture 1) is proved again in Interlude F by a branch and bound in exact rational arithmetic, using a Lagrange multiplier that is the force balance of Lectures 10–11. The same chapter treats the case analyses that the Tammes and Kepler problems need, and the certificates (intervals, sums of squares, formal proof) that make them trustworthy.

Part I

Packing bounds

Chapter 1

Lecture 1: Packing problems

1.1 Course format

1.0 "Admin": contact, room, breaks omitted

Evaluation is by short quizzes and a project: a talk of at least 20 minutes together with a written summary. Updates are posted by email, through TU Online, and on the course page `wkusner.github.io/MAT670/`.

1.2 Mathematics of packings, lattices and configurations

The subject of this course sits at the meeting point of several areas:

- classical convex and discrete geometry, and the geometry of numbers;
- applied topology and Morse theory;
- statistical mechanics, combinatorics and information theory.

p. 1: a diagram linking these areas; "Rigidity" crossed out there

We will take various ideas from each of these, all of them dealing with *configurations* of points or bodies. Related topics include rigidity theory and configuration spaces of linkages, information theory (codes), and the geometry of numbers. Packing problems are my main motivation.

Packing problems are typically *easy to state but hard to solve*. The same is true of the geometric inequalities and the configuration spaces we will meet along the way.

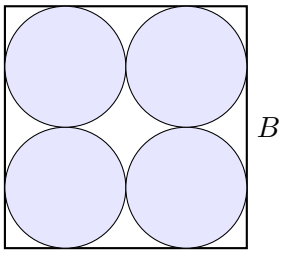
Remark 1.1. Some landmarks, for orientation: the densest packing of congruent balls in $\mathbb{R}^3$ (the Kepler conjecture, density $\pi/\sqrt{18}$) was proved by Hales (announced 1998, published 2005; formal verification by the Flyspeck project 2014/2017). In dimension 8 the E_8 lattice packing was proved optimal by Viazovska (2016), and in dimension 24 the Leech lattice packing by Cohn–Kumar–Miller–Radchenko–Viazovska (2016/2017). These dimension 8 and 24 results appeared during the course itself. In every other dimension $d \geq 4$ the problem is still open.

1.3 Packing problems

1.3.1 Packing congruent bodies in a box

We start in two dimensions, with the problem of packing circles (disks) in a box. Let $B \subset \mathbb{R}^2$ be a box (a square, say) and K a disk. We want to place N congruent copies of K in B so as to maximize the *density*, or volume fraction,

$$\frac{\operatorname{vol}\left(\bigcup_{i=1}^{N} K_i\right)}{\operatorname{vol}(B)}.$$



Four congruent disks packed in a square B (this 2×2 arrangement is optimal for $N = 4$).

Definition 1.2 (Packing in a box)**.** A collection $K_1, \dots, K_N$ of subsets of $\mathbb{R}^n$ is a *packing of K in B* if

1. the K_i are congruent to K: $K_i = \varphi_i K$ with $\varphi_i \in \mathbb{R}^n \rtimes \mathrm{SO}(n)$ a rigid motion;
2. $K_i \subset B$ for all i;
3. the interiors are pairwise disjoint: $\operatorname{int} K_i \cap \operatorname{int} K_j = \emptyset$ for $i \neq j$.

p. 3 writes $\mathbb{R}^2 \rtimes \mathrm{SO}(2)$

Example 1.3. Two equivalent formulations of the disk-in-a-box problem:

1. Find the largest scale r such that $\bigcup_{i=1}^N \varphi_i(rK) \subset B$ is a packing of B, where K is the unit disk. (Here rK is in Minkowski notation: $rK = \{rx : x \in K\}$.)
2. Find a configuration $\{x_1, \dots, x_N\}$ of N points in B such that

$$d(x_i, x_j) \;\geq\; 2r \quad \text{for } i \neq j, \qquad \text{and} \qquad d(x_i, \partial B) \;\geq\; r \quad \text{for all } i,$$

with r as large as possible.

The points x_i are the centres of the disks $\varphi_i(rK)$.

Fix: notes: $d(x_i, x_j) > r$; disks of radius r need centres $2r$ apart

These are hard problems, and there is an infinite variety of them: we can change N, K and B, the dimension, add other constraints, and so on.

One case of particular interest is $B = \mathbb{R}^2$: the circle packing problem in the plane. Other natural choices are $B = \mathbb{R}^d$, or $B = \mathbb{S}^2$. (For $B = \mathbb{R}^d$ the number of bodies is infinite and "density" must be defined by a limit; this is done next.)

1.3.2 Packings of the plane

numbered "1.2" a second time in the notes

Now take $B = \mathbb{R}^2$. We have an infinite collection of congruent disks K_i, now normalized to have unit radius, with $\operatorname{int} K_i \cap \operatorname{int} K_j = \emptyset$ for $i \neq j$. Write $P = \bigcup_i K_i$.

Definition 1.4 (Upper and lower density)**.** The *upper density* of the packing P is

$$\delta^+(P) \;=\; \limsup_{R \to \infty} \frac{\mathrm{vol}(P \cap R\,\mathbb{D}^2)}{\mathrm{vol}(R\,\mathbb{D}^2)},$$

where $\mathbb{D}^2$ is the closed unit disk centred at the origin. The *lower density* $\delta^-(P)$ is defined in the same way with $\liminf$. In general dimension,

$$\delta^+(P) \;=\; \limsup_{R \to \infty} \frac{\mathrm{vol}(P \cap R\,\mathbb{D}^n)}{\mathrm{vol}(R\,\mathbb{D}^n)}$$

with $\mathbb{D}^n$ the unit ball in $\mathbb{R}^n$.

Remark 1.5. These quantities need not be nice in general (e.g. $\delta^+(P) \neq \delta^-(P)$ is possible, and then "the" density of P is not defined). However:

- they do *not* depend on the choice of origin (moving the centre changes the numerator only in a boundary layer of volume $O(R^{n-1})$);
- they *do* depend, in general, on the shape of the averaging region $\mathbb{D}^n$ (for periodic packings, though, any reasonable shape gives the same value).

Does a packing of maximal density exist? We might look at this in more detail later. (It does: the supremum of δ^+ over all packings is attained, by a compactness argument; see e.g. Groemer's theorem in Conway–Sloane, *Sphere Packings, Lattices and Groups*, Ch. 1.)

? existence deferred in the notes

1.3.3 Bounds on the density

How can we compute bounds on the density? Clearly $\delta^+(P) \leq 1$, which is a bad bound.

We can also construct lower bounds using highly structured packings. For example, the unit balls centred at the points of $(2\mathbb{Z})^d$ form a packing. Since this structure is periodic, we can work with the fundamental domain.

Exercise 1.6 (Easy)**.** Why can we compute the density on a fundamental domain? Show that for a periodic packing, with period lattice Λ and m balls of radius 1 centred in a fundamental domain F of Λ, the density exists (that is, $\delta^+ = \delta^-$) and equals $m \operatorname{vol}(B^d)/\operatorname{vol}(F)$.

Recall the volume of the ball of radius r in $\mathbb{R}^d$:

$$V_d(r) \;=\; \operatorname{vol}\bigl(B^d(r)\bigr) \;=\; \frac{\pi^{d/2}}{\Gamma\bigl(\frac{d}{2}+1\bigr)}\, r^d.$$

This is easier to remember in even dimension: $V_{2k}(r) = \dfrac{\pi^k}{k!}\, r^{2k}$. For $(2\mathbb{Z})^d$ the fundamental domain is the cube $[0,2)^d$ of volume 2^d, containing one ball, so the density is

$$\delta\bigl((2\mathbb{Z})^d\bigr) \;=\; \frac{V_d(1)}{2^d} \;=\; \frac{\pi^{d/2}}{\Gamma\bigl(\frac{d}{2}+1\bigr)\,2^d} \overset{d=2k}{=} \frac{\pi^k}{k!\,2^{2k}}.$$

notes write $(\frac{d}{2})!$ for $\Gamma(\frac{d}{2}+1)$

We could improve this by considering better periodic systems.

Exercise 1.7 (Easy)**.** Show that periodic packings can approximate a densest packing: the supremum of $\delta^+(P)$ over all packings P equals the supremum over periodic packings. (Hint: cut a large cube out of a near-optimal packing, remove the balls that meet its boundary, and repeat periodically.)

But which periodic systems should we use?

Exercise 1.8. Consider the lattices coming from the root systems $A_1 \times \cdots \times A_1$ (d factors), …, A_d. Compute their packing densities and compare. (For reference: A_1^d gives $\mathbb{Z}^d$, which is $(2\mathbb{Z})^d$ after scaling, with density $V_d(1)/2^d$. The lattice A_d has minimal norm $\sqrt{2}$ and covolume $\sqrt{d+1}$, hence density $V_d(1)\,2^{-d/2}/\sqrt{d+1}$. This gives $\pi/\sqrt{12}$ for $d=2$ and $\pi/\sqrt{18}$ (FCC) for $d=3$.)

? p. 6: "Consider A^1, $A_1 \times^d A_1$ … A_d? root systems"

1.4 A better bound (non-constructive)

§1.3 in the notes

Definition 1.9 (Saturated packing)**.** A packing $P = \bigcup_i K_i$ of congruent copies of K in $\mathbb{R}^n$ is *saturated* if $P \cup \varphi K$ is not a packing for any $\varphi \in \mathbb{R}^n \rtimes \mathrm{SO}(n)$; that is, no additional disk (ball) can be added.

- Saturated packings certainly exist: start from any packing and keep adding balls. This is an "infinite algorithm". It can be made rigorous with Zorn's lemma, or by adding balls greedily along an enumeration of a countable dense set of centres.
- Saturation does not decrease $\delta^+(P)$ since we only add balls.

So for lower bounds on the best possible density we may assume that P is saturated.

This is a surprisingly useful idea. Let K be the unit ball, $K_i = \varphi_i K$ with centre c_i.

Proposition 1.10. *If P is a saturated packing of unit balls in $\mathbb{R}^d$, then $\delta^+(P) \geq 2^{-d}$.*

Proof. If P is saturated, then for every $x \in \mathbb{R}^d$ there is an i with $d(x, K_i) < 1$. Otherwise $|x - c_i| \geq 2$ for all i, and the unit ball centred at x could be added. Hence the doubled balls

$$2K_i \;:=\; \varphi_i\big(2\,\varphi_i^{-1}(K_i)\big) \;=\; B(c_i, 2)$$

form a covering of $\mathbb{R}^d$. Each $2K_i$ has 2^d times the volume of K_i, so the fraction of $R\,\mathbb{D}^d$ covered by $\bigcup_i 2K_i$ is, up to a boundary term that vanishes as $R \to \infty$, at most 2^d times the fraction covered by P. Since the doubled balls cover everything,

$$2^d\,\delta^+(P) \;\geq\; 1, \qquad \text{i.e.} \qquad \delta^+(P) \;\geq\; \frac{1}{2^d}. \qquad \square$$

boxed correction: dilate about the centre, $(2\varphi_i^{-1}K)\varphi_i$

Remark 1.11. Compare with $(2\mathbb{Z})^d$, of density $V_d(1)/2^d$. Since $V_d(1) < 1$ exactly for $d \geq 13$, and $V_d(1) \to 0$ superexponentially, the non-constructive bound 2^{-d} is the better one in high dimensions.

We may go over some slight improvements in the future, but none seem satisfying.

Remark 1.12. Known improvements of the 2^{-d} lower bound include the bound of Minkowski and Hlawka ($2\zeta(d)\,2^{-d}$), Rogers and Ball ($c\,d\,2^{-d}$), Venkatesh 2013 ($c\,d\log\log d\,2^{-d}$ along a sparse sequence of d), and, after the course, Campos, Jenssen, Michelen and Sahasrabudhe 2023 ($c\,d\log d\,2^{-d}$) and Klartag 2025 ($c\,d^2\,2^{-d}$). All of them are only polynomial improvements over 2^{-d}.

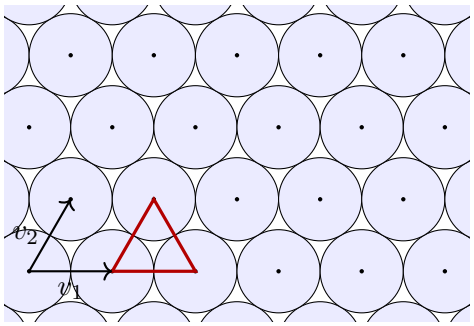
The problem is that there is much more freedom in dimensions $d > 3$! For example, take $(2\mathbb{Z})^4$. The main diagonal of the cube $[0,2]^4$ has length $2\sqrt{4} = 4$, so the centre $(1,1,1,1)$ is at distance 2 from every vertex. So a unit ball fits into every such hole. Adding them gives the D_4 packing $(2\mathbb{Z})^4 \cup \big((2\mathbb{Z})^4 + (1,1,1,1)\big)$, which has twice the density, $\pi^2/16$.

? "$d > 3$! even.": perhaps "even $d = 4$"

1.5 Upper bounds in dimension 2

The lower bound in $d = 2$ is constructive: it comes from the hexagonal (triangular) lattice packing.

§1.4



Hexagonal packing of unit disks, basis $v_1 = (2, 0)$, $v_2 = (1, \sqrt{3})$ (angle $\pi/3$), and an equilateral triangle of side 2.

Example 1.13 (Hexagonal packing)**.** Consider the equilateral triangle of side 2 spanned by three mutually tangent unit disks. Its height is $\sqrt{2^2 - 1^2} = \sqrt{3}$, so its area is $\sqrt{3}$. It contains three sectors of angle $\pi/3$, i.e. half a unit disk, of area $\pi/2$. The plane is tiled by such triangles, so

$$\delta_{\text{hex}} \;=\; \frac{\pi/2}{\sqrt{3}} \;=\; \frac{\pi}{2\sqrt{3}} \;=\; \frac{\pi}{\sqrt{12}} \;\approx\; 0.9069.$$

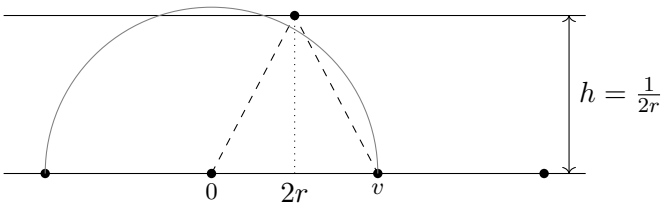
1.5.1 Lattice packings

For lattices, we want to maximize the minimum distance between two points of a unit lattice (a lattice $\Lambda \subset \mathbb{R}^2$ of covolume 1). This minimum distance can be arbitrarily small, but it is bounded above.

crossed out: "minimize the det of the span for a unit lattice"

Theorem 1.14 (Lagrange)**.** *Every lattice packing of congruent disks in $\mathbb{R}^2$ has density at most $\pi/\sqrt{12}$. Equality holds only for the hexagonal lattice.*

Proof. Scale so that Λ has covolume 1, and let $2r$ be its minimal distance, so that disks of radius r centred at Λ form a packing of density πr^2. Let v be a shortest vector, $|v| = 2r$. The lattice is the union of translates of the row $\mathbb{Z}v$, which lie on parallel lines. Since the covolume is 1, adjacent lines are at distance $h = 1/(2r)$. Note that this is the height of a fundamental parallelogram; the second basis vector need not be orthogonal to v.



Translating along the row by multiples of v, we may take a lattice point w in the adjacent row whose orthogonal projection onto the first line lies within distance r of some lattice point there. Since $|w - u| \geq 2r$ for all lattice points $u \neq w$,

$$h^2 + r^2 \;\geq\; (2r)^2, \qquad \text{so} \qquad \frac{1}{2r} \;=\; h \;\geq\; \frac{\sqrt{3}}{2}\,(2r) \;=\; \sqrt{3}\,r.$$

Hence $r^2 \le \frac{1}{2\sqrt{3}}$, i.e. $2r \le \sqrt{2/\sqrt{3}}$, and

$$\delta \;=\; \frac{\pi r^2}{1} \;\le\; \frac{\pi}{2\sqrt{3}} \;=\; \frac{\pi}{\sqrt{12}}.$$

Equality forces $h = \sqrt{3}r$ with offset exactly r, which is the hexagonal lattice. □

p. 9 is an unnumbered duplicate of this computation

1.6 Partitioning space: Voronoi cells and Delaunay triangulations

§1.5

For packings that are not lattices, we need some analogous method to partition space. Classically, Dirichlet–Voronoi diagrams work well.

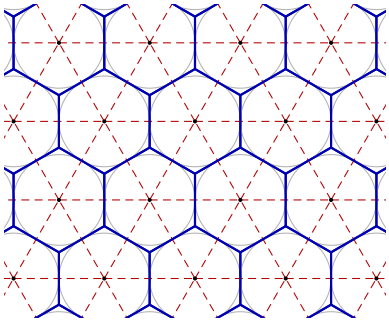
Definition 1.15 (Voronoi cell)**.** Given a packing with bodies P_i, the *(Dirichlet–)Voronoi cell* associated with P_i is

$$D_i \;=\; \{x \in \mathbb{R}^2 : d(x, P_i) < d(x, P_j) \text{ for all } j \neq i\}.$$

When the P_i are congruent disks and d is the standard distance, this is the same as using the distance to the centres of the disks. Indeed $d(x, P_i) = |x - c_i| - 1$ for $x \notin P_i$.

Voronoi cells have nice properties:

- each cell is an intersection of half-spaces determined by the centres ($D_i = \bigcap_{j \neq i} \{x : |x - c_i| < |x - c_j|\}$), so it is convex;
- the cells partition space up to a boundary term (a set of measure zero).



Hexagonal packing: Voronoi cells (solid hexagons) and Delaunay triangles (dashed).

The dual notion is the *Delaunay triangulation*.

- It is not unique in general (e.g. four points at the vertices of a square).
- It is harder to define directly. Its characterization: the circumcircles of the Delaunay triangles are empty of other points (vertices).
- The circumcentre of a Delaunay triangle is a Voronoi vertex.

[Added] Precisely: for a locally finite set $X \subset \mathbb{R}^2$ (here, the centres), a triangle with vertices in X is *Delaunay* if its open circumdisk contains no point of X. A Delaunay triangulation is a triangulation of (the convex hull of) X by Delaunay triangles. It is unique when no four points of X are concyclic.

1.6.1 Existence via lifting

Consider a collection of points in the plane and lift them to the paraboloid $z = x^2 + y^2$.

Lemma 1.16. *Let a, b, c, d be points in the (x, y)-plane, and lift them to $\hat{a}, \hat{b}, \hat{c}, \hat{d}$ on $z = x^2 + y^2$ (i.e. $\hat{p} = (p, |p|^2)$). Then d lies inside the circumcircle of a, b, c if and only if $\hat{d}$ lies in the lower half-space of the plane $\langle \hat{a}, \hat{b}, \hat{c} \rangle$ spanned by $\hat{a}, \hat{b}, \hat{c}$.*

Proof. A plane tangent to the paraboloid at the point above (p, q) has the form

$$z \;=\; 2px + 2qy - (p^2 + q^2).$$

Shifting it up by h^2 gives

$$z \;=\; 2px + 2qy - (p^2 + q^2) + h^2.$$

It meets the paraboloid where

$$x^2 + y^2 \;=\; 2px + 2qy - (p^2 + q^2) + h^2 \quad\Longleftrightarrow\quad (x - p)^2 + (y - q)^2 \;=\; h^2.$$

Fix: signs: notes have $+(p^2 + q^2)$ and $(x + p)^2$, $(y + q)^2$

So the plane through $(p, q, p^2 + q^2)$, parallel to the tangent plane there, at height h^2 above it, projects its intersection with the paraboloid onto the circle of radius h about (p, q) in the (x, y)-plane.

[Added] Every non-vertical plane $z = \alpha x + \beta y + \gamma$ meeting the paraboloid in more than one point is of this form, with $p = \alpha/2$, $q = \beta/2$ and $h^2 = \gamma + p^2 + q^2 > 0$. Apply this to the plane through $\hat{a}, \hat{b}, \hat{c}$. The corresponding circle passes through a, b, c, so it is their circumcircle. A point $\hat{x}$ of the paraboloid lies strictly below this plane iff $x^2 + y^2 < 2px + 2qy - (p^2 + q^2) + h^2$, i.e. iff $(x - p)^2 + (y - q)^2 < h^2$, i.e. iff x lies strictly inside the circumcircle.

□

Corollary 1.17 (Existence of Delaunay triangulations)**.** *The existence of the convex hull of the lifted points implies the existence of a Delaunay triangulation. More precisely: project the faces of the lower convex hull of $\{\hat{x} : x \in X\}$ to the plane. By Lemma 1.16, a lower face has no lifted point below its plane, so its projection has an empty circumcircle. Triangulating any non-triangular faces (which come from concyclic points) gives a Delaunay triangulation. For the infinite, locally finite point sets of a packing, apply this to large finite pieces; the triangles near a fixed point stabilize.*

1.7 Thue's theorem

§1.6; a first "1.6 Thm" on p. 13 is abandoned

Let P be a saturated packing of unit disks in $\mathbb{R}^2$ and DT a Delaunay triangulation of its set of centres. Every edge of DT has length at least 2, since the disks do not overlap. Moreover, every triangle of DT has circumradius $R < 2$. Its open circumdisk contains no centres, so the circumcentre is at distance $\geq R$ from every centre. If $R \geq 2$, a unit disk centred there could be added, contradicting saturation.

Lemma 1.18. *The largest angle θ of a triangle $ABC \in DT$ of a saturated packing satisfies*

$$\frac{\pi}{3} \;\leq\; \theta \;<\; \frac{2\pi}{3}.$$

? upper inequality written ambiguously; the proof gives strict "<"

Proof. The lower bound is obvious, since θ is the largest angle of a triangle. Assume $\theta \geq 2\pi/3$. Let A be the smallest angle. Then $A \leq \frac{1}{2}(\pi - \theta) \leq \pi/6$, so $\sin A \leq \frac{1}{2}$. By the circumradius formula, and since the side BC opposite A has length at least 2,

$$R \;=\; \frac{1}{2}\,\frac{|BC|}{\sin A} \;\geq\; \frac{1}{2}\cdot\frac{2}{1/2} \;=\; 2.$$

This contradicts $R < 2$. □

Fix: notes: "$= 4$"; $\frac{1}{2}\cdot\frac{2}{1/2} = 2$

Lemma 1.19. *The density in a single Delaunay triangle is at most $\pi/\sqrt{12}$, with equality if and only if the triangle is equilateral (of side 2).*

[Added] Here the *density* of a triangle $T \in DT$ means $\frac{\pi/2}{\text{area}(T)}$: the three sectors of the unit disks at the vertices of T have angles summing to π, hence total area $\pi/2$.

Proof. Let B be the largest angle of ABC. Then

$$\text{area}(ABC) \;=\; \tfrac{1}{2}\,|AB|\,|CB|\,\sin B \;\geq\; \tfrac{1}{2}\cdot 2\cdot 2\cdot \min_{B\in[\pi/3,\,2\pi/3)} \sin B \;=\; \tfrac{1}{2}\cdot 2\cdot 2\cdot \tfrac{\sqrt{3}}{2} \;=\; \sqrt{3},$$

using Lemma 1.18, and the minimum is attained at $B = \pi/3$. So $\text{area}(T) \geq \sqrt{3}$ for every $T \in DT$, and the density of T is at most $\frac{\pi/2}{\sqrt{3}} = \frac{\pi}{2\sqrt{3}} = \frac{\pi}{\sqrt{12}}$. Equality requires $|AB| = |CB| = 2$ and $B = \pi/3$, i.e. an equilateral triangle of side 2. □

Theorem 1.20 (Thue; Fejes Tóth)**.** *Every packing P of unit disks in the plane satisfies $\delta^+(P) \leq \pi/\sqrt{12}$.*

Proof. Saturating P does not decrease δ^+ (§1.4), so we may assume P is saturated. For a finite union of Delaunay triangles, the density is

$$\sum_{T\in DT} \frac{\text{area}(T)}{\text{area}(DT)}\cdot\delta(T) \;\leq\; \frac{\pi}{\sqrt{12}},$$

a weighted average of the densities of the triangles, each at most $\pi/\sqrt{12}$ by Lemma 1.19. Hence any finite union of Delaunay triangles has density at most $\pi/\sqrt{12}$.

For a large disk $R\,\mathbb{D}^2$, most of its area is covered by the Delaunay triangles inside it, so the error in the limit is a boundary term.

[Added] Every Delaunay triangle has circumradius < 2, so diameter < 4. Let U_R be the union of the triangles meeting $R\,\mathbb{D}^2$. Then $R\,\mathbb{D}^2 \subset U_R \subset (R+4)\mathbb{D}^2$. The sectors in the triangles of U_R account for every disk centred in $R\,\mathbb{D}^2$. The disks meeting $R\,\mathbb{D}^2$ but centred outside it lie in the annulus $(R+1)\mathbb{D}^2 \setminus (R-1)\mathbb{D}^2$, of area $O(R)$. So

$$\text{vol}(P \cap R\,\mathbb{D}^2) \;\leq\; \frac{\pi}{\sqrt{12}}\,\text{vol}(U_R) + O(R) \;\leq\; \frac{\pi}{\sqrt{12}}\,\pi(R+4)^2 + O(R).$$

Dividing by πR^2 and letting $R \to \infty$ gives $\delta^+(P) \leq \pi/\sqrt{12}$.

□

p. 16 sketch: large concentric disks

Remark 1.21. Combined with Example 1.13, this shows that the hexagonal packing is the densest packing of congruent disks in the plane. The statement goes back to Thue (1892, 1910). The first proof generally accepted as complete is due to L. Fejes Tóth (1940); the argument above follows his Delaunay-triangle approach. For lattice packings the result is Lagrange's (Theorem 1.14).

Chapter 2

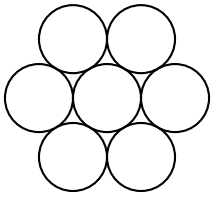
Lecture 2: The Fejes Tóth inequality and Blichfeldt's bound

2.1 Last time: Thue's theorem in the plane

Notes pp. 17–31 (MAT670_2).

Last time we gave a modern take on Thue's result in $\mathbb{R}^2$:

Theorem 2.1 (Thue). *The density of a packing of the plane by congruent circles is bounded by the density of the regular hexagonal packing, namely* $\pi/\sqrt{12} \approx 0.9069$.



The idea of the proof was to use *saturation* and the *Delaunay triangulation*.

Remark 2.2. In the proof we left out the error analysis of the limits, etc.

Remark 2.3. Rogers showed a density bound coming from the regular simplex. Precisely (C. A. Rogers, 1958): the density of any packing of unit balls in $\mathbb{R}^n$ is at most the fraction of a regular simplex of edge length 2 that is covered by the $n+1$ unit balls centred at its vertices. For $n = 2$ this is exactly $\pi/\sqrt{12}$, so Rogers' bound generalizes Thue's theorem. See Section 2.7 for a sketch.

2.2 The Fejes Tóth inequality for $\mathbb{S}^2$

All distances on $\mathbb{S}^2$ are spherical (angular) distances.

Theorem 2.4 (Fejes Tóth, 1943). *For any $n \geq 3$ points on $\mathbb{S}^2$ there exists a pair with spherical distance*

$$d \;\leq\; \arccos\!\left(\frac{\cot^2(\omega) - 1}{2}\right), \qquad \omega = \frac{n}{n-2} \cdot \frac{\pi}{6}. \tag{$*$}$$

Fix: Notes have $\cot(\omega)$; must be $\cot^2(\omega)$.

[Added] Check: for $n = 12$, $\omega = \pi/5$, and $(\cot^2(\pi/5) - 1)/2 = 1/\sqrt{5}$, so $d \leq \arccos(1/\sqrt{5}) \approx 63.43°$, the edge of the regular icosahedron. Similarly $n = 3, 4, 6$ give $120°$, $\arccos(-1/3) \approx 109.47°$ and $90°$ (equilateral triangle on a great circle, regular tetrahedron, regular octahedron). These are exactly the cases in which the bound is attained.

Remark 2.5. The inequality can be rewritten as a density result for spherical caps of diameter d (i.e. angular radius $d/2$): a cap of angular radius r has area $2\pi(1-\cos r)$, so the density of n non-overlapping caps of diameter d is

$$\frac{n\cdot 2\pi\big(1-\cos(d/2)\big)}{4\pi} \le \frac{n}{2}\Big(1-\cos\tfrac{d_n}{2}\Big) \longrightarrow \frac{\pi}{\sqrt{12}} \quad (n\to\infty),$$

where d_n is the right-hand side of $(*)$. So in the limit we recover Thue's bound.

Exercise 2.6. Verify the limit in Remark 2.5. (Hint: as $n\to\infty$, $d_n\to 0$ and $\frac{\sqrt{3}}{4}d_n^2 \sim \frac{4\pi}{2n}$; see also Exercise 2.9.)

The "flat" values also converge, but it is not so nice to take the limit.

? "Flat values": presumably the planar triangles of the convex hull instead of spherical ones.

A natural question: is there a good *lower* bound, i.e. can we construct a sequence of sets of points that achieves this bound?

[Added] Equality in $(*)$ holds only for $n=3,4,6,12$. Asymptotically the bound is sharp: there are configurations (e.g. suitable subdivisions of the icosahedron) whose cap density tends to $\pi/\sqrt{12}$.

2.2.1 A lemma on spherical triangles

To prove the Fejes Tóth theorem we need the following lemma.

Lemma 2.7. *Let ABC be a spherical triangle and let $\overline{AB}$ be its shortest side. Let ABC' be the equilateral spherical triangle drawn on $\overline{AB}$, with C' on the same side of the great circle AB as C. If*

$$\mathrm{Area}(ABC) < \mathrm{Area}(ABC'),$$

then the spherical radius of the circumcircle of ABC (the circle in which the plane through A,B,C meets $\mathbb{S}^2$) is greater than the length of $\overline{AB}$.

Proof. Write a for the length of $\overline{AB}$.

Step 1: draw the picture. Let C' be such that ABC' is equilateral, with C' on the same side of the great circle AB as C. Let α,β,γ be the circles of radius a centred at A,B,C' respectively; thus α passes through B and C', β through A and C', and γ through A and B. Let A' be the second intersection point of α and γ (so that $AC'A'$ is equilateral), and B' the second intersection point of β and γ. Let $-A,-B$ denote the antipodes of A,B.

Hand sketch p. 19 has the great circle AB as a straight line; redrawn exactly here.

Step 2: show the picture is "correct". Note that the spherical triangles ABA', ABB' and ABC' all have equal area. (For ABA': it is isosceles with legs a and apex angle 2θ, where θ is the angle of ABC'; a direct computation with the formula $\tan\frac{E}{2} = \frac{\sin\varphi\tan\frac{b}{2}\tan\frac{c}{2}}{1+\cos\varphi\tan\frac{b}{2}\tan\frac{c}{2}}$ for the area E of a triangle with sides b,c and included angle φ, together with $\tan^2\frac{a}{2} = 1-2\cos\theta$, shows that both areas agree.) Hence the circle in which the plane through $\{A',B',C'\}$ meets the sphere is the locus of points X with $\mathrm{Area}(ABX) = \mathrm{Area}(ABC')$ (on that side of AB): this is the *Lexell circle theorem.* (Lexell: for fixed A,B, the locus of apexes X of triangles ABX of fixed area is an arc of a small circle through $-A$ and $-B$.)

By assumption, $\mathrm{Area}(ABC) < \mathrm{Area}(ABC')$, so C lies between the Lexell circle and the great circle AB. As $\overline{AB}$ is the shortest edge of ABC, C does not lie in the open discs bounded by α or β. It follows (see Figure 2.1) that C does not lie in the disc bounded by γ. (The part of the disc of γ on the C' side of AB and below the Lexell circle is covered by the discs of α and β.)

? This covering step is read off the picture in the notes.

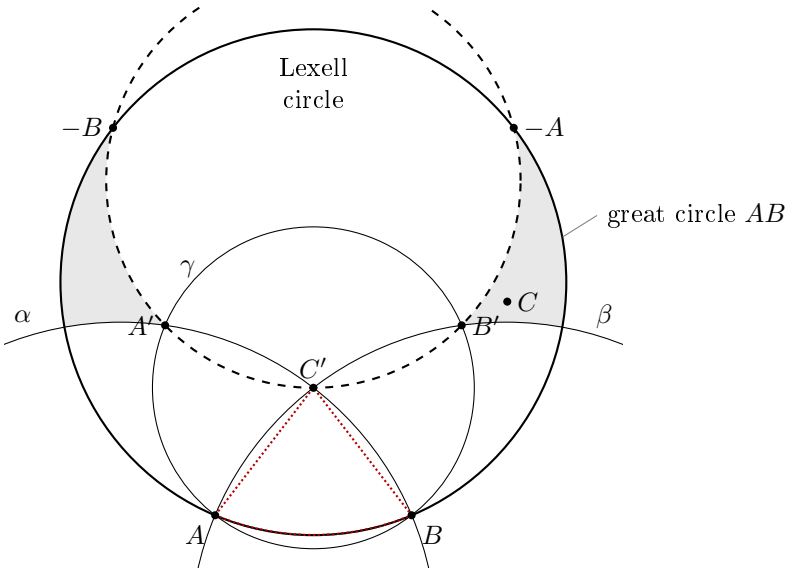


Figure 2.1: The configuration of Lemma 2.7 (side $a = 70°$), drawn in stereographic projection centred at C', so circles on $\mathbb{S}^2$ appear as circles. Shaded: the points between the great circle AB and the Lexell circle that lie outside α and β (the possible positions of C); they avoid the disc bounded by γ.

But C is on the C' side of AB. The circles through A and B form a pencil whose centres lie on the perpendicular bisector of AB; on the C' side, the region enclosed by such a circle grows with its radius. Since γ is the member of the pencil centred at C' and C lies outside it, the circumcircle of ABC has radius greater than the radius of γ, which is the length of $\overline{AB}$. □

2.2.2 Proof of the Fejes Tóth theorem

The case $n = 3$ is trivial: the three sides of a spherical triangle have total length at most 2π, so one of them is at most $2\pi/3 = \arccos(-1/2)$. So let $n \geq 4$ and let $p_1, \dots, p_n \in \mathbb{S}^2$.

Without loss of generality the convex hull $\operatorname{conv}\{p_1, \dots, p_n\}$ contains the origin 0 in its interior. Otherwise we could flow the points from the pole towards the equator of a hemisphere containing them and increase all distances.

[Added] If all points lie in an open hemisphere with pole N, replace the colatitude ϕ (angle from N) of each point by $\lambda\phi$, $\lambda > 1$ close to 1. From $\cos d = \cos\phi_1 \cos\phi_2 + \sin\phi_1 \sin\phi_2 \cos\Delta$ one checks $\frac{\partial}{\partial\lambda} \cos d \leq -(\phi_1 - \phi_2)\sin(\phi_1 - \phi_2) \leq 0$, with strict inequality for distinct points, so all mutual distances increase.

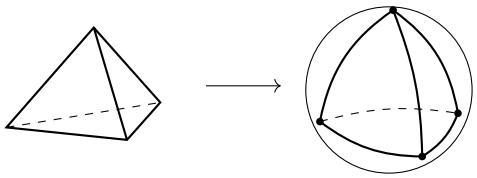
? Degenerate case (points in a closed, not open, hemisphere) needs a small perturbation argument.

Step 1: triangulate the convex hull. The convex hull is a polytope whose vertices are among the p_i; after a small perturbation (which changes distances only slightly) we may assume it is simplicial, with all n points as vertices. By the Euler characteristic, it has $2n - 4$ faces.

Exercise 2.8. Show that a triangulated convex polytope with n vertices has $2n - 4$ faces. *Solution sketch:* $v - e + f = 2$ and $3f = 2e$ for triangulations; hence $2n - 2e + 2f = 4$, so $2n - 3f + 2f = 4$, i.e. $f = 2n - 4$.

The triangulation of $\operatorname{conv}\{p_1, \dots, p_n\}$ gives a *spherical net* by radial projection to the sphere: the edges become geodesic arcs, and since 0 is in the interior of the hull the $2n - 4$ spherical

triangles tile $\mathbb{S}^2$.



Step 2. Now, for $\{p_1, \dots, p_n\}$, suppose that

$$\text{length}\big(\widehat{p_i p_j}\big) > d \qquad \text{for all } i \neq j,$$

where $d = d_n$ is the right-hand side of $(*)$.

Exercise 2.9 (L'Huilier)**.** Show that d in $(*)$ is the side length of an equilateral spherical triangle of area $\dfrac{4\pi}{2n-4}$. (One can use L'Huilier's formula, the spherical analogue of Heron's formula. Alternatively: the angle θ of such a triangle satisfies $3\theta - \pi = \frac{2\pi}{n-2}$, i.e. $\theta = 2\omega$, and the spherical law of cosines for angles gives $\cos d = \frac{\cos\theta}{1-\cos\theta} = \frac{\cot^2\omega - 1}{2}$.)

Then the triangle $p_i p_j p_k$ of minimal area in the spherical net satisfies the hypotheses of the Lemma: its area is small, but its edges are all longer than d. Here "small" means: less than or equal to the area of the equilateral triangle of side length d, since $2n-4$ triangles of total area 4π tile the sphere. As the area of an equilateral triangle increases with its side, and the shortest side of $p_i p_j p_k$ is longer than d, the hypothesis of Lemma 2.7 holds with strict inequality.

Hence the circumcircle of $p_i p_j p_k$ has radius larger than d. But the net construction implies that the circumcircle is empty: the face $p_i p_j p_k$ lies in a supporting plane of the convex hull, so no p_l lies in the open cap cut off by that plane (and this cap is smaller than a hemisphere, since 0 is on the other side).

So we may place a new point at the centre of this cap; it has distance greater than d from every p_l. This new collection of $n+1$ points satisfies the same inequalities for the *same* d! Moreover the new hull still contains 0 in its interior and has $2(n+1)-4$ faces, so its smallest triangle again has area less than that of the equilateral triangle of side d, and we can repeat indefinitely.

This is absurd, e.g. by a volume bound: $\text{vol}(\mathbb{S}^2) = 4\pi$, and the caps of radius $d/2$ around the points are pairwise disjoint, so we would have

$$\sum_{i=1}^{N} \text{area}\big(\text{cap}(d/2)\big) < 4\pi \qquad \text{for all } N > 0,$$

which is impossible since each cap has positive area. □

2.3 Higher-dimensional sphere packing: Blichfeldt's method

Consider an early method of Blichfeldt (1929) to get density bounds in higher dimensions. (H. F. Blichfeldt, *The minimum value of quadratic forms, and the closest packing of spheres*, Math. Ann. 101 (1929).)

Definition 2.10. A *convex body* C in $\mathbb{R}^n$ is a compact convex subset of $\mathbb{R}^n$ with non-empty interior.

Also consider collections of isometries $\{\varphi_i\}_{i=1}^{\infty}$ of $\mathbb{R}^n$ such that $\{\varphi_i C\}_{i=1}^{\infty}$ is a packing (i.e. the interiors of the $\varphi_i C$ are pairwise disjoint). Crossed out: "(so think $C = \mathbb{B}^n$)".

For now, $C = B^n$ is the closed unit ball, and a packing is of the form $\{\varphi_i B^n\}_{i=1}^{\infty}$.

Consider replacing the ball with its characteristic function

$$\chi(x) = \begin{cases} 1, & |x| \le 1, \\ 0, & |x| > 1. \end{cases}$$

So it has mass

$$I(\chi) = \int_{\mathbb{R}^n} \chi(x)\, dx = \mathrm{vol}(B^n),$$

and a packing corresponds to the function

$$x \longmapsto \sum_{i=1}^{\infty} \chi(\varphi_i^{-1} x)$$

(the characteristic function of $\bigcup_i \varphi_i B^n$, up to the measure-zero set of boundary points). For a general convex body C this works as well.

Furthermore, we can now replace χ with some other function f, and in general associate to a packing the function

$$x \longmapsto \sum_{i=1}^{\infty} f(\varphi_i^{-1} x).$$

A density may be defined similarly to the packing density, where each object now has mass

$$I(f) = \int_{\mathbb{R}^n} f(x)\, dx$$

spread over $\mathrm{supp}(f)$. So:

$$\Delta_f \;=\; \delta_{\text{packing}} \cdot \frac{I(f)}{\mathrm{vol}(C)}.$$

Here δ_{packing} is the density of the packing $\{\varphi_i C\}$ (Lecture 1), i.e. the average number of copies per unit volume times $\mathrm{vol}(C)$; Δ_f is the average of $\sum_i f(\varphi_i^{-1} x)$ over $x \in \mathbb{R}^n$.

We can think of this as spreading out and renormalizing the characteristic function in some (nice, hopefully) way: "cut up the sphere and spread it out."

Insight. There are functions f for which the sums $\sum_{i=1}^{\infty} f(\varphi_i^{-1} x)$ are uniformly pointwise bounded over $\mathbb{R}^n$ *and* over all collections of isometries $\{\varphi_i\}$ such that $\{\varphi_i C\}$ is a packing!

2.4 Blichfeldt gauges

Definition 2.11. Given a convex body C in $\mathbb{R}^n$, a non-negative, integrable function f on $\mathbb{R}^n$ is a *Blichfeldt gauge* for C if for any collection $\{\varphi_i\}_{i=1}^{\infty}$ of isometries of $\mathbb{R}^n$ such that $\{\varphi_i C\}_{i=1}^{\infty}$ is a packing,

$$\sum_{i=1}^{\infty} f(\varphi_i^{-1} x) \le 1 \qquad \text{for all } x \in \mathbb{R}^n.$$

Fix: Notes: "isometries of $\mathbb{R}^{d}$"; should be $\mathbb{R}^n$.
Margin: "add non-negative condition".

Lemma 2.12. *If f is a Blichfeldt gauge for C, then the density of any packing of $\mathbb{R}^n$ by copies of C satisfies*

$$\delta(C) \;\le\; \frac{\operatorname{vol}(C)}{I(f)}. \tag{$**$}$$

Proof sketch. By the definition, $\Delta_f \le 1$ (the average of a function bounded by 1 is at most 1). Since $\Delta_f = \delta \cdot I(f)/\operatorname{vol}(C)$, this gives $(**)$. The limiting argument that makes this precise is carried out for balls in Section 2.6; the same argument works for any C and any gauge with bounded support. □

Fix: Notes have strict $<$; equality can occur (e.g. $f = \chi_{\operatorname{int} C}$, C a cube).

2.5 Back to $C = B^n$

We can consider radially symmetric functions by an averaging argument: there is no benefit to anisotropy, since $\mathrm{SO}(n)$ kills it.
[Added] If f is a gauge for B^n, so is $\bar f(x) = \int_{\mathrm{SO}(n)} f(gx)\,dg$ (Haar probability measure): for each g, $\{\varphi_i g^{-1} B^n\} = \{\varphi_i B^n\}$ is the same packing, so $\sum_i f(g\varphi_i^{-1}x) \le 1$, and averaging over g gives $\sum_i \bar f(\varphi_i^{-1}x) \le 1$. Moreover $I(\bar f) = I(f)$, so $\bar f$ gives the same bound.

So it suffices to consider functions of the distance $|x|$! Rankin does a complicated study to get a good function (R. A. Rankin, *On the closest packing of spheres in n dimensions*, Ann. of Math. 48 (1947)). We consider

$$f_0(x) = \begin{cases} 1 - \tfrac{1}{2}|x|^2, & |x| \le \sqrt{2}, \\ 0, & |x| > \sqrt{2}. \end{cases}$$

We can show the following for a packing of unit radius balls.

Lemma 2.13. *f_0 is a Blichfeldt gauge for B^n.*

Proof. Consider m unit balls in a packing with centres $x_1, \dots, x_m \in \mathbb{R}^n$. Then clearly, for all pairs,

$$|x_i - x_j|^2 \ge 4. \tag{$\dagger$}$$

Summing $(\dagger)$ over $1 \le i < j \le m$ (the cross terms cancel, via the identity $\sum_{i<j} |x_i - x_j|^2 = m\sum_i |x_i|^2 - \left|\sum_i x_i\right|^2$), we get

$$m\sum_{i=1}^{m} |x_i|^2 - \Big|\sum_{i=1}^{m} x_i\Big|^2 \;\ge\; 2m(m-1),$$

and therefore

$$\sum_{i=1}^{m} |x_i|^2 \;\ge\; 2(m-1).$$

The origin here is an arbitrary point. So, given $x \in \mathbb{R}^n$, consider the collection of the m balls whose centres lie within $\sqrt{2}$ of x (the only ones contributing to $\sum_i f_0(x - x_i)$), with masses defined by f_0 centred at the x_i. Writing $r_i = |x - x_i|$, we see that

Notes write the centres in coordinates $(a_i, b_i, \dots, n_i)$.

$$\sum_{i=1}^{m} \frac{2 - r_i^2}{2} \;\le\; \frac{2m - 2(m-1)}{2} \;=\; 1,$$

and so f_0 is a Blichfeldt gauge. □

2.6 Blichfeldt's bound

So we can integrate this, using spherical shells:

$$I(f_0) = \int_0^{\sqrt{2}} \frac{2-r^2}{2}\, d\big(\mathrm{vol}(B^n)\, r^n\big) = \frac{2^{\frac{n}{2}+1}}{n+2}\,\mathrm{vol}(B^n).$$

Fix: Notes first had $\frac{4\cdot 2^{n/2}}{n+2}$; corrected in red to $\frac{2^{(n+2)/2}}{n+2}$.

Exercise 2.14. Verify this integral using spherical shells. ($I(f_0) = n\,\mathrm{vol}(B^n)\int_0^{\sqrt{2}}(1-\frac{r^2}{2})r^{n-1}dr = \mathrm{vol}(B^n)\,2^{n/2}\big(1-\frac{n}{n+2}\big)$.)

Limit argument for a box

Notes write $|I|$ for K.

Take a large box of side length t with K balls in it. Their centres lie in the concentric box of side $t-2$, so the supports of the gauges $f_0(x-x_i)$ lie in the concentric box of side $t+2\sqrt{2}-2$. Since $f_0 \geq 0$ and $\sum_i f_0(x-x_i) \leq 1$ everywhere, integrating over this larger box gives

$$(t+2\sqrt{2}-2)^n \;\geq\; K\int_{\mathbb{R}^n} f_0(x)\,dx \;=\; \frac{K\,\mathrm{vol}(B^n)\,2^{\frac{n}{2}+1}}{n+2}.$$

Hence the density of these K balls in the box $[0,t]^n$ satisfies

$$\frac{K\,\mathrm{vol}(B^n)}{t^n} \;\leq\; \frac{n+2}{2^{\frac{n}{2}+1}}\Big(1+\frac{2\sqrt{2}-2}{t}\Big)^n,$$

and letting $t\to\infty$:

Theorem 2.15 (Blichfeldt's bound)**.** *For any packing of $\mathbb{R}^n$ by unit balls,*

$$\delta \;\leq\; \frac{n+2}{2^{\frac{n}{2}+1}} \;=\; \frac{n+2}{2}\,2^{-n/2}.$$

Fix: Notes had $\frac{n+2}{4\cdot 2^{n/2}}$ (red correction in margin).

? Probably: balls meeting the boundary of the box; they contribute $O(t^{n-1})$, negligible as $t\to\infty$.

One can do this explicitly for balls too; we are also still missing a boundary term.

Remark 2.16. For $n=2$ the bound is the trivial $\delta\leq 1$; for $n=3$ it gives $5/(4\sqrt{2})\approx 0.884$, compared with $\pi/\sqrt{18}\approx 0.7405$ for the FCC packing (Rogers' simplex bound gives ≈ 0.7796). Blichfeldt's bound decays like $2^{-0.5n}$ up to polynomial factors; Rankin (1947) improved the constant, and Kabatiansky–Levenshtein (1978) improved the exponent to $2^{-0.599n}$.

2.7 Extension: Rogers' simplex bound

[Added] This section is an added extension, not in the handwritten notes. It is set in black for readability.

The remark in Section 2.1 mentioned Rogers' bound. Here is what it says and why it is true, at the level of a sketch. It is the natural sequel to Blichfeldt: instead of spreading the mass of each ball over a larger region, we cut space into pieces adapted to the packing and bound the density in each piece.

The tightest local cluster. The densest possible arrangement of $n+1$ unit balls is a regular simplex of edge 2 with a ball at each vertex: every pair touches. Let

$$\sigma_n = \frac{\operatorname{vol}\left(S \cap \bigcup_v B(v,1)\right)}{\operatorname{vol} S}$$

be the fraction of such a simplex S that is covered by its $n+1$ vertex balls. If ω_n is the fraction of the unit ball lying in the cone at a vertex of S (its normalized solid angle), then

$$\sigma_n = \frac{(n+1)\,\omega_n \operatorname{vol} B^n}{\operatorname{vol} S}, \qquad \operatorname{vol} S = \frac{2^{n/2}\sqrt{n+1}}{n!}.$$

Theorem 2.17 (Rogers 1958)**.** *The upper density of any packing of unit balls in $\mathbb{R}^n$ is at most σ_n.*

If regular simplices tiled space, Rogers' bound would be attained. They do for $n = 2$, where $\sigma_2 = \pi/\sqrt{12}$ and the theorem is Thue's. They do not for $n \geq 3$: already in $\mathbb{R}^3$ the dihedral angle $\arccos\frac{1}{3}$ does not divide 2π, and twenty regular tetrahedra do not make an icosahedron (Interlude E). So for $n \geq 3$ the bound is not attained (and the remark below records better bounds).

n	Blichfeldt $\frac{n+2}{2^{n/2+1}}$	Rogers σ_n	best known packing	ratio best/σ_n
2	1.0000	0.9069	0.9069 (A_2)	1.000
3	0.8839	0.7796	0.7405 (A_3)	0.950
4	0.7500	0.6478	0.6169 (D_4)	0.952
5	0.6187	0.5257	0.4653 (D_5)	0.885
6	0.5000	0.4192	0.3729 (E_6)	0.890
7	0.3977	0.3299	0.2953 (E_7)	0.895
8	0.3125	0.2568	0.2537 (E_8)	0.988

(The values of σ_n were computed from the formula above, with ω_n evaluated as a Gaussian orthant probability: ω_n is the probability that $n+1$ independent standard normals $Y_0, \dots, Y_n$ satisfy $Y_i > \bar{Y}$ for $i = 1, \dots, n$.) Note how close the root lattices come, especially A_3, D_4 and E_8; the end of Lecture 6 takes up why.

Sketch of the proof.

(1) *Reduce to Voronoi cells.* As in Lecture 1, we may assume the packing is saturated, and the density is (in the limit) an average of $\operatorname{vol} B / \operatorname{vol} V(x)$ over the Voronoi cells $V(x)$. So it is enough to cut each cell into pieces, each of which is covered by the ball $B(x,1)$ in a fraction at most σ_n.

(2) *Cut each cell into orthoschemes.* Pick a flag of faces of $V(x)$, $F_{n-1} \supset F_{n-2} \supset \cdots \supset F_0$, with $\dim F_k = k$. Let p_k be the foot of the perpendicular from x to the affine hull of F_k. The simplex $\operatorname{conv}(x, p_{n-1}, \dots, p_0)$ is an *orthoscheme*: the steps $x \to p_{n-1} \to p_{n-2} \to \cdots \to p_0$ are mutually orthogonal. Over all flags these simplices tile $V(x)$, at least when each foot p_k lies in its face. (In general some pieces must be counted with signs, and handling this is part of Rogers' work.)

(3) *Each step is long.* A k-face F_k consists of points equidistant from x and at least $n-k$ further centres $y_1, \dots, y_{n-k}$, which can be chosen with $y_i - x$ linearly independent so that $\operatorname{aff} F_k$ is exactly the set of points equidistant from $x, y_1, \dots, y_{n-k}$. The foot p_k is the circumcentre of the $(n-k)$-simplex $x, y_1, \dots, y_{n-k}$, so $|x - p_k|$ is its circumradius. All its edges are at least 2, and this forces the circumradius to be large. If $j+1$ points $v_0, \dots, v_j$ lie on a sphere of radius r about the origin, then $\sum_{a<b} |v_a - v_b|^2 = (j+1)\sum_a |v_a|^2 - |\sum_a v_a|^2 \le (j+1)^2 r^2$. There are $j(j+1)/2$ pairs, so some distance satisfies $|v_a - v_b|^2 \le 2(j+1)r^2/j$. Since every distance is at least 2,

$$|x - p_k| \;\ge\; R_{n-k}, \qquad R_j = \sqrt{\frac{2j}{j+1}},$$

with equality for the regular simplex. Here R_j is the circumradius of the regular j-simplex of edge 2. In particular $R_1 = 1$, since a facet is at distance at least 1, half the minimum distance 2.

(4) *Monotonicity (the heart of the matter).* Among all orthoschemes with $|x - p_k| \ge R_{n-k}$ for every k, the fraction covered by $B(x,1)$ is largest when all of these are equalities. That extremal orthoscheme is one of the $(n+1)!$ congruent pieces of the barycentric subdivision of the regular simplex of edge 2, so its covered fraction is exactly σ_n. Pushing a vertex of the orthoscheme outward adds volume that is farther from x, where the ball is thinner. Making this precise is Rogers' lemma, the technical core of his paper.

For $n = 2$ the picture is Thue's proof from Lecture 1, cut more finely. The faces are an edge at distance ≥ 1 and a vertex at distance $\ge 2/\sqrt{3}$, the extremal orthoscheme is the 30–60–90 triangle that is one sixth of the equilateral triangle, and its covered fraction is $\pi/\sqrt{12}$.

Remark 2.18.

1. Asymptotically $\sigma_n \sim \frac{n}{e}\,2^{-n/2}$ (Rogers; Daniels). This beats Blichfeldt's $\frac{n+2}{2}2^{-n/2}$ only by the constant factor $2/e$. The Kabatiansky–Levenshtein bound of Lecture 3 is exponentially better in high dimensions.
2. The Fejes Tóth inequality of Section 2.2 is the same idea on the sphere $\mathbb{S}^2$. Böröczky (Acta Math. Acad. Sci. Hungar. 32 (1978)) extended the simplex bound to spheres and hyperbolic spaces of every dimension.
3. The Cohn–Elkies linear programming bound (Interlude B) was, when it appeared (2003), the best known bound in dimensions 4 through 36, so in particular it improves on Rogers there. In dimensions 8 and 24 it is sharp (Viazovska; Cohn–Kumar–Miller–Radchenko–Viazovska).
4. References: C. A. Rogers, *The packing of equal spheres*, Proc. London Math. Soc. (3) 8 (1958) 609–620; Conway–Sloane, *Sphere Packings, Lattices and Groups*, Ch. 1.

Chapter 3

Lecture 3: Blichfeldt gauges and the Kabatjanskii–Levenshtein bound

Lecture of 18 April 2016. Last time we proved the Fejes Tóth inequality for point configurations on the sphere and Blichfeldt's sphere packing bound. Today we make some remarks on both, look at the landscape of known bounds for the packing density, prove the Kabatjanskii–Levenshtein-type bound via a Blichfeldt gauge, and then extend the notion of a Blichfeldt gauge from balls to general convex bodies.

Notes pp. 32–48.

Throughout, "a packing of unit balls" means a family of balls of radius 1 in $\mathbb{R}^n$ with pairwise disjoint interiors, i.e. with centers at mutual distance at least 2.

3.1 Some remarks

Where is the Fejes Tóth inequality sharp?

Recall the Fejes Tóth inequality: if $n \geq 3$ points are placed on the unit sphere $\mathbb{S}^2 \subset \mathbb{R}^3$, then the minimal angular distance d between two of them satisfies

$$d \;\leq\; \arccos\!\left(\frac{\cot^2\omega - 1}{2}\right), \qquad \omega = \frac{n}{n-2}\cdot\frac{\pi}{6}. \tag{3.1}$$

The right-hand side is the angular edge length of the equilateral spherical triangle with area $\frac{4\pi}{2n-4}$ on the unit sphere. (A triangulation of the sphere with n vertices has $2n-4$ triangular faces, so this is the area of a face when the 4π of total area is split evenly.)

Fix: Notes have $\cot(\omega)$; the square is needed (check $n = 12$).

So the bound should be sharp exactly when the points are the vertices of an equilateral deltahedron (a polytope all of whose faces are congruent equilateral triangles) inscribed in the sphere, i.e. for

$$n = 3,\ 4,\ 6,\ 12,$$

and perhaps asymptotically as $n \to \infty$.

[Added] Check: $n = 3$ gives $\omega = \pi/2$, $d \leq \arccos(-\frac{1}{2}) = 120^\circ$ (an equilateral triangle on a great circle, the degenerate case); $n = 4$ gives $\cot^2\omega = \frac{1}{3}$, $d \leq \arccos(-\frac{1}{3}) \approx 109.47^\circ$ (regular tetrahedron); $n = 6$ gives $\omega = \pi/4$, $d \leq 90^\circ$ (regular octahedron); $n = 12$ gives $\omega = \pi/5$, $d \leq \arccos(1/\sqrt{5}) \approx 63.43^\circ$ (regular icosahedron). Equality requires a triangulation by congruent equilateral triangles, so these are the only cases of equality; as $n \to \infty$ the bound is asymptotically sharp.

$n = 3$ underlined in the notes.

There are other proofs of the Fejes Tóth inequality, but they are longer.

3.2 Aside: Blichfeldt's gauge revisited

Numbered "3.2 aside". Crossed out above: "Again, there exists e.g. a trivial Blichfeldt gauge for C, given by χ_C."

We saw last time that

$$f_0(x) = \begin{cases} 1 - \frac{1}{2}|x|^2, & |x| \le \sqrt{2}, \\ 0, & |x| > \sqrt{2}, \end{cases} \tag{3.2}$$

is a Blichfeldt gauge for B^n: for every packing of unit balls with centers $c_1, c_2, \dots$, $\sum_i f_0(x - c_i) \le 1$ for all $x \in \mathbb{R}^n$, and consequently the packing density satisfies $\delta \le \mathrm{vol}(B^n) / \int_{\mathbb{R}^n} f_0$. It is non-trivial for our purposes, that is,

$$\int_{\mathbb{R}^n} f_0(x)\, dV \;>\; \int_{\mathbb{R}^n} \chi_{B^n}\, dV = \mathrm{vol}(B^n) \qquad \text{(at least for } n > 2\text{)}.$$

Indeed,

$$\int_{\mathbb{R}^n} f_0(x)\, dV = \frac{2^{1+\frac{n}{2}}}{n+2}\, \mathrm{vol}(B^n). \tag{3.3}$$

[Added] Proof: in polar coordinates, $dV = n\,\mathrm{vol}(B^n)\, t^{n-1}\, dt\, d\sigma$ (with σ the normalized surface measure), so

$$\int f_0 = n\,\mathrm{vol}(B^n) \int_0^{\sqrt{2}} \Big(1 - \frac{t^2}{2}\Big) t^{n-1}\, dt = \mathrm{vol}(B^n)\Big(2^{n/2} - \frac{n}{2(n+2)} 2^{(n+2)/2}\Big) = \frac{2 \cdot 2^{n/2}}{n+2}\, \mathrm{vol}(B^n).$$

Hence $\int f_0 > \mathrm{vol}(B^n)$ iff $2^{1+n/2} > n+2$, which holds exactly for $n \ge 3$; for $n = 2$ there is equality and for $n = 1$ the gauge is worse than χ_{B^1}.

From the gauge to the density bound

Consider a packing P of unit balls and a large cube $\mathrm{Box}(t)$ of side length t.
[Added] Split the balls meeting $\mathrm{Box}(t)$ into the *inner* ones, whose centers lie in the concentric cube of side $t - 2\sqrt{2}$, and the *boundary* ones. The support of $f_0(\cdot - c_i)$ for an inner center lies in $\mathrm{Box}(t)$, so integrating $\sum_i f_0(x - c_i) \le 1$ over $\mathrm{Box}(t)$ gives $N_{\text{in}} \int f_0 \le t^n$. The boundary centers lie in a shell of bounded width around $\partial\,\mathrm{Box}(t)$, so there are at most $C\, t^{n-1}$ of them (by a volume argument, C depending only on n).

Therefore, for all t,

$$\frac{\mathrm{vol}\big(P \cap \mathrm{Box}(t)\big)}{\mathrm{vol}\big(\mathrm{Box}(t)\big)} \;\le\; \frac{n+2}{2^{1+\frac{n}{2}}} \;+\; \frac{C\, t^{n-1}}{t^n},$$

and letting $t \to \infty$,

$$\delta(B^n) \;\le\; \limsup_{t\to\infty} \frac{\mathrm{vol}(P \cap \mathrm{Box}(t))}{\mathrm{vol}(\mathrm{Box}(t))} \;\le\; \frac{n+2}{2^{1+\frac{n}{2}}}. \tag{3.4}$$

Fix: Notes: main term times $(1 + \frac{\sqrt{2}-2}{t})^n$ (Lecture 2: $2\sqrt{2} - 2$); not needed here, tends to 1.

Remark 3.1. We could equally well use

$$\delta = \limsup_{t\to\infty} \frac{\mathrm{vol} \bigcup \{\, B \in P :\; B \subseteq \mathrm{Box}(t) \,\}}{\mathrm{vol}\big(\mathrm{Box}(t)\big)},$$

counting only the balls entirely inside the box; since the balls meeting $\partial\,\mathrm{Box}(t)$ have total volume $O(t^{n-1})$, this gives the same limit.

3.3 What does this look like?

Section number overwritten: "3.3" / "4".

The bound (3.4) coming from f_0 gives the following density bounds (two-digit values as in the notes):

n	1	2	3	4	5	6	7	8	9	10
$\frac{n+2}{2^{1+n/2}}$	1.06	1	.88	.75	.61	.5	.39	.31	.24	.19

For comparison, the true optimal densities are 1, $\pi/\sqrt{12} \approx .9069$, $\pi/\sqrt{18} \approx .7405$ in dimensions $1, 2, 3$, and $\pi^4/384 \approx .2537$ in dimension 8.

Asymptotic bounds

Lower bounds	**Upper bounds**
$\frac{1}{2^n}$ (saturation)	$\frac{n+2}{2^{\frac{n}{2}+1}}$ (Blichfeldt)
$\vdots$	$\vdots$
Ball (1992): $\frac{2n}{2^n}$	$2^{-0.599n}$ (Kabatjanskii–Levenshtein)
Vance, $n \equiv 0 \bmod 4$: $\frac{6}{e} \cdot \frac{n}{2^n}$	(Cohn, Elkies, Kumar, ...)
Venkatesh (sparse) (2013): $\frac{1}{2} \cdot \frac{n \log\log n}{2^n}$ (!!!)	

[Added] Comments. (1) *Saturation:* if a packing of unit balls is saturated (no further ball can be added), then the balls of radius 2 with the same centers cover $\mathbb{R}^n$, so $2^n\delta \geq 1$; saturated packings exist (e.g. by Zorn's lemma or a greedy construction), hence $\delta(B^n) \geq 2^{-n}$. (2) Ball (1992) proved $\delta \geq 2(n-1)\zeta(n)2^{-n}$ for lattice packings; Vance (2011) proved $\delta \geq \frac{6n}{e\,2^n}$ for lattice packings when $4 \mid n$; Venkatesh (*A note on sphere packings in high dimension*, IMRN 2013) proved $\delta \geq 65963\, n\, 2^{-n}$ for all large n and $\delta \geq \frac{1}{2}\, n \log\log n\, 2^{-n}$ for infinitely many n (the "!!!": the first improvement over linear growth of the numerator). The $\log\log n$ comes from cyclotomic lattices: by Mertens' theorem $m/\varphi(m)$ can be as large as about $e^\gamma \log\log m$. (3) *Update after 2016:* Campos, Jenssen, Michelen and Sahasrabudhe (2023) proved $\delta \geq (\frac{1}{2} - o(1))\, n \log n\, 2^{-n}$ for all n, and Klartag (2025) proved $\delta \geq c n^2 2^{-n}$ (even for lattice packings). On the upper side, the exponent 0.599 of Kabatjanskii–Levenshtein (1978) has only been improved by subexponential factors (e.g. Cohn–Zhao 2014).

Fix: Notes: $e^{-\gamma}/2$; Venkatesh proves $\Delta_n \geq n \log\log n/2^{n+1}$, i.e. 1/2.

Known good constructions

There are known good constructions in low dimensions and special ones (e.g. D_4, E_8, the Leech lattice Λ_{24}). We need to talk about these at some point.

Margin doodle (sad face) omitted.

Exact results (upper bounds that are attained)

- $d = 1$: trivial.

? Notes: "(1892...???) Thue".

- $d = 2$: we proved it. Thue (1892, 1910); the first complete proof is due to L. Fejes Tóth (1940). The optimal lattice packing was known to Lagrange and Gauss.
- $d = 3$: Hales, twice, around 2000; a complicated geometric analysis. Hales (with Ferguson), announced 1998, published in Ann. of Math. 2005; the formal computer verification (Flyspeck project) was completed in 2014.
- $d = 8$ and $d = 24$: one month ago, via linear programming duality: one constructs a special ("magic") auxiliary function. Viazovska (dimension 8, March 2016) and Cohn–Kumar–Miller–Radchenko–Viazovska (dimension 24, a week later); both published in Ann. of Math. 2017. The LP bound itself is due to Cohn–Elkies (2003).

Remark 3.2. Remember that the cubic lattice $\mathbb{Z}^n$ gives a packing of balls of radius $\frac{1}{2}$ with density

$$\delta(\mathbb{Z}^n) = \operatorname{vol}\left(\tfrac{1}{2}B^n\right) = \frac{\pi^{n/2}}{2^n\,(\frac{n}{2})!},$$

where $(\frac{n}{2})! := \Gamma(\frac{n}{2}+1)$. This decays superexponentially, much faster than the lower bound 2^{-n}.

3.4 Kabatjanskii and Levenshtein

Definition 3.3. For $\varphi \in (0,\pi]$, let $M(d,\varphi)$ be the maximum number of points on $\mathbb{S}^{d-1}$ with pairwise angular distance at least φ. Equivalently, $M(d,\varphi)$ is the maximum number of points in $\mathbb{R}^d \setminus \{0\}$ such that the angle at the origin spanned by any two of them is at least φ (project radially onto the sphere).

For example $M(d,\pi) = 2$, and $M(d,\pi/3) = \tau_d$ is the kissing number ($\tau_2 = 6$, $\tau_3 = 12$, $\tau_4 = 24$, $\tau_8 = 240$, $\tau_{24} = 196560$).

Notes: "in high dimensions".

Theorem 3.4. *In any dimension d and for $1 < r < 2$, the function*

$$f_d(x) = \begin{cases} M\big(d, \arccos(1-\frac{2}{r^2})\big)^{-1}, & |x| < r, \\ 0, & |x| \geq r, \end{cases}$$

is a Blichfeldt gauge for B^d.

The key geometric fact is the following.

Lemma 3.5. *Let $r < 2$, let $x \in \mathbb{R}^d$ and let $y, z \in \mathbb{R}^d$ with $|y-z| \geq 2$ and $|x-y|, |x-z| \leq r$. Then the angle θ at x spanned by the segment $[y,z]$ satisfies $\cos\theta \leq 1 - \frac{2}{r^2}$, i.e. $\theta \geq \arccos(1-\frac{2}{r^2})$.*

Proof. First suppose both points are at distance exactly r from x (the worst case, see Figure 3.1). Put $a = (y-x)/r$, $b = (z-x)/r$, unit vectors with $a \cdot b = \cos\theta$, and let $\ell = |y-z| \geq 2$. Then

$$\left|\frac{2}{r}\right|^2 \leq \left|\frac{\ell}{r}\right|^2 = |a-b|^2 = 2 - 2\,a\cdot b = 2 - 2\cos\theta,$$

so $\frac{4}{r^2} \leq 2 - 2\cos\theta$, i.e. $\frac{2}{r^2} \leq 1 - \cos\theta$, which is

$$\boxed{\cos\theta \leq 1 - \frac{2}{r^2}.}$$

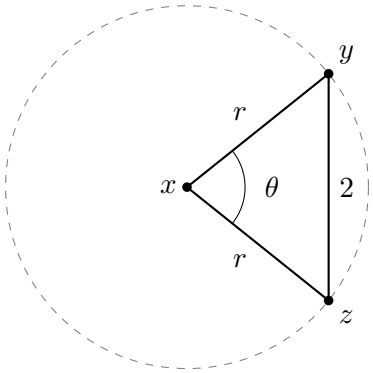


Figure 3.1: The extremal case of Lemma 3.5: both centers at distance exactly r from x and at distance 2 from each other; then $\sin(\theta/2) = 1/r$.

[Added] Why this is the worst case: in general, with $\alpha = |x-y|$, $\beta = |x-z| \in (0, r]$, the law of cosines gives $\cos\theta = \frac{\alpha^2+\beta^2-\ell^2}{2\alpha\beta} \le h(\alpha,\beta) := \frac{\alpha^2+\beta^2-4}{2\alpha\beta}$, and $\partial_\alpha h = \frac{\alpha^2-\beta^2+4}{2\alpha^2\beta} > 0$ because $\beta \le r < 2$. By symmetry h is increasing in each variable, so $h(\alpha,\beta) \le h(r,r) = 1-\frac{2}{r^2}$. Equivalently, the apex angle of the isosceles triangle with legs r and base 2 is $2\arcsin(1/r)$, and $\cos(2\arcsin\frac{1}{r}) = 1-\frac{2}{r^2}$. □

Fix: Side sketch says apex angle $2\arccos(1/r)$; it is $2\arcsin(1/r)$.

Proof of Theorem 3.4. Take any packing of unit balls in $\mathbb{R}^d$ and $r < 2$. For every $x \in \mathbb{R}^d$ there are at most $M\big(d, \arccos(1-2/r^2)\big)$ centers at distance less than r from x: by Lemma 3.5, the angle at x spanned by any segment of length ≥ 2 whose ends are within r of x is at least $\arccos(1-2/r^2)$, so the directions from x to these centers form a configuration as in Definition 3.3. (If x is itself a center, all other centers are at distance $\ge 2 > r$, and the count is 1.) Hence $\sum_i f_d(x-c_i) \le 1$. □

Corollary 3.6. *For $1 \le r < 2$,*

$$\delta(B^d) \le r^{-d}\, M\big(d, \arccos(1-\tfrac{2}{r^2})\big).$$

Equivalently, substituting $r = 1/\sin(\varphi/2)$,

$$\delta(B^d) \le \sin\big(\tfrac{\varphi}{2}\big)^d\, M(d,\varphi), \qquad \frac{\pi}{3} < \varphi \le \pi.$$

Proof. $\int f_d = r^d \operatorname{vol}(B^d)/M$, so $\delta \le \operatorname{vol}(B^d)/\int f_d = r^{-d}M$. For the substitution: $1-\frac{2}{r^2} = 1-2\sin^2\frac{\varphi}{2} = \cos\varphi$, and $r \in [1,2)$ corresponds to $\varphi \in (\pi/3, \pi]$. □

Letting $r \to 2$ recovers the simple bound $\delta(B^d) \le \tau_d\, 2^{-d}$.

For d large, take $\varphi \approx 63°$; the Kabatjanskii–Levenshtein bound on spherical codes gives

$$\sin\big(\tfrac{\varphi}{2}\big)^d M(d,\varphi) \le 2^{-0.599d+o(d)},$$

and hence

$$\delta(B^d) \le 2^{-0.599d+o(d)}.$$

[Added] More precisely, Kabatjanskii and Levenshtein (1978), using linear programming bounds for spherical codes, proved $M(d,\varphi) \le 2^{d(c(\varphi)+o(1))}$ for $0 < \varphi < \pi/2$, where with $s = \sin\varphi$,

$$c(\varphi) = \frac{1+s}{2s}\log_2\frac{1+s}{2s} - \frac{1-s}{2s}\log_2\frac{1-s}{2s}.$$

The exponent $c(\varphi) + \log_2\sin(\varphi/2)$ is minimized at $\varphi^* \approx 62.9974°$, where it equals $-0.5990\ldots$ (see Conway–Sloane, *Sphere Packings, Lattices and Groups*, Ch. 1 and 9).

3.5 Blichfeldt gauges for general bodies

Section number overwritten, presumably 3.5.

So we defined a Blichfeldt gauge for B^n. Can we extend this? Abstractly:

Definition 3.7. Given a (convex) body C in $\mathbb{R}^n$, a function $f : \mathbb{R}^n \to \mathbb{R}$ is a *Blichfeldt gauge for* C if for any collection of isometries $\{\varphi_i\}_{i=1}^{\infty}$ of $\mathbb{R}^n$ such that $\{\varphi_i C\}_{i=1}^{\infty}$ is a packing,

$$\sum_{i=1}^{\infty} f(\varphi_i^{-1} x) \le 1 \qquad \forall x \in \mathbb{R}^n.$$

As for balls, integrating over a large box shows that if f is integrable and compactly supported, then every packing of congruent copies of C has density at most $\mathrm{vol}(C)/I(f)$, where $I(f) := \int_{\mathbb{R}^n} f$.

Remark 3.8. One could add a non-negativity condition on f; what we really want is that the mass of f is large: $I(f) \ge I(\chi_C)$.

Remark 3.9. The index set could also be finite.

Remark 3.10. The function f could be much stranger than what we allow here.

? Rest of the remark illegible: "...accuracy first"?

Exercise 3.11 $(***)$**.** Think about such functions distributionally.

There always exists a Blichfeldt gauge for C, e.g. χ_C (the copies $\varphi_i C$ have disjoint interiors, so a point lies in the interior of at most one of them; here one should take χ of the interior, or note that boundaries have measure zero).

Lemma 3.12. *If $f(x)$ is a Blichfeldt gauge for C, then $f(x/\lambda)$ is a Blichfeldt gauge for λC, the λ-scaling of C.*

Indeed, a packing $\{\varphi_i(\lambda C)\}$ is the image under $x \mapsto \lambda x$ of the packing $\{\psi_i C\}$ with $\psi_i(y) = \lambda^{-1}\varphi_i(\lambda y)$. For instance, f_0 (supported on the ball of radius $\sqrt{2}$) is a gauge for B^n, and $f_0(x/\lambda)$ (supported on the ball of radius $\lambda\sqrt{2}$) is a gauge for λB^n.

Diagram of this scaling omitted.

Definition 3.13. For a body C define its *inradius* $r(C)$ to be the radius of the largest sphere (ball) contained in C. For $0 \le \gamma \le r(C)$ define C_γ, the *inner parallel body at depth* γ, to be the set of points

$$C_\gamma := \{x \in C : \ x + \gamma B^n \subseteq C\}.$$

If C is convex and compact, then C_γ is convex, compact and non-empty for $\gamma \le r(C)$. We can define the straight-line distance from a point x to a body C by $d(x, C) = \min_{y \in C} |x - y|$.

Theorem 3.14. *Let C be a convex body, $0 < \gamma \le r(C)$, and let f be a radial Blichfeldt gauge for B^n, i.e. $f(x) = F(|x|)$. Define*

$$g(x) = F\Big(\frac{d(x, C_\gamma)}{\gamma}\Big).$$

Then g is a Blichfeldt gauge for C.

This is fairly easy to see from a picture (Figure 3.2).

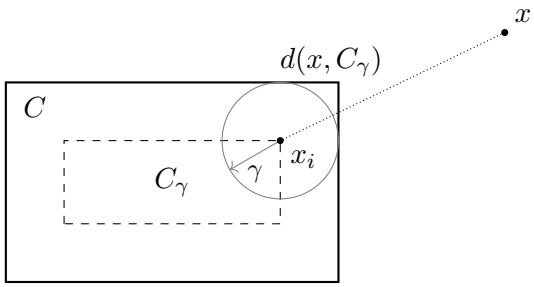


Figure 3.2: The inner parallel body C_γ and the nearest point $x_i \in C_\gamma$ to x; the ball $x_i + \gamma B^n$ lies in C.

Proof. We would like to show that

$$\sum_{i=1}^{\infty} g(\varphi_i^{-1}x) \le 1 \qquad \forall x \in \mathbb{R}^n, \text{ for all packings } \{\varphi_i C\}_{i=1}^{\infty} \text{ of } C.$$

Consider a point $x \in \mathbb{R}^n$ and, for each i, a point $x_i \in \varphi_i C_\gamma$ at distance $d(x, \varphi_i C_\gamma)$ from x (it exists since C_γ is compact). Then $x_i + \gamma B^n$ is contained in $\varphi_i C$ (note $\varphi_i C_\gamma = (\varphi_i C)_\gamma$), and so $\{x_i + \gamma B^n\}_i$ is a packing. Since $F(|y|/\gamma)$ is a Blichfeldt gauge for γB^n (Lemma 3.12), and x is at distance $|x - x_i| = d(x, \varphi_i C_\gamma)$ from x_i, we get

Parenthetical: "since C is convex, things are well defined".

$$\sum_i g(\varphi_i^{-1}x) = \sum_i F\Big(\frac{d(x, \varphi_i C_\gamma)}{\gamma}\Big) = \sum_i F\Big(\frac{|x - x_i|}{\gamma}\Big) \le 1,$$

using $d(\varphi_i^{-1}x, C_\gamma) = d(x, \varphi_i C_\gamma)$ because φ_i is an isometry. □

Is this useful?

As with Blichfeldt gauges and other approaches: is this useful? If we have a convex body like $A + B^n$ (a Minkowski sum with a ball), it could be, provided there is enough high-dimensional mass.

? "is this useful?" tentative; could be "is there a catch?"

A better gauge for balls. Consider

$$f(x) = \begin{cases} f_0(x), & |x| > 1, \\ 1 - f_0\big(2 - |x|\big), & |x| \le 1, \end{cases}$$

with f_0 from (3.2), viewed as a function of the radius. Explicitly, $f = 1$ for $|x| \le 2 - \sqrt{2}$, $f = \frac{1}{2}(2 - |x|)^2$ for $2 - \sqrt{2} \le |x| \le 1$, and $f = f_0$ beyond; f is continuous and $f \ge f_0$ everywhere (Figure 3.3). This leads to

$$\delta \le \Big(\frac{2}{d+2}\sqrt{2}^{\,d}\,(1 + b_d)\Big)^{-1}, \qquad b_d = \frac{1}{(\sqrt{2})^d(d+1)} - (\sqrt{2} - 1)^{d+1}\Big(1 + \frac{\sqrt{2}}{d+1}\Big).$$

[Added] The factor $\frac{2}{d+2}\sqrt{2}^{\,d} = \int f_0 / \operatorname{vol}(B^d)$ is Blichfeldt's, and $b_d = \int (f - f_0) / \int f_0$ (we checked this integral numerically against the formula for $d \le 10$). The notes do not prove that f is a

? Notes do not verify that f is a gauge; see the added paragraph below.

Blichfeldt gauge. The idea: a point x lies in at most one ball of the packing, say at distance $s = |x - c_0| \le 1$ from its center, and all other centers are at distance $\ge 2 - s$ from x; one has to show that their total f_0-contribution is at most $f_0(2-s)$, so that it is compensated by replacing $f_0(s)$ with $1 - f_0(2-s)$. Blichfeldt's inequality $\sum_i |x - c_i|^2 \ge 2(m-1)$ alone only gives the weaker bound $s^2/2$ for this contribution, so a finer argument is needed. (A numerical search in dimensions $2, 3, 4$ found no violation.)

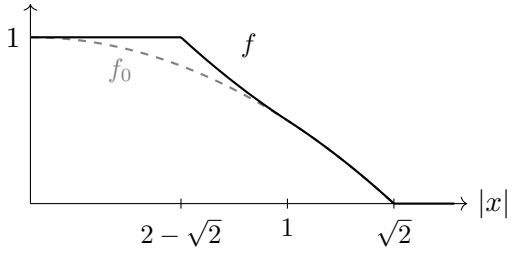


Figure 3.3: Radial profiles of Blichfeldt's gauge f_0 (dashed) and the improved function f (solid); they agree for $|x| \ge 1$.

The resulting bounds compared with those from f_0 (truncated to two digits):

d	1	2	3	4	5	6	7	8	9	10
bound from f	1	.94	.84	.72	.60	.49	.39	.31	.24	.18
bound from f_0	1.06	1	.88	.75	.61	.5	.39	.31	.24	.19

An arrow on p. 47 links .94 (f, $d = 2$) with 1 (f_0, $d = 2$).

Bodies with a lot of 2-dimensional mass. So this works for things with a lot of "2-dimensional" (or higher-dimensional) mass. For a long capsule in $\mathbb{R}^3$ (a segment plus a ball), the cross-sections are discs and Theorem 3.14 gives a bound of .94 + error. By contrast, for a long box in the plane the cross-sections are 1-dimensional and we only get 1+error: the method gives nothing, since there is no nontrivial 1-dimensional Blichfeldt gauge. (In dimension 1, intervals tile the line, so every gauge has $I(f) \le \mathrm{vol}(B^1)$.)

? Reading of the box sketch and of "hence doesn't work as there is no 1d Blichfeldt gauge" is tentative.

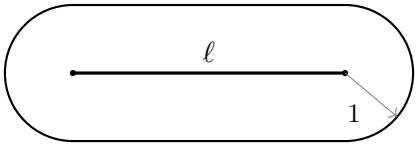


Figure 3.4: An ℓ-capsule: a segment of length ℓ plus a unit ball, drawn in a plane through the axis.

Example 3.15 (ℓ-capsules)**.** Let $C = S_\ell + B^d$, where S_ℓ is a segment of length ℓ (the notes write "$\ell \times B^n$"). Then $r(C) = 1$ and $C_1 = S_\ell$. Take a radial gauge $F(|x|)$ for B^d and $\gamma = 1$, so $g(x) = F(d(x, S_\ell))$. Write $I_k(F) = \int_{\mathbb{R}^k} F(|y|)\, dy$. Points whose nearest point on S_ℓ is interior contribute a cylinder over S_ℓ, and the two ends together contribute a full ball's worth, so

$$I(g) = \ell \cdot \frac{I_{d-1}(F)}{\mathrm{vol}(B^{d-1})}\,\mathrm{vol}(B^{d-1}) + \frac{I_d(F)}{\mathrm{vol}(B^d)}\,\mathrm{vol}(B^d) = \ell\, I_{d-1}(F) + I_d(F),$$

and

$$\mathrm{vol}(\ell\text{-capsule}) = \ell\,\mathrm{vol}(B^{d-1}) + \mathrm{vol}(B^d).$$

Hence

$$\delta(\ell\text{-capsule}) \le \frac{\ell\,\mathrm{vol}(B^{d-1}) + \mathrm{vol}(B^d)}{\ell\,I_{d-1}(F) + I_d(F)}.$$

As $\ell \to \infty$ this tends to $\mathrm{vol}(B^{d-1})/I_{d-1}(F)$, the $(d-1)$-dimensional ball bound for the same profile; for $d = 3$ and the improved f above this is the value .94 from the table (assuming f is a gauge in $\mathbb{R}^3$).

[Added] This extension of Blichfeldt's method to "longish" bodies, such as long cylinders $B^{n-1} \times [0,h]$ and sausage-like solids $B^n + [0,h]$, is due to G. Fejes Tóth and W. Kuperberg, *Blichfeldt's density bound revisited*, Math. Ann. 295 (1993), 721–727. For circular cylinders see also A. Bezdek and W. Kuperberg, Mathematika 37 (1990) (infinite cylinders: density $\pi/\sqrt{12}$), and W. Kusner, Discrete Comput. Geom. 51 (2014) (finite cylinders: $\pi/\sqrt{12} + 10/h$). Whether the improved radial gauge for the ball above is also taken from the 1993 paper (or from Blichfeldt 1929 or Rankin 1947) has not been checked against the original.

Remark 3.16. For a genuine cylinder $S_\ell \times B^{d-1}$ (flat ends) the same computation works up to end errors.

Part II

Lattices and root systems

Chapter 4

Lecture 4: The classification of (crystallographic) root systems

4.1 Motivation and lattice basics

Heading reads "Lecture 4) (and 5?)".

In this lecture we examine *root lattices*, coming from *crystallographic* root systems. One motivation is to find nice constructions of good packings in low dimensions. Another is to show that such simple constructions do not really extend to high dimensions in a good way.

crossed out: "there are not such simple constructions"

The best known packings up through $\mathbb{R}^8$ are root lattices. In fact A_1, A_2, A_3 and E_8 are optimal among *all* packings in their dimensions. (A_1: trivial; A_2: Thue, Fejes Tóth 1940; A_3, the FCC packing: Hales 1998/2005, the Kepler conjecture; E_8: Viazovska 2016. In dimension 24 the Leech lattice, which is not a root lattice, is optimal by Cohn–Kumar–Miller–Radchenko–Viazovska 2016.)

? Box reads "A_1, A_2, A_3, E_8 optimal as gen. packings"; "A_3" read from a blurred subscript.

Remark 4.1. The classification we are about to give covers, in effect, half of the classification of complex semisimple Lie algebras (equivalently, of complex semisimple Lie groups): the irreducible root systems correspond to the simple ones. (See e.g. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Ch. III, or Fulton–Harris, *Representation Theory*, Lecture 21.)

Definition 4.2. A *lattice* Λ is a discrete additive subgroup of $\mathbb{R}^n$. For our purposes it will always be co-compact, i.e. have full rank. Such a lattice has a $\mathbb{Z}$-basis $\{a_1, \dots, a_n\}$, $\Lambda = \mathbb{Z}a_1 \oplus \cdots \oplus \mathbb{Z}a_n$, that is at the same time an $\mathbb{R}$-basis of $\mathbb{R}^n$.

Messy boxed aside, partly scribbled over; a book reference is unreadable.

Example 4.3. $\{(1,0)^T, (0,1)^T\}$ is a basis of the lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$.

A lattice can have many different bases. However, two bases $\{a_i\}$, $\{a_i'\}$ differ by an *integer* change of basis $K = (k_{ij})$ with integral inverse:

$$a_i' = \sum_{j=1}^{n} k_{ij} a_j, \qquad k_{ij} \in \mathbb{Z} \quad \Longrightarrow \quad \det K = \pm 1$$

(since $\det K$ and $\det K^{-1}$ are both integers whose product is 1). So the volume of the torus $\mathbb{R}^n/\Lambda$,

$$\operatorname{vol}(\mathbb{R}^n/\Lambda) = |\det \Lambda| = \operatorname{vol}\left(\left\{ \textstyle\sum_i t_i a_i : 0 \le t_i \le 1 \right\}\right),$$

i.e. the volume of the parallelepiped spanned by $a_1, \dots, a_n$, where $\det \Lambda$ denotes the determinant of the matrix with columns $a_1, \dots, a_n$, is a clear invariant of Λ.

The largest balls that can be placed at the lattice points without overlapping have radius $\frac{1}{2}\min_{\lambda\in\Lambda\setminus\{0\}}|\lambda|$, so each lattice gives a packing

$$\Big\{\varphi_i + B\big(\tfrac{1}{2}\min_{\lambda\in\Lambda\setminus\{0\}}|\lambda|\big) \;:\; \varphi_i\in\Lambda\Big\}.$$

[Added] Its density is $\delta(\Lambda) = \operatorname{vol} B^n\big(\frac{1}{2}\lambda_1(\Lambda)\big)\big/\,|\det\Lambda|$, where $\lambda_1(\Lambda)=\min_{\lambda\neq 0}|\lambda|$.

Since scaling does not change the density, we can normalize to $|\det\Lambda|=1$ (i.e. consider lattices $g\mathbb{Z}^n$ with $g\in \mathrm{SL}_n(\mathbb{R})$). The space $\mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$ of such lattices is *not* compact (e.g. the lattices $\operatorname{diag}(t,t^{-1},1,\dots,1)\mathbb{Z}^n$, $t\to 0$, leave every compact set). What is true is Mahler's compactness criterion: for each $\varepsilon>0$ the set of unimodular lattices with $\lambda_1\geq\varepsilon$ is compact. Then there is clearly an upper bound on the length of the shortest vector among lattices with $|\det\Lambda|=1$, and so a bound on the density. (Trivially, the balls are disjoint, so $\operatorname{vol}B^n(\lambda_1/2)\leq 1$; Minkowski's convex body theorem gives the sharper $\operatorname{vol}B^n(\lambda_1)\leq 2^n$. Since lattices with very small λ_1 have small density, the supremum of the density can be taken over a set $\{\lambda_1\geq\varepsilon\}$, which is compact by Mahler; so an optimal lattice exists in each dimension.)

Fix: notes: "(to SL_n, compact)"; the space is not compact

So... what is the optimal lattice? The optimal lattice packings are known up to $n=8$ (Blichfeldt 1935 for $n=6,7,8$), and they are

$$A_1,\ A_2,\ A_3,\ D_4,\ D_5,\ E_6,\ E_7,\ E_8.$$

These are root lattices. (Earlier cases: Lagrange for $n=2$, Gauss for $n=3$, Korkine–Zolotarev for $n=4,5$. The only other dimension where the optimal lattice is known is $n=24$: the Leech lattice, Cohn–Kumar 2009.)

crossed out: "still open"

4.2 Good candidates in low dimensions

Definition 4.4. A lattice Λ is *integral* if $\langle x,y\rangle\in\mathbb{Z}$ for all $x,y\in\Lambda$. A *root lattice* is an integral lattice generated by roots, i.e. by elements of a root system (Definition 4.5).

[Added] In lattice language one usually calls a vector v of an integral lattice a root if $\langle v,v\rangle\in\{1,2\}$; then $2\langle x,v\rangle/\langle v,v\rangle\in\mathbb{Z}$ for all $x\in\Lambda$, so the reflection s_v (below) maps Λ to itself, and the roots of Λ form a root system. The irreducible lattices generated by vectors of norm 2 are exactly A_n $(n\geq 1)$, D_n $(n\geq 4)$, E_6, E_7, E_8 (Witt); see Conway–Sloane, *Sphere Packings, Lattices and Groups*, Ch. 4.

Definition 4.5. A *root system* is a subset $\Delta\subset\mathbb{R}^n\setminus\{0\}$ satisfying:

1) Δ is finite and spans $\mathbb{R}^n$;

2) if $\alpha\in\Delta$, then $n\alpha\in\Delta \iff n=\pm 1$ (*reduced*);

3) Δ is invariant under the orthogonal reflection in the hyperplane $\alpha^\perp$, for any $\alpha\in\Delta$: writing

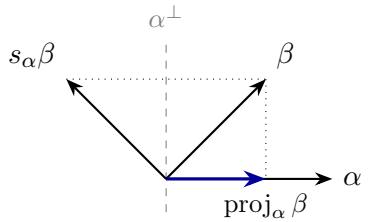
$$\operatorname{proj}_\alpha\beta = \alpha\,\frac{\langle\beta,\alpha\rangle}{\langle\alpha,\alpha\rangle},\qquad s_\alpha\beta := \beta - 2\operatorname{proj}_\alpha\beta,$$

we require $s_\alpha\beta\in\Delta$ for all $\alpha,\beta\in\Delta$;

4) (*crystallographic*) for all $\alpha, \beta \in \Delta$,

$$n_{\beta\alpha} := \frac{2\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}.$$

The elements of Δ are called *roots*.



Property 4) is the restriction that the projection of β onto α is an integer or half-integer multiple of α; but it really forces Δ to give rise to a lattice (the $\mathbb{Z}$-span of Δ is discrete: it is the $\mathbb{Z}$-span of a fundamental system, which is a basis of $\mathbb{R}^n$, see §4.3).

Exercise 4.6. Try building some other root systems, i.e. sets satisfying 1)–3), which are *not* crystallographic. (Hint: the 10 unit vectors at angles $k\pi/5$ in $\mathbb{R}^2$; the 30 vertices of an icosidodecahedron in $\mathbb{R}^3$, i.e. type H_3.)

Property 4) is strong. We have

$$n_{\beta\alpha} = \frac{2\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} = \frac{2\,|\beta|}{|\alpha|} \cos\theta,$$

where θ is the angle between α and β. Since both $n_{\beta\alpha}$ and $n_{\alpha\beta}$ are integers,

$$\boxed{n_{\beta\alpha}\, n_{\alpha\beta} = 4\cos^2\theta \in \mathbb{Z}} \qquad \Longrightarrow \qquad 4\cos^2\theta \in \{0, 1, 2, 3, 4\}.$$

If $4\cos^2\theta = 4$, then α and β are parallel, so $\alpha = \beta$ or $\alpha = -\beta$ by the reducedness axiom 2).

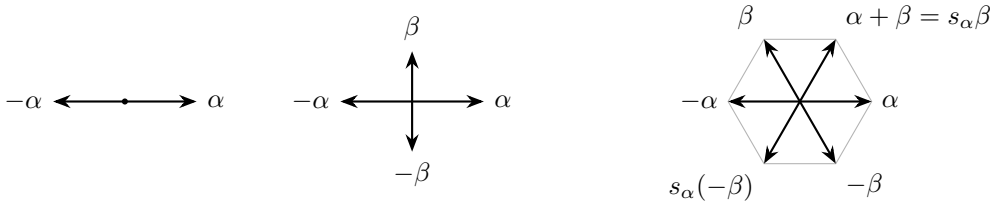
Assuming $|\alpha| \geq |\beta|$ (and $\alpha \neq \pm\beta$), this leaves the following table.

$4\cos^2\theta$	$n_{\beta\alpha}$	$n_{\alpha\beta}$	$\lvert\alpha\rvert \,/\, \lvert\beta\rvert$	$\cos\theta$	θ
3	+1	+3	$\sqrt{3}$	$+\sqrt{3}/2$	$\pi/6$
3	−1	−3	$\sqrt{3}$	$-\sqrt{3}/2$	$5\pi/6$
2	+1	+2	$\sqrt{2}$	$+\sqrt{2}/2$	$\pi/4$
2	−1	−2	$\sqrt{2}$	$-\sqrt{2}/2$	$3\pi/4$
1	+1	+1	1	$+1/2$	$\pi/3$
1	−1	−1	1	$-1/2$	$2\pi/3$
0	0	0	(undetermined)	0	$\pi/2$

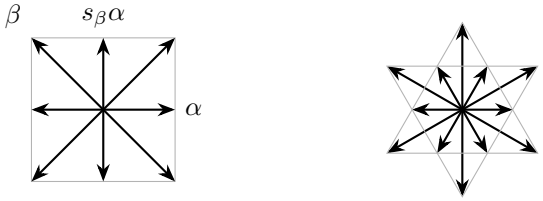
(For $4\cos^2\theta \neq 0$ the ratio follows from $n_{\alpha\beta}/n_{\beta\alpha} = |\alpha|^2 \,/\, |\beta|^2$.)

Definition 4.7. A root system is *reducible* (*decomposable*) if $\Delta = \Delta_1 \cup \Delta_2$ with Δ_1, Δ_2 nonempty such that $\langle \alpha_1, \alpha_2 \rangle = 0$ for all $\alpha_1 \in \Delta_1$, $\alpha_2 \in \Delta_2$; i.e. all the elements of one part are orthogonal to all the elements of the other. Otherwise we call it *irreducible* (*indecomposable*).

Example 4.8 (Root systems in dimensions 1 and 2)**.**



A_1 $(n = 1)$ $A_1 \times A_1$ (reducible), $\theta = \frac{\pi}{2}$ A_2, $\theta = \frac{\pi}{3}$



B_2, $\theta = \frac{\pi}{4}$ G_2, $\theta = \frac{\pi}{6}$

(Note: Δ is not a basis!) In B_2 the roots are $\pm e_1, \pm e_2, \pm e_1 \pm e_2$; in G_2 there are 6 short roots and 6 long roots of length ratio $\sqrt{3}$. In each case θ is the smallest angle between two roots.

4.3 Classification of root systems: preliminaries

From a root system Δ we can choose a (non-unique) subset, the *simple roots*.

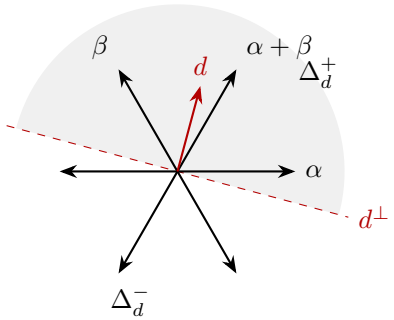
crossed out: "there is a unique $\perp$ hyperplane"

For each $\alpha \in \Delta$ consider $\alpha^\perp$, its (linear) orthogonal complement. Since $|\Delta|$ is finite, there exists $d \in \mathbb{R}^n$ such that $\langle \alpha, d \rangle \neq 0$ for all $\alpha \in \Delta$ (avoid the finitely many hyperplanes $\alpha^\perp$). Hence there is a partition of Δ into

$$\text{positive roots } \Delta_d^+ = \{\alpha \in \Delta : \langle \alpha, d \rangle > 0\}, \qquad \text{negative roots } \Delta_d^- = \{\alpha \in \Delta : \langle \alpha, d \rangle < 0\}$$

(note $\Delta_d^- = -\Delta_d^+$).

Definition 4.9. A root $\alpha \in \Delta_d^+$ is *simple* if it is not the sum of two other positive roots. A set of simple roots is a *fundamental system* of Δ (we write $\Pi = \Pi_d$).



Sketch on p. 8, redrawn for A_2: simple roots α, β.

The choice of d affects the fundamental system, but such systems are equivalent under the action of the group of reflections $\langle s_\alpha \rangle_{\alpha \in \Delta}$ (the *Weyl group* W; see Proposition 4.11).

Theorem 4.10. *A fundamental system of Δ is an $\mathbb{R}$-basis for $\mathbb{R}^n$. Moreover every positive root is a linear combination of simple roots with nonnegative integer coefficients.*

Sketch. Linear independence. Suppose $\sum_i c_i \gamma_i = 0$ with $\gamma_i \in \Pi$. Partition the indices into those with $c_i > 0$ (+), those with $c_i < 0$ (−), and those with $c_i = 0$, and put $x := \sum_{c_i>0} c_i \gamma_i = \sum_{c_j<0} |c_j| \gamma_j$. Then

$$\langle x, x \rangle = \sum_{c_i>0,\ c_j<0} c_i \, |c_j| \, \langle \gamma_i, \gamma_j \rangle \leq 0$$

by Lemma 4.13 (distinct simple roots make non-acute angles), so $x = 0$. But then $0 = \langle x, d \rangle = \sum_{c_i>0} c_i \langle \gamma_i, d \rangle$ with every $\langle \gamma_i, d \rangle > 0$, so no c_i is positive; likewise none is negative.

Span. Let γ be orthogonal to all simple roots. Then by the orbit argument (Proposition 4.11: every root is the image of a simple root under the group generated by the simple reflections s_α, $\alpha \in \Pi$, each of which fixes γ), γ is orthogonal to all roots, hence $\gamma = 0$ since Δ spans.

[Added] A more elementary route: if $\alpha \in \Delta_d^+$ is not simple, then $\alpha = \beta + \beta'$ with $\beta, \beta' \in \Delta_d^+$, and $\langle \beta, d \rangle, \langle \beta', d \rangle < \langle \alpha, d \rangle$. Since Δ is finite, induction on $\langle \alpha, d \rangle$ shows every positive root is a sum of simple roots. Hence Π spans $\operatorname{span} \Delta = \mathbb{R}^n$, and the integrality claim follows too.

□

Proposition 4.11. *Δ can be reconstructed from a fundamental system via reflections. Precisely: if W_Π denotes the group generated by s_γ, $\gamma \in \Pi$, then $\Delta = W_\Pi \cdot \Pi$, and any two fundamental systems are related by an element of W_Π.*

Sketch. Consider the *Weyl chamber* of the fundamental system Π,

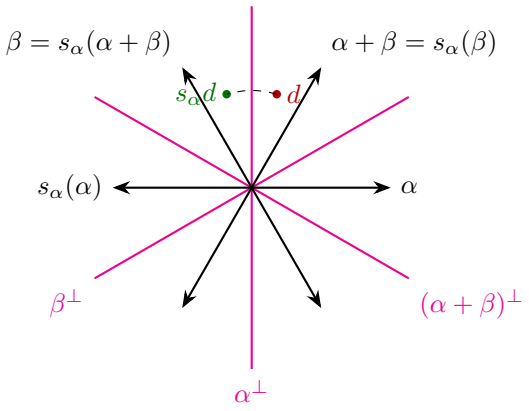
$$C(\Pi) = \{x \in \mathbb{R}^n : \langle x, \gamma \rangle > 0 \text{ for all } \gamma \in \Pi\}.$$

Then $d \in C(\Pi)$, and moving d within its chamber does not change the fundamental system. Any other fundamental system is based on some d' in another chamber, and there are finitely many chambers (the connected components of $\mathbb{R}^n \setminus \bigcup_\alpha \alpha^\perp$). Take a path $d \to d'$ in general position; the walls it crosses give reflections s_{walls}, whose composition takes d to (the chamber of) d', and hence takes the fundamental system of d to that of d'.

crossed out: "orbit d to each..."

[Added] Details: if d and $d' = s_\alpha d$ lie in adjacent chambers separated by the wall $\alpha^\perp$, then $\Pi_{d'} = s_\alpha \Pi_d$, since s_α is an isometry permuting Δ with $\langle s_\alpha \beta, s_\alpha d \rangle = \langle \beta, d \rangle$. The walls of $C(\Pi)$ are exactly the hyperplanes $\gamma^\perp$, $\gamma \in \Pi$; writing each wall-crossing reflection in terms of the previous ones ($s_{w\gamma} = w s_\gamma w^{-1}$) shows the composite lies in W_Π. Finally every root α is simple for a suitable d' (take d' very close to a generic point of $\alpha^\perp$, on the side where $\langle \alpha, d' \rangle > 0$), so $\alpha = w\gamma$ for some $w \in W_\Pi$, $\gamma \in \Pi$. See Humphreys, *Introduction to Lie Algebras and Representation Theory*, §10.3.

□



Colour sketch on p. 11, redrawn.

For A_2: with d as shown, $\Pi_d = \{\alpha, \beta\}$; reflecting across $\alpha^\perp$ gives $s_\alpha d$, with fundamental system $\Pi_{s_\alpha d} = \{s_\alpha(\alpha), s_\alpha(\beta)\} = \{-\alpha, \alpha+\beta\}$.

Lemma 4.12. *If $\alpha, \beta \in \Delta$ are not parallel and $\langle \alpha, \beta\rangle > 0$, then $\alpha - \beta \in \Delta$.*

Proof. We have $n_{\alpha\beta} = 2\langle\alpha,\beta\rangle/\langle\beta,\beta\rangle > 0$, and by the table ($n_{\alpha\beta}n_{\beta\alpha} \in \{1,2,3\}$ with both factors positive integers) $n_{\alpha\beta} = 1$ or $n_{\beta\alpha} = 1$. If $n_{\alpha\beta} = 1$, then

$$s_\beta(\alpha) = \alpha - 2\beta\,\frac{\langle\alpha,\beta\rangle}{\langle\beta,\beta\rangle} = \alpha - n_{\alpha\beta}\,\beta = \alpha - \beta \in \Delta.$$

Fix: written $m_{\beta\alpha}$, $s_\beta(-\beta)$ (typos)

Similarly, if $n_{\beta\alpha} = 1$, then $s_\alpha(-\beta) = -\beta + n_{\beta\alpha}\alpha = \alpha - \beta \in \Delta$. □

Lemma 4.13. *If α, β are distinct simple roots, then $\langle\alpha,\beta\rangle \le 0$.*

Proof. If $\langle\alpha,\beta\rangle > 0$, then (as $\alpha \neq \pm\beta$, both being positive) Lemma 4.12 gives $\gamma := \alpha - \beta \in \Delta$. So γ or $-\gamma$ lies in Δ_d^+, and then $\alpha = \beta + \gamma$ or $\beta = \alpha + (-\gamma)$ is a sum of two positive roots, contradicting simplicity. □

Proposition 4.14. *A fundamental system is decomposable (into two nonempty mutually orthogonal subsets) if and only if the root system is decomposable.*

Sketch. Similar arguments.
[Added] If $\Delta = \Delta_1 \cup \Delta_2$ orthogonally, then $\Pi = (\Pi\cap\Delta_1)\cup(\Pi\cap\Delta_2)$, and both parts are nonempty because Π spans $\mathbb{R}^n$. Conversely, if $\Pi = \Pi_1 \cup \Pi_2$ orthogonally, the reflections s_γ, $\gamma \in \Pi_1$, fix Π_2 pointwise and preserve $\operatorname{span}\Pi_1$, and vice versa; so by Proposition 4.11 every root lies in $W_\Pi\Pi_1 \subset \operatorname{span}\Pi_1$ or in $W_\Pi\Pi_2 \subset \operatorname{span}\Pi_2$, which are orthogonal.

□

4.4 Classification via Coxeter graphs

Simple roots determine root systems (Proposition 4.11), so it is enough to classify fundamental systems.

Elements α, β of a fundamental system are orthogonal or make an obtuse angle (Lemma 4.13). By the table, therefore,

$$\theta \in \left\{\frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{6}\right\}, \qquad \text{that is,} \qquad n_{\alpha\beta}n_{\beta\alpha} = 4\cos^2\theta \in \{0,1,2,3\}.$$

Definition 4.15. The *Coxeter graph* of Δ has a vertex for each simple root, and an edge of weight $n_{\alpha\beta} \cdot n_{\beta\alpha}$ between α and β (drawn as that many edges; no edge when $\alpha \perp \beta$).

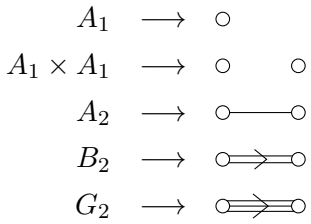
It is well defined (independent of the choice of fundamental system) by the Weyl chamber argument, since fundamental systems are carried to fundamental systems by the reflections $\langle s_\alpha \rangle$, which are isometries and so preserve all the angles.

Definition 4.16. The *Dynkin diagram* is the Coxeter graph with arrows on the double and triple edges, pointing to the shorter vector.

Remark 4.17. In an irreducible system, the possible lengths are determined by the angles (by the table, the ratio of the lengths of two non-orthogonal roots is fixed by their angle; connecting through the diagram fixes all lengths up to one overall scale). For a reducible system the components can be rescaled independently.

? Margin note by $A_1 \times A_1$, probably "unrelated scale!".

Example 4.18.



End of Lecture 4 (p. 15).

Chapter 5

Lecture 5: Classification of crystallographic root systems (continued)

Recall. A root system $\Delta \subset \mathbb{R}^n \setminus \{0\}$ satisfies

1) Δ is finite and spans $\mathbb{R}^n$;

2) $\alpha \in \Delta \implies (n\alpha \in \Delta \iff n = \pm 1)$;

3) Δ is invariant under the reflections in the hyperplanes $\alpha^\perp$: $s_\alpha(\beta) \in \Delta$ for all $\alpha, \beta \in \Delta$;

4) $n_{\beta\alpha} := 2\langle \beta, \alpha \rangle / \langle \alpha, \alpha \rangle \in \mathbb{Z}$,

and 4) implies $4\cos^2\theta_{\alpha\beta} \in \mathbb{Z}$, really $\in \{0, 1, 2, 3, 4\}$.

The plan: root lattices $\to$ root systems $\longleftrightarrow$ simple roots (last time) $\longleftrightarrow$ Dynkin diagrams (today).

From §4.4: the *Coxeter graph* has a vertex for each simple root and an edge of weight $n_{\alpha\beta} \cdot n_{\beta\alpha}$ between α and β. The *Dynkin diagram* is the Coxeter graph with arrows on the multi-edges pointing at the shorter root.

Page has "(longer)" with "shorter" written below.

5.1 Classification of Coxeter graphs

We ignore lengths.

Definition 5.1. A linearly independent set of n unit vectors $\{v_1, \ldots, v_n\}$ spanning $\mathbb{R}^n$ is *admissible* if for all $i \neq j$

$$\langle v_i, v_j \rangle \leq 0 \qquad \text{and} \qquad 4\langle v_i, v_j \rangle^2 = 4\cos^2\theta_{ij} \in \{0, 1, 2, 3\}.$$

An *admissible diagram* is the Coxeter graph of an admissible set (vertices v_i, with $4\langle v_i, v_j \rangle^2$ edges between v_i and v_j).

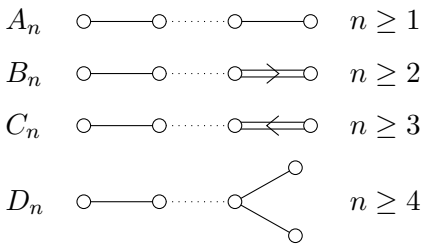
A normalized set of simple roots is admissible. (Note: the angles are the determining factor.)

This follows from Theorem 4.10, Lemma 4.13 and the table.

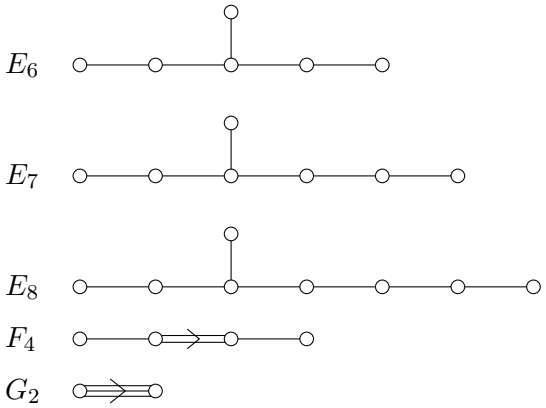
The simple roots of an irreducible Δ are not all orthogonal — more precisely, they cannot be split into two mutually orthogonal sets (Proposition 4.14) — so the Coxeter graph is connected. So we only need to classify connected admissible diagrams.

Theorem 5.2. *The Dynkin diagram of an irreducible (crystallographic) root system is of one of the following types (n = number of vertices).*

Infinite families:



Exceptional types:



Conversely, each of these diagrams is the Dynkin diagram of an irreducible root system, which is unique up to isomorphism. (The ranges of n avoid repetitions: $B_2 = C_2$, $D_3 = A_3$.)

? Parenthetical under D_n reads roughly "(each also unique)".

It will suffice to classify admissible diagrams. By definition, subsets of the vectors of an admissible set satisfy the admissibility condition inside their span (the resulting diagram could be disconnected); that is, any subdiagram of an admissible diagram, obtained by deleting vertices, is admissible.

Lemma 5.3. *A connected admissible diagram is a tree.*

Proof. Consider $v = \sum_{i=1}^{n} v_i$, where $\{v_i\}$ is admissible. The v_i are linearly independent, so $v \neq 0$, and

$$0 < \langle v, v\rangle = \sum_i \langle v_i, v_i\rangle + \sum_{i<j} 2\langle v_i, v_j\rangle = n + \sum_{i<j} 2\langle v_i, v_j\rangle.$$

If v_i is connected to v_j in the Coxeter graph, then by the table $2\langle v_i, v_j\rangle \in \{-1, -\sqrt{2}, -\sqrt{3}\}$, in particular $2\langle v_i, v_j\rangle \leq -1$. Hence $\sum_{i<j} 2\langle v_i, v_j\rangle$ has at most $n-1$ terms not equal to 0, i.e. there are at most $n-1$ connected pairs of vertices. But by assumption the diagram is connected, so there are exactly $n-1$ pairs of connected vertices (a connected graph on n vertices has at least $n-1$ edges), and it is a tree (maybe with multiple edges, but not too many). □

a scribbled-out remark at the bottom of p. 20

Lemma 5.4. *The degree of each vertex, counted with multiplicity, is at most* 3.

Fix: written "fix vertex v_1", then used as v

Proof. Fix a vertex v, connected to the vertices $v_1, \dots, v_k$. Since the diagram is a tree (Lemma 5.3, applied to the subdiagram on $v, v_1, \dots, v_k$), $\langle v_i, v_j \rangle = 0$ for $i \neq j$, so $\{v_1, \dots, v_k\}$ is orthonormal. Also, $\{v_1, \dots, v_k, v\}$ is linearly independent (by admissibility; for simple roots this is Theorem 4.10), so the normalized projection

$$v_0 := \frac{\operatorname{proj}_{\langle v_1,\dots,v_k\rangle^\perp}(v)}{\left\|\operatorname{proj}_{\langle v_1,\dots,v_k\rangle^\perp}(v)\right\|}$$

is defined, and $\langle v, v_0 \rangle \neq 0$. Now $\{v_0, \dots, v_k\}$ is an orthonormal set, and v lies in its span, so

$$v = \sum_{i=0}^{k} \langle v, v_i \rangle v_i \qquad \text{and} \qquad \langle v, v \rangle = \sum_{i=0}^{k} \langle v, v_i \rangle^2 = 1.$$

Since $\langle v, v_0 \rangle^2 \neq 0$,

$$\sum_{i=1}^{k} 4\langle v, v_i \rangle^2 < 4.$$

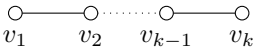
But $4\cos^2\theta = 4\langle v, v_i \rangle^2$ is the number of edges from v to v_i. So the degree of v, counted with multiplicity, is at most 3. □

Corollary 5.5.  *is the only connected admissible diagram with a triple edge.*

Proof. If v and w are joined by a triple edge, both already have degree 3, so by Lemma 5.4 neither is joined to any other vertex; by connectedness there are no other vertices. □

So from now on we consider only double and single edges.

Definition 5.6. A *simple chain* is a collection of vertices connected by single edges:



Lemma 5.7 (Simple chain collapse)**.** *In an admissible diagram, a simple chain represented by* $\{v_1, \dots, v_k\}$ *can be replaced by the single vector* $v = \sum_{i=1}^{k} v_i$*, giving an admissible diagram.*

Proof. We want to show that v is a unit vector and that the collapsed diagram is admissible. (Linear independence of the new set is clear, since the v_i and the remaining vectors are linearly independent.) We have

$$\langle v, v \rangle = k + \sum_{i<j} 2\langle v_i, v_j \rangle,$$

and $\langle v_i, v_j \rangle = 0$ for all $i < j$ except $j = i + 1$ (a tree has no other edges among the v_i). For a single edge, recall $2\langle v_i, v_{i+1} \rangle = -1$. Hence

$$\langle v, v \rangle = k - (k - 1) = 1.$$

Now consider a vertex u not represented in the chain. It connects to at most one vertex in the chain (by the tree property), say v_j (if none, $\langle u, v \rangle = 0$). Then

$$\langle u, v \rangle = \sum_{i=1}^{k} \langle u, v_i \rangle = \langle u, v_j \rangle,$$

so the angles $\theta_{u,v}$ and θ_{u,v_j} are the same, and the collapsed set is admissible. □

scribbled-out "recall admissible..." at the foot of p. 23

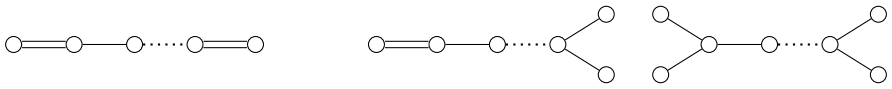
5.2 Classification of Coxeter graphs (continued)

continues Lecture 5 from p. 23

Recall from the previous pages: a connected admissible diagram is a tree; each vertex meets at most three edges counted with multiplicity; the only admissible diagram with a triple edge is ⟹ (G_2); and a simple chain $\{v_1, \dots, v_k\}$ may be collapsed to the single unit vector $v = \sum_i v_i$, leaving an admissible diagram. From now on only single and double edges occur.

5.2.1 Forbidden configurations

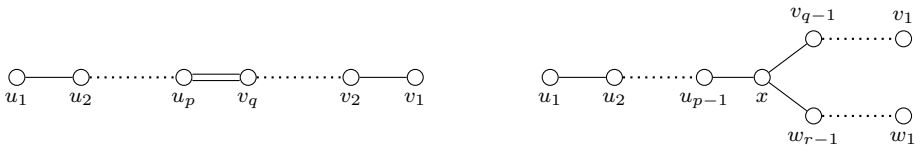
Proposition 5.8. *Coxeter graphs of admissible diagrams containing any of the following subgraphs are forbidden:*



Hence a connected admissible diagram (with no triple edge) contains at most one branch point or *at most one double edge, but not both.*

Proof. In each case collapse the simple chain joining the two special features (possibly of length zero) to a single vertex. By the collapsing lemma the result is again admissible, but the new vertex meets edges of total multiplicity $2+2$, $2+1+1$, or $1+1+1+1$, i.e. 4, contradicting the bound of 3 edges (with multiplicity) at each vertex. The same argument rules out two double edges, two branch points, or a double edge and a branch point anywhere in the tree, since they are always joined by a simple chain. □

So, besides G_2, the only possible connected Coxeter graphs are:



together with the simple chains ○—○·····○—○, which give A_n.

5.2.2 One double edge: B_n, C_n, F_4

Label the diagram with one double edge as above: $u_1, \dots, u_p$, then the double edge $u_p = v_q$, then $v_q, \dots, v_1$. Put

$$u = \sum_{i=1}^{p} i\, u_i, \qquad v = \sum_{j=1}^{q} j\, v_j.$$

Since the u_i form a simple chain, $2\langle u_i, u_{i+1}\rangle = -1$ for $1 \le i \le p-1$ and all other pairs are orthogonal. Hence

$$\langle u, u\rangle = \sum_{i=1}^{p} i^2 \langle u_i, u_i\rangle + \sum_{i<j} 2ij\, \langle u_i, u_j\rangle = \sum_{i=1}^{p} i^2 + \sum_{i=1}^{p-1} i(i+1) \cdot \underbrace{2\langle u_i, u_{i+1}\rangle}_{=-1}$$

$$= \sum_{i=1}^{p} i^2 - \sum_{i=1}^{p-1} i(i+1) = p^2 - \sum_{i=1}^{p-1} i = \frac{p(p+1)}{2},$$

and similarly $\langle v, v\rangle = q(q+1)/2$.

Since the double edge $u_p = v_q$ is the only non-orthogonal relation between the two chains, $\langle u, v\rangle = pq\,\langle u_p, v_q\rangle$, and $4\langle u_p, v_q\rangle^2 = 2$. Thus

$$\langle u, v\rangle^2 = \frac{p^2q^2}{2}.$$

Since u and v are not parallel (the u_i, v_j are linearly independent), the Cauchy–Schwarz inequality is strict: $\langle u, v\rangle^2 < \langle u, u\rangle\langle v, v\rangle$, i.e.

$$\frac{p^2q^2}{2} < \frac{p(p+1)}{2}\cdot\frac{q(q+1)}{2}, \qquad \text{so} \qquad 2pq < (p+1)(q+1),$$

which, for integers $p, q \geq 1$, is

$$\boxed{(p-1)(q-1) < 2.}$$

Hence either $p = q = 2$, which is F_4, or $p = 1$ (or $q = 1$) with the other arbitrary, which is B_n and C_n ($n = p + q$; the Coxeter graphs of B_n and C_n coincide, and they are distinguished only by the direction of the arrow in the Dynkin diagram).

5.2.3 One branch point: D_n, E_6, E_7, E_8

Finally, take the branched diagram above, with branch vertex x and three arms $u_1, \dots, u_{p-1}$, $v_1, \dots, v_{q-1}$, $w_1, \dots, w_{r-1}$ ($p, q, r \geq 2$), the indices increasing towards x. As before put

$$u = \sum_{i=1}^{p-1} i\,u_i, \qquad v = \sum_{i=1}^{q-1} i\,v_i, \qquad w = \sum_{i=1}^{r-1} i\,w_i.$$

Note that u, v, w are mutually orthogonal vectors (no edges between different arms), and x is not in their span $\langle u, v, w\rangle$ (by linear independence of all the vertices). By the previous computation with p replaced by $p-1$, $\langle u, u\rangle = p(p-1)/2$, and similarly for v, w.

Since x is a unit vector not in $\langle u, v, w\rangle$, and $u/\|u\|, v/\|v\|, w/\|w\|$ is an orthonormal basis of that space, the squared length of the projection of x onto it is strictly less than 1:

$$1 = \langle x, x\rangle > \left\langle x, \frac{u}{\|u\|}\right\rangle^2 + \left\langle x, \frac{v}{\|v\|}\right\rangle^2 + \left\langle x, \frac{w}{\|w\|}\right\rangle^2.$$

Also $\langle x, u_i\rangle = 0$ unless $i = p-1$, and $4\langle x, u_{p-1}\rangle^2 = 1$ (single edge). Then

$$\langle x, u\rangle^2 = \Big(\sum_{i=1}^{p-1} i\,\langle x, u_i\rangle\Big)^2 = (p-1)^2\langle x, u_{p-1}\rangle^2 = \frac{(p-1)^2}{4}$$

and therefore

Fix: notes: $\sum i^2\langle x, u_i\rangle$, square misplaced

$$\left\langle x, \frac{u}{\|u\|}\right\rangle^2 = \frac{(p-1)^2}{4}\cdot\frac{2}{p(p-1)} = \frac{1}{2}\Big(1 - \frac{1}{p}\Big).$$

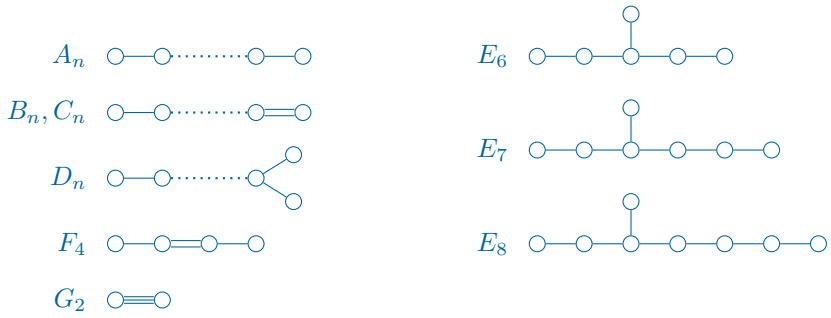
The same holds for v (with q) and w (with r). Then

$$2 > \Big(1 - \frac{1}{p}\Big) + \Big(1 - \frac{1}{q}\Big) + \Big(1 - \frac{1}{r}\Big) \quad\Longrightarrow\quad \frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1,$$

where $p, q, r \geq 2$ are integers. Assume $p \geq q \geq r \geq 2$. Then $3/r > 1$ forces $r = 2$, and:

- $q = 2$: any $p \geq 2$ works — this is D_{p+2};
- $q = 3$: $\frac{1}{q} + \frac{1}{r} = \frac{5}{6}$, so $\frac{1}{p} > \frac{1}{6}$, i.e. $3 \leq p < 6$ — $p = 3, 4, 5$ give E_6, E_7, E_8;
- $q \geq 4$: $\frac{1}{p} + \frac{1}{q} + \frac{1}{2} \leq \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$, so there are no solutions.

(The diagram has $p + q + r - 2$ vertices, which gives the subscripts.)
[Added] **Summary.** The connected Coxeter graphs of admissible diagrams (equivalently, of irreducible crystallographic root systems) are A_n ($n \geq 1$), $B_n = C_n$ as graphs ($n \geq 2$), D_n ($n \geq 4$), E_6, E_7, E_8, F_4, G_2.



Remark 5.9. It remains to show that each of A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, G_2 actually exists, i.e. to construct the corresponding root systems. This is done in the next lecture; G_2 is left as an exercise (its 12 roots were drawn in Lecture 4: check axioms 1)–4) directly).

p. 31: "Existence of $A_n, \ldots, F_4$: construct next time; G_2 – ex."

Chapter 6

Lecture 6: Root systems and root lattices

6.1 Construction of crystallographic root systems

A_n

In A_n the simple roots are pairwise orthogonal or at angle $2\pi/3$.

Consider $x \in \mathbb{Z}^{n+1}$ such that $\sum_{i=1}^{n+1} x_i = 0$. These form an n-dimensional sublattice S of $\mathbb{Z}^{n+1}$. Let

$$\Delta = \{x \in S : \|x\| = \sqrt{2}\}.$$

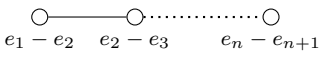
That is, all but two coordinates are 0, one coordinate is $+1$ and one is -1: the vectors are the permutations of $(1, -1, 0, \dots, 0)$, so $|\Delta| = n(n+1)$. Choose the simple roots

$$\alpha_i = e_i - e_{i+1}, \qquad i = 1, \dots, n,$$

with matrix of coordinates and Dynkin diagram

diagram labelled $e_{i+1} - e_i$

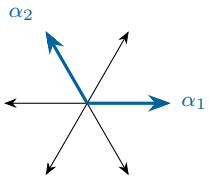
$$\begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix}$$



To check the Dynkin diagram: the number of edges joining α_i and α_j is $4\langle \alpha_i, \alpha_j \rangle^2 / (\|\alpha_i\|^2 \|\alpha_j\|^2) = \langle \alpha_i, \alpha_j \rangle^2 \in \{1, 0\}$, and it is 1 exactly when $j = i \pm 1$.

Note that these roots generate the full sublattice $\{x \in \mathbb{Z}^{n+1} : \sum x_i = 0\}$ (any such x equals $\sum_{k=1}^{n} (x_1 + \cdots + x_k)\, \alpha_k$). So the A_n root lattice is all of S. This description is maybe not so useful for visualizing, since S sits inside $\mathbb{R}^{n+1}$.

Example 6.1 (A_2)**.** Simple roots $(-1, 1, 0)$ and $(0, -1, 1)$, whose sum is $(-1, 0, 1)$; together with the negatives $(1, -1, 0)$, $(0, 1, -1)$, $(1, 0, -1)$ these are the six roots, lying in the plane $x_1 + x_2 + x_3 = 0$. In that plane they form the familiar hexagonal configuration:



p. 33: pink star in the plane $\sum x_i = 0$

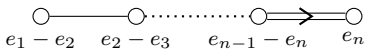
Example 6.2 (A_3)**.** The 12 roots of A_3 are the 12 nearest neighbours of the origin in the FCC lattice ("the FCC system") — the A_3 root lattice is the face-centred cubic lattice, with kissing number 12.

B_n

Δ is the set of all integer vectors $x \in \mathbb{Z}^n$ of norm 1 or norm $\sqrt{2}$, i.e. $\pm e_i$ and $\pm e_i \pm e_j$. So $|\Delta| = 2n + 4\binom{n}{2} = 2n^2$. Simple roots:

$$\alpha_i = e_i - e_{i+1}\ (1 \le i \le n-1), \qquad \text{short root } \alpha_n = e_n,$$

$$\begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \\ & & & & 1 \end{bmatrix}$$



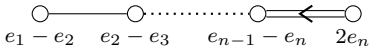
Sadly, from this construction the associated root lattice is all of $\mathbb{Z}^n$ (it contains every e_i), so we get nothing new.

C_n

Δ is the set of all integer vectors of the form $2x$ with $x \in \mathbb{Z}^n$ of length 1, or $x \in \mathbb{Z}^n$ of length $\sqrt{2}$; i.e. $\pm 2e_i$ and $\pm e_i \pm e_j$. Again $|\Delta| = 2n + 4\binom{n}{2} = 2n^2$. Simple roots:

$$\alpha_i = e_i - e_{i+1}\ (1 \le i \le n-1), \qquad \text{long root } \alpha_n = 2e_n,$$

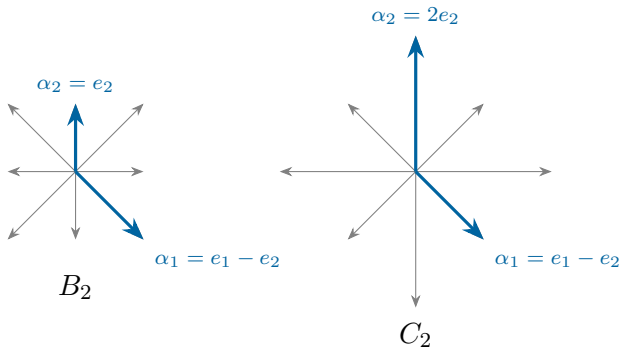
$$\begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \\ & & & & 2 \end{bmatrix}$$



So the C_n root lattice is the set of all integer vectors with *even* coordinate sum.

Fix: diagram drawn with a single last edge

Example 6.3 (B_2 and C_2)**.** The two root systems, with the simple roots highlighted:



In both, the simple roots meet at angle $3\pi/4$; C_2 is B_2 rotated by $\pi/4$ and rescaled.

redrawn; sketch showed only α_1, α_2

D_n

Δ is the set of all integer vectors $x \in \mathbb{Z}^n$ of length $\sqrt{2}$, i.e. $\pm e_i \pm e_j$, so

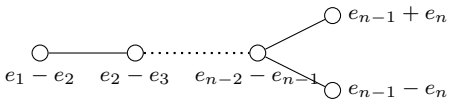
$$|\Delta| = 4\binom{n}{2} = 2n(n-1).$$

Fix: notes: $2\binom{n}{2}$

Simple roots:

$$\alpha_i = e_i - e_{i+1} \ (1 \le i \le n-1) \qquad \text{and} \qquad \alpha_n = e_{n-1} + e_n,$$

$$\begin{bmatrix} 1 & -1 & & & \\ & \ddots & \ddots & & \\ & & 1 & -1 & \\ & & & 1 & -1 \\ & & & 1 & 1 \end{bmatrix}$$

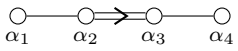


This also generates the lattice $\{x \in \mathbb{Z}^n : \text{the coordinate sum is even}\}$, the same as for C_n. (I can't draw D_4, sadly.)

F_4

Δ is the set of vectors $x \in \mathbb{R}^4$ of length 1 or $\sqrt{2}$ such that the coordinates of $2x$ are all even or all odd integers. Explicitly: $\pm e_i$ (8), $\pm e_i \pm e_j$ (24) and $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$ (16), so $|\Delta| = 48$. Simple roots:

$$\alpha_1 = (1,-1,0,0), \quad \alpha_2 = (0,1,-1,0), \quad \alpha_3 = (0,0,1,0), \quad \alpha_4 = -\tfrac{1}{2}(1,1,1,1),$$



(check by hand). Indeed $\langle \alpha_1, \alpha_2 \rangle = -1$ with both of length $\sqrt{2}$ (angle $2\pi/3$); $\langle \alpha_2, \alpha_3 \rangle = -1$ with lengths $\sqrt{2}, 1$ (angle $3\pi/4$, double edge); $\langle \alpha_3, \alpha_4 \rangle = -\frac{1}{2}$ with both of length 1 (angle $2\pi/3$); all other pairs are orthogonal.

The F_4 root lattice: all coordinates are integers, or all coordinates are halves of odd integers — no mixing. That is, $\mathbb{Z}^4 \cup (\mathbb{Z}^4 + \frac{1}{2}(1,1,1,1))$, which is D_4^*, a scaled copy of D_4.

G_2

We already drew this one (the 12 roots form a "star of David", two hexagons of lengths 1 and $\sqrt{3}$). The G_2 root lattice coincides with the A_2 root lattice (the hexagonal lattice).

star sketch, see Lecture 4

6.2 The E_8 root system

Definition 6.4. The E_8 root system consists of the $|\Delta| = 240$ vectors in $\mathbb{R}^8$ of the forms

$$\pm e_i \pm e_j \ (i \ne j) \qquad \text{and} \qquad \tfrac{1}{2}(\pm e_1 \pm \cdots \pm e_8) \text{ with an even number of minus signs.}$$

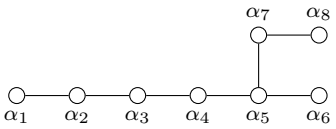
There are $4\binom{8}{2} = 112$ of the first kind and $2^7 = 128$ of the second.

Fix: added the even-sign condition

Simple roots:

$$\alpha_i = e_i - e_{i+1}\ (1 \le i \le 6), \qquad \alpha_7 = e_6 + e_7, \qquad \alpha_8 = -\tfrac{1}{2}(1,\dots,1),$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ & & 1 & -1 & 0 & 0 & 0 & 0 \\ & & & 1 & -1 & 0 & 0 & 0 \\ & & & & 1 & -1 & 0 & 0 \\ & & & & & 1 & -1 & 0 \\ & & & & & 1 & 1 & 0 \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$



(Check: $\langle \alpha_7, \alpha_5 \rangle = -1$, $\langle \alpha_7, \alpha_6 \rangle = 0$, $\langle \alpha_8, \alpha_7 \rangle = -1$ and $\langle \alpha_8, \alpha_i \rangle = 0$ for $i \le 6$; the arms at α_5 give $(p, q, r) = (5, 3, 2)$.)

This generates the lattice E_8 of all $x \in \mathbb{R}^8$ such that

1) the coordinates are all integers or all half-integers (halves of odd integers), and

2) the sum of the coordinates is even.

This is clearly related to D_8. Consider the "deep holes" of the D_n lattice, i.e. the half-integer points such as $h = \frac{1}{2}(1,\dots,1)$. In dimension 8,

$$\left\| \tfrac{1}{2}(1,\dots,1) \right\| = \sqrt{2},$$

and a typical shortest lattice vector has $\|(1,1,0,\dots,0)\| = \sqrt{2}$. Similarly,

$$\left\| (1,1,0,\dots,0) - \tfrac{1}{2}(1,\dots,1) \right\| = \sqrt{2}.$$

So if we have D_8, there is another copy $D_8 + h$ that fits into the holes of D_8 without decreasing the minimal distance: $E_8 = D_8 \cup (D_8 + h)$. (For general n, h is at distance $\sqrt{n}/2$ from D_n; this equals the minimal distance $\sqrt{2}$ exactly when $n = 8$.)

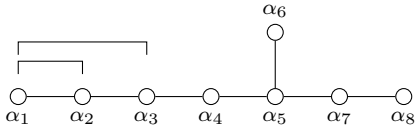
E_7, E_6

Let $v = (1,1,\dots,1)$ and $w = -e_7 - e_8$. Then

$$E_7 = E_8 \cap v^\perp, \qquad E_6 = E_8 \cap \langle w, v \rangle^\perp.$$

Here $\frac{1}{2}v = -\alpha_8$ and w are roots with $\langle \frac{1}{2}v, w \rangle = -1$, so they span an A_2; thus E_7 and E_6 are the orthogonal complements of an A_1 and an A_2 in E_8.

These descriptions are not so satisfying, but perhaps there is an easy-to-see construction given by collapsing simple chains: replace α_1, α_2 by $\alpha_1 + \alpha_2$, then $\alpha_1 + \alpha_2, \alpha_3$ by $\alpha_1 + \alpha_2 + \alpha_3$, and so on.



By the collapsing lemma, $\alpha_1+\alpha_2, \alpha_3, \ldots, \alpha_8$ is a simple system of type E_7, and $\alpha_1+\alpha_2+\alpha_3, \alpha_4, \ldots, \alpha_8$ one of type E_6, so the lattices they generate are copies of the E_7, E_6 root lattices inside E_8. This continues (collapsing suitable chains) to

$$D_5, \quad D_4, \quad A_3, \quad A_2, \quad A_1.$$

Exercise 6.5. Find other constructions of E_8. (For instance via the extended Hamming code and Construction A, §6.4.)

6.3 Laminated lattices

Before (Lecture 4) we had the lattices

$$A_1,\ A_2,\ A_3,\ D_4,\ D_5,\ E_6,\ E_7,\ E_8$$

as the best lattice packings, and best known packings, in dimensions 1 to 8. Somehow these are all related to E_8 (as we just saw, they all sit inside E_8).

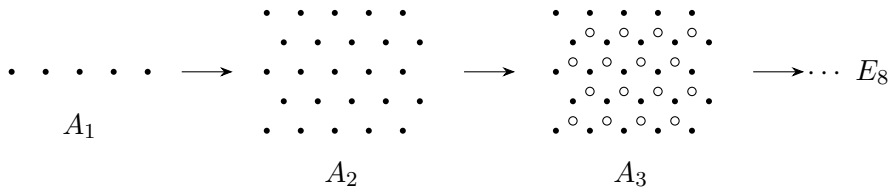
[Added] Optimality among lattices is proven in dimensions 1–8 (Lagrange $n=2$, Gauss $n=3$, Korkine–Zolotarev $n=4,5$, Blichfeldt $n=6,7,8$); among *all* packings it is proven for $n=2$ (Thue, Fejes Tóth), $n=3$ (Hales) and $n=8$ (Viazovska, posted March 2016, during this course). Similarly the Leech lattice is optimal in dimension 24 (Cohn–Kumar among lattices; Cohn–Kumar–Miller–Radchenko–Viazovska 2016/17 among all packings).

Dimension 9 is, as far as I know, still unknown. (This is still the case, to the editor's knowledge.)

The best constructions for lattices come from an "inductive" process. Start with the even integers $\Lambda_1 = 2\mathbb{Z} \subset \mathbb{R}^1$.

Definition 6.6. The *n-dimensional laminated lattice* Λ_n is a lattice of maximal density among all lattices in $\mathbb{R}^n$ with shortest vector of length $\|x\| = 2$ that contain Λ_{n-1} as a sublattice.

So Λ_n is obtained by gluing: $\Lambda_n = \langle \Lambda_{n-1}, g\rangle$, i.e. by stacking layers $\Lambda_{n-1} + kg$, $k \in \mathbb{Z}$, with each new layer placed over the deepest holes of the previous one.



p. 40 sketch; circles = next layer

The laminated lattices $\Lambda_1, \ldots, \Lambda_8$ coincide with the root lattices above (scaled to minimal length 2).

Remark 6.7. Dimension 9 is unusual: there is a continuous family of packings built from layers of Λ_8, all of the same density as Λ_9. These are the "fluid" packings; in general they are *not* lattice packings, so "family of lattices" should be read as "family of packings" (see Conway–Sloane, *Sphere Packings, Lattices and Groups*, Ch. 1 and Ch. 6).

Fix: notes: "cts family of lattices"

6.4 Dimension 10: codes and Construction A

In dimension 10, a *non-lattice* packing beats the best known lattices.

Definition 6.8. A *binary code* of length n is a subset $C \subseteq \{0,1\}^n$ of the binary words of length n. The elements of C are the *code words*. The *Hamming distance* between two words is the number of positions in which they differ; $d(C)$ denotes the minimal distance between distinct code words.

Definition 6.9 (Construction A)**.** Given a binary code C of length n, there is a packing

$$P(C) = \{x \in \mathbb{Z}^n : x \bmod 2 \in C\}.$$

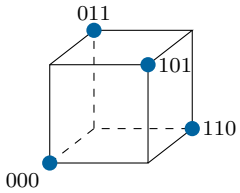
Fix: notes: $|C| = n$

[Added] Two distinct points of $P(C)$ either differ by an element of $2\mathbb{Z}^n$ (distance ≥ 2) or reduce to different code words, hence differ by ± 1 in at least $d(C)$ coordinates (distance $\geq \sqrt{d(C)}$). So $P(C)$ is a packing of balls of radius $\frac{1}{2}\min(2, \sqrt{d(C)})$ with $|C|$ centres per cube $[0,2)^n$, and its centre density is $|C|\,2^{-n}\big(\frac{1}{2}\min(2,\sqrt{d(C)})\big)^n$.

This gives some relation between packing problems in high dimensions and error detecting / error correcting codes. Think of a code word sent through a continuous channel and received with an error of size up to ε: the centres of the ε-spheres of a packing correspond to code words that are at distance $\geq 2\varepsilon$ apart. Dense packings correspond to large collections of code words that are well separated.

sketch: ε-ball around a code word

Example 6.10 (Block codes)**.** The $(6,3)$ repetition code repeats the pattern: $101 \mapsto 101101$. The $(3,2)$ parity check code (mod 2) appends a parity bit: $11 \mapsto 110$. We can see the latter on the Hamming cube: its code words $000, 011, 101, 110$ all have Hamming distance 2, i.e. are 2 bits apart.



In general, parity check codes (all even-weight words) correspond to D_n, the lattice with even coordinate sums.

Construction A thus links binary codes and lattices: $P(C)$ is a lattice if and only if C is a *linear* code, that is, sums of code words (mod 2) are code words. The Hamming codes give E_7 and E_8. More precisely (after scaling by $1/\sqrt{2}$): the extended Hamming code $[8,4,4]$ gives E_8 ($16 + 14 \cdot 16 = 240$ vectors of norm 4 in $P(C)$), and the $[7,3,4]$ simplex code, the dual of the Hamming code $[7,4,3]$, gives E_7 ($14 + 7 \cdot 16 = 126$).

The code C_{10}

So in dimension 10 there is a special binary code C_{10} (Best's code), which is unique (Litsyn–Vardy) and quite dense. It is based on a Gray code, or reflected binary code.

Definition 6.11. A *Gray code* is a map $(\mathbb{Z}_2)^n \to (\mathbb{Z}_2)^n$, a re-ordering of all binary words of length n, where the successor of an element differs by a single bit flip.

binary		Gray
00	0	00
01	1	01
10	2	11
11	3	10

There is a replacement procedure to convert between binary and Gray codes.

Exercise 6.12. Find the conversion binary $\leftrightarrow$ Gray, and iterate it to get the Gray code of length n. (Hint: $g = b \oplus (b \gg 1)$; conversely $b_k = g_1 \oplus \cdots \oplus g_k$ from the leading bit.)

In dimension 10 we look for binary codes with minimum distance at least 4. We construct C_{10} as follows. Take $(a, b, c, d, e) \in (\mathbb{Z}_4)^5$ satisfying

$$b, c, d \in \{\pm 1\}, \qquad a = c - d, \qquad e = b + c,$$

together with all cyclic permutations thereof. Then apply the Gray map

$$0 \mapsto 00, \quad 1 \mapsto 01, \quad 2 \mapsto 11, \quad 3 \mapsto 10.$$

? {±1}: a crossed-out entry; {±1} fits the count 2^3

This gives $8 \cdot 5 = 40$ code words (2^3 choices of (b, c, d) times 5 cyclic shifts) of length 10.

Claim 6.13. *The Hamming distance between distinct code words of C_{10} is at least 4.*

[Added] The Gray map is an isometry from $\mathbb{Z}_4$ with the Lee metric ($d(0,2) = d(1,3) = 2$, other distinct pairs distance 1) to $\mathbb{Z}_2^2$ with the Hamming metric, so it suffices to check Lee distance ≥ 4 in $(\mathbb{Z}_4)^5$. (Editor: confirmed by direct computer check of all $\binom{40}{2}$ pairs; the 40 words are distinct.)

Moreover $A(10, 4) = 40$, i.e. 40 is the maximal size of a binary code of length 10 and minimal distance 4, so C_{10} is optimal for dimension 10.

? notes: "LP bound $\Rightarrow < 40 \ldots$"; $A(10,4) = 40$ is due to Best (c. 1980)

[Added] Construction A applied to C_{10} gives the non-lattice packing P_{10c}: radius 1 and 40 centres per 2^{10} volume, so centre density $40/2^{10} = 5/128 \approx 0.0391$, larger than that of the best known lattice Λ_{10} (centre density $1/(16\sqrt{3}) \approx 0.0361$). See Conway–Sloane, Ch. 5.

6.5 An aside: lattice vs. non-lattice packings of other objects

For other objects than balls, the question of lattice versus non-lattice packings looks different:

- ellipsoid packings;
- polygon packings;
- honeycomb (HCB) and bubble problems.

? "HCB", "Bubble prob."?

For example, in the plane a lattice packing is optimal for any centrally symmetric convex body (Fejes Tóth, Rogers), but not for some non-symmetric ones; and congruent ellipsoids in $\mathbb{R}^3$ can be packed more densely by non-lattice arrangements than by any lattice (Bezdek–Kuperberg). Hales proved the honeycomb conjecture in 1999. Density issues are subtle here, e.g. in 2d versus 1d.

On the same page there is a computation: with $a^2 b = 1$,

$$\begin{bmatrix} 1 & & \\ & a & \\ & & a^2 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ a \\ a^2 \end{pmatrix}, \qquad \begin{bmatrix} 1 & & \\ & a & \\ & & a^2 \end{bmatrix} \begin{pmatrix} a \\ a \\ b \end{pmatrix} = \begin{pmatrix} a \\ a^2 \\ 1 \end{pmatrix}.$$

? purpose unclear; maybe an ellipsoid-packing example

6.6 Extension: why root lattices pack well (in small dimensions)

[Added] This section is an added extension, not in the handwritten notes. It is set in black for readability.

Lecture 4 opened with two motivations: root lattices give good packings in low dimensions, and such simple constructions do not extend to high dimensions. Here are some reasons for both.

1. The density is written in the diagram. Normalize the roots to have norm 2. Then the Gram matrix of the simple roots is the Cartan matrix C, the shortest vectors are the roots, and

$$\delta(\text{root lattice}) = \frac{\operatorname{vol} B^n\left(1/\sqrt{2}\right)}{\sqrt{\det C}}$$

(Interlude A, Section A.9). A dense lattice needs a large minimum and a small covolume. Root lattices have both: the minimum is fixed at $\sqrt{2}$, and $\det C$ is tiny, namely $n+1$ for A_n, 4 for D_n, and $3, 2, 1$ for E_6, E_7, E_8. Against Rogers' simplex bound (Section 2.7):

n	lattice	$\det C$	kissing number $\lvert\Delta\rvert$	density	density/σ_n
2	A_2	3	6	0.9069	1.000
3	A_3	4	12	0.7405	0.950
4	D_4	4	24	0.6169	0.952
5	D_5	4	40	0.4653	0.885
6	E_6	3	72	0.3729	0.890
7	E_7	2	126	0.2953	0.895
8	E_8	1	240	0.2537	0.988

These are the densest lattices in their dimensions, and their kissing numbers $\lvert\Delta\rvert$ are the largest possible for lattices there. Among all arrangements of touching balls, $6, 12, 24, 240$ are known to be optimal in dimensions $2, 3, 4, 8$; for $n = 5, 6, 7$ the values $40, 72, 126$ are only the best known.

2. The roots are balanced: eutaxy from symmetry. Let $S = \sum_{\alpha\in\Delta} \alpha\alpha^T$. For any w in the Weyl group, w permutes Δ, so $wSw^{-1} = S$. For an irreducible root system the Weyl group acts irreducibly on $\mathbb{R}^n$, so each eigenspace of the symmetric matrix S is a nonzero W-invariant subspace and hence all of $\mathbb{R}^n$. Thus $S = cI$, and taking traces gives $c = 2\lvert\Delta\rvert/n$:

$$\sum_{\alpha\in\Delta} \langle\alpha, x\rangle^2 = \frac{2\lvert\Delta\rvert}{n}\,\lvert x\rvert^2 \qquad \text{for all } x.$$

The minimal vectors pull equally in every direction (they form a *spherical 2-design*). In Voronoi's language the lattice is *strongly eutactic*.

3. The roots are plentiful: perfection. A lattice is *perfect* if its minimal vectors determine its Gram matrix: the quadratic forms $x \mapsto \langle\alpha, x\rangle^2$, α minimal, span the whole $\frac{n(n+1)}{2}$-dimensional space of quadratic forms. For A_n, inside the hyperplane $\sum x_i = 0$ of $\mathbb{R}^{n+1}$, this is a pleasant exercise in graph Laplacians. A quadratic form on the hyperplane is the same thing as a symmetric

$(n+1) \times (n+1)$ matrix Q with zero row sums, and every such matrix is a weighted Laplacian,

$$Q = \sum_{i<j} (-q_{ij})\,(e_i - e_j)(e_i - e_j)^T,$$

that is, a combination of the forms $(x_i - x_j)^2 = \langle e_i - e_j, x\rangle^2$ attached to the positive roots. So A_n is perfect, with exactly as many positive roots, $\frac{n(n+1)}{2}$, as the dimension requires. D_n ($n \geq 4$), E_6, E_7 and E_8 are perfect too (Coxeter, *Extreme forms*, 1951).

Theorem 6.14 (Voronoi 1908)**.** *A lattice is* extreme*, that is, a local maximum of the packing density among lattices, if and only if it is perfect and eutactic. (The maximum is then strict up to similarity, since perfect lattices are isolated.)*

So *every irreducible root lattice is a local optimum for lattice packing.* Symmetry (irreducibility of W) gives eutaxy, and abundance of roots gives perfection. Voronoi also showed that each dimension has only finitely many perfect forms up to similarity. There are $1, 1, 1, 2, 3, 7, 33, 10916$ of them in dimensions $1, \ldots, 8$ (the last count is due to Dutour Sikirić, Schürmann and Vallentin, 2007), so in principle the densest lattice can be found by a finite search. In dimension 2 the only perfect form is the hexagonal A_2, which recovers Lagrange's theorem from Lecture 1.

4. Why it stops working. Two numbers are pinned down by the classification, and both are fatal in high dimension.

- *Too few roots.* Root lattices have $O(n^2)$ minimal vectors (D_n has $2n(n-1)$), but the best kissing numbers grow exponentially in n. There is only a finite list of exceptional diagrams, so there is only a finite budget of exceptional symmetry.
- *An integral lattice cannot spread out.* The minimum stays at $\sqrt{2}$ while $\det C$ stays bounded, so Hermite's invariant $\min^2 / \det^{1/n} = 2/(\det C)^{1/n}$ stays near 2. The best lattices have Hermite invariant growing linearly in n (Minkowski–Hlawka; Interlude B). Concretely, $\delta(D_n) = \operatorname{vol} B^n(1/\sqrt{2})/2$ decays like $(\pi e/n)^{n/2}$, up to polynomial factors. From $n = 28$ on it is below even the trivial 2^{-n} of Lecture 1.

5. Filling the holes. The first repair is to fill the deep holes, as in the construction of E_8. The deep holes of D_n include $h = (\frac{1}{2}, \ldots, \frac{1}{2})$, at distance $\sqrt{n}/2$ from the lattice. For even $n \geq 8$, $D_n^+ = D_n \cup (D_n + h)$ is again a lattice with minimum $\sqrt{2}$, now of covolume 1, so

$$\delta(D_n^+) = \operatorname{vol} B^n\big(1/\sqrt{2}\big) \qquad (n \geq 8 \text{ even}).$$

At $n = 8$ this is E_8, and the added balls exactly touch the old ones. Gluing (and its cousin, lamination) is how many of the best known lattices up to dimension 24 are built, culminating in the Leech lattice. But it only postpones the decay: $\delta(D_{10}^+) \approx 0.0797$, while the best known lattice in dimension 10, the laminated lattice Λ_{10}, has density about 0.0920.

Projects and quiz

admin (date, room) omitted

Ideas for projects:

- error correcting codes and information theory;

- other sphere packing bounds;
- entropy of jammed systems;
- translation of an older work on packings.

? last item barely legible

Format of the quiz: definitions, short answers, sketches.

Chapter A

Interlude A: From Dynkin diagrams to Lie algebras and Lie groups

[Added] This interlude is new: it is not in the handwritten notes. It picks up the remark at the start of Lecture 4, that the classification of root systems is "half of the classification of complex semisimple Lie algebras", and follows it. To keep it readable it is set in black; all of it should be read as added material.

In Lectures 4–6 we took the root system axioms as given, classified the possible Dynkin diagrams, and then built each root system by hand from lattices. Two questions were left hanging.

- Where do the axioms come from? Why should anyone care about a finite set of vectors closed under reflections, with the integrality condition $n_{\beta\alpha} \in \mathbb{Z}$?
- The same root system A_{n-1} appears twice in these notes, once as the vectors $e_i - e_j$ in the hyperplane $\sum x_i = 0$ (Lecture 6) and, later, once as the walls $x_i = x_j$ of $\mathrm{Conf}(N, \mathbb{R})$ (Lecture 8). Is there a single object behind both?

The answer to both is *matrices*. In this interlude we find root systems inside matrix algebras, see that the crystallographic axioms are forced by a small piece of linear algebra, and then use the exponential map to pass from the algebra to groups of matrices. At the end the root lattice comes back, this time sitting inside a group.

Throughout, M_n denotes the $n \times n$ complex matrices.

A.1 The roots of A_{n-1} inside matrices

Matrix units

Let E_{ij} be the matrix with a 1 in position (i, j) and zeros elsewhere. For 3×3 matrices, for instance,

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E_{23} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

and multiplying them out,

$$E_{12}E_{23} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = E_{13}, \qquad E_{23}E_{12} = 0.$$

The pattern is easiest to see by watching what E_{ij} does to the standard basis: it sends e_j to e_i and kills every other e_m. Think of E_{ij} as an *arrow from j to i*. A product $E_{ij}E_{kl}$ first follows the arrow $l \to k$ and then the arrow $j \to i$. The arrows compose when the head of the first meets the tail of the second ($k = j$), and otherwise the product is zero:

$$E_{ij}E_{kl} = \delta_{jk}\, E_{il}. \tag{A.1}$$

A diagonal matrix is a combination of the "loops" E_{kk}: $h = \mathrm{diag}(h_1, \ldots, h_n) = \sum_k h_k E_{kk}$. By (A.1),

$$hE_{ij} = \sum_k h_k E_{kk}E_{ij} = h_i E_{ij}, \qquad E_{ij}h = \sum_k h_k E_{ij}E_{kk} = h_j E_{ij}:$$

multiplying by h on the left scales row i, and on the right scales column j. With $h = \mathrm{diag}(a, b, c)$:

$$hE_{12} = \begin{pmatrix} 0 & a & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E_{12}h = \begin{pmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Symmetry first: conjugating by diagonal matrices

What are the symmetries of the matrix algebra M_n? Any invertible g gives a map

$$\mathrm{Ad}_g \colon M_n \to M_n, \qquad X \mapsto gXg^{-1},$$

which respects both sums and products, since $g(XY)g^{-1} = (gXg^{-1})(gYg^{-1})$. This is the *adjoint action* of the group GL_n on its own matrices. ("Adjoint" is used in the old sense of "attached": the group acts on the space it lives in.) Each Ad_g is a linear map on the n^2-dimensional vector space M_n. For a general g it is hard to describe, but for a *diagonal* $t = \mathrm{diag}(t_1, \ldots, t_n)$ it is as simple as possible. Using the two formulas above (with t and $t^{-1} = \mathrm{diag}(t_1^{-1}, \ldots, t_n^{-1})$ in place of h),

$$tE_{ij}t^{-1} = t_i\, t_j^{-1}\, E_{ij}.$$

For example, with $t = \mathrm{diag}(t_1, t_2, t_3)$,

$$tE_{12}t^{-1} = \begin{pmatrix} t_1 & & \\ & t_2 & \\ & & t_3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} t_1^{-1} & & \\ & t_2^{-1} & \\ & & t_3^{-1} \end{pmatrix} = \begin{pmatrix} 0 & t_1/t_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \frac{t_1}{t_2}\, E_{12}.$$

So Ad_t *is a diagonal linear map in the basis* $\{E_{ij}\}$, with eigenvalue t_i/t_j on E_{ij}. Better still, the same basis diagonalizes Ad_t for *every* diagonal t at once. Writing the eigenvalue on E_{ij} in position (i, j), for $n = 3$:

$$\mathrm{Ad}_t \text{ has eigenvalues } \begin{pmatrix} 1 & t_1/t_2 & t_1/t_3 \\ t_2/t_1 & 1 & t_2/t_3 \\ t_3/t_1 & t_3/t_2 & 1 \end{pmatrix}.$$

The diagonal matrices themselves are fixed (eigenvalue 1), since diagonal matrices commute.

Differentiating: the bracket appears

Multiplicative eigenvalues like t_1/t_2 are awkward, so take logarithms. Put

$$t(s) = \operatorname{diag}(e^{sh_1}, \dots, e^{sh_n}),$$

a curve of diagonal matrices with $t(0) = I$ and velocity $t'(0) = h = \operatorname{diag}(h_1, \dots, h_n)$. Then

$$\operatorname{Ad}_{t(s)} E_{ij} = e^{s(h_i - h_j)} E_{ij}.$$

Differentiate at $s = 0$. On the right we get $(h_i - h_j)E_{ij}$. On the left, the product rule applied to $t(s)Xt(s)^{-1}$, with $\frac{d}{ds}\big|_0 t(s)^{-1} = -h$, gives for any matrix X

$$\frac{d}{ds}\Big|_{s=0} t(s)\, X\, t(s)^{-1} = hX - Xh.$$

This is where the *bracket* comes from. Define

$$\operatorname{ad}_h \colon M_n \to M_n, \qquad \operatorname{ad}_h(X) = [h, X] := hX - Xh.$$

Lower-case ad is the *derivative* of upper-case Ad: conjugation seen infinitesimally. We did not invent the commutator; differentiating the symmetry produced it. (In Section A.7 we will integrate this back up: $\operatorname{Ad}_{e^X} = e^{\operatorname{ad}_X}$.) Since $\operatorname{Ad}_{t(s)}$ is diagonal on each E_{ij}, so is its derivative:

$$\operatorname{ad}_h(E_{ij}) = hE_{ij} - E_{ij}h = (h_i - h_j)\, E_{ij},$$

which you can also read off the two 3×3 matrices above: $hE_{12} - E_{12}h = (a - b)E_{12}$.

The eigenvalues are the roots

Now regard the eigenvalue as a function of h. On E_{ij} it is the linear functional $h \mapsto h_i - h_j$. If we identify functionals with vectors using the standard inner product, so that $h \mapsto h_i$ corresponds to e_i, then the eigenvalue on E_{ij} is the vector $e_i - e_j$. So:

the nonzero eigenvalues of ad_h*, as h runs over diagonal matrices, are exactly the roots $e_i - e_j$ of A_{n-1}.*

For $n = 3$, with $\alpha_1 = e_1 - e_2$ and $\alpha_2 = e_2 - e_3$, write the eigenvalue of E_{ij} in position (i, j). The root system A_2 appears in the shape of a 3×3 matrix, with positive roots above the diagonal and negative roots below:

$$\begin{pmatrix} 0 & e_1 - e_2 & e_1 - e_3 \\ e_2 - e_1 & 0 & e_2 - e_3 \\ e_3 - e_1 & e_3 - e_2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \alpha_1 & \alpha_1 + \alpha_2 \\ -\alpha_1 & 0 & \alpha_2 \\ -\alpha_1 - \alpha_2 & -\alpha_2 & 0 \end{pmatrix}.$$

Where is the hyperplane $\sum x_i = 0$? Scalar matrices $h = cI$ commute with everything, so they tell us nothing, and it is natural to discard them. That leaves the *traceless* matrices

$$\mathfrak{sl}_n = \{X \in M_n : \operatorname{tr} X = 0\},$$

whose diagonal part

$$\mathfrak{h} = \{\operatorname{diag}(h_1, \dots, h_n) : h_1 + \cdots + h_n = 0\}$$

is exactly the hyperplane $\sum x_i = 0$ of Lecture 6, of dimension $n-1$. Every traceless matrix splits into its diagonal part and its off-diagonal entries. For $n = 3$:

$$\begin{pmatrix} a & x_{12} & x_{13} \\ x_{21} & b & x_{23} \\ x_{31} & x_{32} & c \end{pmatrix} = \underbrace{\operatorname{diag}(a,b,c)}_{\in \mathfrak{h},\ a+b+c=0} + x_{12}E_{12} + x_{23}E_{23} + x_{13}E_{13} + x_{21}E_{21} + x_{32}E_{32} + x_{31}E_{31},$$

that is,

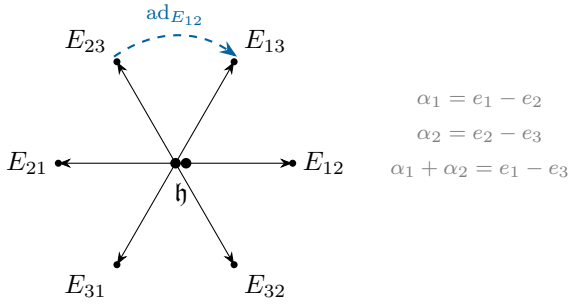
$$\mathfrak{sl}_3 = \mathfrak{h} \oplus \underbrace{\mathbb{C}E_{12}}_{\alpha_1} \oplus \underbrace{\mathbb{C}E_{23}}_{\alpha_2} \oplus \underbrace{\mathbb{C}E_{13}}_{\alpha_1+\alpha_2} \oplus \underbrace{\mathbb{C}E_{21}}_{-\alpha_1} \oplus \underbrace{\mathbb{C}E_{32}}_{-\alpha_2} \oplus \underbrace{\mathbb{C}E_{31}}_{-\alpha_1-\alpha_2}, \qquad 8 = 2 + 6.$$

In general we get the decomposition into eigenspaces of $\operatorname{ad}\mathfrak{h}$,

$$\mathfrak{sl}_n = \mathfrak{h} \ \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha, \qquad \mathfrak{g}_{e_i - e_j} = \mathbb{C}E_{ij}, \tag{A.2}$$

where $\Delta = \{e_i - e_j : i \neq j\}$ is the root system A_{n-1} and $\mathfrak{g}_\alpha$ is the space on which every ad_h acts by the scalar $\alpha(h)$. As a check on dimensions: $n^2 - 1 = (n-1) + n(n-1)$, which is (rank) + (number of roots), using the count $|\Delta| = n(n-1)$ from Lecture 6.

Example A.1 ($\mathfrak{sl}_3$)**.** Here $\mathfrak{h}$ is two-dimensional and the six off-diagonal matrix units sit at the six roots of A_2. Placing each E_{ij} at its root gives the familiar hexagon:



The two dots at the centre stand for the two-dimensional zero eigenspace $\mathfrak{h}$. The dashed arrow anticipates Section A.2: bracketing with E_{12} moves E_{23} to E_{13}, that is, it *adds the root* α_1.

The choices made in Lecture 4 also have matrix meanings. Take the direction $d = (n, n-1, \ldots, 1)$, so that $\langle e_i - e_j, d\rangle = j - i$. Then:

- the positive roots $e_i - e_j$ with $i < j$ are the matrix units *above* the diagonal, and the negative roots are those below it;
- the simple roots $\alpha_i = e_i - e_{i+1}$ are the entries just above the diagonal, $E_{12}, E_{23}, \ldots, E_{n-1,n}$;
- every positive root is a sum of simple roots: $e_i - e_j = \alpha_i + \alpha_{i+1} + \cdots + \alpha_{j-1}$. On the matrix side this is the statement that E_{ij} is an iterated product (indeed an iterated commutator) of superdiagonal units.

The Weyl group, generated by the reflections $s_{e_i - e_j}$, acts on $\mathfrak{h}$ by swapping h_i and h_j. It is the symmetric group Σ_n permuting the diagonal entries, which is conjugation by permutation matrices. The Weyl chamber containing d is $\{h_1 > h_2 > \cdots > h_n\}$, one of the $n!$ chambers of $\operatorname{Conf}(n, \mathbb{R})$ from Lecture 8.

A.2 The bracket and the definition of a Lie algebra

In Section A.1 the bracket appeared as the derivative of conjugation by diagonal matrices. Nothing there needed h to be diagonal: if $g(s)$ is any curve of invertible matrices with $g(0) = I$ and velocity $g'(0) = X$, the same product-rule computation gives

$$\frac{d}{ds}\Big|_{s=0} g(s)\, Y\, g(s)^{-1} = XY - YX = [X, Y] = \mathrm{ad}_X(Y).$$

So every matrix X has its linear map $\mathrm{ad}_X \colon M_n \to M_n$, the infinitesimal conjugation in the direction X. Diagonal X gave diagonal ad_X; for a matrix unit, for instance, $\mathrm{ad}_{E_{12}}$ is nilpotent and moves matrix units around, as in $\mathrm{ad}_{E_{12}}(E_{23}) = E_{12}E_{23} - E_{23}E_{12} = E_{13}$ by (A.1).

The operation $[X, Y] = XY - YX$ is the *commutator*. It has three properties that are easy to check:

(i) it is bilinear;

(ii) it is antisymmetric, $[X, Y] = -[Y, X]$;

(iii) it satisfies the *Jacobi identity*

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]],$$

which says that $\mathrm{ad}_X = [X, \cdot]$ obeys the product rule (it is a derivation of the bracket). Expanding both sides into products of three matrices proves it.

Definition A.2. A *Lie algebra* over a field k is a vector space $\mathfrak{g}$ with a bilinear map $[\cdot,\cdot] \colon \mathfrak{g}\times\mathfrak{g} \to \mathfrak{g}$ that is antisymmetric and satisfies the Jacobi identity. A *subalgebra* is a subspace closed under the bracket; an *ideal* is a subspace I with $[\mathfrak{g}, I] \subseteq I$. A Lie algebra is *simple* if it is not abelian and has no ideals other than 0 and $\mathfrak{g}$. A *representation* of $\mathfrak{g}$ on a vector space V is a linear map $\rho \colon \mathfrak{g} \to \mathrm{End}(V)$ with $\rho([X, Y]) = \rho(X)\rho(Y) - \rho(Y)\rho(X)$.

Example A.3.

1. $\mathfrak{gl}_n = M_n$ with the commutator.
2. $\mathfrak{sl}_n$ is a subalgebra, since $\mathrm{tr}(XY - YX) = 0$ for all X, Y. It is simple for $n \geq 2$.
3. $\mathfrak{so}_n = \{X \in M_n : X^T = -X\}$ is a subalgebra: if X, Y are antisymmetric, then $[X, Y]^T = Y^T X^T - X^T Y^T = YX - XY = -[X, Y]$.
4. $\mathbb{R}^3$ with the cross product is a Lie algebra (Jacobi is the identity $a \times (b \times c) = (a \times b) \times c + b \times (a \times c)$). Sending $\omega \in \mathbb{R}^3$ to the matrix X_ω with $X_\omega v = \omega \times v$ gives an isomorphism $(\mathbb{R}^3, \times) \cong \mathfrak{so}_3(\mathbb{R})$.
5. By the Jacobi identity, $X \mapsto \mathrm{ad}_X$ is a representation of $\mathfrak{g}$ on itself, the *adjoint representation*. Section A.1 was a study of the adjoint representation of $\mathfrak{sl}_n$, restricted to $\mathfrak{h}$.

The Jacobi identity also explains the dashed arrow in Example A.1. Suppose $[h, x] = \alpha(h)x$ and $[h, y] = \beta(h)y$ for all $h \in \mathfrak{h}$. Then

$$[h, [x, y]] = [[h, x], y] + [x, [h, y]] = \big(\alpha(h) + \beta(h)\big)[x, y].$$

Lemma A.4. *$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$, where $\mathfrak{g}_{\alpha+\beta} = 0$ if $\alpha + \beta$ is neither a root nor 0, and $\mathfrak{g}_0 = \mathfrak{h}$.*

So the bracket moves around the root diagram by *adding roots*. In $\mathfrak{sl}_3$, $[E_{12}, E_{23}] = E_{13}$ is the statement $\alpha_1 + \alpha_2 = e_1 - e_3$, while $[E_{12}, E_{13}] = 0$ because $2e_1 - e_2 - e_3$ is not a root.

A.3 The atom: $\mathfrak{sl}_2$

Fix a root $\alpha = e_i - e_j$ of A_{n-1} and put

$$e_\alpha = E_{ij}, \qquad f_\alpha = E_{ji}, \qquad h_\alpha = [e_\alpha, f_\alpha] = E_{ii} - E_{jj} \in \mathfrak{h}.$$

(The middle equality is (A.1): $E_{ij}E_{ji} - E_{ji}E_{ij} = E_{ii} - E_{jj}$.) The smallest case is $n = 2$, where

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are a basis of $\mathfrak{sl}_2$. Multiplying out,

$$he = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad eh = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}, \quad ef = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad fe = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so $[h, e] = 2e$, $[e, f] = h$, and similarly $[h, f] = -2f$. The same computation with (A.1) works for any root:

$$[h_\alpha, e_\alpha] = 2e_\alpha, \qquad [h_\alpha, f_\alpha] = -2f_\alpha, \qquad [e_\alpha, f_\alpha] = h_\alpha. \tag{A.3}$$

(The first is $\mathrm{ad}_{h_\alpha}(E_{ij}) = (h_i - h_j)E_{ij}$ with $h_i = 1$, $h_j = -1$.) So every root α comes with a copy $\mathfrak{s}_\alpha = \mathrm{span}(e_\alpha, f_\alpha, h_\alpha) \cong \mathfrak{sl}_2$ sitting inside $\mathfrak{sl}_n$.

Now look at how h_α acts on the other root spaces. For a root $\beta = e_k - e_l$,

$$\beta(h_\alpha) = (\delta_{ki} - \delta_{kj}) - (\delta_{li} - \delta_{lj}) = \langle \beta, \alpha \rangle = \frac{2\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} = n_{\beta\alpha},$$

since $\langle \alpha, \alpha \rangle = 2$. So *the Cartan integer $n_{\beta\alpha}$ is the eigenvalue of h_α on $\mathfrak{g}_\beta$.* (h_α is called the *coroot* of α.) To see why these eigenvalues have to be integers, we need one theorem about $\mathfrak{sl}_2$.

Theorem A.5 (Representations of $\mathfrak{sl}_2$)**.** *Let V be a finite-dimensional complex representation of $\mathfrak{sl}_2$. Then h acts diagonalizably on V, its eigenvalues are integers, and for every $k \in \mathbb{Z}$ the eigenspaces satisfy $\dim V_k = \dim V_{-k}$. If V is irreducible of dimension $m+1$, then the eigenvalues of h are $m, m-2, \ldots, -m$, each with multiplicity one.*

Proof for irreducible V. The key observation is that e raises h-eigenvalues by 2 and f lowers them by 2. If $hv = \lambda v$, then by (A.3)

$$h(ev) = e(hv) + [h, e]v = (\lambda + 2)\, ev, \qquad h(fv) = (\lambda - 2)\, fv.$$

Take any eigenvector of h and apply e repeatedly. The vectors obtained are eigenvectors for distinct eigenvalues while they are nonzero, and V is finite-dimensional, so eventually we reach a *highest weight vector*: $v \neq 0$ with $hv = \lambda v$ and $ev = 0$. Put $v_k = f^k v$. Then $hv_k = (\lambda - 2k)v_k$, and induction on k using $ef = fe + h$ gives

$$e\, v_k = k(\lambda - k + 1)\, v_{k-1}.$$

Let m be the largest index with $v_m \neq 0$. Then $0 = e\, v_{m+1} = (m+1)(\lambda - m)\, v_m$, so $\lambda = m$. In particular λ is a non-negative integer. The span of $v_0, \ldots, v_m$ is closed under e, f and h, so by irreducibility it is all of V. The eigenvalues are therefore $m, m-2, \ldots, -m$, as claimed. $\square$

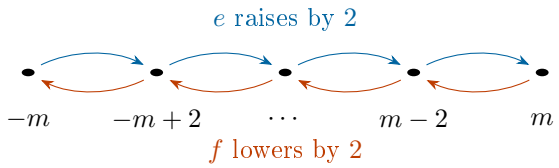


Figure A.1: The $(m+1)$-dimensional irreducible representation of $\mathfrak{sl}_2$: a string of h-eigenvalues, symmetric about 0.

For a general finite-dimensional V one uses *Weyl's complete reducibility theorem*: V is a direct sum of irreducible representations (see Humphreys, *Introduction to Lie Algebras and Representation Theory*, §6.3, or Hall, *Lie Groups, Lie Algebras, and Representations*, Chs. 4 and 10). The statements about eigenvalues then follow from the irreducible case.

Example A.6 (An $\mathfrak{sl}_2$ inside $\mathfrak{sl}_3$)**.** Take $\alpha = \alpha_1 = e_1 - e_2$ in $\mathfrak{sl}_3$, so $e_\alpha = E_{12}$, $f_\alpha = E_{21}$ and $h_\alpha = \operatorname{diag}(1,-1,0)$. Let it act by brackets on the two matrix units E_{23} (root α_2) and E_{13} (root $\alpha_1+\alpha_2$). By (A.1),

$$[E_{12},E_{23}] = E_{13}, \quad [E_{12},E_{13}] = 0, \quad [E_{21},E_{13}] = E_{23}, \quad [E_{21},E_{23}] = 0,$$

and h_α acts on E_{23} by $h_2 - h_3 = -1$ and on E_{13} by $h_1 - h_3 = +1$. So $\operatorname{span}(E_{23},E_{13})$ is the two-dimensional irreducible representation of $\mathfrak{sl}_2$ ($m=1$): a string of eigenvalues $-1,+1$, with E_{13} the highest weight vector. The eigenvalue -1 on E_{23} is the Cartan integer $n_{\alpha_2\alpha_1} = 2\langle\alpha_2,\alpha_1\rangle/\langle\alpha_1,\alpha_1\rangle = -1$, and the string $\alpha_2,\ \alpha_2+\alpha_1$ runs from α_2 to its mirror image $s_{\alpha_1}(\alpha_2) = \alpha_2+\alpha_1$. That is the reflection axiom happening in front of us.

A.4 Why the root system axioms hold

Now apply Theorem A.5 to the copy $\mathfrak{s}_\alpha \cong \mathfrak{sl}_2$, acting by brackets. Fix roots $\beta \neq \pm\alpha$ and form the *α-string through β*,

$$V = \bigoplus_{k\in\mathbb{Z}} \mathfrak{g}_{\beta+k\alpha}.$$

By Lemma A.4, bracketing with e_α or f_α shifts k by ±1, so V is a representation of $\mathfrak{s}_\alpha$. On the summand $\mathfrak{g}_{\beta+k\alpha}$, h_α acts by the scalar

$$(\beta+k\alpha)(h_\alpha) = n_{\beta\alpha} + 2k.$$

- **Axiom 4 (crystallographic).** $n_{\beta\alpha}$ is an eigenvalue of h_α on V (take $k=0$), so by Theorem A.5 it is an integer.

- **Axiom 3 (reflections).** The eigenvalues are symmetric about 0, so $-n_{\beta\alpha}$ is also an eigenvalue: $n_{\beta\alpha} + 2k = -n_{\beta\alpha}$ for some k with $\mathfrak{g}_{\beta+k\alpha} \neq 0$. That is, $k = -n_{\beta\alpha}$, and

  $$\beta - n_{\beta\alpha}\,\alpha = s_\alpha(\beta)$$

  is a root. The reflection s_α maps Δ to itself.

- **Strings have no gaps.** Each irreducible piece of V has eigenvalues forming an unbroken string with step 2. In fact V is irreducible here, since the root spaces turn out to be one-dimensional. So the roots $\beta + k\alpha$ form an unbroken string $\beta - p\alpha, \dots, \beta + q\alpha$ with $p - q = n_{\beta\alpha}$. This is the "$\alpha - \beta$" phenomenon behind Lemma 4.12 of Lecture 4.

For $\mathfrak{sl}_n$ all of this can of course be checked directly. The point is that the argument used only three ingredients: a commutative subalgebra $\mathfrak{h}$ acting diagonalizably, the copies of $\mathfrak{sl}_2$ attached to the roots, and finite-dimensionality. These are exactly what a *complex semisimple Lie algebra* provides. In such an algebra a maximal commutative subalgebra of diagonalizable elements (a *Cartan subalgebra*) always exists, and its nonzero eigenvalues form a reduced, crystallographic root system. (Reducedness, axiom 2, takes one more argument; see Humphreys §8.4.) *The root system axioms are the shadow of $\mathfrak{sl}_2$ representation theory.*

A.5 The other classical families

The families B_n, C_n, D_n from Lecture 6 come from the Lie algebras that preserve a bilinear form. Let J be an invertible $N \times N$ matrix and put

$$\mathfrak{g}_J = \{X \in M_N : X^T J + JX = 0\}.$$

This is a subalgebra (the same computation as for $\mathfrak{so}_n$), and in Section A.7 we will see that it consists of the *infinitesimal symmetries* of the form $\langle v, w\rangle_J = v^T J w$. For $J = I$ this is $\mathfrak{so}_N$.

The trick is to choose J so that a large family of *diagonal* matrices lies in $\mathfrak{g}_J$. Write $i' = N + 1 - i$ and let J be the antidiagonal matrix with $J_{i,i'} = 1$. For $X \in M_N$ one computes $(JX)_{ij} = X_{i'j}$ and $(X^T J)_{ij} = X_{j'i}$, so

$$X \in \mathfrak{g}_J \iff X_{ij} = -X_{j'i'} \text{ for all } i, j.$$

(Over $\mathbb{C}$ any two nondegenerate symmetric forms are equivalent, so $\mathfrak{g}_J \cong \mathfrak{so}_N(\mathbb{C})$.)

- The diagonal matrices in $\mathfrak{g}_J$ are $h = \mathrm{diag}(h_1, \dots, h_n, [0], -h_n, \dots, -h_1)$, where the middle 0 is present only when $N = 2n + 1$ is odd. These form a commutative subalgebra $\mathfrak{h}$ of dimension n.

- The matrices $E_{ij} - E_{j'i'}$, for pairs with $i \neq j$ and $j \neq i'$, span the rest of $\mathfrak{g}_J$ (for $j = i'$ the expression is 0: the antidiagonal of X vanishes). Each is an eigenvector of ad_h with eigenvalue $\tilde{h}_i - \tilde{h}_j$, where $\tilde{h}$ is the diagonal of h. (The two matrix units give the same eigenvalue, because $\tilde{h}_{k'} = -\tilde{h}_k$.)

- Reading off the eigenvalues: $i, j \leq n$ gives $\pm(e_i - e_j)$; $i \leq n < j$ gives $\pm(e_i + e_k)$ with $j = k'$ and $k \neq i$ (the case $k = i$ is excluded since then $j = i'$); and when N is odd the middle index gives $\pm e_i$.

So $\mathfrak{so}_{2n}$ has root system $\{\pm e_i \pm e_j\} = D_n$ and $\mathfrak{so}_{2n+1}$ has $\{\pm e_i \pm e_j\} \cup \{\pm e_i\} = B_n$. If instead $J = \left(\begin{smallmatrix} 0 & K \\ -K & 0 \end{smallmatrix}\right)$, with K the $n \times n$ antidiagonal matrix of ones, the form is *symplectic*. Now signs enter: with $\varepsilon_i = 1$ for $i \leq n$ and $\varepsilon_i = -1$ for $i > n$, the condition becomes $X_{ij} = -\varepsilon_i \varepsilon_j X_{j'i'}$, the diagonal part is the same $\mathfrak{h}$ as before, and the off-diagonal basis is $E_{ij} - \varepsilon_i \varepsilon_j E_{j'i'}$. The same bookkeeping gives the symplectic algebra $\mathfrak{sp}_{2n}$, whose root system is $\{\pm e_i \pm e_j\} \cup \{\pm 2e_i\} = C_n$: now the case $k = i$ is allowed, since for $j = i'$ the basis element is $2E_{ii'} \neq 0$, and it produces the long roots $\pm 2e_i$.

type	Lie algebra	condition	roots	rank $+\,\lvert\Delta\rvert$	$\dim \mathfrak{g}$
A_n	$\mathfrak{sl}_{n+1}$	$\operatorname{tr} X = 0$	$e_i - e_j$	$n + n(n+1)$	$(n+1)^2 - 1$
B_n	$\mathfrak{so}_{2n+1}$	$X^T J + JX = 0$	$\pm e_i \pm e_j,\ \pm e_i$	$n + 2n^2$	$n(2n+1)$
C_n	$\mathfrak{sp}_{2n}$	$X^T J + JX = 0$ (J skew)	$\pm e_i \pm e_j,\ \pm 2e_i$	$n + 2n^2$	$n(2n+1)$
D_n	$\mathfrak{so}_{2n}$	$X^T J + JX = 0$	$\pm e_i \pm e_j$	$n + 2n(n-1)$	$n(2n-1)$

In every row, $\dim \mathfrak{g} = \text{rank} + |\Delta|$, as it must be by the root decomposition (A.2). The root counts are the ones computed in Lecture 6.

Remark A.7 (The "repetitions" in the classification). Theorem 5.2 restricted the ranges of n (B_n for $n \geq 2$, C_n for $n \geq 3$, D_n for $n \geq 4$) to avoid repeating diagrams. Each coincidence of small diagrams is an isomorphism of Lie algebras: $B_1 = C_1 = A_1$ gives $\mathfrak{so}_3 \cong \mathfrak{sp}_2 = \mathfrak{sl}_2$; $B_2 = C_2$ gives $\mathfrak{so}_5 \cong \mathfrak{sp}_4$; $D_2 = A_1 \times A_1$ gives $\mathfrak{so}_4 \cong \mathfrak{sl}_2 \times \mathfrak{sl}_2$; and $D_3 = A_3$ gives $\mathfrak{so}_6 \cong \mathfrak{sl}_4$.

A.6 The classification of simple Lie algebras

Everything so far points to a dictionary between Dynkin diagrams and Lie algebras. The result is due to Killing (1888–90) and É. Cartan (thesis, 1894); the diagrams themselves are due to Dynkin (1946).

Theorem A.8 (Killing–Cartan classification)**.** *Let $\mathfrak{g}$ be a complex semisimple Lie algebra.*

1. *$\mathfrak{g}$ has a Cartan subalgebra $\mathfrak{h}$, unique up to automorphisms of $\mathfrak{g}$, and the nonzero eigenvalues of $\operatorname{ad}\mathfrak{h}$ form a root system $\Delta(\mathfrak{g})$ in (the real span of) $\mathfrak{h}^*$, with one-dimensional root spaces.*
2. *$\mathfrak{g}$ is simple if and only if $\Delta(\mathfrak{g})$ is irreducible.*
3. *$\mathfrak{g} \cong \mathfrak{g}'$ if and only if $\Delta(\mathfrak{g}) \cong \Delta(\mathfrak{g}')$.*
4. *(Existence; Serre's theorem) Every root system arises this way.*

Consequently the complex simple Lie algebras correspond exactly to the connected Dynkin diagrams A_n ($n \geq 1$), B_n ($n \geq 2$), C_n ($n \geq 3$), D_n ($n \geq 4$), E_6, E_7, E_8, F_4, G_2.

Serre's theorem even writes the algebra down from the diagram. Let $\alpha_1, \dots, \alpha_r$ be the simple roots and $a_{ij} = n_{\alpha_j \alpha_i}$ the *Cartan matrix*. Then $\mathfrak{g}$ is generated by e_i, f_i, h_i ($1 \leq i \leq r$), which are exactly the $\mathfrak{sl}_2$-triples attached to the simple roots, subject to the relations

$$[h_i, h_j] = 0, \quad [h_i, e_j] = a_{ij} e_j, \quad [h_i, f_j] = -a_{ij} f_j, \quad [e_i, f_j] = \delta_{ij} h_i,$$
$$(\operatorname{ad} e_i)^{1-a_{ij}}(e_j) = 0, \qquad (\operatorname{ad} f_i)^{1-a_{ij}}(f_j) = 0 \qquad (i \neq j).$$

The last two relations say that an α_i-string through α_j has exactly the length the Cartan integer predicts. For $\mathfrak{sl}_n$ this recovers $e_i = E_{i,i+1}$, $f_i = E_{i+1,i}$ and $h_i = E_{ii} - E_{i+1,i+1}$.

The dimension formula $\dim \mathfrak{g} = \text{rank} + |\Delta|$ and the root counts (12 for G_2 from Lecture 4; 48, 126, 240 from Lecture 6; and 72 for E_6, which one checks by counting the roots of E_8 orthogonal to an A_2) give the dimensions of the exceptional algebras:

	G_2	F_4	E_6	E_7	E_8
rank	2	4	6	7	8
$\lvert\Delta\rvert$	12	48	72	126	240
$\dim \mathfrak{g}$	14	52	78	133	248

In particular, the 240 minimal vectors of the E_8 lattice, the kissing configuration in dimension 8, *literally* label the root spaces of the largest exceptional Lie algebra, $\mathfrak{e}_8 = \mathfrak{h} \oplus \bigoplus_{\alpha \in E_8} \mathfrak{g}_\alpha$, $248 = 8 + 240$. The smallest one, $\mathfrak{g}_2$, is the algebra of derivations of the octonions (É. Cartan).

A.7 From the algebra to the group: the exponential map

The name "Lie algebra" comes from Sophus Lie's idea that a group of transformations is determined, near the identity, by its *infinitesimal* transformations. For matrices this can be made completely explicit.

Definition A.9. For $X \in M_n$ (real or complex) the *matrix exponential* is

$$\exp X = e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

The series converges absolutely, since $\|X^k\| \le \|X\|^k$ in the operator norm.

Proposition A.10. *For $X, Y \in M_n$ and $g \in \mathrm{GL}_n$:*

1. *if $XY = YX$, then $e^{X+Y} = e^X e^Y$; in particular $e^{(s+t)X} = e^{sX}e^{tX}$ and $(e^X)^{-1} = e^{-X}$;*
2. *$\frac{d}{dt} e^{tX} = X e^{tX} = e^{tX} X$;*
3. *$e^{gXg^{-1}} = g\, e^X g^{-1}$ and $(e^X)^T = e^{X^T}$;*
4. *$\det e^X = e^{\operatorname{tr} X}$;*
5. *$\exp$ maps a neighbourhood of 0 diffeomorphically onto a neighbourhood of I, with inverse $\log(I + A) = \sum_{k \ge 1} (-1)^{k+1} A^k / k$ for $\|A\| < 1$.*

Sketch. (1) For commuting X and Y the binomial theorem applies to $(X+Y)^k$, and the Cauchy product of the two series gives the result. (2) Differentiate the series term by term. (3) Apply the identities $(gXg^{-1})^k = gX^kg^{-1}$ and $(X^k)^T = (X^T)^k$ to each term. (4) Over $\mathbb{C}$, write $X = gTg^{-1}$ with T upper triangular (Schur). Then e^T is upper triangular with diagonal entries $e^{t_{ii}}$, so $\det e^X = \det e^T = e^{\sum t_{ii}} = e^{\operatorname{tr} X}$. (5) The derivative of exp at 0 is the identity, so the inverse function theorem applies; the series for log inverts exp near I as a formal identity of power series. □

By (1), $t \mapsto e^{tX}$ is a homomorphism $(\mathbb{R}, +) \to \mathrm{GL}_n$: a *one-parameter subgroup*. This is the matrix version of the one-parameter groups of diffeomorphisms of Lecture 9. The linear vector field $v \mapsto Xv$ on $\mathbb{R}^n$ generates the one-parameter group of diffeomorphisms $\varphi_t(v) = e^{tX}v$, because $\frac{d}{dt} e^{tX} v = X(e^{tX}v)$ by (2). So X is the "velocity" of the subgroup at the identity.

This also closes the loop on the names Ad and ad from Section A.1.

Proposition A.11. *For all $X, Y \in M_n$, $e^X Y e^{-X} = e^{\mathrm{ad}_X}(Y) = Y + [X,Y] + \frac{1}{2!}[X,[X,Y]] + \cdots$, that is, $\mathrm{Ad}_{e^X} = e^{\mathrm{ad}_X}$.*

Proof. Both $A(t) = \mathrm{Ad}_{e^{tX}}$ and $B(t) = e^{t\,\mathrm{ad}_X}$ are curves of linear maps on M_n with value id at $t = 0$. Both satisfy $\frac{d}{dt} C(t) = C(t) \circ \mathrm{ad}_X$: for A, $\frac{d}{dt} e^{tX} Y e^{-tX} = e^{tX}(XY - YX)e^{-tX}$ (because X commutes with e^{tX}), and for B this is property (2) of Proposition A.10, applied to the linear map ad_X of the n^2-dimensional space M_n. By uniqueness of solutions of linear ODEs, $A = B$; now put $t = 1$. □

For diagonal $X = h$ this is the formula we started from: ad_h has eigenvalue $h_i - h_j$ on E_{ij}, so e^{ad_h} has eigenvalue $e^{h_i - h_j} = t_i / t_j$, where $t = e^h$.

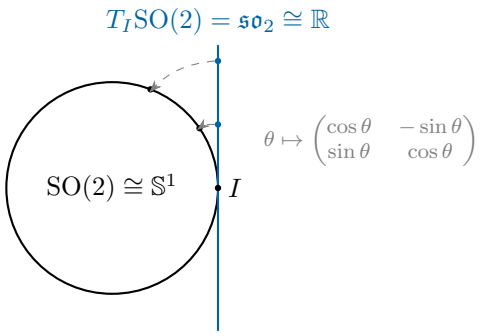


Figure A.2: The exponential map wraps the tangent line $\mathfrak{so}_2$ around the circle $\mathrm{SO}(2)$. Its kernel is $2\pi\mathbb{Z}$, a lattice.

Matrix groups and their Lie algebras

Definition A.12. A *matrix group* is a closed subgroup $G \subseteq \mathrm{GL}_n(\mathbb{R})$ or $\mathrm{GL}_n(\mathbb{C})$. Its *Lie algebra* is

$$\mathrm{Lie}(G) = \{X \in M_n : e^{tX} \in G \text{ for all } t \in \mathbb{R}\}.$$

Theorem A.13 (von Neumann 1929; Cartan 1930)**.** *Let G be a matrix group. Then* $\mathrm{Lie}(G)$ *is a real vector space closed under the commutator, so it is a Lie algebra. Moreover G is a smooth submanifold of M_n (a* Lie group*) with tangent space $T_I G = \mathrm{Lie}(G)$, and* exp *maps a neighbourhood of 0 in* $\mathrm{Lie}(G)$ *diffeomorphically onto a neighbourhood of I in G.*

We will not prove that G is a manifold (see Hall, Ch. 3, or Stillwell, *Naive Lie Theory*, Ch. 7), but it is worth seeing where the bracket comes from. Closure under addition follows from the *Lie product formula* $e^{X+Y} = \lim_{k\to\infty} \left(e^{X/k} e^{Y/k}\right)^k$. For the bracket, take $X, Y \in \mathrm{Lie}(G)$. For each t the matrix $e^{tX} Y e^{-tX}$ lies in $\mathrm{Lie}(G)$, because $\exp\!\big(s\, e^{tX} Y e^{-tX}\big) = e^{tX} e^{sY} e^{-tX} \in G$. Since $\mathrm{Lie}(G)$ is a closed subspace, it also contains the derivative

$$\frac{d}{dt}\Big|_{t=0} e^{tX} Y e^{-tX} = XY - YX = [X, Y].$$

Equivalently, expanding the group commutator gives

$$e^{sX} e^{tY} e^{-sX} e^{-tY} = I + st\,[X, Y] + (\text{terms of total degree} \geq 3 \text{ in } s, t).$$

The bracket is the infinitesimal shadow of the failure of the group to commute.

Example A.14 (The classical groups recover the classical algebras)**.**

- $\mathrm{Lie}(\mathrm{GL}_n) = M_n = \mathfrak{gl}_n$.
- $\mathrm{Lie}(\mathrm{SL}_n) = \mathfrak{sl}_n$. If $\operatorname{tr} X = 0$ then $\det e^{tX} = 1$ by Proposition A.10(4); conversely, $\det e^{tX} = e^{t \operatorname{tr} X} = 1$ for all t forces $\operatorname{tr} X = 0$.
- $\mathrm{Lie}(G_J) = \mathfrak{g}_J$ for the isometry group $G_J = \{g : g^T J g = J\}$ of the form J. If $X^T J + JX = 0$, then $X^T = -JXJ^{-1}$, so $(e^X)^T J e^X = J e^{-X} J^{-1} J e^X = J$; applying this to tX shows $\mathfrak{g}_J \subseteq \mathrm{Lie}(G_J)$. Conversely, differentiating $(e^{tX})^T J e^{tX} = J$ at $t = 0$ gives $X^T J + JX = 0$. Thus $\mathfrak{so}_N$ belongs to $\mathrm{O}(N)$ and $\mathrm{SO}(N)$, and $\mathfrak{sp}_{2n}$ to Sp_{2n}. This is why $\mathfrak{g}_J$ deserves the name "infinitesimal isometries".

- The unitary group $\mathrm{U}(n) = \{g \in \mathrm{GL}_n(\mathbb{C}) : g^* g = I\}$ has Lie algebra $\mathfrak{u}(n) = \{X : X^* = -X\}$ (skew-Hermitian matrices), and $\mathrm{SU}(n)$ has $\mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}_n$. Note that $\mathfrak{su}(n) \otimes \mathbb{C} = \mathfrak{sl}_n(\mathbb{C})$: the compact group $\mathrm{SU}(n)$ is a *real form* of $\mathfrak{sl}_n(\mathbb{C})$.

Example A.15 ($\mathrm{SO}(2)$ and $\mathrm{SO}(3)$)**.** For $X = \theta \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ we have $X^2 = -\theta^2 I$, and splitting the series into even and odd terms gives

$$e^X = \cos\theta\, I + \sin\theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

which is rotation by θ (Figure A.2). In $\mathrm{SO}(3)$, write $X_\omega v = \omega \times v$ with $\theta = |\omega|$. Using $X_\omega^3 = -\theta^2 X_\omega$, the same splitting gives *Rodrigues' formula*

$$e^{X_\omega} = I + \frac{\sin\theta}{\theta} X_\omega + \frac{1 - \cos\theta}{\theta^2} X_\omega^2,$$

which is rotation by angle θ about the axis ω. Every rotation has an axis, so exp is onto $\mathrm{SO}(3)$. It maps the closed ball $|\omega| \le \pi$ onto $\mathrm{SO}(3)$, identifying only antipodal points of the boundary sphere, so $\mathrm{SO}(3) \cong \mathbb{R}P^3$.

Remark A.16 (Same algebra, different groups). The group $\mathrm{SU}(2)$ of unit quaternions is a 3-sphere. It acts on the purely imaginary quaternions $\cong \mathbb{R}^3$ by conjugation, and this gives a two-to-one homomorphism $\mathrm{SU}(2) \to \mathrm{SO}(3)$ with kernel $\{\pm I\}$. The two groups have isomorphic Lie algebras, $\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong (\mathbb{R}^3, \times)$, but they are not isomorphic: one is simply connected, the other has $\pi_1 = \mathbb{Z}/2$. The Lie algebra determines a connected group only up to coverings. The missing global information turns out to be a *lattice*, as the next section shows.

A.8 Tori, Weyl groups, and the lattice inside the group

Return to A_{n-1}, now in the compact group $\mathrm{SU}(n)$. The diagonal matrices in $\mathrm{SU}(n)$ form the *maximal torus*

$$T = \big\{ \operatorname{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}) : \ \theta_1 + \cdots + \theta_n \equiv 0 \pmod{2\pi} \big\},$$

with Lie algebra

$$\mathfrak{t} = \{\operatorname{diag}(i\theta_1, \dots, i\theta_n) : \ \theta_1 + \cdots + \theta_n = 0\} \cong \{\theta \in \mathbb{R}^n : \textstyle\sum \theta_j = 0\},$$

which is again the hyperplane of Lecture 6. Since $\mathfrak{t}$ is commutative, $\exp\colon \mathfrak{t} \to T$ is a surjective homomorphism, and its kernel consists of the θ with every $\theta_j \in 2\pi\mathbb{Z}$ and $\sum \theta_j = 0$:

$$\ker\big(\exp|_{\mathfrak{t}}\big) = 2\pi \cdot \{x \in \mathbb{Z}^n : \textstyle\sum x_j = 0\} = 2\pi \cdot A_{n-1},$$

the A_{n-1} root lattice from Lecture 6. So

$$T \ \cong \ \mathfrak{t} \,/\, 2\pi A_{n-1} :$$

the maximal torus of $\mathrm{SU}(n)$ *is Euclidean space modulo the* A_{n-1} *root lattice.* For $\mathrm{SU}(3)$ it is the plane modulo the hexagonal lattice, the lattice of the densest disk packing from Lecture 1.

The Weyl group. Permutation matrices (with one sign changed if necessary, to make the determinant 1) normalize T and permute the θ_j. The resulting group

$$W = N(T)/T \cong \Sigma_n$$

is the Weyl group of A_{n-1}, generated by the reflections $s_{e_i-e_j}$, which swap θ_i and θ_j. By the spectral theorem every element of $\mathrm{SU}(n)$ is conjugate to an element of T, uniquely up to the action of W, so

$$\{\text{conjugacy classes of } \mathrm{SU}(n)\} \longleftrightarrow T/W \longleftrightarrow \left\{\begin{matrix}\text{unordered } n\text{-tuples of points} \\ \text{on } \mathbb{S}^1 \text{ with product } 1\end{matrix}\right\}.$$

Remark A.17 (Back to configuration spaces). An element of $\mathrm{SU}(n)$ is *regular* when its eigenvalues are distinct, that is, when its n eigenvalues form a configuration of distinct points on the circle. Choose real eigenvalue angles with $\sum\theta_j = 0$, ordered so that $\theta_1 \geq \theta_2 \geq \cdots \geq \theta_n \geq \theta_1 - 2\pi$ (this is always possible, in exactly one way), and record the gaps

$$g_k = \theta_k - \theta_{k+1}\ (1 \leq k < n), \qquad g_n = 2\pi - (\theta_1 - \theta_n).$$

The gaps are non-negative and sum to 2π, and together with $\sum\theta_j = 0$ they determine the θ_j. So T/W is a closed simplex of dimension $n-1$, the *Weyl alcove*. Its interior, the regular classes, is identified through the gaps with the reduced configuration space of n points on a circle in a fixed cyclic order (one of the open simplices of gaps in Lecture 7). For $n = 3$ it is the same open triangle of gap coordinates $(x, y, 1-x-y)$, up to the scale factor 2π. Its boundary walls, where two eigenvalues collide, are the walls $\theta_i = \theta_j$ of $\mathrm{Conf}(n, \mathbb{R})$ from Lecture 8, wrapped around the circle.

All of this generalizes.

Theorem A.18 (Maximal tori; Weyl, Cartan)**.** *Let G be a compact connected Lie group. Then G has a maximal torus T, unique up to conjugacy, and every element of G is conjugate into T. The Weyl group $W = N(T)/T$ is finite and is the Weyl group of the root system of $\mathfrak{g} \otimes \mathbb{C}$. The kernel of* $\exp\colon \mathfrak{t} \to T$ *is a lattice containing 2π times the coroot lattice (the coroots h_α placed in $\mathfrak{t}$ as ih_α, as in the* $\mathrm{SU}(n)$ *example), with equality exactly when G is simply connected; in general the quotient is $\pi_1(G)$. The compact simply connected simple Lie groups correspond to the connected Dynkin diagrams:*

$$\mathrm{SU}(n+1)\ (A_n), \quad \mathrm{Spin}(2n+1)\ (B_n), \quad \mathrm{Sp}(n)\ (C_n), \quad \mathrm{Spin}(2n)\ (D_n), \quad G_2,\ F_4,\ E_6,\ E_7,\ E_8.$$

For example, $\mathrm{SU}(2)$ has $T = \{\mathrm{diag}(e^{i\theta}, e^{-i\theta})\}$ and kernel $\theta \in 2\pi\mathbb{Z}$, while $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ has the finer kernel $\theta \in \pi\mathbb{Z}$. The covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ of the previous section is recorded by an index-two inclusion of lattices.

A.9 What else a Dynkin diagram knows

A Dynkin diagram is a tiny picture: a few dots, some single, double or triple edges, and a few arrows. It is remarkable how much can be read off from it. We collect the main items, with the $\mathfrak{sl}_n$ case as the running check.

1. The Cartan matrix. Number the nodes $1,\dots,r$. The Cartan matrix $a_{ij} = n_{\alpha_j\alpha_i} = 2\langle\alpha_j,\alpha_i\rangle/\langle\alpha_i,\alpha_i\rangle$ has 2 on the diagonal, 0 for non-adjacent nodes, and $-1,-1$ for a single edge. For a double (triple) edge the entry in the row of the *short* root is -2 (-3) and the other is -1. For example, with α_1 long and α_2 short (for B_2: $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2$, so $a_{21} = 2\langle\alpha_1,\alpha_2\rangle/\langle\alpha_2,\alpha_2\rangle = -2$),

$$A_2:\ \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad B_2:\ \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}, \qquad G_2:\ \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

This is the convention that makes the Serre relations read $[h_i, e_j] = a_{ij}e_j$ (it is Kac's); Humphreys and Bourbaki use the transpose, so their B_2 matrix has the -2 in the first row. From the Cartan matrix the whole root system, and by Serre's theorem the whole Lie algebra, can be rebuilt (Section A.6).

2. The Weyl group, by generators and relations. The Coxeter graph gives a presentation

$$W = \big\langle s_1,\dots,s_r \;\big|\; s_i^2 = 1,\ (s_is_j)^{m_{ij}} = 1\big\rangle, \qquad m_{ij} = 2,3,4,6 \text{ for } 0,1,2,3 \text{ edges.}$$

For A_{n-1} this is the classical presentation of Σ_n by adjacent transpositions: $s_is_j = s_js_i$ for $|i-j| \geq 2$ and $s_is_{i+1}s_i = s_{i+1}s_is_{i+1}$. Dropping the relations $s_i^2 = 1$ gives Artin's *braid group* (Interlude D, Section D.4).

3. Counting: roots, Coxeter number, exponents, $|W|$. Let $c = s_1s_2\cdots s_r$ be the product of all the simple reflections (a *Coxeter element*). Its order h is the *Coxeter number*, and its eigenvalues are $e^{2\pi i m_j/h}$ for integers $1 = m_1 \leq \cdots \leq m_r = h-1$, the *exponents*. Then

$$|\Delta| = rh, \qquad \#\{\text{positive roots}\} = m_1 + \cdots + m_r, \qquad |W| = \prod_{j=1}^{r}(m_j+1).$$

The numbers $d_j = m_j + 1$ are the degrees of the basic W-invariant polynomials (Chevalley). For A_{n-1} the invariants of Σ_n are the elementary symmetric polynomials of degrees $2,\dots,n$ (degree 1 vanishes on the hyperplane), so $|W| = 2\cdot 3\cdots n = n!$, and $h = n$. Moreover c permutes the roots in r orbits of size h. Projecting onto the plane where c acts as a rotation by $2\pi/h$ (the *Coxeter plane*), the roots land on r concentric regular h-gons. For E_8 these are the famous eight rings of 30 points.

4. The determinant: the density of the root lattice. For the simply laced diagrams (A, D, E), normalize all roots to have $\langle\alpha,\alpha\rangle = 2$. Then the Gram matrix of the simple roots *is* the Cartan matrix, so the root lattice has covolume $\sqrt{\det C}$. Its shortest vectors are the roots, of length $\sqrt{2}$, so the lattice packing has balls of radius $1/\sqrt{2}$ and

$$\delta(\text{root lattice}) = \frac{\operatorname{vol} B^r\big(1/\sqrt{2}\big)}{\sqrt{\det C}}.$$

The packing density of Lectures 1, 4 and 6 *can be read off the diagram*: $\det C = 3,4,4,3,2,1$ for $A_2, A_3, D_4, E_6, E_7, E_8$ gives

$$\frac{\pi}{\sqrt{12}}, \quad \frac{\pi}{\sqrt{18}}, \quad \frac{\pi^2}{16}, \quad \frac{\pi^3}{48\sqrt{3}}, \quad \frac{\pi^3}{105}, \quad \frac{\pi^4}{384}.$$

The same determinant has a Lie-theoretic meaning. It is the index of the root lattice in the weight lattice, which equals the order of the centre of the simply connected compact group, or equivalently $|\pi_1|$ of the adjoint group. For $\mathrm{SU}(n)$ (A_{n-1}, $\det C = n$) the centre is the n-th roots of unity times I. For E_8 ($\det C = 1$) the root lattice is unimodular and the group is simultaneously simply connected and adjoint.

5. Symmetries of the diagram: outer automorphisms. Automorphisms of the diagram give the automorphisms of $\mathfrak{g}$ that are not conjugations. A_n ($n \geq 2$), D_n ($n \geq 5$) and E_6 have a flip; D_4 has the symmetry group Σ_3 permuting its three outer nodes, whose elements of order three are called *triality*. For $\mathfrak{sl}_n$ the flip is, up to an inner automorphism, $X \mapsto -X^T$.

6. Folding. Fold a diagram along a symmetry, identifying the nodes in each orbit, and you get a non-simply-laced diagram. The corresponding Lie algebra is the subalgebra fixed by the symmetry:

$$A_{2n-1} \rightsquigarrow C_n, \qquad D_{n+1} \rightsquigarrow B_n, \qquad E_6 \rightsquigarrow F_4, \qquad D_4 \rightsquigarrow G_2 \text{ (by triality)}.$$

Here the symmetry σ has to be lifted to $\mathfrak{g}$ by permuting the Serre generators, $e_i \mapsto e_{\sigma(i)}$, $f_i \mapsto f_{\sigma(i)}$. For $\mathfrak{sl}_{2n}$ a lift (up to signs, which can be absorbed into the generators) is $X \mapsto -JX^TJ^{-1}$ with the skew J of Section A.5, and its fixed points are exactly $\mathfrak{sp}_{2n}$; the plain flip $X \mapsto -X^T$ differs from it by an inner automorphism and fixes $\mathfrak{so}_{2n}$ instead. Which node becomes short is visible on the diagram: averaging the roots in an orbit of two orthogonal simple roots halves the squared length. So in A_{2n-1} the one fixed (middle) node stays long, giving C_n, while in D_{n+1} the two swapped end nodes become one short node, giving B_n. So the multiple edges remember a hidden symmetry of a simply laced diagram. Folding with the golden ratio instead leads to the non-crystallographic $E_8 \rightsquigarrow H_4$, $D_6 \rightsquigarrow H_3$, $A_4 \rightsquigarrow H_2$ and to quasicrystals (Interlude C).

7. The extended diagram and McKay's correspondence. Adding the negative of the highest root as an extra node gives the *extended* (affine) Dynkin diagram, the diagram of the affine Weyl group that tiles $\mathfrak{t}$ by alcoves (Section A.8). For the simply laced types something unexpected happens. The finite subgroups $\Gamma \subset \mathrm{SU}(2)$ (cyclic, binary dihedral, binary tetrahedral, octahedral, icosahedral) correspond exactly to the extended diagrams $\tilde{A}_{n-1}, \tilde{D}_n, \tilde{E}_6, \tilde{E}_7, \tilde{E}_8$ (McKay, 1980). The nodes are the irreducible representations of Γ, and the edges record tensoring with the two-dimensional representation. So the binary icosahedral group, the 120 unit quaternions that are the vertices of the 600-cell and the double cover of the rotation group of the icosahedron of Interlude C, *is* $\tilde{E}_8$. The same ADE list, now with the ordinary (unextended) diagrams, classifies the simple surface singularities $\mathbb{C}^2/\Gamma$, whose minimal resolutions have the Dynkin diagram as dual graph (du Val, 1934; Arnold, 1972), and the quivers of finite representation type (Gabriel, 1972).

8. Representations: label the nodes. The irreducible finite-dimensional representations of $\mathfrak{g}$ correspond to labellings of the nodes by non-negative integers $(n_1, \ldots, n_r)$, the coordinates of the highest weight in the basis of fundamental weights. Their dimensions are given by Weyl's dimension formula. For $\mathfrak{sl}_2$ the label m gives dimension $m+1$, as in Theorem A.5. For $\mathfrak{sl}_3$ the labels (a, b) give

$$\dim V_{(a,b)} = \tfrac{1}{2}(a+1)(b+1)(a+b+2),$$

so $(1,0)$ is $\mathbb{C}^3$ (dimension 3) and $(1,1)$ is the adjoint representation (dimension 8).

type	$\lvert\Delta\rvert$	h	exponents	$\lvert W\rvert$	$\det C$	$\dim \mathfrak{g}$
A_n	$n(n+1)$	$n+1$	$1, 2, \ldots, n$	$(n+1)!$	$n+1$	$n(n+2)$
B_n, C_n	$2n^2$	$2n$	$1, 3, \ldots, 2n-1$	$2^n n!$	2	$n(2n+1)$
D_n	$2n(n-1)$	$2n-2$	$1, 3, \ldots, 2n-3;\ n-1$	$2^{n-1}n!$	4	$n(2n-1)$
E_6	72	12	$1, 4, 5, 7, 8, 11$	51 840	3	78
E_7	126	18	$1, 5, 7, 9, 11, 13, 17$	2 903 040	2	133
E_8	240	30	$1, 7, 11, 13, 17, 19, 23, 29$	696 729 600	1	248
F_4	48	12	$1, 5, 7, 11$	1 152	1	52
G_2	12	6	$1, 5$	12	1	14

(References: Humphreys, *Reflection Groups and Coxeter Groups*, §§3.7–3.9 and 3.16–3.20; Bourbaki, *Lie Groups and Lie Algebras*, Ch. VI, Plates; McKay, *Graphs, singularities, and finite groups*, Proc. Sympos. Pure Math. 37 (1980).)

A.10 Summary: a dictionary

Packings and lattices (Lectures 4–8)	Lie theory
root system Δ	nonzero eigenvalues of $\operatorname{ad}\mathfrak{h}$ on $\mathfrak{g}$
Cartan integer $n_{\beta\alpha} \in \mathbb{Z}$	eigenvalue of the coroot h_α; integral by $\mathfrak{sl}_2$ theory
reflection s_α, Weyl group W	symmetry of α-strings; $W = N(T)/T$
choice of d, positive roots, Weyl chamber	upper triangular matrices (a Borel subalgebra)
simple roots, Dynkin diagram	generators e_i, f_i, h_i and Serre relations
rank $+\lvert\Delta\rvert$	$\dim \mathfrak{g}$
det of the Cartan matrix; density of the root lattice	order of the centre of the simply connected group
symmetries and foldings of the diagram	outer automorphisms; fixed subalgebras
root lattice $\mathbb{Z}\Delta$ (coroot lattice, in general)	$\frac{1}{2\pi}$ kernel of exp on the maximal torus (simply connected, compact)
E_8 kissing configuration (240 vectors)	root spaces of $\mathfrak{e}_8$, $248 = 8 + 240$
walls $x_i = x_j$ of $\operatorname{Conf}(N, \mathbb{R})$	eigenvalue collisions; non-regular elements of $\mathrm{SU}(N)$
one-parameter groups of diffeomorphisms (Lecture 9)	$t \mapsto e^{tX}$

Exercises

Exercise A.19. Write out the root decomposition of $\mathfrak{sl}_3$ explicitly. Check that $[E_{12}, E_{23}] = E_{13}$, $[E_{13}, E_{32}] = E_{12}$ and $[E_{12}, E_{13}] = 0$, and interpret each identity on the hexagon of Example A.1.

Exercise A.20. Prove the Jacobi identity for the commutator, and show that $\omega \mapsto X_\omega$ is an isomorphism $(\mathbb{R}^3, \times) \to \mathfrak{so}_3(\mathbb{R})$.

Exercise A.21. Expand $e^{sX}e^{tY}e^{-sX}e^{-tY}$ to second order and confirm that the result is $I + st[X,Y]$ plus higher-order terms.

Exercise A.22. Show that $\exp\colon \mathfrak{sl}_2(\mathbb{R}) \to \mathrm{SL}_2(\mathbb{R})$ is not surjective: the matrix $\left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$ is not in the image. *Hint:* a real traceless 2×2 matrix has eigenvalues $\pm\lambda$ or $\pm i\mu$. Compute the possible traces of e^X, and decide when the trace can equal -2.

Exercise A.23. Using the antidiagonal form J, find the roots of $\mathfrak{so}_6$ and check that they form a copy of A_3 (twelve roots, simply laced, rank 3). Do the same for $\mathfrak{so}_4$ and $A_1 \times A_1$, and for $\mathfrak{sp}_4$ and $\mathfrak{so}_5$ (show that C_2 and B_2 differ by a rotation of 45° and a scaling).

Exercise A.24. Prove Rodrigues' formula, and show that exp maps the closed ball of radius π in $\mathfrak{so}_3$ onto $\mathrm{SO}(3)$, identifying exactly the antipodal points of the boundary.

Exercise A.25. Draw the kernel of exp on the maximal torus of $\mathrm{SU}(3)$ inside the plane $\theta_1 + \theta_2 + \theta_3 = 0$, together with the Weyl alcove. Check that the kernel is a hexagonal lattice and that the alcove is an equilateral triangle. How many alcoves fit into a fundamental domain of the lattice?

Exercise A.26. Using only the Dynkin diagrams, compute $\det C$ for D_n and E_7 and recover the densities of the lattices D_5 and E_7. Compare with the laminated lattices of Lecture 6 (Section 6.3).

Exercise A.27. Check $|W(E_8)| = 696\,729\,600$ from the exponents, and verify that the exponents of A_{n-1} give $|\Sigma_n| = n!$. Find a Coxeter element of Σ_4 and show that it has order $h = 4$.

Further reading

- J. Stillwell, *Naive Lie Theory*, Springer UTM, 2008. Matrix groups first, with minimal prerequisites.
- B. C. Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., Springer GTM 222, 2015. Matrix groups, the exponential map, and a full treatment of the classification.
- J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Springer GTM 9, 1972. The algebraic route from $\mathfrak{sl}_2$ to root systems and Serre's theorem.
- W. Fulton and J. Harris, *Representation Theory: A First Course*, Springer GTM 129, 1991. Lectures 11–13 work through $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ by hand, and Lecture 21 does the classification.
- J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Ch. 4 and 8, for the lattice side of E_8.

Chapter B

Interlude B: Lattices, quadratic forms and the geometry of numbers

[Added] This interlude is new: it is not in the handwritten notes. It starts from the title of Blichfeldt's 1929 paper quoted in Lecture 2, *"The minimum value of quadratic forms, and the closest packing of spheres"*, and follows the first half of that title into number theory. To keep it readable it is set in black; all of it should be read as added material.

Blichfeldt's title puts two problems side by side, and in Lectures 1–6 we only looked at the second one. This interlude is about the first. We shall see that

- a lattice *is* a positive definite quadratic form, so the lattice packing problem is the question of how large the minimum of a form of given determinant can be (Hermite's problem);
- counting lattice points in balls is the classical problem of counting representations of integers by forms, and the leading term of the count is the covolume, which is what density measures;
- Blichfeldt's averaging trick from Lecture 2 has a famous cousin, Blichfeldt's lemma of 1914, which gives Minkowski's convex body theorem, and through it Fermat's and Lagrange's theorems on sums of squares;
- the generating function of the lattice point counts, the *theta series*, is a modular form. Poisson summation turns this into strong restrictions on very good lattices, and the same Poisson summation is the engine of the Cohn–Elkies bound behind Viazovska's theorem.

Throughout, $\Lambda \subset \mathbb{R}^n$ is a full-rank lattice, $\operatorname{covol}(\Lambda) = |\det \Lambda|$ as in Lecture 4, $\lambda_1(\Lambda)$ is the length of a shortest nonzero vector, $V_n = \operatorname{vol}(B^n)$, and

$$\delta(\Lambda) = \frac{V_n \, (\lambda_1/2)^n}{\operatorname{covol}(\Lambda)}$$

is the density of the associated packing (Lecture 4). To avoid a clash with the centre density θ, the theta series will be written Θ_Λ.

B.1 Lattices are quadratic forms

Let B be a basis matrix of Λ, with the basis vectors $b_1, \dots, b_n$ as its columns. A lattice vector is Bx with $x \in \mathbb{Z}^n$, and its squared length is

$$|Bx|^2 = x^T B^T B x = x^T G x =: Q(x), \qquad G = B^T B = \big(\langle b_i, b_j \rangle\big)_{i,j}.$$

The *Gram matrix* G is symmetric and positive definite, and $\det G = \operatorname{covol}(\Lambda)^2$. Conversely, every positive definite symmetric G factors as $G = B^T B$ (Cholesky), and B is unique up to $B \mapsto UB$ with $U \in \mathrm{O}(n)$. Changing the basis replaces B by BK with $K \in \mathrm{GL}_n(\mathbb{Z})$ (Lecture 4), hence G by $K^T G K$. So:

$$\frac{\{\text{lattices in } \mathbb{R}^n\}}{\text{rotations}} \quad \longleftrightarrow \quad \frac{\{\text{positive definite } n \times n \text{ symmetric matrices}\}}{G \sim K^T G K,\ K \in \mathrm{GL}_n(\mathbb{Z})}.$$

Under this dictionary

$$\lambda_1(\Lambda)^2 = \min_{x \in \mathbb{Z}^n \setminus 0} Q(x) =: \min Q, \qquad \operatorname{covol}(\Lambda) = \sqrt{\det G},$$

and the number of lattice vectors of squared length m is the number of *representations* of m by Q. Integral lattices (Lecture 4) are the forms with $G \in M_n(\mathbb{Z})$. For example $\mathbb{Z}^n$ is the sum of squares $x_1^2 + \cdots + x_n^2$, and the hexagonal lattice with basis $(1,0)$, $(\frac{1}{2}, \frac{\sqrt{3}}{2})$ is

$$Q_{\mathrm{hex}}(x,y) = x^2 + xy + y^2, \qquad G = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix},$$

while the root lattice A_2 (Lecture 6, minimal norm 2) is $2Q_{\mathrm{hex}}$.

Definition B.1 (Hermite's constant)**.** For a positive definite form put $\gamma(Q) = \min Q / (\det G)^{1/n}$; this is invariant under scaling and $\mathrm{GL}_n(\mathbb{Z})$. *Hermite's constant* is $\gamma_n = \sup_Q \gamma(Q)$, the supremum over all positive definite n-ary forms. Equivalently, $\gamma_n = \sup_\Lambda \lambda_1(\Lambda)^2 / \operatorname{covol}(\Lambda)^{2/n}$.

Comparing with the formula for $\delta(\Lambda)$ gives

$$\delta(\Lambda) = \frac{V_n}{2^n}\, \gamma(\Lambda)^{n/2}, \qquad \text{so} \qquad \sup_\Lambda \delta(\Lambda) = \frac{V_n}{2^n}\, \gamma_n^{n/2}. \tag{B.1}$$

Finding the densest lattice packing is the same problem as computing Hermite's constant. The known values are below. They are attained exactly by the lattices of Lectures 4–6, and the reader can recompute each from minimal norm 2 and the determinants $\det G = n+1$ for A_n, 4 for D_n, and $3, 2, 1$ for E_6, E_7, E_8. For the Leech lattice the minimal norm is 4 and $\det G = 1$.

n	1	2	3	4	5	6	7	8	24
γ_n^n	1	$\frac{4}{3}$	2	4	8	$\frac{64}{3}$	64	2^8	4^{24}
optimal lattice	$\mathbb{Z}$	A_2	A_3	D_4	D_5	E_6	E_7	E_8	Λ_{24}
density	1	.9069	.7405	.6169	.4653	.3729	.2953	.2537	.00193

(Lagrange for $n = 2$; Gauss for $n = 3$; Korkine–Zolotarev for $n = 4, 5$; Blichfeldt for $n = 6, 7, 8$; Cohn–Kumar for $n = 24$. See the Gallery for references.)

Binary forms: Lagrange's reduction

For $n = 2$ write $Q(x, y) = ax^2 + bxy + cy^2$, so $G = \left(\begin{smallmatrix} a & b/2 \\ b/2 & c \end{smallmatrix}\right)$ and $\det G = ac - b^2/4$. Following Lagrange (1773) and Gauss (*Disquisitiones*, 1801), call Q *reduced* if

$$|b| \le a \le c.$$

Proposition B.2 (Lagrange reduction)**.** *Every positive definite binary form is equivalent under* $\mathrm{SL}_2(\mathbb{Z})$ *to a reduced form, and if Q is reduced then* $\min Q = a$.

Proof. Let v_1 be a shortest nonzero vector of the lattice Λ of Q. It is primitive, since $v_1 = kw$ with $k \ge 2$ would make w shorter. So v_1 extends to a basis (v_1, w), and every other completion of v_1 to a basis is $\pm w + kv_1$, $k \in \mathbb{Z}$. Among the vectors $w + kv_1$ choose a shortest one, v_2, and replace v_2 by $-v_2$ if needed to make the determinant positive. In the basis (v_1, v_2) we have $a = |v_1|^2 = \min Q$, $c = |v_2|^2 \ge a$, and $b = 2\langle v_1, v_2\rangle$. Minimality of v_2 gives $c \le |v_2 \pm v_1|^2 = c \pm b + a$, that is, $|b| \le a$. For the second statement: if $|b| \le a \le c$ and $|x| \ge |y| > 0$, then $Q(x, y) \ge ax^2 - a\,|x|\,|y| + cy^2 \ge cy^2 \ge a$; the case $|y| \ge |x| > 0$ is symmetric (use $|b| \le c$), and $Q(x, 0) = ax^2 \ge a$, $Q(0, y) = cy^2 \ge a$. □

Theorem B.3 (Lagrange; $\gamma_2 = 2/\sqrt{3}$)**.** *For every positive definite binary form,* $\min Q \le \frac{2}{\sqrt{3}}\sqrt{\det G}$*, with equality only for forms equivalent to a multiple of* $x^2 + xy + y^2$*. Consequently every lattice packing of disks has density at most* $\pi/\sqrt{12}$*, with equality only for the hexagonal lattice.*

Proof. Take Q reduced. Then $\det G = ac - \frac{b^2}{4} \ge a^2 - \frac{a^2}{4} = \frac{3}{4}a^2$, so $\min Q = a \le \sqrt{4/3}\sqrt{\det G}$. Equality forces $c = a$ and $|b| = a$, i.e. $Q = a(x^2 \pm xy + y^2)$, and $x^2 - xy + y^2$ is equivalent to $x^2 + xy + y^2$ via $y \mapsto -y$. The density statement is (B.1): $\frac{\pi}{4} \cdot \frac{2}{\sqrt{3}} = \frac{\pi}{2\sqrt{3}}$. □

This is Theorem 1.14 of Lecture 1 again. In fact the proof there, which stacked the rows $\mathbb{Z}v$ at height h and slid the next row so that one of its points sits over the gap, is exactly the choice of v_1 and v_2 above. That proof generalizes to all n as Hermite's inequality (Exercise B.40).

The picture: the modular figure

Rotate and scale so that $v_1 = 1 \in \mathbb{C}$, and write $v_2 = \tau$ with $\operatorname{Im}\tau > 0$. Then

$$Q(x, y) = |v_1|^2\,|x + y\tau|^2 = a\big(x^2 + 2\operatorname{Re}\tau\, xy + |\tau|^2\, y^2\big),$$

so $b/a = 2\operatorname{Re}\tau$ and $c/a = |\tau|^2$. The reduction conditions become

$$|\operatorname{Re}\tau| \le \tfrac{1}{2}, \qquad |\tau| \ge 1,$$

which describe the classical fundamental domain $\mathcal{F}$ of $\mathrm{SL}_2(\mathbb{Z})$ acting on the upper half plane $\mathbb{H}$ by Möbius transformations. A change of basis in $\mathrm{SL}_2(\mathbb{Z})$ acts on $\tau = v_2/v_1$ exactly by $\tau \mapsto \frac{a\tau + b}{c\tau + d}$. So the space of planar lattices up to rotation and scaling is $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, and $\mathcal{F}$ is a map of it (Figure B.1). Allowing $\mathrm{GL}_2(\mathbb{Z})$ also identifies τ with $-\bar\tau$, which folds $\mathcal{F}$ in half.

In these coordinates the density is very simple. The covolume is $|v_1|^2 \operatorname{Im}\tau$ and, for $\tau \in \mathcal{F}$, the minimum is $|v_1|^2$, so

$$\delta(\tau) = \frac{\pi\,|v_1|^2/4}{|v_1|^2 \operatorname{Im}\tau} = \frac{\pi}{4\operatorname{Im}\tau} \qquad (\tau \in \mathcal{F}).$$

The level sets of the density are horizontal lines. The densest lattice is the *lowest point* of $\mathcal{F}$, the corner $\rho = e^{\pi i/3}$ with $\operatorname{Im}\rho = \frac{\sqrt{3}}{2}$: the hexagonal lattice. The square lattice is $\tau = i$, with $\delta = \pi/4$. Lattices high up in the cusp are long and thin and have small density; this is Mahler's criterion (Lecture 4) in its simplest form.

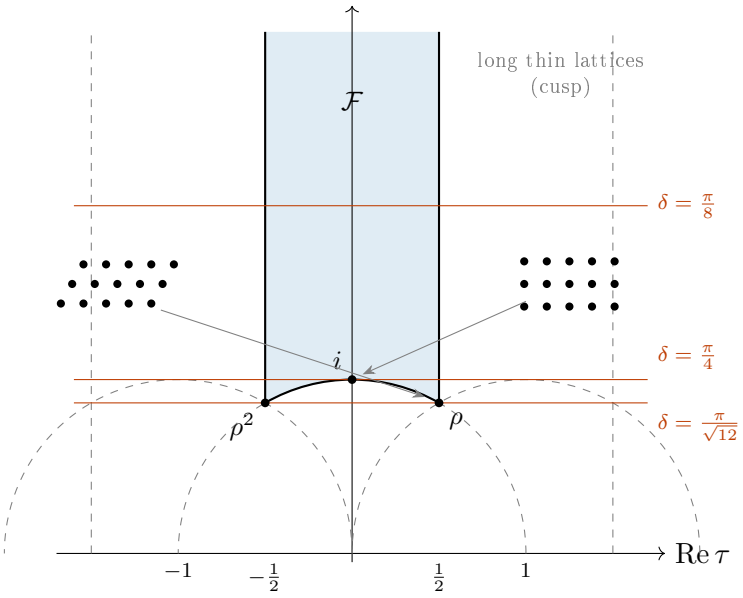


Figure B.1: The reduced domain $\mathcal{F} = \{|\operatorname{Re}\tau| \le \frac{1}{2},\ |\tau| \ge 1\}$ for binary forms, with some of its $\mathrm{SL}_2(\mathbb{Z})$-translates dashed. Each point is a planar lattice with basis $1, \tau$, up to rotation and scaling. The density $\pi/(4\operatorname{Im}\tau)$ is constant on horizontal lines and is largest at the corners ρ, ρ^2, the hexagonal lattice.

Remark B.4 (Class numbers). For integral forms ($a, b, c \in \mathbb{Z}$) the discriminant $D = b^2 - 4ac < 0$ is an invariant, and reduced forms satisfy $3a^2 \le -D$. So there are only finitely many reduced forms of a given discriminant. The number of *primitive* ones ($\gcd(a,b,c) = 1$), with the usual boundary convention ($b \ge 0$ if $|b| = a$ or $a = c$), is the *class number* $h(D)$. For instance, $h(-4) = 1$ (only $x^2 + y^2$) and $h(-20) = 2$ ($x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$). Gauss asked which D have $h(D) = 1$; see the Gallery.

B.2 Counting lattice points: the Gauss circle problem

The density of a lattice packing is a count in disguise: by Exercise 1.6 of Lecture 1, the number of lattice points in a large ball is

$$N_\Lambda(r) = \#\{x \in \Lambda : |x| \le r\} \ \sim\ \frac{V_n r^n}{\operatorname{covol}(\Lambda)}, \qquad \text{so} \qquad \delta(\Lambda) = \lim_{r\to\infty} \frac{N_\Lambda(r)\, V_n(\lambda_1/2)^n}{V_n r^n}.$$

How good is the approximation? For $\Lambda = \mathbb{Z}^2$ this is *Gauss's circle problem.* Write

$$N(r) = \#\{(x,y) \in \mathbb{Z}^2 : x^2 + y^2 \le r^2\} = \pi r^2 + E(r).$$

Proposition B.5 (Gauss). *$|E(r)| \le \pi\big(\sqrt{2}\, r + \frac{1}{2}\big)$ for $r \ge 1$; in particular $E(r) = O(r)$.*

Proof. Attach to each lattice point p the unit square $p + [-\frac{1}{2}, \frac{1}{2}]^2$; these squares tile the plane and each has diameter $\sqrt{2}$. The squares of the $N(r)$ points in the closed disk of radius r lie in the disk of radius $r + \frac{\sqrt{2}}{2}$. Conversely, every point of the disk of radius $r - \frac{\sqrt{2}}{2}$ lies in a square whose centre is in the disk of radius r. Comparing areas,

$$\pi\big(r - \tfrac{\sqrt{2}}{2}\big)^2 \le N(r) \le \pi\big(r + \tfrac{\sqrt{2}}{2}\big)^2,$$

and $\pi(r \pm \frac{\sqrt{2}}{2})^2 - \pi r^2 = \pm\pi\sqrt{2}\, r + \frac{\pi}{2}$. □

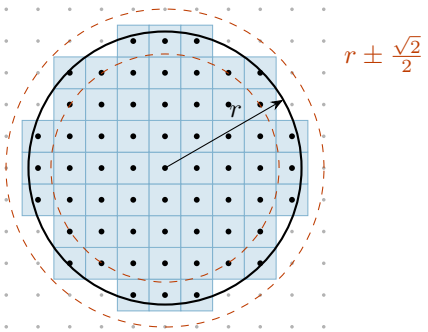


Figure B.2: Gauss's argument for $r = 4.3$: the $N(r) = 61$ unit squares centred at the lattice points in the disk are trapped between the disks of radii $r \pm \frac{\sqrt{2}}{2}$, so $\big|N(r) - \pi r^2\big| = O(r)$.

The same argument, with a fundamental parallelepiped in place of the square, gives $N_\Lambda(r) = V_n r^n / \operatorname{covol}(\Lambda) + O(r^{n-1})$ for every lattice and every dimension, and also for any ellipsoid $\{Q(x) \le r^2\}$ in place of the ball. So the error term is a "boundary layer", as in the density remarks of Lecture 1. The question is how much the lattice points conspire along the boundary.

Representation numbers. Let $r_2(m) = \#\{(x, y) \in \mathbb{Z}^2 : x^2 + y^2 = m\}$. Then $N(\sqrt{X}) = 1 + \sum_{1 \le m \le X} r_2(m)$, so Proposition B.5 says that *the average value of* $r_2(m)$ *is* π. At the same time $r_2(m)$ is given exactly by a formula of Jacobi (1829): with $\chi(d) = 0, 1, -1$ for d even, $d \equiv 1$, $d \equiv 3 \pmod 4$,

$$r_2(m) = 4 \sum_{d \mid m} \chi(d) = 4\big(d_1(m) - d_3(m)\big), \tag{B.2}$$

where $d_i(m)$ counts the divisors of m that are $\equiv i \pmod 4$. (For example $r_2(5) = 8$ and $r_2(3) = 0$; it was checked numerically for all $m < 2000$ in preparing these notes.) Combining the geometric count with the arithmetic formula gives a lattice point proof of Leibniz's series.

Proposition B.6 (Lattice points prove Leibniz)**.** $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

Proof. Put $L = \sum_{d \ge 1} \chi(d)/d$ and $S(t) = \sum_{d \le t} \chi(d) \in \{0, 1\}$. By (B.2), $N(\sqrt{X}) - 1 = 4 \sum_{dm \le X} \chi(d)$, a sum over the lattice points under the hyperbola $dm = X$. With $y = \sqrt{X}$, Dirichlet's hyperbola method splits it as

$$\sum_{dm \le X} \chi(d) = \sum_{d \le y} \chi(d) \Big\lfloor \frac{X}{d} \Big\rfloor + \sum_{m \le y} S\Big(\frac{X}{m}\Big) - S(y) \lfloor y \rfloor.$$

The last two terms are $O(y)$. In the first, $\lfloor X/d \rfloor = X/d + O(1)$, and the alternating series gives $\sum_{d \le y} \chi(d)/d = L + O(1/y)$. Hence $N(\sqrt{X}) = 4LX + O(\sqrt{X})$. Proposition B.5 gives $N(\sqrt{X}) = \pi X + O(\sqrt{X})$. Divide by X and let $X \to \infty$. □

How big is $E(r)$? Gauss's $O(r)$ uses nothing about circles except that they are curves of finite length. Curvature helps: Sierpiński (1906) proved $E(r) = O(r^{2/3})$, and the exponent has been lowered slowly since. The conjecture, usually attributed to Hardy, is that

$$E(r) = O_\varepsilon\big(r^{1/2+\varepsilon}\big) \qquad \text{for every } \varepsilon > 0,$$

and this would be optimal: Hardy and, independently, Landau showed in 1915 that $E(r)$ is not $o(r^{1/2})$, and Hardy (1916) that it is *not* $o(r^{1/2}(\log r)^{1/4})$, so the exponent $\frac{1}{2}$ cannot be lowered. Sharper logarithmic factors are due to Hafner (1981) and Soundararajan (2003). The exponent $\frac{1}{2}$ is also the right one on average: Cramér (Math. Z. 15, 1922) showed that $\frac{1}{R}\int_0^R E(r)^2\,dr$ grows like a constant times R, so a typical $|E(r)|$ is of size $r^{1/2}$. The current upper bounds, as exponents of r, are

Gauss (c. 1800)	1	elementary
Sierpiński (1906)	$2/3 \approx 0.6667$	
Huxley (2003)	$131/208 \approx 0.62981$	Proc. London Math. Soc. 87
Bourgain–Watt (2017)	$517/824 \approx 0.62743$	arXiv:1709.04340 (claimed)
Li–Yang (2023)	≈ 0.628966	arXiv:2308.14859
conjecture	$1/2 + \varepsilon$	

Huxley's exponent is the best one published in a refereed journal: as of 2026 both later papers are unpublished arXiv preprints. Both build on the Bombieri–Iwaniec method, and Bourgain–Watt add Bourgain's decoupling theory. The Bourgain–Watt preprint claims the smaller exponent, but Li and Yang describe Huxley's bound as the best known before their work and present theirs as its first improvement. So the record is not settled, and should be quoted with care.

Remark B.7 (Why a packing person should care). The circle problem is the cleanest model of a phenomenon that appears all through these notes: the bulk term of a count is a volume, and hence a density, while the correction is a boundary term whose size depends on fine arithmetic. In Lecture 2 we removed the boundary term by letting a box grow. Here it is the main object of study.

B.3 Blichfeldt again: Minkowski's convex body theorem

In Lecture 2 Blichfeldt's method compared a pointwise bound with an average: if $\sum_i f(\varphi_i^{-1}x) \le 1$ everywhere, then its average, which is proportional to the density, is at most 1. Fifteen years earlier Blichfeldt had used the same comparison the other way round.

Lemma B.8 (Blichfeldt 1914)**.** *Let $S \subset \mathbb{R}^n$ be measurable with $\mathrm{vol}(S) > \mathrm{covol}(\Lambda)$. Then S contains two distinct points s, s' with $s - s' \in \Lambda$.*

Proof. Consider the periodization $F(x) = \sum_{\lambda\in\Lambda} \chi_S(x+\lambda)$, a Λ-periodic function. Integrating over a fundamental domain P,

$$\int_P F(x)\,dx = \sum_{\lambda\in\Lambda} \mathrm{vol}\,\big(S \cap (P+\lambda)\big) = \mathrm{vol}(S) > \mathrm{vol}(P).$$

So F cannot be ≤ 1 everywhere: there is an x with $F(x) \ge 2$, that is, two different $\lambda, \lambda' \in \Lambda$ with $s = x+\lambda$ and $s' = x+\lambda'$ in S. Then $s - s' = \lambda - \lambda' \in \Lambda \setminus 0$. □

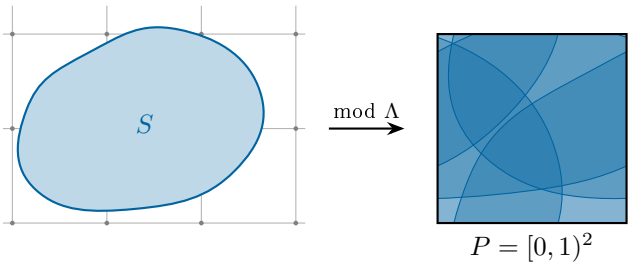


Figure B.3: Blichfeldt's lemma for $\Lambda = \mathbb{Z}^2$. The pieces of S in the different unit squares are translated into one fundamental square P. Their total area is $\mathrm{vol}(S) > 1 = \mathrm{vol}(P)$, so two pieces overlap (the darker regions). A point of overlap comes from two points of S that differ by a lattice vector.

Notice the parallel. In Lecture 2 the periodized function was bounded by 1 (packing condition), and the conclusion was that its average, and so the density, is at most 1. Here the average exceeds 1, so the function is somewhere at least 2: the translates $S + \lambda$ are *not* a packing. Both are pigeonhole arguments for the periodization of a function.

Theorem B.9 (Minkowski's convex body theorem, 1889/1896)**.** *Let $K \subset \mathbb{R}^n$ be convex and symmetric ($K = -K$). If $\mathrm{vol}(K) > 2^n \,\mathrm{covol}(\Lambda)$, then K contains a nonzero point of Λ. If K is also compact, $\mathrm{vol}(K) \geq 2^n \,\mathrm{covol}(\Lambda)$ suffices.*

Proof. Apply Lemma B.8 to $S = \frac{1}{2}K$, of volume $2^{-n}\,\mathrm{vol}(K) > \mathrm{covol}(\Lambda)$. This gives $s \neq s'$ in $\frac{1}{2}K$ with $s - s' \in \Lambda$. By symmetry $-s' \in \frac{1}{2}K$, and by convexity $\frac{1}{2}(2s) + \frac{1}{2}(-2s') = s - s' \in K$. For compact K apply the first part to $(1+\varepsilon)K$ for each $\varepsilon > 0$. Each of these contains a nonzero lattice point, and they all lie in the bounded set $2K$, which contains only finitely many lattice points. So one nonzero point lies in $(1+\varepsilon)K$ for arbitrarily small ε, hence in K. □

Remark B.10 (Minkowski and the trivial bound). Applied to the ball $K = B^n(\lambda_1)$, Minkowski gives $V_n \lambda_1^n \leq 2^n \,\mathrm{covol}(\Lambda)$, which is $\delta(\Lambda) \leq 1$, the trivial bound that the balls of radius $\lambda_1/2$ are disjoint. (So the two estimates quoted in Lecture 4 are actually the same inequality.) For sphere packing Minkowski's theorem is therefore weak. Its power lies in two other directions: in arithmetic, where the lattice is cut out by congruences, and in the bodies K other than balls. Here are the two classic examples.

Theorem B.11 (Fermat)**.** *Every prime $p \equiv 1 \pmod 4$ is a sum of two squares.*

Proof. Since $p \equiv 1 \pmod 4$, -1 is a square modulo p; for instance $u = \left(\frac{p-1}{2}\right)!$ satisfies $u^2 \equiv -1$ by Wilson's theorem. Let

$$\Lambda = \{(x, y) \in \mathbb{Z}^2 : \; y \equiv ux \pmod p\},$$

a sublattice of index p in $\mathbb{Z}^2$, so $\mathrm{covol}(\Lambda) = p$. The open disk of radius $\sqrt{2p}$ has area $2\pi p > 4p$, so by Theorem B.9 it contains some $(x, y) \in \Lambda \backslash 0$. Then $0 < x^2 + y^2 < 2p$ and $x^2 + y^2 \equiv x^2(1+u^2) \equiv 0 \pmod p$. Hence $x^2 + y^2 = p$. □

Theorem B.12 (Lagrange's four square theorem, 1770)**.** *Every positive integer is a sum of four squares.*

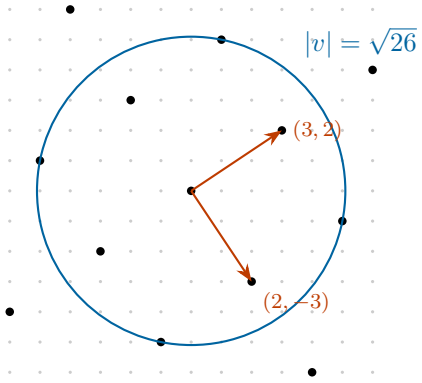


Figure B.4: Fermat by Minkowski for $p = 13$, $u = 5$ ($5^2 \equiv -1$). The lattice $\{y \equiv 5x \bmod 13\}$ (black) has covolume 13. The disk of area $26\pi > 4 \cdot 13$ must contain lattice vectors, and they have $x^2 + y^2 = 13$. Here the lattice is a square lattice of side $\sqrt{13}$, with reduced basis $(3, 2)$, $(2, -3)$.

Proof. By Euler's identity, which expresses $(a_1^2 + \cdots + a_4^2)(b_1^2 + \cdots + b_4^2)$ as a sum of four squares (it is the multiplicativity of the quaternion norm), it suffices to treat primes, and $2 = 1^2 + 1^2$. Let p be an odd prime. The sets $\{a^2 \bmod p\}$ and $\{-1 - b^2 \bmod p\}$ each have $\frac{p+1}{2}$ elements, so they meet: $a^2 + b^2 + 1 \equiv 0 \pmod p$ for some a, b. Let

$$\Lambda = \{x \in \mathbb{Z}^4 : \ x_3 \equiv ax_1 + bx_2, \quad x_4 \equiv bx_1 - ax_2 \pmod p\},$$

of index p^2 in $\mathbb{Z}^4$. For $x \in \Lambda$,

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 \equiv (x_1^2 + x_2^2)(1 + a^2 + b^2) \equiv 0 \pmod p.$$

The open ball of radius $\sqrt{2p}$ in $\mathbb{R}^4$ has volume $\frac{\pi^2}{2}(2p)^2 = 2\pi^2 p^2 > 16p^2$, so it contains a nonzero $x \in \Lambda$, and then $0 < |x|^2 < 2p$ and $p \mid |x|^2$ force $|x|^2 = p$. □

Three squares are more delicate. There is no product identity for sums of three squares, and the known geometric proof (Ankeny, 1957) needs Dirichlet's theorem on primes in arithmetic progressions in addition to Minkowski's theorem.

Theorem B.13 (Legendre 1798; Gauss 1801)**.** *A positive integer is a sum of three squares if and only if it is not of the form* $4^a(8b + 7)$.

The "only if" half is a computation mod 8. The "if" half needs the theory of ternary forms (or Dirichlet's theorem on primes in progressions). Which other forms represent *every* positive integer? Ramanujan (1917) listed 55 diagonal quaternary forms $ax^2 + by^2 + cz^2 + dw^2$ with this property. One of them, $x^2 + 2y^2 + 5z^2 + 5w^2$, fails to represent 15, and the other 54 are exactly the universal ones. The definitive answer is surprisingly finite; see the 15 and 290 theorems in the Gallery.

B.4 Theta series and modular forms

All the representation numbers of a lattice can be packaged into one function.

Definition B.14. The *theta series* of Λ is

$$\Theta_\Lambda(\tau) = \sum_{x\in\Lambda} e^{\pi i\tau|x|^2} = \sum_{t\geq 0} N_t\, e^{\pi i\tau t}, \qquad \tau\in\mathbb{H},$$

where N_t is the number of vectors of squared length t. The series converges absolutely and locally uniformly on $\mathbb{H}$, because $\left|e^{\pi i\tau|x|^2}\right| = e^{-\pi \operatorname{Im}\tau|x|^2}$.

For example $\Theta_\mathbb{Z}(\tau) = \sum_{k\in\mathbb{Z}} q^{k^2}$ with $q = e^{\pi i\tau}$ is Jacobi's theta function, and $\Theta_{\Lambda\oplus\Lambda'} = \Theta_\Lambda\Theta_{\Lambda'}$, so $\Theta_{\mathbb{Z}^2} = \Theta_\mathbb{Z}^2 = \sum_m r_2(m)q^m$. Formula (B.2) and Jacobi's four square formula

$$r_4(m) = 8\sum_{d|m,\ 4\nmid d} d \tag{B.3}$$

are identities between q-series (both were checked numerically for small m). The tool that makes theta series powerful is the following.

Theorem B.15 (Poisson summation over a lattice)**.** *Let $f\colon\mathbb{R}^n\to\mathbb{C}$ be continuous and satisfy $|f(x)|, \left|\hat f(x)\right| \leq C(1+|x|)^{-n-\eta}$ for some $C,\eta>0$, where $\hat f(y) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\langle x,y\rangle}dx$. Then for every $v\in\mathbb{R}^n$,*

$$\sum_{x\in\Lambda} f(x+v) = \frac{1}{\operatorname{covol}(\Lambda)}\sum_{y\in\Lambda^*}\hat f(y)\,e^{2\pi i\langle y,v\rangle}, \qquad \Lambda^* = \{y\in\mathbb{R}^n : \langle x,y\rangle\in\mathbb{Z} \text{ for all } x\in\Lambda\}.$$

Both sides converge absolutely.

Sketch. The left side $F(v)$ is a continuous Λ-periodic function, so it is a function on the torus $\mathbb{R}^n/\Lambda$, whose characters are $v\mapsto e^{2\pi i\langle y,v\rangle}$, $y\in\Lambda^*$. Its Fourier coefficients are

$$\frac{1}{\operatorname{covol}(\Lambda)}\int_{\mathbb{R}^n/\Lambda} F(v)e^{-2\pi i\langle y,v\rangle}dv = \frac{1}{\operatorname{covol}(\Lambda)}\int_{\mathbb{R}^n} f(v)e^{-2\pi i\langle y,v\rangle}dv = \frac{\hat f(y)}{\operatorname{covol}(\Lambda)},$$

by unfolding the sum against the integral. These coefficients are absolutely summable by the decay of $\hat f$, so the Fourier series converges uniformly to F. □

The *dual lattice* Λ^* has $\operatorname{covol}(\Lambda^*) = 1/\operatorname{covol}(\Lambda)$ and Gram matrix G^{-1}. Λ is integral iff $\Lambda\subseteq\Lambda^*$, and *unimodular* iff $\Lambda=\Lambda^*$ (equivalently, it is integral with covol $= 1$). It is *even* if all $|x|^2$ are even. $\mathbb{Z}^n$ is unimodular but odd; E_8 is even unimodular: it is integral and has determinant 1 (Lecture 6).

Apply Theorem B.15 to a Gaussian. For $\tau\in\mathbb{H}$ the function $f(x) = e^{\pi i\tau|x|^2}$ has $\hat f(y) = (\tau/i)^{-n/2}e^{\pi i(-1/\tau)|y|^2}$ (principal branch; for $\tau = it$ this is the familiar fact that $e^{-\pi t|x|^2}$ has Fourier transform $t^{-n/2}e^{-\pi|y|^2/t}$, and both sides are holomorphic in τ). With $v=0$ this gives $\Theta_\Lambda(\tau) = \operatorname{covol}(\Lambda)^{-1}(i/\tau)^{n/2}\,\Theta_{\Lambda^*}(-1/\tau)$; now replace τ by $-1/\tau$.

Corollary B.16 (Theta transformation formula)**.** $\Theta_\Lambda\Big(-\dfrac{1}{\tau}\Big) = \Big(\dfrac{\tau}{i}\Big)^{n/2}\dfrac{1}{\operatorname{covol}(\Lambda)}\,\Theta_{\Lambda^*}(\tau)$.

So the theta series of a lattice controls that of its dual, through the substitution $\tau\mapsto -1/\tau$. For an even unimodular lattice we get two symmetries:

$$\Theta_\Lambda(\tau+1) = \Theta_\Lambda(\tau), \qquad \Theta_\Lambda\Big(-\frac{1}{\tau}\Big) = \Big(\frac{\tau}{i}\Big)^{n/2}\Theta_\Lambda(\tau). \tag{B.4}$$

The maps $T\colon \tau \mapsto \tau+1$ and $S\colon \tau \mapsto -1/\tau$ generate $\mathrm{SL}_2(\mathbb{Z})$, the group whose fundamental domain we drew in Figure B.1. The fact that *lattices in the plane* are parametrized by $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ and the fact that *theta series of lattices in any dimension* transform under $\mathrm{SL}_2(\mathbb{Z})$ are two separate appearances of the same group.

Theorem B.17. *If $\mathbb{R}^n$ contains an even unimodular lattice, then $8 \mid n$.*

Proof. Let $f = \Theta_\Lambda$. Since $\Theta_{\Lambda\oplus\Lambda} = f^2$ and $\Lambda \oplus \Lambda$ is again even unimodular, we may pass to $\Lambda \oplus \Lambda$ or to $\Lambda^{\oplus 4}$. So if $8 \nmid n$ we may assume $n \equiv 4 \pmod 8$. Put $k = n/2 \equiv 2 \pmod 4$. Then $(\tau/i)^k = i^{-k}\tau^k = -\tau^k$, and (B.4) reads $f(\tau+1) = f(\tau)$, $f(-1/\tau) = -\tau^k f(\tau)$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ define $(f|\gamma)(\tau) = (c\tau+d)^{-k} f(\gamma\tau)$. The chain rule for $\gamma\tau$ shows that $(f|\gamma)|\gamma' = f|(\gamma\gamma')$, and $f|(-I) = f$ since k is even. With $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ we have $f|T = f$ and $f|S = -f$. A direct multiplication gives $(ST)^3 = -I$. Therefore

$$f = f|(-I) = f|STSTST = (-1)^3 f = -f,$$

so $f \equiv 0$. This contradicts $f(\tau) \to 1$ as $\operatorname{Im}\tau \to \infty$ (only $x = 0$ survives). □

Conversely, even unimodular lattices exist in every dimension $8k$: $E_8^{\oplus k}$, or $D_{8k}^+ = D_{8k} \cup (D_{8k} + \frac{1}{2}\mathbf{1})$ (Exercise B.44; $E_8 = D_8^+$ by Lecture 6).

The theta series of E_8

For $8 \mid n$ the factor $(\tau/i)^{n/2}$ equals $\tau^{n/2}$, and (B.4) says that Θ_Λ is a *modular form of weight $n/2$* for $\mathrm{SL}_2(\mathbb{Z})$: a holomorphic function on $\mathbb{H}$, bounded as $\operatorname{Im}\tau \to \infty$, with $f(\gamma\tau) = (c\tau+d)^{n/2} f(\tau)$. Write $\mathsf{q} = e^{2\pi i\tau}$ (so $\mathsf{q} = q^2$). The basic theorem on such forms (see Serre, *A Course in Arithmetic*, Ch. VII) is that the space M_k of modular forms of weight k is finite-dimensional, with

$$M_4 = \mathbb{C}E_4, \qquad M_8 = \mathbb{C}E_4^2, \qquad M_{12} = \mathbb{C}E_4^3 \oplus \mathbb{C}\Delta,$$

where $E_4 = 1 + 240\sum_{m\ge1}\sigma_3(m)\mathsf{q}^m$ is the Eisenstein series ($\sigma_3(m) = \sum_{d\mid m} d^3$) and $\Delta = \mathsf{q}\prod_{m\ge1}(1-\mathsf{q}^m)^{24}$. Since $\Theta_{E_8} = \sum_m N_{2m}\mathsf{q}^m$ lies in M_4 and has constant term 1,

$$\Theta_{E_8} = E_4, \qquad \text{i.e.} \qquad \#\{x \in E_8 : |x|^2 = 2m\} = 240\,\sigma_3(m). \tag{B.5}$$

For $m = 1$ this is 240, the root count from Lecture 6 and the dimension 8 kissing number. For $m = 2$ it predicts $240 \cdot 9 = 2160$ vectors of norm 4, and for $m = 3$, $240 \cdot 28 = 6720$. A brute-force count of E_8 vectors confirms 240, 2160, 6720. *A single modular form knows every shell of E_8.*

Hearing the shape of a drum, and Niemeier's lattices

The flat torus $\mathbb{R}^n/\Lambda$ has Laplace eigenfunctions $e^{2\pi i\langle y,x\rangle}$, $y \in \Lambda^*$, with eigenvalues $4\pi^2|y|^2$. So its spectrum, with multiplicities, is recorded by Θ_{Λ^*}.

Theorem B.18 (Milnor 1964). *The 16-dimensional flat tori $\mathbb{R}^{16}/(E_8 \oplus E_8)$ and $\mathbb{R}^{16}/D_{16}^+$ have the same Laplace spectrum but are not isometric.*

Proof. Both lattices are even unimodular, so both theta series lie in $M_8 = \mathbb{C}E_4^2$ and have constant term 1. So they are equal, and the lattices are self-dual, which makes the spectra equal. (Numerically, both have 480 vectors of norm 2 and 61920 of norm 4.) An isometry of the tori would lift to an isometry of $\mathbb{R}^{16}$ carrying one lattice onto the other. But the 480 norm-2 vectors of $E_8 \oplus E_8$ span it, while those of D_{16}^+ (the vectors $\pm e_i \pm e_j$) span only the index-2 sublattice D_{16}. □

This was the first counterexample to the question "can one hear the shape of a drum?", two years before Kac named the question. In dimension 24, M_{12} is two-dimensional, so Θ_Λ is determined by its constant term and by the number of roots. Taking zero roots gives

$$\Theta_{\Lambda_{24}} = E_4^3 - 720\Delta = 1 + 196560\,\mathsf{q}^2 + 16773120\,\mathsf{q}^3 + \cdots,$$

and 196560 is the kissing number of the Leech lattice. Niemeier (1973) classified the even unimodular lattices in dimension 24. There are exactly 24 of them, determined by their root systems (each one of rank 24 or empty), and the Leech lattice is the unique one without roots (Conway 1969).

B.5 Poisson summation as the engine: LP bounds

Lecture 3 mentioned that the solutions in dimensions 8 and 24 use "linear programming duality" and a "magic" auxiliary function. With Poisson summation in hand, the argument for lattices takes only a few lines.

Definition B.19. Call $f\colon \mathbb{R}^n \to \mathbb{R}$ *admissible* if f and $\hat f$ satisfy the decay hypothesis of Theorem B.15, and f is even (so $\hat f$ is real).

Theorem B.20 (Cohn–Elkies bound for lattices)**.** *Let f be admissible with $f(0) = \hat f(0) > 0$, and suppose*

(i) $f(x) \le 0$ for $|x| \ge r$, and
(ii) $\hat f(y) \ge 0$ for all y.

Then every lattice packing in $\mathbb{R}^n$ has density at most $V_n(r/2)^n$.

Proof. Scale the lattice so that $\lambda_1(\Lambda) = r$. Apply Poisson summation with $v = 0$:

$$f(0) \;\ge\; f(0) + \sum_{x \in \Lambda\setminus 0} f(x) \;=\; \frac{1}{\operatorname{covol}(\Lambda)} \sum_{y \in \Lambda^*} \hat f(y) \;\ge\; \frac{\hat f(0)}{\operatorname{covol}(\Lambda)}.$$

The first inequality is (i), since every $x \neq 0$ has $|x| \ge r$, and the last is (ii), dropping all $y \neq 0$. Hence $\operatorname{covol}(\Lambda) \ge \hat f(0)/f(0) = 1$, and $\delta(\Lambda) = V_n(r/2)^n/\operatorname{covol}(\Lambda) \le V_n(r/2)^n$. □

The full theorem drops the lattice assumption at almost no cost.

Theorem B.21 (Cohn–Elkies 2003)**.** *Under the same hypotheses, every sphere packing in $\mathbb{R}^n$ has density at most $V_n(r/2)^n$.*

Proof. By Exercise 1.7 of Lecture 1 it suffices to treat periodic packings, with centres $\bigcup_{j=1}^N (\Lambda + v_j)$, the v_j pairwise incongruent modulo Λ, scaled to have minimal distance r. Sum Poisson summation over all pairs (j,k) with $v = v_j - v_k$:

$$\sum_{j,k=1}^{N} \sum_{x \in \Lambda} f(x + v_j - v_k) = \frac{1}{\operatorname{covol}(\Lambda)} \sum_{y \in \Lambda^*} \hat f(y) \left| \sum_{j=1}^{N} e^{2\pi i \langle y, v_j \rangle} \right|^2.$$

On the left, $x + v_j - v_k$ is the difference of two centres. It is 0 exactly when $j = k$ and $x = 0$, and otherwise it has length at least r. So the left side is at most $Nf(0)$. On the right every term is ≥ 0, and the $y = 0$ term is $N^2 \hat f(0)/\operatorname{covol}(\Lambda)$. So $N/\operatorname{covol}(\Lambda) \le f(0)/\hat f(0) = 1$. The density is $(N/\operatorname{covol}(\Lambda))V_n(r/2)^n$. □

Remark B.22 (Blichfeldt's gauge versus Cohn–Elkies). Recall the proof that $f_0 = 1 - \frac{1}{2}|x|^2$ is a gauge (Lemma 2.13). It rested on the identity

$$\sum_{i<j} |x_i - x_j|^2 = m \sum_i |x_i|^2 - \Big|\sum_i x_i\Big|^2,$$

and the inequality came from *throwing away a square*, $|\sum_i x_i|^2 \geq 0$. The proof of Theorem B.21 throws away the squares $\hat{f}(y)\left|\sum_j e^{2\pi i\langle y, v_j\rangle}\right|^2 \geq 0$, one for each frequency $y \neq 0$. Blichfeldt uses one quadratic "moment", while Cohn–Elkies use all the exponentials. There is also a difference in what the auxiliary function must satisfy:

- a Blichfeldt gauge is a condition in physical space only: $f \geq 0$ and $\sum_i f(\varphi_i^{-1}x) \leq 1$ for every packing;
- a Cohn–Elkies function has a sign condition in *both* spaces: $f \leq 0$ beyond r, and $\hat{f} \geq 0$ everywhere.

The uncertainty principle makes the second kind of condition hard to satisfy, and this is why the optimal functions are rare and remarkable.

When is the bound sharp? Suppose Λ is unimodular, $\lambda_1(\Lambda) = r$, and f gives equality. Then both inequalities in the proof of Theorem B.20 are equalities, so

$$f(x) = 0 \ \text{ for all } x \in \Lambda \setminus 0, \qquad \hat{f}(y) = 0 \ \text{ for all } y \in \Lambda^* \setminus 0.$$

For E_8 ($r = \sqrt{2}$) this says that f and $\hat{f}$ vanish on all spheres $|x| = \sqrt{2m}$, $m \geq 1$. Since $f \leq 0$ beyond $\sqrt{2}$ and $\hat{f} \geq 0$ everywhere, these zeros must have even order (they are double roots), except that f may change sign, with a simple root, at $\sqrt{2}$ itself. Cohn and Elkies found numerically that such functions seem to exist in dimensions 8 and 24 (and 2), but could not construct them.

Viazovska's idea. Look for f radial and built out of Gaussians:

$$f(x) = \int_\Gamma \phi(z)\, e^{\pi i z|x|^2}\, dz$$

along a path Γ in $\mathbb{H}$. In dimension 8 the Fourier transform of $e^{\pi i z|x|^2}$ is $z^{-4}e^{\pi i(-1/z)|y|^2}$ (the computation before Corollary B.16). Substituting $w = -1/z$ shows that $\hat{f}$ is given by the same kind of integral with weight $\phi(-1/w)\,w^2$ in place of $\phi(w)$, over the image path. So the Fourier transform acts on the *weight* ϕ through $z \mapsto -1/z$, and f is a Fourier eigenfunction when ϕ transforms like a modular form of weight -2 (up to the change of path). Viazovska (2016) built two eigenfunctions of the Fourier transform, with eigenvalues ± 1: one from a weakly holomorphic *quasimodular* form for $\mathrm{SL}_2(\mathbb{Z})$ (a rational expression in E_2, E_4, E_6 and Δ), and one from a weakly holomorphic modular form for the subgroup $\Gamma(2)$ (built from Jacobi theta functions). A factor $\sin^2(\pi|x|^2/2)$ supplies the double roots at $|x|^2 \in 2\mathbb{Z}$. A suitable linear combination of the two is the magic function, and it proves that E_8 is the densest packing in $\mathbb{R}^8$. A week later, Cohn, Kumar, Miller, Radchenko and Viazovska did the same in dimension 24 for the Leech lattice. The theta series of Section B.4 and the magic functions use the same symmetry $\tau \mapsto -1/\tau$, once applied to a lattice and once to an auxiliary function.

Universal optimality. The same circle of ideas gives a much stronger statement (Gallery, Theorem B.36): E_8 and Leech minimize energy for every completely monotone function of squared distance, Gaussians $e^{-\alpha|x|^2}$ included. For Gaussian potentials, Poisson summation turns the energy of a lattice into the energy of its dual. The proof needs a *Fourier interpolation formula*: a radial Schwartz function on $\mathbb{R}^8$ is determined by the values of f and $\hat{f}$ (and their radial derivatives) on the spheres $|x| = \sqrt{2m}$.

B.6 Averaging over all lattices: Siegel and Minkowski–Hlawka

So far we have looked at special lattices. At the opposite extreme, take a *random* lattice. Let $X_n = \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$ be the space of lattices of covolume 1 in $\mathbb{R}^n$. It carries a unique $\mathrm{SL}_n(\mathbb{R})$-invariant probability measure μ (Minkowski, Siegel). For $n = 2$, after quotienting by rotations, X_2 becomes the modular surface of Figure B.1, and μ becomes the hyperbolic area $\frac{3}{\pi}\frac{dx\,dy}{y^2}$ restricted to $\mathcal{F}$.

Theorem B.23 (Siegel's mean value theorem, 1945)**.** *For $n \geq 2$ and every integrable $f\colon \mathbb{R}^n \to \mathbb{R}$,*

$$\int_{X_n} \sum_{x \in \Lambda \setminus 0} f(x)\, d\mu(\Lambda) = \int_{\mathbb{R}^n} f(x)\, dx, \qquad \int_{X_n} \sum_{x \in \Lambda \ \text{primitive}} f(x)\, d\mu(\Lambda) = \frac{1}{\zeta(n)} \int_{\mathbb{R}^n} f(x)\, dx.$$

The two formulas are equivalent, because every nonzero vector is uniquely kx with $k \geq 1$ and x primitive, and $\sum_k k^{-n} = \zeta(n)$. In words: *on average, a lattice looks like Lebesgue measure.* From this one gets a lower bound in a few lines.

Corollary B.24 (Minkowski–Hlawka)**.** *For $n \geq 2$, $\sup_\Lambda \delta(\Lambda) \geq \dfrac{2\zeta(n)}{2^n}$.*

Proof. Let B be a ball centred at 0 with $\mathrm{vol}(B) < 2\zeta(n)$, and apply the primitive form of Theorem B.23 to $f = \chi_B$. The average number of primitive vectors in B is $\mathrm{vol}(B)/\zeta(n) < 2$. Primitive vectors come in pairs $\pm x$, so some $\Lambda \in X_n$ has no primitive vector in B, and hence no nonzero vector at all (if $kx \in B$ with $k \geq 1$, then $x \in B$). Balls of half the radius centred at Λ form a packing of density $\mathrm{vol}(B)/2^n$. Let $\mathrm{vol}(B) \uparrow 2\zeta(n)$. □

This is the "$2\zeta(d)2^{-d}$" of the remark after Proposition 1.10 in Lecture 1, and it is still within a polynomial factor of the best known lower bounds (for lattices, Klartag's 2025 bound $c\,n^2\,2^{-n}$). Upper bounds are exponentially larger (Kabatiansky–Levenshtein, Lecture 3). It is still unknown whether the best lattices in high dimension are exponentially better than random ones.

Siegel's theorem treats X_n as a probability space. Interlude C treats it as a dynamical system: the diagonal flow on X_n, Mahler's criterion as a statement about the cusp, and equidistribution results that sharpen Siegel's average.

B.7 A dictionary

Packings and lattices	Number theory
lattice up to rotation	positive definite form up to $\mathrm{GL}_n(\mathbb{Z})$
λ_1^2, covol^2	$\min Q$, $\det G$
densest lattice packing	Hermite's constant γ_n
planar lattices, Figure B.1	$\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, reduced binary forms
number of points in a ball	$\sum_{m\le r^2} r_Q(m)$; Gauss circle problem
kissing number of Λ	number of representations of $\min Q$
Blichfeldt's gauge (Lecture 2)	Blichfeldt's lemma, Minkowski's theorem
shells of E_8, Leech	coefficients of E_4, $E_4^3 - 720\Delta$
dual lattice Λ^*	G^{-1}; $\tau \mapsto -1/\tau$
LP bound, magic function	Poisson summation, (quasi)modular forms
random lattice	Siegel's mean value theorem

B.8 Gallery of theorems

Classical results first, then recent ones. Each comes with a sentence on why it belongs in this course.

Theorem B.25 (Hermite's inequality; Hermite 1850)**.** *$\gamma_n \le (4/3)^{(n-1)/2}$ for all $n \ge 1$.*

Published in Hermite's letters to Jacobi, J. reine angew. Math. 40 (1850). For $n = 2$ it is Theorem B.3. The proof (Exercise B.40) is the row-stacking argument of Lecture 1, run by induction on the dimension.

Theorem B.26 (The known Hermite constants)**.** *The values of γ_n in the table of Section B.1 are correct, and in each of these dimensions the optimal lattice is unique up to similarity.*

Lagrange (1773) and Gauss (1831, in his review of Seeber) for $n = 2, 3$; Korkine and Zolotarev, Math. Ann. 6 (1873) and 11 (1877), for $n = 4, 5$; Blichfeldt, Math. Z. 39 (1935), for $n = 6, 7, 8$ (uniqueness there is due to Vetčinkin, 1980); H. Cohn and A. Kumar, *Optimality and uniqueness of the Leech lattice among lattices*, Ann. of Math. 170 (2009), for $n = 24$. The case $n = 24$ uses the Cohn–Elkies bound of Section B.5 together with an argument special to lattices. These are exactly the lattices of Lectures 4–6.

Theorem B.27 (Voronoi 1908)**.** *A positive definite form Q is a local maximum of $\gamma(Q)$ (an* extreme *form) if and only if it is* perfect *(it is the only form with its minimum value and its set of minimal vectors; equivalently, the matrices xx^T, x minimal, span the space of symmetric matrices) and* eutactic *(G^{-1} is a combination of the matrices xx^T, x minimal, with strictly positive coefficients). Up to equivalence and scaling there are finitely many perfect forms in each dimension.*

G. Voronoi, J. reine angew. Math. 133 (1908). The numbers of perfect forms in dimensions $2, \dots, 8$ are $1, 1, 2, 3, 7, 33, 10916$ (dimension 7: Jaquet-Chiffelle, Ann. Inst. Fourier 43 (1993);

dimension 8: Dutour Sikirić, Schürmann and Vallentin, Electron. Res. Announc. AMS 13 (2007)). This is a finite algorithm for the lattice packing problem in each fixed dimension. Eutaxy is the lattice analogue of the force balanced contact graphs of Lecture 10: the minimal vectors are the contacts, and the positive coefficients are the stresses.

Theorem B.28 (Class number one; Heegner 1952, Baker 1966, Stark 1967)**.** *The imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$ with class number 1 are exactly those with $d = 1, 2, 3, 7, 11, 19, 43, 67, 163$.*

Gauss conjectured the list. Heegner's proof (Math. Z. 56, 1952) was not accepted at first. Baker (Mathematika 13, 1966) and Stark (Michigan Math. J. 14, 1967) gave independent proofs, and Heegner's argument was later found to be essentially correct. In the language of Remark B.4: for the corresponding discriminants $D = -3, -4, -7, -8, -11, -19, -43, -67, -163$ there is exactly one reduced primitive form. For $D = -3$ and $D = -4$ it is the hexagonal and the square lattice. For $D = -163$ it is $x^2 + xy + 41y^2$, and Euler's observation that $k^2 + k + 41$ is prime for $0 \le k \le 39$ is a shadow of $h(-163) = 1$ (Rabinowitsch, 1913).

Theorem B.29 (The 15 theorem; Conway–Schneeberger 1993, Bhargava 2000)**.** *A positive definite quadratic form with integer Gram matrix represents every positive integer if and only if it represents $1, 2, 3, 5, 6, 7, 10, 14, 15$.*

Conway and Schneeberger did not publish their proof. M. Bhargava, *On the Conway–Schneeberger fifteen theorem*, in *Quadratic Forms and Their Applications* (Dublin 1999), Contemp. Math. 272, AMS (2000), 27–37, gave a short proof, built by "escalating" lattices: one adds a vector whose norm is the smallest integer not yet represented. This is the arithmetic version of the laminated lattices of Lecture 6. None of the nine numbers can be omitted.

Theorem B.30 (The 290 theorem; Bhargava–Hanke)**.** *A positive definite integer-valued quadratic form represents every positive integer if and only if it represents the 29 numbers 1, 2, 3, 5, 6, 7, 10, 13, 14, 15, 17, 19, 21, 22, 23, 26, 29, 30, 31, 34, 35, 37, 42, 58, 93, 110, 145, 203, 290.*

Conjectured by Conway. M. Bhargava and J. Hanke, *Universal quadratic forms and the 290-theorem*, preprint (2011). It is widely cited, but as of 2026 it has not appeared in a journal. "Integer-valued" allows odd off-diagonal coefficients, such as $x^2 + xy + y^2$; that is why the list is longer.

Theorem B.31 (Hardy 1916; Huxley 2003)**.** *In the circle problem, $E(r)$ is not $o\big(r^{1/2}(\log r)^{1/4}\big)$. On the other hand, $E(r) = O_\varepsilon(r^{131/208+\varepsilon})$ for every $\varepsilon > 0$.*

The lower bound is in G. H. Hardy, *On Dirichlet's divisor problem*, Proc. London Math. Soc. (2) 15 (1916). The upper bound is in M. N. Huxley, *Exponential sums and lattice points III*, Proc. London Math. Soc. (3) 87 (2003), 591–609. The logarithmic factors in the lower bound were sharpened by Hafner (Invent. Math. 63, 1981) and by Soundararajan (Int. Math. Res. Not. 2003, no. 36). For the arXiv improvements of the upper bound, see Section B.2. This is the prototype of a boundary-layer error in a density count.

Theorem B.32 (Milnor 1964; Niemeier 1973)**.** *There are non-isometric isospectral flat tori in dimension 16. There are exactly 24 even unimodular lattices in $\mathbb{R}^{24}$, and the Leech lattice is the only one with no vectors of norm 2.*

J. Milnor, *Eigenvalues of the Laplace operator on certain manifolds*, Proc. Nat. Acad. Sci. USA 51 (1964), 542; H.-V. Niemeier, J. Number Theory 5 (1973), 142–178; J. H. Conway, Invent. Math. 7 (1969), 137–142, for the characterization of Leech. Both are consequences of the theta series being a modular form (Section B.4).

Theorem B.33 (Cohn–Elkies 2003)**.** *Theorem B.21 holds.*

H. Cohn and N. Elkies, *New upper bounds on sphere packings I*, Ann. of Math. 157 (2003), 689–714. Their numerical bounds were the best known in many dimensions, and in dimensions 8 and 24 they came extremely close to the densities of E_8 and Leech. This is the bridge from Blichfeldt's method to the solution in dimensions 8 and 24.

Conjecture B.34 (Cohn–Elkies)**.** *The linear programming bound of Theorem B.21 is sharp in dimension 2, that is, its optimum equals $\pi/\sqrt{12}$.*

Numerical optimization supports it, but no exact auxiliary function is known. Thue's theorem is known by other means (Lecture 1), so what is open is whether *this method* reproves it.

Theorem B.35 (Viazovska 2016; Cohn–Kumar–Miller–Radchenko–Viazovska 2016)**.** *The E_8 lattice packing is the densest sphere packing in $\mathbb{R}^8$, and the Leech lattice packing is the densest in $\mathbb{R}^{24}$.*

M. Viazovska, *The sphere packing problem in dimension 8*, Ann. of Math. 185 (2017), 991–1015; H. Cohn, A. Kumar, S. D. Miller, D. Radchenko and M. Viazovska, *The sphere packing problem in dimension 24*, Ann. of Math. 185 (2017), 1017–1033. In both cases the magic function is built from (quasi)modular forms, as sketched in Section B.5.

Theorem B.36 (Universal optimality; Cohn–Kumar–Miller–Radchenko–Viazovska 2022)**.** *Let $p(r) = g(r^2)$ with $g\colon (0,\infty) \to \mathbb{R}$ completely monotone. Among all point configurations in $\mathbb{R}^8$ (resp. $\mathbb{R}^{24}$) of density one point per unit volume, E_8 (resp. the Leech lattice) minimizes the lower p-energy.*

H. Cohn, A. Kumar, S. D. Miller, D. Radchenko and M. Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, Ann. of Math. 196 (2022), no. 3, 983–1082. Sphere packing is a limiting case (very steep potentials). This connects the lattice problem to the Boltzmann weights and energy landscapes of Lectures 7–11: E_8 and Leech are ground states for a whole class of interactions, and not only optimal hard spheres.

Theorem B.37 (Siegel 1945; Minkowski–Hlawka 1943)**.** *Theorem B.23 and Corollary B.24 hold.*

C. L. Siegel, *A mean value theorem in geometry of numbers*, Ann. of Math. 46 (1945), 340–347; E. Hlawka, *Zur Geometrie der Zahlen*, Math. Z. 49 (1943/44), 285–312. Minkowski had stated the bound without a complete proof. The lattice analogue of the saturation bound of Lecture 1.

Theorem B.38 (Klartag 2025)**.** *There is a constant $c > 0$ such that for all large n there is a lattice packing in $\mathbb{R}^n$ of density at least $c\, n^2\, 2^{-n}$.*

B. Klartag, *Lattice packing of spheres in high dimensions using a stochastically evolving ellipsoid*, Invent. Math. (2026), doi:10.1007/s00222-026-01412-w; arXiv:2504.05042 (2025). It is the first lower bound, for lattices or for general packings, that beats $n\,2^{-n}$ by more than a logarithmic factor. The proof grows an ellipsoid that contains no nonzero point of a lattice, by a random process. Compare the Minkowski–Hlawka bound $2\zeta(n)2^{-n}$ above and the table in Lecture 3.

Exercises

Exercise B.39. Reduce the form $26x^2 + 130xy + 169y^2$ (the Gram matrix of the basis $(1,5)$, $(0,13)$ of the lattice in Figure B.4). Check that you get $13(x^2+y^2)$, and read off a representation $13 = a^2 + b^2$. Show that $h(-20) = 2$ by listing the reduced forms of discriminant -20.

Exercise B.40. (Hermite's inequality.) Let v_1 be a shortest vector of $\Lambda \subset \mathbb{R}^n$, and let Λ' be the orthogonal projection of Λ onto $v_1^\perp$.

1. Show that Λ' is a lattice with $\operatorname{covol}(\Lambda') = \operatorname{covol}(\Lambda)/\,|v_1|$.
2. Show that every $w' \in \Lambda'$ lifts to $w \in \Lambda$ with $|w|^2 \le |w'|^2 + \frac{1}{4}\,|v_1|^2$, and deduce $\frac{3}{4}\lambda_1(\Lambda)^2 \le \lambda_1(\Lambda')^2$.
3. Conclude that $\gamma_n \le \left(\frac{4}{3}\gamma_{n-1}\right)^{(n-1)/n}$, and hence $\gamma_n \le (4/3)^{(n-1)/2}$. Compare with the proof of Theorem 1.14.

Exercise B.41. Show that $N_\Lambda(r) = V_n r^n/\operatorname{covol}(\Lambda) + O(r^{n-1})$ for every lattice $\Lambda \subset \mathbb{R}^n$, by Gauss's argument with a fundamental parallelepiped. Then use it to show that the average of $r_2(m)$ for $m \le X$ tends to π, and that the average of $r_4(m)$ over $m \le X$ is asymptotic to $\frac{\pi^2}{2}X$.

Exercise B.42. Use Minkowski's theorem to show that every prime $p \equiv 1$ or $3 \pmod 8$ can be written as x^2+2y^2. (Use that -2 is a square mod p for these primes, and the ellipse $x^2+2y^2 < 2p$, of area $\sqrt{2}\,\pi p > 4p$.)

Exercise B.43. Prove Corollary B.16 for $\Lambda = \mathbb{Z}$ directly, by applying Poisson summation to $e^{-\pi t x^2}$, $t > 0$. Deduce that $t^{1/4}\,\Theta_{\mathbb{Z}}(it)$ is invariant under $t \mapsto 1/t$, and use this to show that $\Theta_{\mathbb{Z}}(it) \sim t^{-1/2}$ as $t \to 0^+$. Interpret this limit as a lattice point count.

Exercise B.44. Show that $D_n^+ = D_n \cup (D_n + \frac{1}{2}\mathbf{1})$ is a lattice if and only if n is even, that it is unimodular when it is integral, and that it is even unimodular if and only if $8 \mid n$. Count the vectors of norm 2 and 4 in D_{16}^+ and compare with $E_8 \oplus E_8$.

Exercise B.45. Using $\Theta_{E_8} = E_4$, show that the number of vectors of norm $2m$ in E_8 is divisible by 240 for every $m \ge 1$. Then verify by hand that E_8 has 2160 vectors of norm 4. (They are $\pm 2e_i$, the vectors $\pm e_i \pm e_j \pm e_k \pm e_l$, and the half-integral vectors with one coordinate $\pm\frac{3}{2}$.)

Exercise B.46. (The LP bound in dimension one.) Let $f(x) = \max(0, 1 - |x|)$. Compute $\hat{f}(y) = \left(\frac{\sin \pi y}{\pi y}\right)^2$, check that f satisfies the hypotheses of Theorem B.21 with $r = 1$, and conclude that the bound is sharp in dimension 1. Which lattice packings achieve it, and how does this show up in the equality analysis of Section B.5?

Exercise B.47. Using Siegel's theorem, compute the average over X_n of the number of nonzero lattice vectors in a ball of volume V centred at 0. Deduce that some unimodular lattice has $\lambda_1 \ge (2/V_n)^{1/n}$, which is asymptotically $\sqrt{n/(2\pi e)}$. Compare with the upper bound $\lambda_1 \le 2V_n^{-1/n}$ coming from Minkowski's theorem, and translate both into bounds on γ_n.

Further reading

- J.-P. Serre, *A Course in Arithmetic*, Springer GTM 7, 1973. Chapter V: quadratic forms over $\mathbb{Z}$ (including $8 \mid n$ for even unimodular lattices); Chapter VII: modular forms and theta series.

- J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999. Ch. 2 (theta series), Ch. 16–18 (Niemeier lattices, Leech).
- J. W. S. Cassels, *An Introduction to the Geometry of Numbers*, Springer, 1959. Blichfeldt, Minkowski, Siegel and Hlawka.
- J. Martinet, *Perfect Lattices in Euclidean Spaces*, Springer, 2003. Hermite constants and Voronoi's theory.
- J. H. Conway (with F. Fung), *The Sensual (Quadratic) Form*, MAA, 1997. Binary forms and the topograph, and the 15 theorem.
- H. Cohn, *A conceptual breakthrough in sphere packing*, Notices AMS 64 (2017), 102–115. The LP bound and Viazovska's construction, for non-specialists.
- M. N. Huxley, *Area, Lattice Points, and Exponential Sums*, Oxford, 1996. For the circle problem.

Chapter C

Interlude C: The torus, Diophantine approximation and aperiodic order

[Added] This interlude is new: it is not in the handwritten notes. It starts from the torus $\mathbb{R}^n/\Lambda$ of Lecture 4 and asks what a single straight line does on it. The answer leads through Diophantine approximation, the points on a circle of Lecture 7, the Fibonacci words of Lecture 11, and finally to quasicrystals, which are obtained by *folding* the Dynkin diagrams of Lectures 4–6. To keep it readable it is set in black; all of it should be read as added material.

Several objects in these notes have quietly been tori. Lecture 4 attached to a lattice $\Lambda \subset \mathbb{R}^n$ the torus $\mathbb{R}^n/\Lambda$, whose volume is $|\det \Lambda|$. Interlude A found the maximal torus of $\mathrm{SU}(n)$ to be $\mathfrak{t}/2\pi A_{n-1}$ (Section A.8). And the circle $\mathbb{S}^1 = \mathbb{R}/\mathbb{Z}$ carrying the points of Lecture 7 is the one-dimensional torus. The simplest dynamical question we can ask about a torus is:

draw a straight line on $\mathbb{R}^2/\mathbb{Z}^2$. Where does it go?

If the slope is rational, the line closes up. If the slope is irrational, the line never closes and *shreds* the torus: it comes arbitrarily close to every point. How fast it does so is the subject of Diophantine approximation, and the answer singles out the golden ratio

$$\varphi = \tau = \frac{1+\sqrt{5}}{2} = 1.618\ldots, \qquad \varphi^2 = \varphi + 1.$$

(Both letters are standard. We write φ in the number-theoretic sections and τ in the Coxeter group sections, following each literature; $\tau' = -1/\tau = \frac{1-\sqrt{5}}{2}$ is its Galois conjugate. Here τ is *not* the kissing number.) Cutting the line out of the torus in a different way produces the Fibonacci chain, the simplest quasicrystal. Its higher-dimensional relatives come from the lattices A_4, D_6 and E_8 of Lecture 6.

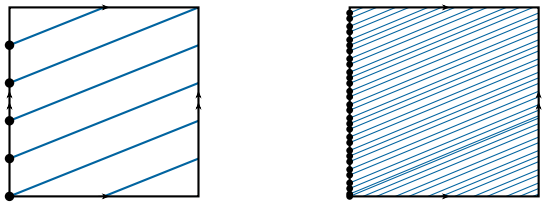
Throughout, $\{x\} = x - \lfloor x \rfloor$ is the fractional part, $\|x\|_{\mathbb{Z}} = \min_{p\in\mathbb{Z}} |x-p|$ is the distance to the nearest integer, and $e(x) = e^{2\pi i x}$.

C.1 Shredding the torus

Let ℓ_α be the image in $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ of the line $y = \alpha x$. Each time the line crosses the vertical circle $\{x = 0\}$ (that is, at $x = k \in \mathbb{Z}$) its height is $k\alpha$ modulo 1. So the *first return map* to this circle is the rotation

$$R_\alpha \colon \mathbb{T}^1 \to \mathbb{T}^1, \qquad t \mapsto t + \alpha \pmod 1,$$

and ℓ_α is the suspension of R_α: it is dense in $\mathbb{T}^2$ exactly when the orbit $\{k\alpha\}$ is dense in $\mathbb{T}^1$.



slope $\frac{2}{5}$: a closed curve slope $\sqrt{2}-1$: 30 windings

Figure C.1: The line $y = \alpha x$ wrapped around the torus $\mathbb{R}^2/\mathbb{Z}^2$ (opposite edges identified as marked). The dots on the left edge are the returns $\{k\alpha\}$ to the circle $x = 0$. For rational $\alpha = p/q$ the line closes after q turns; for irrational α it shreds the torus.

If $\alpha = p/q$ in lowest terms, the line closes up after q horizontal and p vertical turns, a closed geodesic of length $\sqrt{p^2+q^2}$. The irrational case is Kronecker's theorem, and the proof is the pigeonhole principle.

Proposition C.1 (Dirichlet 1842; Kronecker 1884)**.** *Let $\alpha \in \mathbb{R}$ and let $Q \geq 1$ be an integer. There is an integer q with $1 \leq q \leq Q$ and $\|q\alpha\|_{\mathbb{Z}} < 1/Q$. If α is irrational, the orbit $\{k\alpha\}_{k\geq 0}$ is dense in $[0,1)$, and so ℓ_α is dense in $\mathbb{T}^2$.*

Proof. The $Q+1$ numbers $\{k\alpha\}$, $0 \leq k \leq Q$, lie in the Q intervals $[i/Q, (i+1)/Q)$, so two of them, say for $j < k$, lie in the same interval. Then $q = k - j$ satisfies $\|q\alpha\|_{\mathbb{Z}} < 1/Q$. If α is irrational then $\delta = \|q\alpha\|_{\mathbb{Z}} > 0$, and the multiples of $q\alpha$ march around the circle in steps of length $\delta < 1/Q$ (all in the same direction), so every interval of length $1/Q$ contains a point of the orbit. Since Q was arbitrary, the orbit is dense. □

Density says nothing about *how often* the orbit visits a given arc. The right notion is the one behind the Boltzmann distribution of Lecture 7: time averages should equal space averages.

Definition C.2. A sequence (x_k) in $[0,1)$ is *equidistributed* if for every interval $I \subseteq [0,1)$

$$\lim_{N\to\infty} \frac{1}{N} \#\{0 \leq k < N : x_k \in I\} = |I|\,.$$

Theorem C.3 (Weyl's criterion; Weyl 1916)**.** *(x_k) is equidistributed if and only if*

$$\frac{1}{N}\sum_{k<N} e(mx_k) \to 0 \qquad \text{for every integer } m \neq 0.$$

Equivalently, $\frac{1}{N}\sum_{k<N} f(x_k) \to \int_0^1 f$ for every continuous 1-periodic f.

Sketch. If (x_k) is equidistributed, then the averages of step functions converge, and continuous functions are uniform limits of step functions. Conversely, the condition gives convergence for trigonometric polynomials, which are uniformly dense in the continuous periodic functions (Fejér, or Stone–Weierstrass). For an interval I, choose continuous $f_- \leq \mathbf{1}_I \leq f_+$ with $\int (f_+ - f_-) < \varepsilon$ and squeeze. □

Theorem C.4 (Weyl 1916; also Bohl and Sierpiński 1909–10)**.** *If α is irrational, $(\{k\alpha\})_{k\geq 0}$ is equidistributed.*

Proof. For $m \neq 0$ we have $m\alpha \notin \mathbb{Z}$, so $e(m\alpha) \neq 1$ and the geometric series gives

$$\Big|\frac{1}{N}\sum_{k=0}^{N-1} e(mk\alpha)\Big| = \frac{|e(mN\alpha)-1|}{N\,|e(m\alpha)-1|} \leq \frac{1}{N\,|\sin \pi m\alpha|} \longrightarrow 0. \qquad \square$$

The same Fourier argument gives a statement about measures rather than orbits.

Proposition C.5 (Unique ergodicity)**.** *If α is irrational, Lebesgue measure is the only Borel probability measure on $\mathbb{T}^1$ invariant under R_α.*

Proof. Let μ be invariant and $\hat{\mu}(m) = \int e(-mt)\,d\mu(t)$. Invariance gives $\hat{\mu}(m) = \int e(-m(t+\alpha))\,d\mu = e(-m\alpha)\hat{\mu}(m)$, so $\hat{\mu}(m) = 0$ for $m \neq 0$. A measure on the circle is determined by its Fourier coefficients, and these are those of Lebesgue measure. $\square$

For a general measure-preserving system there is no such uniformity, and the best one can say is *Birkhoff's ergodic theorem* (1931): if T preserves a probability measure μ and $f \in L^1(\mu)$, then $\frac{1}{N}\sum_{k<N} f(T^k x)$ converges for μ-almost every x, and the limit is $\int f\,d\mu$ when T is *ergodic* (every invariant set has measure 0 or 1). Unique ergodicity upgrades "almost every x" to "every x, uniformly".

Higher dimensions, and the torus inside a Lie group

Now let $\Lambda \subset \mathbb{R}^n$ be a lattice and $\mathbb{T}_\Lambda = \mathbb{R}^n/\Lambda$ the torus of Lecture 4. Its characters are $x \mapsto e(\langle w, x\rangle)$ with w in the *dual lattice* $\Lambda^* = \{w : \langle w, \lambda\rangle \in \mathbb{Z}\ \forall \lambda \in \Lambda\}$ (the same dual lattice that appears in the Poisson summation formula of Interlude B). Weyl's criterion holds on $\mathbb{T}_\Lambda$ with these characters, and the geometric series argument gives:

Theorem C.6 (Kronecker–Weyl)**.** *Let $v \in \mathbb{R}^n$.*

1. *The sequence $kv \bmod \Lambda$ ($k \in \mathbb{N}$) is equidistributed in $\mathbb{T}_\Lambda$ if and only if $\langle w, v\rangle \notin \mathbb{Z}$ for every $w \in \Lambda^* \setminus \{0\}$.*
2. *The line $tv \bmod \Lambda$ ($t \in \mathbb{R}$) is equidistributed (in particular dense) if and only if $\langle w, v\rangle \neq 0$ for every $w \in \Lambda^* \setminus \{0\}$, i.e. v lies in no rational hyperplane of Λ.*

For $\Lambda = \mathbb{Z}^n$, (1) says that $1, v_1, \dots, v_n$ are linearly independent over $\mathbb{Q}$.

If the condition fails for some w, the orbit stays in the closed subgroup $\{x : \langle w, x\rangle \in \mathbb{Z}\}$, a finite union of parallel subtori. So orbit closures of linear flows on tori are always subtori. This is the abelian ancestor of Ratner's theorem in Section C.8.

Remark C.7 (Generic elements of a maximal torus). In Interlude A the maximal torus of a compact group is $T = \mathfrak{t}/\Lambda_0$, with Λ_0 the kernel of exp (Section A.8). By Theorem C.6(1), an element $t = \exp(H)$ generates a dense subgroup of T if and only if $\chi(t) \neq 1$ for every nontrivial character χ of T. Such *topological generators* form a dense set of full measure. They are the standard tool for deducing from Theorem A.18 (every element is conjugate into T) that any two maximal tori are conjugate: conjugate a generator t' of T' into T, and its closure T' goes along with it.

C.2 How close does the line come? Diophantine approximation

The line ℓ_α passes within $\|q\alpha\|_{\mathbb{Z}}$ of the lattice point (q, p), measured vertically. Dirichlet's bound $\|q\alpha\|_{\mathbb{Z}} < 1/q$ for infinitely many q is a statement about lattice points near a line, and it has a second proof from the geometry of numbers of Lecture 4.

Second proof of Dirichlet, via Minkowski. The parallelogram $K_Q = \{(x,y) : |x| \le Q,\ |x\alpha - y| \le 1/Q\}$ is convex, symmetric and compact, of area $2Q \cdot 2/Q = 4$. By Minkowski's convex body theorem it contains a lattice point $(q,p) \neq 0$. If $q = 0$ then $|p| \le 1/Q < 1$ forces $p = 0$; so $1 \le |q| \le Q$ and $|q\alpha - p| \le 1/Q$. $\square$

Continued fractions

Write $\alpha = [a_0; a_1, a_2, \dots] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots}}$ with $a_i \in \mathbb{Z}$, $a_i \ge 1$ for $i \ge 1$, and let $p_n/q_n = [a_0; a_1, \dots, a_n]$ be the convergents, so $q_{n+1} = a_{n+1}q_n + q_{n-1}$. The following facts are classical (see e.g. Khinchin's *Continued Fractions*, or Cassels' *Introduction to Diophantine Approximation*, Ch. I).

(i) $\frac{1}{(a_{n+1}+2)} < q_n\,\|q_n\alpha\|_{\mathbb{Z}} < \frac{1}{a_{n+1}}$ for $n \ge 1$, and $q_n\alpha - p_n$ alternates in sign.

(ii) (Best approximation, Lagrange) If $1 \le q < q_{n+1}$, then $\|q\alpha\|_{\mathbb{Z}} \ge \|q_n\alpha\|_{\mathbb{Z}}$. So the record holders for $\|q\alpha\|_{\mathbb{Z}}$ are exactly the q_n.

(iii) For $\varphi = [1; 1, 1, 1, \dots]$ the convergents are F_{n+1}/F_n (Fibonacci numbers, $F_1 = F_2 = 1$), and $F_n\varphi - F_{n+1} = -(\tau')^n = (-1)^{n+1}\varphi^{-n}$.

Fact (iii) is Binet's formula $\varphi^n = F_n\varphi + F_{n-1}$ applied to τ', which satisfies the same quadratic equation. By (i), $\inf_q q\,\|q\alpha\|_{\mathbb{Z}} > 0$ if and only if the partial quotients a_n are bounded; such α are called *badly approximable*. The golden ratio has the smallest possible partial quotients, and it turns out to be the worst approximable number of all.

Theorem C.8 (Hurwitz 1891)**.** *For every irrational α there are infinitely many fractions p/q with $|\alpha - p/q| < \frac{1}{\sqrt{5}\,q^2}$. The constant $\sqrt{5}$ cannot be increased: for $\alpha = \varphi$ and any $A > \sqrt{5}$ only finitely many p/q satisfy $|\alpha - p/q| < \frac{1}{Aq^2}$.*

Proof of the sharpness. The existence part follows by showing that of any three consecutive convergents one satisfies the inequality (Hardy–Wright, Thm. 195). For the optimality, note that $(p - q\varphi)(p - q\tau') = p^2 - pq - q^2$ is a nonzero integer for $(p,q) \neq (0,0)$, because φ is irrational. Hence

$$q^2\Big|\varphi - \frac{p}{q}\Big| = \frac{q\,|p^2 - pq - q^2|}{|p - q\tau'|} \;\ge\; \frac{1}{|p/q - \tau'|} \longrightarrow \frac{1}{\varphi - \tau'} = \frac{1}{\sqrt{5}}$$

as $p/q \to \varphi$. So $\liminf q^2\,|\varphi - p/q| \ge 1/\sqrt{5}$, and the convergents F_{n+1}/F_n attain this value in the limit. $\square$

The proof is really about the indefinite quadratic form $x^2 - xy - y^2$, which takes no small nonzero values on $\mathbb{Z}^2$. Markov turned this into a complete description of the top of the spectrum.

Theorem C.9 (Markov 1879–80)**.** *For a real indefinite binary quadratic form f of discriminant $d > 0$, put $\mu(f) = \inf_{(x,y)\in\mathbb{Z}^2\setminus 0} |f(x,y)|\,/\sqrt{d}$. The values of $\mu(f)$ that exceed $\frac{1}{3}$ are exactly*

$$\frac{m}{\sqrt{9m^2 - 4}}, \qquad m \in \{1, 2, 5, 13, 29, 34, 89, 169, 194, \dots\},$$

where m runs over the Markov numbers*, the integers occurring in positive solutions of $x^2 + y^2 + z^2 = 3xyz$. The value for m is attained only by forms equivalent to a multiple of an explicit integral form f_m. For $m = 1, 2$ these are $x^2 - xy - y^2$ ($\mu = 1/\sqrt{5}$) and $x^2 - 2y^2$ ($\mu = 1/\sqrt{8}$).*

The dual statement for numbers is that the *Lagrange spectrum* $\{\limsup_q (q\,\|q\alpha\|_{\mathbb{Z}})^{-1}\}$ begins $\sqrt{5}, \sqrt{8}, \sqrt{221}/5, \ldots$ and accumulates only at 3. All Markov triples arise from $(1,1,1)$ by the moves $(x,y,z) \mapsto (x,y,3xy-z)$. Whether each Markov number is the largest entry of only one triple (Frobenius 1913) is the *Markov uniqueness conjecture*, still open.

Badly approximable numbers are bounded orbits

Here the torus meets the space of lattices $X_2 = \mathrm{SL}_2(\mathbb{R})/\mathrm{SL}_2(\mathbb{Z})$ of Lecture 4. Put

$$\Lambda_\alpha = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \mathbb{Z}^2 = \{(q,\, q\alpha + p) : p, q \in \mathbb{Z}\}, \qquad g_t = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{t} \end{pmatrix}.$$

The flow g_t squeezes the first coordinate and stretches the second. A lattice point with small $q\,|q\alpha + p|$ becomes, at the right time t, a short vector of $g_t\Lambda_\alpha$.

Proposition C.10 (Dani correspondence, simplest case)**.** *α is badly approximable if and only if the forward orbit $\{g_t\Lambda_\alpha : t \geq 0\}$ is bounded in X_2.*

Proof. By Mahler's criterion (Lecture 4) a set of unimodular lattices is bounded exactly when the shortest vector length λ_1 is bounded below on it. We use the sup norm, which changes λ_1 by at most a factor $\sqrt{2}$. Suppose $|q|\,|q\alpha + p| \geq c$ for all $q \neq 0$, where we may take $c \leq 1$. A vector $v = (e^{-t}q, e^{t}(q\alpha + p))$ of $g_t\Lambda_\alpha$ with $q = 0$ has $\|v\|_\infty = e^{t}\,|p| \geq 1$. If $q \neq 0$ then $\|v\|_\infty^2 \geq |q|\,|q\alpha + p| \geq c$. So $\lambda_1 \geq \sqrt{c}$ along the orbit. Conversely, suppose $\|v\|_\infty \geq \varepsilon$ for all $t \geq 0$ and $v \neq 0$, where $\varepsilon \leq 1$. Given $q \geq 1$ and p, choose $t \geq 0$ with $e^{-t}q = \varepsilon/2$. Then the first coordinate of v is less than ε, so $e^{t}\,|q\alpha + p| \geq \varepsilon$, that is, $q\,|q\alpha + p| \geq \varepsilon^2/2$. □

This dictionary, due to Dani (1985) in all dimensions, turns questions about numbers into questions about orbits on the space of lattices. We return to it in Section C.8.

C.3 Three gaps

Return to the circle of Lecture 7 (Section 7.3.3), which we take to have length 1, and place on it the first N points of an irrational rotation,

$$x_k = \{k\alpha\}, \qquad 0 \leq k \leq N-1.$$

They cut the circle into N arcs, a point $(g_1, \ldots, g_N)$ of the open gap simplex of Lecture 7. One would expect N different gap lengths. There are at most three.

Theorem C.11 (Three gap theorem; conjectured by Steinhaus, proved by Sós 1957/58, Surányi 1958, Świerczkowski 1959)**.** *Let α be irrational and $N \geq 2$. Let $u, v \in \{1, \ldots, N-1\}$ be the indices of the points closest to 0 from the right and from the left, and put $a = \{u\alpha\}$, $b = 1 - \{v\alpha\}$. Then every gap between neighbouring points has length a, b or $a+b$. More precisely, the right neighbour of x_k is*

$$x_{k+u} \ \ (\text{gap } a) \text{ if } k < N-u, \qquad x_{k-v} \ \ (\text{gap } b) \text{ if } k \geq v, \qquad x_{k+u-v} \ \ (\text{gap } a+b) \text{ otherwise.}$$

There are $N-u$ gaps of length a, $N-v$ of length b, and $u+v-N$ of length $a+b$.

Proof. All the indices named are in $[0, N-1]$, so each claimed neighbour is one of the points, at the stated distance to the right of x_k (because $x_{k+u} = x_k + a$ and $x_{k-v} = x_k + b$ modulo 1). We must show that no point x_j lies strictly inside the arc. We use only the minimality of a and b: no $1 \le m \le N-1$ has $\{m\alpha\} \in (0, a)$ or $\{m\alpha\} \in (1-b, 1)$.

Gap a ($k+u \le N-1$). If $x_j \in (x_k, x_k + a)$ with $j > k$, then $\{(j-k)\alpha\} \in (0, a)$, contradicting minimality of a. If $j < k$, then $x_{k+u} = x_k + a$ lies in $(x_j, x_j + a)$, so $m = k + u - j \in [1, N-1]$ has $\{m\alpha\} \in (0, a)$, again a contradiction.

Gap b ($k \ge v$). If $x_j \in (x_k, x_k + b)$ with $j < k$, then $\{(j-k)\alpha\} \in (0, b)$, i.e. $m = k - j$ has $\{m\alpha\} \in (1-b, 1)$. If $j > k$, then $x_j \in (x_{k-v} - b, x_{k-v})$, and $m = j - k + v$ (which lies in $[1, N-1]$) has $\{m\alpha\} \in (1-b, 1)$. Both contradict minimality of b.

Gap $a+b$ ($N - u \le k \le v - 1$). Suppose $x_j \in (x_k, x_k + a + b)$. If $j > k$, then $\{(j-k)\alpha\} \in (0, a+b)$, and in fact in $(a, a+b)$ by minimality of a. (It cannot equal a, since then $j = k+u \ge N$.) Then $m = k+u-j$ lies in $[1, N-1]$ and $\{-m\alpha\} = \{(j-k)\alpha\} - a \in (0, b)$, so $\{m\alpha\} \in (1-b, 1)$, which is impossible. If $j < k$, then $s = k - j \in [1, N-1]$ has $1 - \{s\alpha\} \in (b, a+b)$, and $m = v - s \in [1, N-1]$ has $\{m\alpha\} = \{v\alpha\} - \{s\alpha\} = (1 - \{s\alpha\}) - b \in (0, a)$, which is impossible.

The three ranges of k are disjoint: a k in the first two would give $a = b$, hence $(u+v)\alpha \in \mathbb{Z}$. Counting the k in each range gives the multiplicities. □

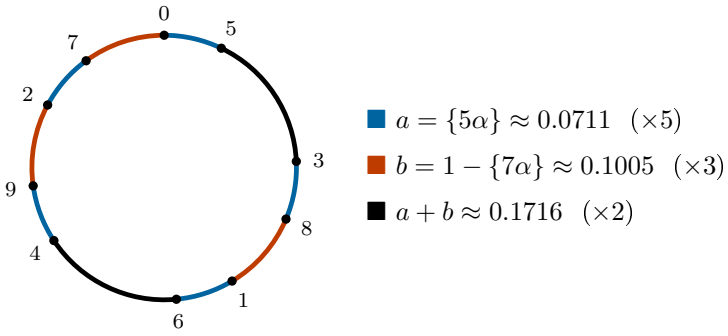


Figure C.2: The points $\{k\alpha\}$, $0 \le k \le 9$, for $\alpha = \sqrt{2} - 1$, drawn clockwise from the top. Here $u = 5$, $v = 7$, $N = 10$, and the multiplicities are $N - u = 5$, $N - v = 3$ and $u + v - N = 2$.

Example C.12 (Fibonacci: two gaps in ratio φ)**.** Take $\alpha = \varphi^{-1} = \varphi - 1$ and $N = F_n$ with $n \ge 4$. By facts (ii) and (iii) above, the closest returns to 0 among $1 \le k < F_n$ occur at $k = F_{n-1}$ and $k = F_{n-2}$. (Since all partial quotients equal 1, there are no intermediate fractions, so the one-sided best approximations are also convergents.) They are at distances $\varphi^{-(n-1)}$ and $\varphi^{-(n-2)}$ on opposite sides (the convergents of φ^{-1} are F_{k-1}/F_k). So $\{u, v\} = \{F_{n-1}, F_{n-2}\}$ and $u + v - N = 0$: *there are only two gap lengths, and their ratio is φ.* There are F_{n-2} short gaps and F_{n-1} long ones, and indeed $F_{n-2}\varphi^{-(n-1)} + F_{n-1}\varphi^{-(n-2)} = 1$. Adding the next point splits one long gap into a long and a short gap, in the ratio $\varphi : 1$. This is the arrangement of leaves ("phyllotaxis") that a plant produces when it turns each new leaf by the golden angle $2\pi/\varphi^2 \approx 137.5°$. (Checked numerically for $N \le 40$.)

Proposition C.13 (How well spread is an orbit?)**.** *Let $m_N(\alpha)$ be the smallest gap of $\{k\alpha\}_{0 \le k < N}$, so that the injectivity radius of Lecture 7 is $\rho = m_N/2$. The regular N-gon has $N m_N = 1$. For every irrational α,*

$$\liminf_{N \to \infty} N\, m_N(\alpha) = \liminf_{q \to \infty} q\, \|q\alpha\|_{\mathbb{Z}} \le \frac{1}{\sqrt{5}},$$

with equality for $\alpha = \varphi$. Moreover $\inf_N Nm_N(\alpha) > 0$ if and only if α is badly approximable.

Proof. The smallest gap is the smallest circular distance between two of the points, so $m_N = \min_{1\le k<N} \|k\alpha\|_{\mathbb{Z}}$. Taking $N = q+1$ gives $Nm_N \le (q+1)\,\|q\alpha\|_{\mathbb{Z}}$. Conversely $m_N = \|k\alpha\|_{\mathbb{Z}}$ for some $k < N$, so $Nm_N \ge k\,\|k\alpha\|_{\mathbb{Z}}$, and $k \to \infty$ as $N \to \infty$ because $m_N \to 0$. This proves the equality of the two $\liminf$s, and the same two inequalities give the last statement. The bound $1/\sqrt{5}$ and the equality case are Hurwitz's Theorem C.8. □

So for *every* rotation, infinitely often the injectivity radius falls short of the optimal value $1/(2N)$ of Lecture 7 by a factor of (almost) $\sqrt{5}$. The golden rotation does no worse than this asymptotically. Rotations by numbers not equivalent to φ do worse (by a factor $\sqrt{8}$ or more), by Markov's Theorem C.9.

Remark C.14 (Gaps as a function on the space of lattices). The vector of gaps lives in the gap simplex of Lecture 7. In Interlude A the same simplex appeared as the Weyl alcove T/W of $\mathrm{SU}(N)$ (the eigenvalues of a unitary matrix are N points on a circle). Marklof and Strömbergsson (2017) observed that the gap lengths of $\{k\alpha\}_{k<N}$ are values of a simple function on the space of lattices X_2, evaluated at a lattice obtained from Λ_α by the diagonal flow at time $t \approx \log N$. The three gap theorem then becomes an elementary statement about two-dimensional lattices.

C.4 Fibonacci points and discrepancy

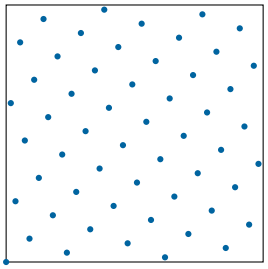
How uniform can N points in the unit square be? Measure the error with the *star discrepancy*

$$D^*(P) = \sup_{0\le x,y\le 1} \Big| \frac{\#\big(P \cap [0,x)\times[0,y)\big)}{N} - xy \Big|.$$

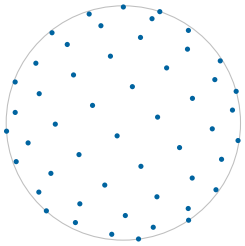
Random points have $D^* \approx N^{-1/2}$. The three gap structure suggests a better choice. The *Fibonacci lattice* with $N = F_m$ points is

$$P_m = \Big\{ \Big(\frac{k}{F_m}, \Big\{ \frac{kF_{m-1}}{F_m} \Big\} \Big) : 0 \le k < F_m \Big\},$$

the points of the line of slope $F_{m-1}/F_m \approx \varphi^{-1}$ on the torus (Figure C.1), sampled at the F_m times it crosses $x = k/F_m$. It is a lattice in $\mathbb{T}^2$, namely $\mathbb{Z}^2 + \mathbb{Z}\,(1, F_{m-1})/F_m$.



$F_{10} = 55$ Fibonacci lattice points



100 spherical Fibonacci points

Figure C.3: Left: the Fibonacci lattice P_{10}. Right: the front half of the spherical Fibonacci point set (C.1) with $N = 100$, viewed from the side.

Theorem C.15 (Discrepancy in the square)**.**

1. *(Schmidt 1972) There is $c > 0$ such that every N-point set $P \subset [0,1]^2$ has $D^*(P) \geq c\,\frac{\log N}{N}$. (Roth 1954 had proved the weaker bound $c\sqrt{\log N}/N$. It is a bound for the L^2 average of the discrepancy, and in that form it is sharp, by Davenport 1956.)*
2. *The Fibonacci lattice has $D^*(P_m) = O\big(\frac{\log F_m}{F_m}\big)$. More generally, a two-dimensional lattice $\{(k/N, \{kg/N\})\}$ has $D^* = O(K \log N/N)$ if the partial quotients of g/N are at most K (see Niederreiter,* Random Number Generation and Quasi-Monte Carlo Methods*, 1992, Ch. 5).*

So the Fibonacci lattice is optimal up to a constant, because all partial quotients of $F_{m-1}/F_m = [0; 1, 1, \ldots, 1]$ equal 1. The same mechanism appears in one dimension: for badly approximable α the discrepancy of $\{k\alpha\}_{k<N}$ is $O(\log N/N)$ (see Kuipers–Niederreiter, Ch. 2), while Schmidt showed that every infinite sequence in $[0,1)$ has discrepancy at least $c \log N/N$ for infinitely many N.

On the sphere. Pull the Fibonacci lattice back to $\mathbb{S}^2$ by the area-preserving cylindrical (Lambert) projection $(\theta, z) \mapsto (\sqrt{1-z^2}\cos\theta, \sqrt{1-z^2}\sin\theta, z)$. Up to small variations this gives the *spherical Fibonacci points*

$$z_k = 1 - \frac{2k+1}{N}, \qquad \theta_k = 2\pi k \varphi^{-1}, \qquad 0 \leq k < N, \tag{C.1}$$

a golden spiral from pole to pole. The relevant measure of uniformity is the *spherical cap discrepancy* D_{cap}, the supremum over spherical caps C of $|\#(P \cap C)/N - \sigma(C)|$, with σ the normalized area. Beck (1984) showed that every N-point set has $D_{\text{cap}} \geq cN^{-3/4}$, and that some N-point sets have $D_{\text{cap}} \leq CN^{-3/4}\sqrt{\log N}$. For the spherical Fibonacci lattice, Aistleitner, Brauchart and Dick (2012) proved $D_{\text{cap}} \leq 44\sqrt{8/F_m}$, that is $O(N^{-1/2})$. Their numerical experiments suggest that the true order is close to $N^{-3/4}$, but this is *not* proved. Spherical Fibonacci points are also a standard starting configuration for numerical searches in the Tammes and Thomson problems of Lectures 2 and 11. This is an empirical observation; they are not optimal.

C.5 Sturmian words and the Fibonacci chain

Now *code* the irrational line. Record, at each integer time k, whether the rotation $t \mapsto t + \alpha$ wraps around:

$$s_k = \lfloor (k+1)\alpha + \rho \rfloor - \lfloor k\alpha + \rho \rfloor \in \{0, 1\} \qquad (0 < \alpha < 1).$$

Equivalently, $s_k = 1$ if and only if $\{k\alpha + \rho\} \in [1-\alpha, 1)$. Such a word is called a *mechanical* (or *Sturmian*) word of slope α. It is also the *cutting sequence* of the line $y = \alpha x + \rho$: the record of which horizontal grid lines it crosses between consecutive vertical ones. The *complexity* $p(n)$ of an infinite word is its number of distinct factors (blocks) of length n.

Theorem C.16 (Morse–Hedlund 1938/1940; Coven–Hedlund 1973)**.** *An infinite word with $p(n) \leq n$ for some n is eventually periodic. The one-sided infinite words of the smallest remaining complexity, $p(n) = n + 1$ for all n, are exactly the mechanical words of irrational slope. (For two-sided words this needs recurrence: $\cdots aaabaaa \cdots$ also has $p(n) = n + 1$.)*

Sketch that a mechanical word has $p(n) = n + 1$. The factor $s_k \cdots s_{k+n-1}$ is determined by which of the arcs $[1-\alpha, 1) - j\alpha$, $0 \leq j < n$, contain $y = \{k\alpha + \rho\}$. As y moves around the circle, the

factor can change only when y crosses one of the endpoints of these arcs, which are the $n+1$ points $\{-j\alpha\}$, $0 \le j \le n$. These cut the circle into $n+1$ arcs, so $p(n) \le n+1$. The word is not eventually periodic (its frequency of 1's is the irrational number α), so $p(n) \ge n+1$ by the first half of the theorem (Exercise C.31). Hence $p(n) = n+1$, and the $n+1$ arcs correspond bijectively to the factors. □

By equidistribution, the frequency of a factor is the length of its arc, and by the three gap theorem the $n+1$ arcs have at most three different lengths. So *the factors of length n of a Sturmian word occur with at most three different frequencies.*

The Fibonacci word. Let σ be the substitution $a \mapsto ab$, $b \mapsto a$. Then $\sigma^n(a)$ is a prefix of $\sigma^{n+1}(a)$, and the limit is the *Fibonacci word*

$$f = abaababaabaababaababaabaababaabaab\cdots, \qquad |\sigma^n(a)| = F_{n+2}.$$

The lengths obey $\left|\sigma^{n+1}(a)\right| = |\sigma^n(a)| + \left|\sigma^{n-1}(a)\right|$, because $\sigma^{n+1}(a) = \sigma^n(a)\,\sigma^{n-1}(a)$. The letter b never appears twice in a row, since every b comes from an $a \mapsto ab$ and the next block begins with a. The frequency of b is $\lim F_n/F_{n+2} = \varphi^{-2} \approx 0.382$. In fact f is the mechanical word of slope φ^{-2} with $\rho = \varphi^{-2}$ and $b \leftrightarrow 1$ (the word with $\rho = 0$ is af): its k-th letter ($k \ge 0$) is b exactly when $\lfloor (k+2)\varphi^{-2} \rfloor - \lfloor (k+1)\varphi^{-2} \rfloor = 1$ (Lothaire, *Algebraic Combinatorics on Words*, Ch. 2; we checked the first 987 letters numerically).

Remark C.17 (Back to the channel of Lecture 11). With $b = 1$ and $a = 0$, the Fibonacci word is a binary word with no two adjacent 1's, which is exactly the jamming condition for discs in the narrow channel of Section 11.4. It is a jammed configuration that is neither periodic nor random. Its fraction of 1's is $\varphi^{-2} \approx 0.382$. This lies strictly between the typical value $\frac{1}{2} - \frac{\sqrt{5}}{10} \approx 0.276$ of Exercise 11.10 and the maximum $\frac{1}{2}$. Its length per contact is $2\varphi^{-2} + L\varphi^{-1}$. Every finite block of it occurs in it with bounded gaps, so it is "as ordered as an aperiodic jammed state can be". The count $F_{\ell+2}$ of all such words and the lengths $|\sigma^n(a)| = F_{n+2}$ obey the same Fibonacci recursion.

C.6 Cut and project

Here is the Fibonacci chain as a geometric object. In $\mathbb{R}^2$ take the line $E_\parallel$ of slope φ^{-1} (direction $(\varphi, 1)$), the perpendicular line $E_\perp$, and the strip $S = E_\parallel + W$, where the *window* W is the projection of the unit square to $E_\perp$, shifted generically. Select the lattice points of $\mathbb{Z}^2$ in the strip and project them to $E_\parallel$.

Proposition C.18. *Let $w(m,k) = \varphi k - m + s$ (this is $\sqrt{1+\varphi^2}$ times the $E_\perp$-coordinate, shifted by s), and select the points with $w \in (-1, \varphi]$. Each selected point x is followed in the strip by exactly one of $x+(1,0)$ (if $w(x) > 0$) or $x+(0,1)$ (if $w(x) \le 0$). The steps project to intervals of lengths $\varphi/\sqrt{1+\varphi^2}$ and $1/\sqrt{1+\varphi^2}$. In the normalized coordinate $y = (w+1)/\varphi^2 \in (0,1]$, every step acts as $y \mapsto y - \varphi^{-2} \pmod 1$, and the step is short exactly when $y \le \varphi^{-2}$. So the sequence of tiles is a mechanical word of slope φ^{-2}: a Fibonacci word (for s chosen suitably, exactly f).*

Proof. If $w(x) > 0$ then $w(x+(1,0)) = w(x) - 1 \in (-1, \varphi - 1]$ is in the window, while $w(x + (0,1)) = w(x) + \varphi > \varphi$ is not. If $w(x) \le 0$ the roles are reversed, since $w(x) - 1 \le -1$ and $w(x) + \varphi \in (\varphi - 1, \varphi]$. A step $(1,0)$ changes y by $-1/\varphi^2$, and a step $(0,1)$ by $\varphi/\varphi^2 = 1 - \varphi^{-2}$, and $w > 0$ means $y > \varphi^{-2}$. It remains to see that the staircase through one selected point contains

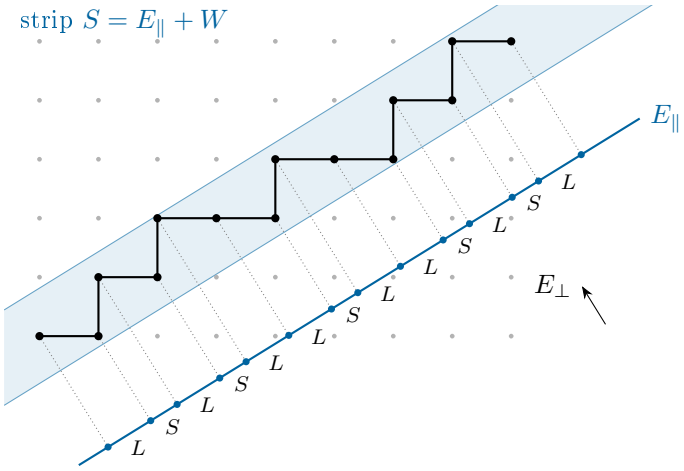


Figure C.4: The Fibonacci chain by cut and project. The lattice points of $\mathbb{Z}^2$ in the strip (black) form a staircase. Projected orthogonally onto $E_\parallel$ they give long (L) and short (S) intervals, of lengths in the ratio $\varphi : 1$, in the order of the Fibonacci word.

all of them. Every selected point lies on such a staircase (run the rule backwards as well), and distinct staircases are disjoint. By Theorem C.4 a staircase has a fraction φ^{-2} of short steps, so it has $1/(\varphi^{-1} \cdot \frac{\varphi}{\sqrt{1+\varphi^2}} + \varphi^{-2}\frac{1}{\sqrt{1+\varphi^2}}) = \varphi^2/\sqrt{1+\varphi^2}$ points per unit length of $E_\parallel$. This is already the density of all lattice points in the strip, which is (strip width)/(covolume) $= \varphi^2/\sqrt{1+\varphi^2}$. So there is only one staircase. □

This is the prototype of a *model set*.

Definition C.19. A *cut and project scheme* is a lattice $\mathcal{L} \subset E_\parallel \times E_\perp$ such that the projection $\pi_\parallel$ is injective on $\mathcal{L}$ and $\pi_\perp(\mathcal{L})$ is dense in $E_\perp$. For a *window* $W \subset E_\perp$, relatively compact with nonempty interior, the *model set* is $\Lambda(W) = \{\pi_\parallel(x) : x \in \mathcal{L},\ \pi_\perp(x) \in W\}$. It is *regular* if ∂W has measure zero.

A model set is a Delone set (uniformly discrete and relatively dense), so it is a packing in the sense of Lecture 1 and has a density. It is in general not periodic. It is a *Meyer set*: $\Lambda - \Lambda \subseteq \Lambda + F$ for a finite set F (Meyer 1972). Its density is $\mathrm{vol}(W)/\operatorname{covol}(\mathcal{L})$, by Weyl equidistribution of $\pi_\perp(\mathcal{L})$ in W. Most importantly for physics, *regular model sets have pure point diffraction*: the Fourier transform of the autocorrelation of the Dirac comb $\sum_{x\in\Lambda} \delta_x$ is a sum of point masses, supported on the projection of the dual lattice $\mathcal{L}^*$ (Hof 1995 for Euclidean internal spaces; Schlottmann 2000 in general). The proof is a Poisson summation formula (Interlude B) for $\mathcal{L}$, cut down by the window.

Penrose tilings and Shechtman's crystals. De Bruijn (1981) showed that the rhombic Penrose tilings (Penrose 1974) are dual to *pentagrids*: superpositions of five families of parallel lines at angles $2\pi j/5$. Equivalently, their vertex sets are obtained by cut and project from $\mathbb{Z}^5$, with $E_\parallel$ the plane on which the cyclic permutation of coordinates acts as rotation by $2\pi/5$. Since the vertices lie in finitely many layers along $(1,1,1,1,1)$, the minimal setting is the four-dimensional root lattice A_4, with a window made of four pentagons in a two-dimensional internal space (Baake–Kramer–Schlottmann–Zeidler 1990). Figure C.5 was drawn from a pentagrid. In 1984

Shechtman, Blech, Gratias and Cahn reported a rapidly solidified Al–Mn alloy whose electron diffraction pattern had sharp spots and icosahedral symmetry. Such symmetry is impossible for a crystal: a rotation of order 5 preserving a lattice in $\mathbb{R}^2$ or $\mathbb{R}^3$ would have integer trace, but its trace is $2\cos\frac{2\pi}{5} = \varphi^{-1}$, or $1+\varphi^{-1}$. The discovery earned Shechtman the 2011 Nobel Prize in Chemistry. Model sets with icosahedral symmetry are the standard mathematical models of such quasicrystals.

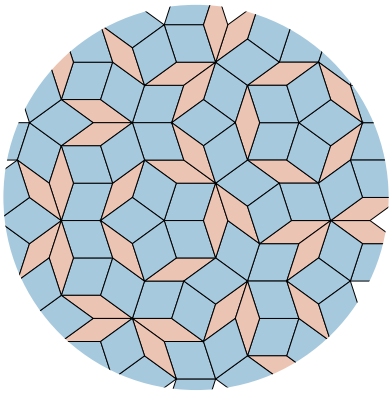


Figure C.5: A patch of a rhombic Penrose tiling, computed from a generic pentagrid (de Bruijn's method). Thick rhombi (72°) blue, thin rhombi (36°) orange.

C.7 Folding Dynkin diagrams: quasicrystals from A_4, D_6, E_8

Where do the icosahedral windows and lattices come from? Return to Lecture 4. Dropping the crystallographic axiom 4) from Definition 4.5 allows more root systems. The exercise after that definition asked for the decagon and the icosidodecahedron. Coxeter's classification of finite reflection groups (1934) adds exactly three types to the list of Lecture 5: the dihedral groups $I_2(m)$ ($m \neq 2,3,4,6$), and two exceptional groups H_3 and H_4, with Coxeter graphs

$$H_2 = I_2(5):\ \circ\!\overset{5}{—}\!\circ, \qquad H_3:\ \circ—\circ\overset{5}{—}\circ, \qquad H_4:\ \circ—\circ—\circ\overset{5}{—}\circ\,.$$

These are the symmetry groups of the regular pentagon (order 10), the icosahedron (order 120) and the 600-cell (order 14400). An edge labelled 5 means an angle of $\pi - \pi/5$ between simple roots, so $\cos\frac{\pi}{5} = \frac{\tau}{2}$ enters, and the Cartan numbers are irrational. These groups preserve no lattice in their own dimension. But they do preserve a lattice of *twice* the dimension, and that is where the quasicrystals come from.

The construction is a *folding* of a simply laced Dynkin diagram (Shcherbak 1988; Moody–Patera 1993). Pair up the $2r$ nodes of $\Gamma = A_4, D_6, E_8$ as (α_i, β_i), $1 \le i \le r$ (so $r = 2,3,4$), as in Figure C.6. The rule is simple. Each plain edge i—j of H_r gives the two edges α_i—α_j and β_i—β_j. Each 5-edge i—j gives the three edges α_i—β_j, β_i—α_j and β_i—β_j. For H_4 this produces the tree with arms of lengths $4,2,1$ from the node β_3, which is E_8. For H_3 it produces D_6, and for H_2 the chain α_1—β_2—β_1—α_2, which is A_4.

Theorem C.20 (Folding). *Let $\Gamma \in \{A_4, D_6, E_8\}$ with simple roots labelled as in Figure C.6, L its root lattice in $V = \mathbb{R}^{2r}$, and define $M\colon L \to L$ on the simple roots by*

$$M\alpha_i = \beta_i, \qquad M\beta_i = \alpha_i + \beta_i.$$

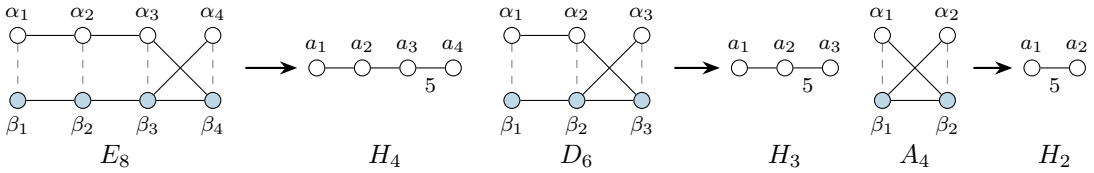


Figure C.6: Folding $E_8 \to H_4$, $D_6 \to H_3$, $A_4 \to H_2$. The dashed lines pair each node α_i with its partner β_i (shaded). These are orthogonal, and under the projection $\beta_i \mapsto \tau a_i$, $\alpha_i \mapsto a_i$. A plain edge of H_r comes from two parallel edges, and a 5-edge from a crossing pair plus the bottom edge. The E_8 diagram here is the chain $\alpha_1\alpha_2\alpha_3\beta_4\beta_3\beta_2\beta_1$ with α_4 attached to β_3.

1. *M is self-adjoint and $M^2 = M + 1$. Hence $V = E_\tau \oplus E_{\tau'}$ is an orthogonal sum of the τ- and τ'-eigenspaces of M, each of dimension r, and L is a module over $\mathbb{Z}[\tau]$ (with τ acting as M).*
2. *Let P be the orthogonal projection onto E_τ and $a_i = P\alpha_i$. Then $P\beta_i = \tau a_i$, and $a_1, \dots, a_r$ are the simple roots of H_r, all of squared length $1 - \frac{1}{\sqrt{5}}$.*
3. *The products $w_i = s_{\alpha_i} s_{\beta_i}$ preserve E_τ and act on it as the reflections s_{a_i}. They generate a subgroup of the Weyl group $W(\Gamma)$ isomorphic to $W(H_r)$.*
4. *P is injective on L, and $P(L) = \mathbb{Z}[\tau]a_1 + \cdots + \mathbb{Z}[\tau]a_r$ is dense in E_τ.*

Proof. (1) M is self-adjoint if and only if $\langle \beta_i, \alpha_j \rangle = \langle \alpha_i, \beta_j \rangle$ and $\langle \beta_i, \beta_j \rangle = \langle \alpha_i, \alpha_j \rangle + \langle \alpha_i, \beta_j \rangle$ for all i, j. Both identities can be read off the edge rule. Each side of the first is -1 exactly for a 5-edge. For the second, take $i = j$ ($2 = 2 + 0$), a plain edge ($-1 = -1 + 0$), a 5-edge ($-1 = 0 - 1$), and no edge ($0 = 0$). Clearly $M^2\alpha_i = M\beta_i = \alpha_i + \beta_i = (M+1)\alpha_i$, and similarly for β_i. A self-adjoint operator with $M^2 = M + 1$ has eigenvalues among τ, τ' and orthogonal eigenspaces. Since M has an integer matrix, the two eigenvalues, being Galois conjugate, have equal multiplicity.

(2) $P = (M - \tau')/\sqrt{5}$ commutes with M, so $P\beta_i = PM\alpha_i = MP\alpha_i = \tau a_i$. Then $\langle a_i, a_j \rangle = \langle P\alpha_i, \alpha_j \rangle = \big(\langle \beta_i, \alpha_j \rangle - \tau' \langle \alpha_i, \alpha_j \rangle\big)/\sqrt{5}$. This gives $-2\tau'/\sqrt{5} = 1 - \frac{1}{\sqrt{5}} =: x$ for $i = j$, $\tau'/\sqrt{5} = -\frac{x}{2}$ for a plain edge, so the angle is $2\pi/3$, and $-1/\sqrt{5} = -x\frac{\tau}{2}$ for a 5-edge, so the angle is $\pi - \pi/5$. Non-adjacent pairs give 0.

(3) The roots α_i, β_i are orthogonal, so $w_i = -1$ on $U_i = \operatorname{span}(\alpha_i, \beta_i)$ and $w_i = 1$ on $U_i^\perp$. The plane U_i is M-invariant, hence so is $U_i^\perp$, so w_i commutes with M and preserves E_τ. On E_τ its (-1)-eigenspace is $U_i \cap E_\tau = \mathbb{R}a_i$. So $w_i|_{E_\tau} = s_{a_i}$, and these generate $W(H_r)$. If some $w \in \langle w_i \rangle$ is trivial on E_τ, then $w - 1$ is a rational matrix vanishing on E_τ, hence also on its Galois conjugate $E_{\tau'}$, so $w = 1$.

(4) A nonzero $v \in L$ with $Pv = 0$ would satisfy $Mv = \tau' v$, impossible for a rational vector v and an integer matrix M. The image is the $\mathbb{Z}[\tau]$-span of the a_i, which is dense because $\mathbb{Z}[\tau]$ is dense in $\mathbb{R}$. □

A direct computation (we did it in python3) shows that P maps the roots of Γ onto $\Delta(H_r) \cup \tau\Delta(H_r)$:

$$A_4 :\ 20 = 10 + 10, \qquad D_6 :\ 60 = 30 + 30, \qquad E_8 :\ 240 = 120 + 120.$$

Here $\Delta(H_r)$ is the root system of H_r: the vertices of a decagon, of an icosidodecahedron, and of a 600-cell. The 120 unit quaternions forming the 600-cell are the *icosians*; their $\mathbb{Z}[\tau]$-span, the

icosian ring, is a $\mathbb{Z}[\tau]$-order in the quaternions, and Theorem C.20(1) is a shadow of the fact that E_8 is the icosian ring, with a suitably modified norm, viewed as a lattice over $\mathbb{Z}$ (Conway–Sloane, Ch. 8, §2; Moody–Patera 1993). The planar case is drawn in Figure C.7.

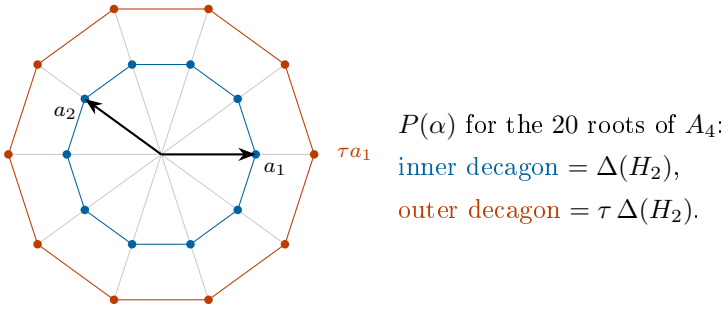


Figure C.7: The 20 roots of A_4 projected to the plane E_τ (the Coxeter plane of A_4). The two concentric decagons have radii in the ratio τ.

Quasicrystals. By Theorem C.20(4), $(L, E_\tau, E_{\tau'})$ is a cut and project scheme whose symmetry group $W(H_r)$ acts on both factors, on E_τ as H_r and on $E_{\tau'}$ as its Galois twin. Choosing an H_r-invariant window in $E_{\tau'}$ gives model sets with decagonal (A_4), icosahedral (D_6; face-centred icosahedral phases such as Al–Cu–Fe are modelled on it, while primitive ones such as Shechtman's Al–Mn use $\mathbb{Z}^6$) or H_4 symmetry (E_8, the four-dimensional quasicrystal of Elser–Sloane 1987). In this way the lattices of Lecture 6 *are* the hidden periodic structures behind the quasicrystals.

Affine extensions and viruses. For a crystallographic diagram, adding the extra node of the extended (affine) Dynkin diagram adds a translation, and the group becomes an infinite discrete affine reflection group, as in Interlude A's alcoves. For H_r, adding a translation produces a group whose orbits are dense, because $\mathbb{Z}[\tau]$ is. Patera and Twarock (2002) constructed an affine extension of each of H_2, H_3, H_4 (unique under their axioms), and used finitely many steps of it to generate finite quasicrystal fragments. Dechant, Boehm and Twarock (2013) showed that some of these extensions are exactly what one gets by folding the *affine* diagrams $\tilde{A}_4, \tilde{D}_6, \tilde{E}_8$ and projecting. Twarock's motivation was biology. Many viral capsids are icosahedral. Caspar–Klug theory explains many of them, but not, for instance, the papovaviruses, whose 72 capsomers are all pentamers. Twarock (2004) explained their architecture by tilings built from the icosahedral group. Keef and Twarock (2009) used affine extensions of the icosahedral group to describe how material at different radii of a virus (capsid, genome) is organised by one common icosahedral blueprint.

C.8 The space of lattices as a dynamical system

Section C.2 used the space $X_n = \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$ of unimodular lattices from Lecture 4. It is not compact (Mahler's criterion characterizes its bounded sets), but it has finite volume. Siegel's mean value theorem (Interlude B) says that, for the normalized invariant probability measure μ_n on X_n and f integrable on $\mathbb{R}^n$,

$$\int_{X_n} \sum_{v \in \Lambda \setminus 0} f(v)\, d\mu_n(\Lambda) = \int_{\mathbb{R}^n} f.$$

Take $f = \mathbf{1}_B$ for a ball B of radius r and volume $V < 2$. The average number of nonzero lattice vectors in B is $V < 2$. Since nonzero lattice vectors come in pairs $\pm v$, *some* lattice has none. Its balls of radius $r/2$ form a packing of density $V2^{-n}$. This gives the Minkowski–Hlawka bound $\delta \geq 2^{1-n}$ for lattice packings, and counting only primitive vectors improves it to $\zeta(n)2^{1-n}$. It is a non-constructive existence statement, like the saturation bound of Lecture 1, but for lattices.

$\mathrm{SL}_n(\mathbb{R})$ acts on X_n, and the dynamics of its subgroups encode arithmetic. The diagonal flow g_t of Proposition C.10 is the geodesic flow on the modular surface. *Unipotent* subgroups behave much more rigidly.

Theorem C.21 (Margulis 1987, the Oppenheim conjecture)**.** *Let Q be a real nondegenerate indefinite quadratic form in $n \geq 3$ variables that is not a multiple of a form with rational coefficients. Then for every $\varepsilon > 0$ there is $x \in \mathbb{Z}^n \setminus 0$ with $|Q(x)| < \varepsilon$. (Dani and Margulis,* Invent. Math. *98 (1989), strengthened this to the density of $Q(x)$, $x \in \mathbb{Z}^n$ primitive, in $\mathbb{R}$.)*

Margulis's proof is a statement about orbits. It suffices to treat $n = 3$. Write $Q = Q_0 \circ g$ with Q_0 a fixed form of signature $(2,1)$, and let $H = \mathrm{SO}(Q_0)^\circ \cong \mathrm{SO}(2,1)^\circ$, which is generated by unipotent elements. Raghunathan observed that it suffices to show: *every bounded H-orbit in X_3 is compact.* Margulis proved this. A compact orbit $Hg\mathbb{Z}^3$ occurs only when Q is a multiple of a rational form. Otherwise the orbit is unbounded, so by Mahler's criterion some $hg\mathbb{Z}^3$ contains a very short vector hgx, and then $|Q(x)| = |Q_0(hgx)|$ is small. Theorem C.21 fails for $n = 2$, and the reason is Section C.2. For $Q(x,y) = (x - \sqrt{2}y)(x - \sqrt{3}y)$ one has $\inf_{\mathbb{Z}^2\setminus 0} |Q| > 0$ (Exercise C.26), because $\sqrt{2}$ and $\sqrt{3}$ are badly approximable. Markov's Theorem C.9 is a detailed study of exactly these binary forms.

Theorem C.22 (Ratner 1991)**.** *Let G be a Lie group, $\Gamma \subset G$ a lattice and $U \subseteq G$ a subgroup generated by one-parameter unipotent subgroups. Then the closure of every orbit $Ux\Gamma$ in G/Γ is* homogeneous*: it is an orbit $Fx\Gamma$ of a closed subgroup $F \supseteq U$, carrying a finite F-invariant measure. Every ergodic U-invariant probability measure is of this algebraic form.*

This is the non-abelian descendant of Theorem C.6, where orbit closures of linear flows were subtori. For diagonal flows nothing like it holds in general (bounded orbits abound, as the badly approximable numbers show). But for higher rank diagonal actions there is a rigidity conjecture of Margulis, of which the Littlewood conjecture in the Gallery is a consequence.

C.9 Gallery of theorems

- **Weyl's polynomial theorem** (Weyl 1916, *Math. Ann.* 77). If a real polynomial p has an irrational coefficient other than the constant term, then $\{p(k)\}$ is equidistributed. This is the first step beyond rotations; the proof uses Weyl differencing.

- **Bounded remainder intervals** (Hecke 1922, Ostrowski; Kesten 1966, *Acta Arith.* 12, 193–212). For irrational α, the discrepancy $\#\{k < N : \{k\alpha\} \in I\} - N\,|I|$ stays bounded if and only if $|I| \in \mathbb{Z} + \alpha\mathbb{Z}$. The window of the Fibonacci chain is such an interval, which is why its tile counts deviate from $N\varphi^{-1}$, $N\varphi^{-2}$ by $O(1)$.

- **Khinchin's theorem** (Khinchin 1924, *Math. Ann.* 92). Let $\psi\colon \mathbb{N} \to [0,\infty)$ be non-increasing. For almost every α, $\|q\alpha\|_{\mathbb{Z}} < \psi(q)$ has infinitely many solutions if $\sum \psi(q) = \infty$, and only finitely many if $\sum \psi(q) < \infty$. In particular almost every number is *not* badly approximable, yet the badly approximable numbers have full Hausdorff dimension (Jarník 1928).

- **The Duffin–Schaeffer conjecture** (Duffin–Schaeffer 1941; proved by Koukoulopoulos and Maynard, *Ann. of Math.* 192 (2020), 251–307). For *any* $\psi\colon \mathbb{N} \to [0,\infty)$, if $\sum_q \psi(q)\phi(q)/q = \infty$ (with ϕ Euler's function), then almost every α has infinitely many reduced fractions a/q with $|\alpha - a/q| \le \psi(q)/q$. This removes monotonicity from Khinchin's theorem. The proof is a delicate study of overlaps of the approximation sets, organised as a graph-theoretic compression argument.

- **Littlewood's conjecture** (c. 1930, open). For all real α, β, $\liminf_{q\to\infty} q\,\|q\alpha\|_{\mathbb{Z}}\,\|q\beta\|_{\mathbb{Z}} = 0$. Einsiedler, Katok and Lindenstrauss (*Ann. of Math.* 164 (2006), 513–560) classified the positive entropy invariant measures for the diagonal group on $\mathrm{SL}_k(\mathbb{R})/\mathrm{SL}_k(\mathbb{Z})$, and deduced that the set of exceptions (α,β) has Hausdorff dimension 0. By the Dani correspondence, Littlewood concerns orbits of the 2-parameter diagonal group on X_3.

- **Three gaps and lattices** (Marklof–Strömbergsson, *Amer. Math. Monthly* 124 (2017), 741–745). A proof of Theorem C.11 in which the gaps are the values of a function on the space of 2-dimensional lattices, as in the remark at the end of Section C.3.

- **Gaps of $\sqrt{n}$ mod 1** (Elkies–McMullen, *Duke Math. J.* 123 (2004), 95–139). The points $\{\sqrt{n}\}$, $n \le N$, also cut the circle (Lecture 7). Their normalized gap distribution exists, is *not* the exponential law of random points, and is given by an explicit piecewise analytic formula. The proof uses Ratner's theorem on a space of affine lattices.

- **Optimal dispersion** (Breneis–Hinrichs, arXiv 2019; in *Discrepancy Theory*, de Gruyter 2020, 117–132). The largest empty axis-parallel box among $n \ge 2$ points in the torus $\mathbb{T}^2$ has area at least $2/n$, and for n a Fibonacci number the Fibonacci lattice attains exactly $2/n$. This is a packing problem for boxes in which the golden lattice is optimal.

- **Spherical Fibonacci points** (Aistleitner–Brauchart–Dick, *Discrete Comput. Geom.* 48 (2012), 990–1024). The spherical cap discrepancy is $O(N^{-1/2})$, with an explicit constant, for the spherical Fibonacci lattice, and the same order for spherical digital nets. Compare Beck's lower bound $cN^{-3/4}$ (*Mathematika* 31, 1984).

- **Pure point diffraction** (Hof, *Comm. Math. Phys.* 169 (1995), 25–43; Schlottmann, in *Directions in Mathematical Quasicrystals*, CRM Monogr. 13, AMS 2000, 143–159). Regular model sets have pure point diffraction. This is the precise sense in which a model set is "crystalline without being periodic".

- **Gauss circle problem for Penrose tilings** (Haynes–Lutsko, arXiv:2512.21444, 2025). The number of vertices of a rhombic Penrose tiling (unit edges) in a disc of radius R is $\pi C_P R^2 + O(R^{2/3}(\log R)^{2/3})$, uniformly over all such tilings, with $C_P \approx 1.231$. The proof combines the cut and project description with Poisson summation in $\mathbb{Q}(\zeta_5)$, a direct bridge to Interlude B. A recent preprint.

- **Undecidability of tiling** (Wang 1961; Berger, *Mem. Amer. Math. Soc.* 66 (1966); Robinson, *Invent. Math.* 12 (1971)). It is undecidable whether a finite set of Wang tiles (unit squares with coloured edges, matched edge to edge, no rotations) tiles the plane. The proof needs, and produces, an *aperiodic* tile set: one that tiles, but only non-periodically. Berger used 20426 tiles, and Robinson used 6 tiles (allowing rotations and reflections). The minimum is 11 Wang tiles, on 4 colours (Jeandel–Rao, *Advances in Combinatorics* 2021).

- **Penrose tilings** (Penrose 1974, *Bull. Inst. Math. Appl.* 10; de Bruijn 1981, *Indag. Math.* 43). Two rhombi with matching rules tile only aperiodically. Every finite patch recurs with bounded gaps, and the vertex sets are model sets (Section C.6).

- **The einstein problem** (Smith–Myers–Kaplan–Goodman-Strauss, "An aperiodic monotile", *Combinatorial Theory* 4(1) (2024); "A chiral aperiodic monotile", *ibid.* 4(2) (2024)). The "hat", a polygon made of 8 kites of the deltoidal trihexagonal tiling, tiles the plane but never periodically. Its tilings use the hat and its mirror image. The "spectre", a modification of a related tile, tiles only aperiodically and needs no mirror images. Earlier, Socolar and Taylor (*J. Combin. Theory Ser. A* 118 (2011), 2207–2231) had given an aperiodic hexagonal monotile whose matching rules are not enforced by shape alone.

- **Keller's conjecture** (Keller 1930). In every tiling of $\mathbb{R}^n$ by unit cubes, some two cubes share a full facet. It is true for $n \le 6$ (Perron 1940). It is false for $n \ge 10$ (Lagarias–Shor, *Bull. Amer. Math. Soc.* 27 (1992), 279–283) and for $n \ge 8$ (Mackey, *Discrete Comput. Geom.* 28 (2002), 275–279). It is true for $n = 7$ (Brakensiek–Heule–Mackey–Narváez, IJCAR 2020; *J. Automated Reasoning* 66 (2022), 277–300, by a SAT solver). By Hajós' 1942 theorem the *lattice* case, Minkowski's conjecture, is true in all dimensions. By a reduction of Szabó (1986) one may assume that the tiling is periodic modulo $2\mathbb{Z}^n$, so that it is a configuration of cubes in the flat torus $\mathbb{R}^n/2\mathbb{Z}^n$.

- **Periodic tiling conjecture** (Grünbaum–Shephard; Lagarias–Wang 1996). If a single finite set $F \subset \mathbb{Z}^d$ tiles $\mathbb{Z}^d$ by translations, does it also tile periodically? Yes for $d = 2$ (Bhattacharya, *Amer. J. Math.* 142 (2020), 255–266), hence tiling by one tile is decidable there. No in general: Greenfeld and Tao (*Ann. of Math.* 200 (2024), 301–363) found a counterexample in sufficiently high dimension, encoding a "Sudoku" by a single tiling equation. So translational tilings by one tile can be forced to be aperiodic.

- **Convex pentagons** (Reinhardt 1918 to Mann–McLoud-Mann–Von Derau, arXiv 2015, *Geom. Dedicata* 194 (2018), 141–167). Fifteen families of convex pentagons are known to tile the plane, the fifteenth found by computer search in 2015. Rao (arXiv 2017, computer-assisted) announced a proof that the list is complete; Hales independently reproduced the main part of the computation, but to our knowledge the proof has not appeared in a refereed journal. As a consequence, no convex polygon tiles the plane only non-periodically.

- **Oppenheim and Ratner** (Margulis, *C. R. Acad. Sci. Paris* 304 (1987); Ratner, *Ann. of Math.* 134 (1991), 545–607, and *Duke Math. J.* 63 (1991), 235–280). See Theorems C.21 and C.22. They are the reason that counting lattice points in thin regions (Interlude B) can be done by dynamics.

- **Quasicrystals and icosians** (Elser–Sloane, *J. Phys. A* 20 (1987), 6161; Moody–Patera, *J. Phys. A* 26 (1993), 2829). These are the $E_8 \to H_4$ projection and its algebraic form, via the icosian ring. Shcherbak (*Russian Math. Surveys* 43 (1988), 149) found the foldings $A_4 \to H_2$, $D_6 \to H_3$, $E_8 \to H_4$ in singularity theory. Affine versions: Patera–Twarock, *J. Phys. A* 35 (2002), 1551–1574; Dechant–Boehm–Twarock, *J. Phys. A* 45 (2012), 285202 and *J. Math. Phys.* 54 (2013), 093508. Viruses: Twarock, *J. Theor. Biol.* 226 (2004), 477–482; Keef–Twarock, *J. Math. Biol.* 59 (2009), 287–313.

- **Quasicrystals in the lab** (Shechtman–Blech–Gratias–Cahn, *Phys. Rev. Lett.* 53 (1984),

1951; Nobel Prize in Chemistry 2011). Sharp icosahedral diffraction in Al–Mn: the experimental counterpart of pure point diffraction of model sets.

Exercises

Exercise C.23. Let $\gcd(p,q)=1$. Show that the closed geodesic $\ell_{p/q}$ on $\mathbb{R}^2/\mathbb{Z}^2$ has length $\sqrt{p^2+q^2}$, and that its complement is a single open annulus of width $1/\sqrt{p^2+q^2}$. (Its lifts to $\mathbb{R}^2$ are the lines $qy-px\in\mathbb{Z}$.) So rational lines are spread evenly, but they leave a gap that shrinks only like 1/length.

Exercise C.24. Prove Theorem C.6 for $\Lambda=\mathbb{Z}^n$ using Weyl's criterion on $\mathbb{T}^n$ (characters $e(\langle m,x\rangle)$, $m\in\mathbb{Z}^n$). Then deduce the general case by a linear change of variables, and identify the dual lattice.

Exercise C.25. For $\alpha=\varphi^{-1}$ and $N=F_n$, list the points $\{k\alpha\}$ for $n=5$ ($N=5$) and $n=6$ ($N=8$), and verify Example C.12 directly. What happens for $N=F_n+1$?

Exercise C.26. Show that $Q(x,y)=(x-\sqrt{2}y)(x-\sqrt{3}y)$ satisfies $\inf_{(x,y)\in\mathbb{Z}^2\setminus 0}|Q(x,y)|>0$. *Hint:* $\sqrt{2}=[1;2,2,\dots]$ and $\sqrt{3}=[1;1,2,1,2,\dots]$ have bounded partial quotients. If $|x-\sqrt{2}y|<\frac{\sqrt{3}-\sqrt{2}}{2}\,|y|$, then the other factor is at least $\frac{\sqrt{3}-\sqrt{2}}{2}\,|y|$.

Exercise C.27. Show that the Fibonacci word f is *balanced*: any two factors of the same length contain numbers of b's differing by at most one. Deduce again that bb does not occur, and show that aaa does not occur either. Interpret both facts for the jammed channel of Lecture 11.

Exercise C.28. Verify the edge rule of Section C.7 for H_3: the resulting graph on the six nodes α_i,β_i is D_6. Then compute the 2×2 Gram matrix of a_1,a_2 in the case $A_4\to H_2$ and check that the angle is $4\pi/5$.

Exercise C.29. Using the explicit A_4 roots $e_i-e_j\in\mathbb{R}^5$, find the plane on which the cyclic shift $e_1\to e_2\to\cdots\to e_5\to e_1$ acts as rotation by $2\pi/5$, project the 20 roots to it, and recover Figure C.7. (The cyclic shift is a Coxeter element of $W(A_4)=\Sigma_5$.)

Exercise C.30. Using only Siegel's mean value theorem, show that for every $\varepsilon>0$ there is a lattice in $\mathbb{R}^n$ with packing density at least $(2-\varepsilon)2^{-n}$. Where exactly is the symmetry $v\leftrightarrow -v$ used?

Exercise C.31. Prove the first half of the Morse–Hedlund theorem. If $p(n)\le n$ for some n, then $p(m)=p(m+1)$ for some $m\le n$. Deduce that every factor of length m has a unique extension to the right, so the word is eventually periodic.

Further reading

- M. Einsiedler and T. Ward, *Ergodic Theory with a view towards Number Theory*, Springer GTM 259, 2011. Rotations, Weyl, the space of lattices, and the Dani correspondence.
- J. W. S. Cassels, *An Introduction to Diophantine Approximation*, Cambridge, 1957; and M. Aigner, *Markov's Theorem and 100 Years of the Uniqueness Conjecture*, Springer, 2013.
- L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley, 1974. Weyl's criterion, discrepancy, $\{k\alpha\}$.

- M. Lothaire, *Algebraic Combinatorics on Words*, Cambridge, 2002, Ch. 2 (Sturmian words).
- M. Baake and U. Grimm, *Aperiodic Order*, Vol. 1, Cambridge, 2013. Substitutions, cut and project, Penrose tilings, diffraction.
- M. Senechal, *Quasicrystals and Geometry*, Cambridge, 1995; B. Grünbaum and G. C. Shephard, *Tilings and Patterns*, Freeman, 1987.
- J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge, 1990, for H_3, H_4 and Coxeter elements. Conway–Sloane, Ch. 8, for the icosians.
- D. W. Morris, *Ratner's Theorems on Unipotent Flows*, Chicago Lectures in Math., 2005.

Part III

Configuration spaces and Morse theory

Chapter 7

Lecture 7: A short introduction to configuration spaces

7.1 Configuration spaces

A *configuration space* is a topological space that records the possible configurations of an object we wish to study. For example:

- the 2^N possible results of N coin flips (a discrete space $\{0,1\}^N$ with 2^N points);
- the tilings of a checkerboard by dominoes (again a finite set);
- the positions and orientations of a polytope in space (for a polytope with no symmetries this is $\mathbb{R}^3 \times \mathrm{SO}(3)$; in general one divides $\mathrm{SO}(3)$ by the rotational symmetry group of the polytope);
- the configurations of a mechanical linkage;
- the positions (and velocities) of particles in a box.

This last example is the one most closely related to what we want to study, and to the first few lectures: positions of points in space.

Definition 7.1 (Configuration space of N points)**.** Let X be a topological space. The *(classical) configuration space of N distinct labelled points in X* is

$$C^N(X) = \mathrm{Conf}(N, X) = \prod_{i=1}^{N} X \setminus \Delta, \qquad \Delta = \{(x_i)_{i=1}^N : x_i = x_j \text{ for some } i \neq j\},$$

with the subspace topology from the product X^N.

crossed out: "statistical mechanics", "also study"

Exercise 7.2. Think about how this might be related to the Weyl group associated to the root system A_n. (Hint: take $X = \mathbb{R}$ and $N = n+1$. The Weyl group of A_n is the symmetric group Σ_{n+1}, acting on $\mathbb{R}^{n+1}$ by permuting coordinates, and its reflecting hyperplanes are exactly the hyperplanes $x_i = x_j$ that make up Δ. This is taken up again in Section 8.1.)

There are also *unlabelled* versions, $\mathrm{Conf}(N,X)/\Sigma_N$, where the symmetric group Σ_N acts by permuting the indices. When X is endowed with a metric we can also consider *reduced* versions, $\mathrm{Conf}(N,X)/\mathrm{Isom}(X)$. (One can of course do both: $\mathrm{Conf}(N,X)/(\Sigma_N \times \mathrm{Isom}(X))$.) So the unlabelled version picks a single representative class for each permutation of the indices, and the reduced version divides out by the symmetries of the ambient space.

Example 7.3 ($\mathrm{Conf}(N,\mathbb{R})$)**.** We have $\mathrm{Conf}(N,\mathbb{R}) = \mathbb{R}^N \setminus \Delta$.

- *Unlabelled:* each point of $\mathrm{Conf}(N,\mathbb{R})/\Sigma_N$ has exactly one representative with $x_1 < x_2 < x_3 < \cdots < x_N$, so we can choose to work in this region of the space. $\mathbb{R}^N \setminus \Delta$ has $N!$ connected components, one for each ordering of the coordinates; each is an open convex cone, hence contractible, and Σ_N permutes them simply transitively.
- *Reduced:* we can shift one of the x_i to 0, for example the smallest. Together with the ordering this gives $\{0 = x_1 < x_2 < \cdots < x_N\}$. (Modulo translations, the ordered region becomes the open cone $\{0 < x_2 < \cdots < x_N\} \subset \mathbb{R}^{N-1}$; if reflections $x \mapsto -x$ are divided out as well, one may in addition fix the order of the two "outer" gaps.)

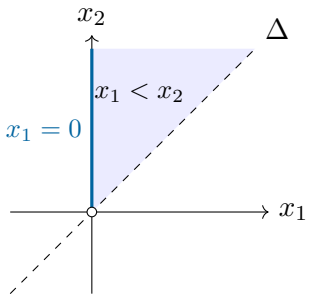


Figure: $\mathrm{Conf}(2,\mathbb{R}) = \mathbb{R}^2 \setminus \Delta$. The shaded half-plane $\{x_1 < x_2\}$ is a copy of the unlabelled space; the thick ray $\{0 = x_1 < x_2\}$ is the reduced unlabelled space.

p. 3 also sketches $N = 3$, labelled "$x_1 = 0$", "$x_1 < x_2$", "$0 < x_2 < x_3$"

Example 7.4 ($\mathrm{Conf}(N,\mathbb{S}^2)$)**.** Consider labelled distinct points on a sphere. The reduced space is the space of N labelled points on $\mathbb{S}^2$ modulo *ambient* rotations. For $N = 4$: two configurations that differ by a rotation of the sphere are the same point of the reduced space; a configuration and its mirror image are in general *not* related by a rotation, but they may become identified in the unlabelled space after relabelling (e.g. swapping the labels 2 and 3). (If one divides by all of $\mathrm{Isom}(\mathbb{S}^2) = \mathrm{O}(3)$ rather than by $\mathrm{SO}(3)$, mirror images are identified as well.)

p. 4: sketches of 4 labelled points on spheres

7.2 Braid groups

These spaces are related to the Artin braid groups, which we may define as follows.

Definition 7.5 (Braid groups)**.** The *braid group* and the *pure braid group* of X (on N strands) are

$$B_N(X) = \pi_1\big(\mathrm{Conf}(N,X)/\Sigma_N\big), \qquad P_N(X) = \pi_1\big(\mathrm{Conf}(N,X)\big).$$

For $X = \mathbb{R}^2$ these are Artin's braid group B_N and pure braid group P_N. Since $\mathrm{Conf}(N,X) \to \mathrm{Conf}(N,X)/\Sigma_N$ is a covering map with deck group Σ_N, for connected $\mathrm{Conf}(N,X)$ there is a short exact sequence $1 \to P_N(X) \to B_N(X) \to \Sigma_N \to 1$.

To see why these are called braid groups, consider what a path is in $\mathrm{Conf}(N, \mathbb{R}^2)$. Fix a base point $x_0 = (x_0^1, \ldots, x_0^N)$. A loop $\gamma\colon [0,1] \to \mathrm{Conf}(N, \mathbb{R}^2)$ consists of N paths $\gamma_1, \ldots, \gamma_N$ in the plane which never collide at any time t. Drawing the graphs $\{(\gamma_i(t), t)\} \subset \mathbb{R}^2 \times [0,1]$ gives N disjoint strands running from the plane $t = 0$ to the plane $t = 1$: a braid. In the unlabelled space a loop may end at a permutation of the starting labels, which is why B_N surjects onto Σ_N; homotopic loops give braids that are isotopic with endpoints fixed.

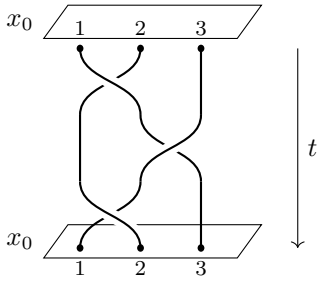


Figure: a loop in $\mathrm{Conf}(3, \mathbb{R}^2)/\Sigma_3$ drawn as a braid on three strands.

p. 5: second sketch (strand graphs) illegible, omitted

Remark 7.6. For $X = \mathbb{R}$ the components of $\mathrm{Conf}(N, \mathbb{R})$ are contractible (Example 7.3), so all braid groups of $\mathbb{R}$ are trivial. For $X = \mathbb{R}^n$ with $n \geq 3$ the diagonal has codimension $n \geq 3$, so $\mathrm{Conf}(N, \mathbb{R}^n)$ is simply connected: $P_N(\mathbb{R}^n) = 1$ and $B_N(\mathbb{R}^n) \cong \Sigma_N$. The plane is the interesting case.

7.3 Functions on configuration spaces

To configurations we would like to associate real-valued functions, based on what they model. For example:

- an electrical potential, or more generally a potential given as a sum of pair interactions $E(x_1, \ldots, x_N) = \sum_{i<j} \phi(x_i, x_j)$;
- a hard shell (hard sphere) potential ($\phi = +\infty$ if two spheres overlap, 0 otherwise);

which can describe different energy levels and can also constrain the system.

We may ask about

- minima and maxima (local and global);
- critical points;
- generic configurations.

This breaks down into a couple of categories:

- small N: by hand;
- large N: by statistical mechanics, or by averaging.

The classic example for large N is the expected value of N coin flips with values in $\{0, 1\}$, which leads to the central limit theorem.

7.3.1 The Boltzmann distribution

So, for example, consider N particles, each in one of K states, with energy E_i associated to the state $i \in \{1, \dots, K\}$. Let N_i be the number of particles in state i. The system is closed, so the total energy is constant, and

$$E = \sum_{i=1}^{K} E_i N_i, \qquad N = \sum_{i=1}^{K} N_i.$$

The number of ways to place the N particles according to the partition $\{N_1, \dots, N_K\}$ is the multinomial coefficient

$$W = \frac{N!}{N_1! \cdots N_K!}.$$

For large N the optimal partition dominates, so we want to determine the partition that maximises W subject to the two constraints above. One can derive the *Boltzmann measure*

$$\frac{N_i}{N} = \frac{e^{-\beta E_i}}{\sum_{j=1}^{K} e^{-\beta E_j}}, \qquad \text{i.e.} \qquad \Pr(\text{state } i) = \frac{e^{-\beta E_i}}{Z},$$

where $Z = \sum_{j=1}^{K} e^{-\beta E_j}$ is the *partition function* and $\beta = 1/(k_B T)$, with k_B Boltzmann's constant and T the temperature.

derivation deferred to "next time" in the notes

[Added] Sketch of the derivation. By Stirling's formula, $\ln W \approx N \ln N - \sum_i N_i \ln N_i$. Maximising this subject to $\sum_i N_i = N$ and $\sum_i E_i N_i = E$ with Lagrange multipliers α, β gives $-\ln N_i - 1 - \alpha - \beta E_i = 0$, i.e. $N_i \propto e^{-\beta E_i}$; normalising gives the formula above. The multiplier β is fixed by the energy constraint, and physics identifies it with $1/(k_B T)$.

Remark 7.7. Note:

- as $T \to \infty$ (so $\beta \to 0$), $\Pr(i) \to 1/K$: all states become equally likely;
- as $T \to 0$ (so $\beta \to \infty$), Pr concentrates on the states of minimal energy (uniformly on the ground states if there are several).

7.3.2 A gas in a box

But let us consider another (sketchy) physics analysis: the distribution of a gas in a box, in the simplest non-interacting setting. Take $2N$ points, i.e. a configuration in

$$\mathrm{Conf}\left(2N, [-1, 1]^3\right),$$

chosen uniformly at random. Let P_m be the probability that exactly $N + m$ particles have positive first coordinate.

There are 2^{2N} choices for the signs of the first coordinates, all equally likely, and $\binom{2N}{N+m}$ of them have exactly $N + m$ particles with first coordinate > 0. (This is really the coin-flip analysis.) So

$$P_m = 2^{-2N} \frac{(2N)!}{(N+m)!\,(N-m)!}.$$

We use a very bad version of Stirling's approximation,

$$n! \sim \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \;\rightsquigarrow\; n! \approx \left(\frac{n}{e}\right)^n.$$

Then

$$P_m \approx 2^{-2N}\Big(\frac{2N}{e}\Big)^{2N} \Big/ \Big[\Big(\frac{N+m}{e}\Big)^{N+m}\Big(\frac{N-m}{e}\Big)^{N-m}\Big] = \frac{N^{2N}}{(N+m)^{N+m}(N-m)^{N-m}}$$
$$= \Big(1+\frac{m}{N}\Big)^{-(N+m)}\Big(1-\frac{m}{N}\Big)^{-(N-m)} = \Big(1-\frac{m^2}{N^2}\Big)^{-N}\Big(1+\frac{m}{N}\Big)^{-m}\Big(1-\frac{m}{N}\Big)^{m}$$
$$\overset{m\ll N}{\approx} \big(e^{-m^2/N^2}\big)^{-N}\big(e^{m/N}\big)^{-m}\big(e^{-m/N}\big)^{m} = e^{-m^2/N}.$$

Since the crude Stirling formula loses the factors $\sqrt{2\pi n}$, we only keep the shape and write

$$P_m \approx P_0\, e^{-m^2/N}$$

(a crude central limit theorem). Normalising,

$$1 = \sum_{m=-N}^{N} P_m \approx \int_{-\infty}^{\infty} P_0\, e^{-m^2/N}\, dm = P_0\sqrt{\pi N} \quad\Longrightarrow\quad P_0 = \frac{1}{\sqrt{\pi N}}, \qquad P_m \approx \frac{1}{\sqrt{\pi N}}\, e^{-m^2/N}.$$

Comparing with $e^{-m^2/(2\sigma^2)}$ we get $\sigma^2 = N/2$, so $\sigma = \sqrt{N/2}$. This agrees with the exact variance $2N\cdot\frac{1}{2}\cdot\frac{1}{2} = N/2$ of the binomial distribution, and $P_0 = 1/\sqrt{\pi N}$ is exactly what the full Stirling formula gives for $\binom{2N}{N}4^{-N}$.

So the number of points with positive first coordinate is approximately Gaussian, centred at the expected value N with standard deviation $\sqrt{N/2}$. For example, for large N, about 95% of configurations are within $2\sigma = 2\sqrt{N/2}$ of the expected value. The relative fluctuation $\sigma/N = 1/\sqrt{2N}$ tends to 0.

Remark 7.8. The approximations $\ln(1\pm x) \approx \pm x - x^2/2$ used above each drop cubic terms of size m^3/N^2, but these cancel between the factors $(1+\frac{m}{N})^{-m}$ and $(1-\frac{m}{N})^{m}$ (the exact exponent is even in m). The first neglected term is $-m^4/(6N^3)$, so the Gaussian approximation is accurate for $m \ll N^{3/4}$; this covers the typical range $m = O(\sqrt{N})$.

Fix: error order was stated as m^3/N^2

So if we pick a point in high dimensions, i.e. a configuration in $[-1,1]^{3\cdot 2N}$ (and the same argument works for the other variables), we find roughly half of the coordinates positive and half negative, just like the coin flips. This makes sense. We can also talk about energy, etc., and generic configurations in this way.

Fix: notes: $[-1,1]^{3N}$; $2N$ points in $\mathbb{R}^3$ give $6N$ coordinates

But we may be interested in smaller configurations, like nearest neighbours, where Stirling's approximation is bad. How do we analyse such things?

? reading of "smaller configurations, like nearest neighbours" uncertain

7.3.3 Points on the circle

Let us look at points on $\mathbb{S}^1$, which we take to have length 1. Consider Reduced$(3,\mathbb{S}^1)$, three points on the circle modulo rotation. Use the rotation to fix one point at 0; going around the circle, the three gaps between consecutive points are x, y and $1-x-y$. So the reduced space is described by

$$\{(x,y) : x>0,\ y>0,\ x+y<1\},$$

an open triangle, whose boundary edges $x=0$, $y=0$, $x+y=1$ are the collisions. (For labelled points there are two such triangles, one for each cyclic order of the labels $1,2,3$; compare Example 8.1 below.) In general, for N points we have a simplex $\{(g_1,\dots,g_N) : g_i>0,\ \sum g_i = 1\}$ of gaps, an open $(N-1)$-simplex (one for each of the $(N-1)!$ cyclic orders of the labels).

Fix: notes write $1-x+y$

We can attach the *injectivity radius* function

$$\rho = \min_{i \neq j} \tfrac{1}{2} \operatorname{geodesic}(x_i, x_j).$$

Since the smallest gap is at most $1/N \leq 1/2$, ρ is half the smallest gap: $\rho = \frac{1}{2}\min(x, y, 1-x-y)$ for $N = 3$. It has a unique maximum $\rho = 1/6$ at the equilateral configuration $x = y = 1/3$; its level sets are triangles, and it fails to be smooth along the three segments from the centre to the vertices. In general the maximum is $1/(2N)$, attained at the regular N-gon.

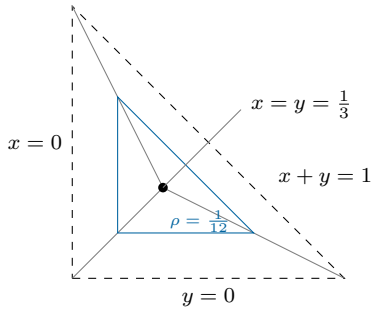


Figure: Reduced$(3, \mathbb{S}^1)$ as an open triangle (dashed boundary = collisions), with a level set of ρ and the lines where ρ is not smooth.

Remark 7.9. The notes end Lecture 7 with a remark about computing with the injectivity radius. The point to keep in mind for the next lecture is that ρ is a minimum of smooth functions, hence only piecewise smooth, so smooth Morse theory does not apply to it directly.

p. 13: closing remark largely illegible

? remark ends "...it's a (box–?)..."

Chapter 8

Lecture 8: (Smooth) Morse theory

8.1 Configuration spaces on a line and Weyl chambers

We started to talk about the spaces

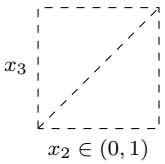
$$\mathrm{Conf}(N, X) = \prod_{i=1}^{N} X \setminus \Delta, \qquad \Delta = \{(x_i)_{i=1}^N : x_i = x_j \text{ for some } i \neq j\}.$$

Example 8.1 ($\mathrm{Conf}(3, \mathbb{S}^1)$)**.** Fixing the first point at 0 by a rotation, the other two points lie in $\mathbb{S}^1 \setminus \{0\} \cong (0,1)$ and must be distinct, so

$$\mathrm{Conf}(3, \mathbb{S}^1) \cong \mathbb{S}^1 \times \big((0,1)^2 \setminus \{x_2 = x_3\}\big),$$

and the reduced space is the open square minus its diagonal: two open triangles, one for each cyclic order (cf. Section 7.3.3).

Fix: notes: "$\mathrm{Conf}(3, \mathbb{S}^1) \approx$ square"; this is the reduced space



Example 8.2 ($\mathrm{Conf}(N, \mathbb{R})$ and the A_{N-1} Weyl chambers)**.** Write $n = N$. Consider the affine subspaces

$$V = V_c = \left\{(x_i)_{i=1}^n : \sum_i x_i = c\right\} \subset \mathbb{R}^n.$$

Translation along $(1, \dots, 1)$ splits off, $\mathrm{Conf}(n, \mathbb{R}) \cong \mathbb{R} \times (V_c \setminus \Delta)$. In these subspaces the hyperplanes $\{x_i = x_j\} \cap V$ are the boundaries of the Weyl chambers in our construction of the A_{n-1} root system in $V \subset \mathbb{R}^n$. Recall $A_{n-1} = \{e_i - e_j : i \neq j\}$, with $n(n-1)$ roots, simple roots $e_i - e_{i+1}$, and Weyl group Σ_n generated by the reflections in the hyperplanes $x_i = x_j$ (i.e. swapping two coordinates). The connected components of $V \setminus \Delta$ are exactly the $n!$ open Weyl chambers.

For $n = 3$ we had the simple roots

Fix: notes: "$n = 2$"; the example is A_2, $n = 3$

$$(1, -1, 0), \qquad (0, 1, -1).$$

So the root system is given by the six vectors

$$\pm(1,-1,0), \qquad \pm(0,1,-1), \qquad \pm(1,0,-1),$$

generated by swapping coordinates, i.e. by reflecting through the hyperplanes $x_i = x_j$, $i \neq j$. To draw them in the plane $V_3 = \{x+y+z=3\}$ we translate by $(1,1,1)$.

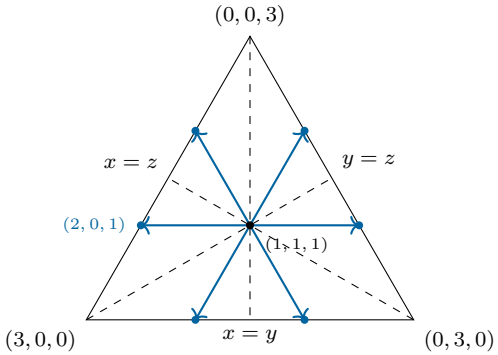


Figure: the plane $x+y+z=3$. The dashed lines are the walls $x_i = x_j$; they cut the plane into the six Weyl chambers (the components of $V_3 \setminus \Delta$). The arrows are the six roots of A_2, translated by $(1,1,1)$; they end at the trisection points of the edges, e.g. $(1,1,1)+(1,-1,0) = (2,0,1)$, and each root is perpendicular to one wall.

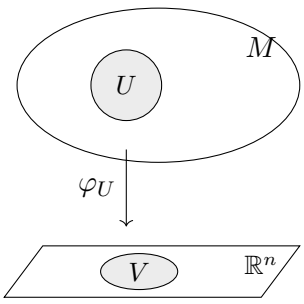
8.1.1 Manifolds

We will consider spaces M that are Hausdorff, second countable topological spaces: the topology separates points, and there is a countable basis for the topology. (Basically this is to eliminate pathological examples.)

Definition 8.3 (Homeomorphism)**.** A *homeomorphism* $\varphi\colon M \to N$ is a continuous bijection with continuous inverse. Equivalently: φ is a bijection, and $U \subset M$ is open if and only if $\varphi(U) \subset N$ is open.

Definition 8.4 (Coordinate chart)**.** A *coordinate chart* for a space M is a homeomorphism from a coordinate neighbourhood U (an open subset of M) onto an open subset of $\mathbb{R}^n$,

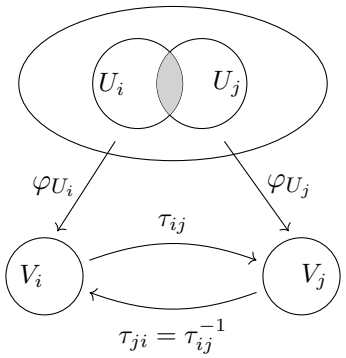
$$\varphi_U\colon U \subset M \longrightarrow V \subset \mathbb{R}^n.$$



A smooth bijection with smooth inverse is a *diffeomorphism*. So in the smooth setting, coordinate changes should be diffeomorphisms.

A *manifold* is a space M with a collection of charts $\{\varphi_{U_i}\}$ (an *atlas*), indexed by a cover $\{U_i\}$ of M, together with *transition maps* that patch the charts together:

$$\tau_{ij} = \tau_{U_iU_j} = \varphi_{U_j} \circ \varphi_{U_i}^{-1} \colon \varphi_{U_i}(U_i \cap U_j) \to \varphi_{U_j}(U_i \cap U_j).$$



Definition 8.5 (Smooth structure)**.** A *smooth differentiable structure* on M is a collection of (smooth) coordinate charts $\varphi_i \colon U_i \to V_i \subset \mathbb{R}^n$, $U_i \subset M$, such that

1. $M = \bigcup_i U_i$;
2. every transition map $\tau_{ij} = \varphi_j \circ \varphi_i^{-1}$ is smooth (on its domain $\varphi_i(U_i \cap U_j)$);
3. it is a *maximal* atlas: if a chart is compatible with all φ_i as above, then it is in the collection.

A topological space with a smooth structure is a *smooth manifold* (of dimension n).

Basically we can ignore maximality for computations, as maximal atlases are "unique": every atlas satisfying (1) and (2) is contained in exactly one maximal atlas, so it suffices to write down any atlas.

8.2 Examples; smooth maps

Example 8.6.

- $\mathbb{R}^n$: choose the identity map as the chart (one chart, $U = \mathbb{R}^n$).
- $\mathbb{C}^n \to \mathbb{R}^{2n}$, using $(\mathrm{Re}, \mathrm{Im})$ on each U_i.
- $\mathbb{S}^n \to \mathbb{R}^n$: stereographic projection on $\mathbb{S}^n \setminus \{N\}$ and on $\mathbb{S}^n \setminus \{S\}$ (north and south poles; the transition map is the inversion $y \mapsto y/|y|^2$ on $\mathbb{R}^n \setminus \{0\}$, which is smooth).
- On tori, by products of the $\mathbb{S}^1$ charts.
- On open subsets of manifolds, by carrying over the subspace topology and restricting the charts.

In particular $\mathrm{Conf}(N, X)$ is an open subset of X^N, so it is a smooth manifold of dimension $N \dim X$ whenever X is.

We can define *smooth maps* between manifolds.

Definition 8.7 (Smooth map)**.** A map $f\colon M \to N$ is *smooth* if for each $p \in M$ and some charts φ on $U \ni p$ for M and ψ on $V \ni f(p)$ for N (and then for all such charts), with $f(U) \subset V$, the induced map

$$\begin{array}{ccc} U & \xrightarrow{f} & V \\ \varphi\downarrow & & \downarrow\psi \\ \varphi(U) & \xrightarrow{\psi f \varphi^{-1}} & \psi(V) \end{array}$$

$\psi \circ f \circ \varphi^{-1}$ is smooth on its domain. ("Some $\Rightarrow$ all" because the transition maps are smooth.)

p. 12 (= scan p. 25) repeats this diagram

Similarly for maps between manifolds of other kinds. So we can define diffeomorphisms (homeomorphisms) of manifolds: a (smooth) bijection $M \to N$ with (smooth) inverse, written $M \cong N$. If we change the underlying conditions on the maps (e.g. continuous, C^k, or piecewise linear transition maps) we can still define coordinate charts, atlases, etc.

crossed out: "Sadly, not all the maps we want to work with will be smooth"
a first "8.3 Morse theory" is crossed out here

8.3 Homotopy type

We will want to talk about *homotopy type* as well, since we may be dealing with things that are not diffeomorphic but are topologically "similar".

Definition 8.8 (Homotopy)**.** Two maps $f, g\colon M \to N$ are *homotopic*, written $f \simeq g$, if there exists a continuous function

$$H\colon M \times I \to N, \qquad I = [0,1],$$

such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for all $x \in M$.

Definition 8.9 (Homotopy equivalence)**.** Two (smooth) spaces M and N are *homotopy equivalent* (have the same homotopy type) if there are maps

$$f\colon M \to N, \qquad g\colon N \to M \qquad \text{such that} \qquad f \circ g \simeq \mathrm{id}_N, \quad g \circ f \simeq \mathrm{id}_M.$$

letters X, Y corrected to M, N in blue on the page

The idea: the spaces can be continuously (smoothly) deformed into each other.

Example 8.10 ($D^2 \simeq \{0\}$)**.** Let $f\colon D^2 \to \{0\}$ be the radial retraction to 0 (the constant map) and $g\colon \{0\} \to D^2$ the inclusion. In polar coordinates put

$$h\big(t,(r,\theta)\big) := \big((1-t)r, \theta\big).$$

Then $h(0,\cdot) = \mathrm{id}_{D^2}$ and $h(1,\cdot) = g \circ f$ (the map $D^2 \to 0$), so $g \circ f \simeq \mathrm{id}_{D^2}$; and $f \circ g = \mathrm{id}_{\{0\}}$. ($h$ is continuous also at $r = 0$, since $(1-t)\cdot 0 = 0$ regardless of θ.)

Definition 8.11 (Relative homotopy; deformation retract)**.** A *homotopy relative to a subspace* is a homotopy which fixes the points of the subspace: for $f, g\colon M \to N$ and $K \subset M$, f and g are *homotopic relative to* K if there is a homotopy $h\colon M \times I \to N$ between f and g such that

$$h(k,t) = f(k) = g(k) \qquad \text{for all } k \in K,\ t \in I.$$

If $f = \mathrm{id}_M$ and g is a *retraction*, i.e. a continuous map $M \to K$ with $g|_K = \mathrm{id}$ (followed by the inclusion $K \hookrightarrow M$), then this homotopy defines a *strong deformation retraction* of M onto K.

See the last example: h is a strong deformation retraction of D^2 onto $\{0\}$.

8.4 An aside about homotopy groups

We consider homotopy classes of maps to pointed spaces (with homotopies relative to the base point). Consider $f\colon \mathbb{S}^1 \to (X, x_0)$, or equivalently paths $f\colon I \to X$ with $f(0) = f(1) = x_0$. These have a group structure under concatenation,

$$(f \cdot g)(t) = \begin{cases} f(2t), & t \in [0, \frac{1}{2}], \\ g(2t-1), & t \in [\frac{1}{2}, 1]. \end{cases}$$

This is the fundamental group $\pi_1(X, x_0)$.

In general one can consider maps $f\colon \mathbb{S}^n \to (X, x_0)$, viewed as maps $f\colon I^n \to X$ with $f(\partial I^n) = x_0$, with concatenation defined appropriately (in the first coordinate); this gives $\pi_n(X, x_0)$. If X is path connected we may ignore the base point x_0 (up to isomorphism).

example left blank on the page

Example 8.12. Draw $(D^2, x_0) \setminus \{0\}$. The punctured disk deformation retracts (radially) onto the circle of radius $|x_0|$, so $\pi_1(D^2 \setminus \{0\}, x_0) \cong \pi_1(\mathbb{S}^1) \cong \mathbb{Z}$, generated by the loop going once around the puncture. Compare $\pi_1(D^2) = 0$ from Example 8.10: removing a point changes the homotopy type.

We will not deal with this too much, but note that this is a nice invariant for spaces: homotopy equivalent spaces have isomorphic homotopy groups. This is also the source of the braid groups of Section 7.2.

8.5 Morse theory

numbered "8.2" in the notes

Morse theory is a fairly general way to study manifolds and functions on them. The idea is that the structure of the sets of critical points of functions and the topological structure of the manifold are related. (We will follow Milnor and others (see J. Milnor, *Morse Theory*, Annals of Math. Studies 51, Part I).)

Example 8.13 (The classic picture)**.** Consider $f\colon \mathbb{T}^2 \to \mathbb{R}$ given by

$$\mathbb{T}^2 \hookrightarrow \mathbb{R}^3 \xrightarrow{\text{proj}_z} \mathbb{R},$$

the height function $h = \text{proj}_z(x, y, z) = z$ on a torus standing upright, normalised so that it takes values in $[0, 1]$. It has four critical points: p (minimum, height 0), q and r (the bottom and top of the hole) and s (maximum, height 1).

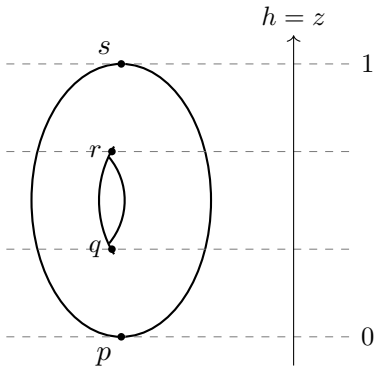


Figure: the torus standing on the plane $z = 0$, seen from the side, with the height function and its critical points.

Consider the *sublevel sets*

$$M_a := \{x \in M : f(x) < a\}.$$

(Milnor uses $M^a = \{f \leq a\}$; for a not a critical value the two have the same homotopy type.)

- $a < 0 = f(p)$: $M_a = \emptyset$.
- $f(p) < a < f(q)$: M_a is a 2-cell, i.e. a 2-ball (a disk).
- $f(q) < a < f(r)$: M_a is a cylinder.
- $f(r) < a < f(s)$: M_a is a torus minus a disk.
- $f(s) < a$: M_a is the whole torus.

Up to homotopy, these are handle or *cell attachments*.

Definition 8.14 (Attaching a cell)**.** Let e^k be a closed k-dimensional ball, with $\partial e^k = \mathbb{S}^{k-1}$, let Y be a topological space and $g\colon \mathbb{S}^{k-1} \to Y$ continuous. Then $Y \cup_g e^k$, the result of *attaching a k-cell to Y by g*, is defined as

$$Y \cup_g e^k = \big(Y \sqcup e^k\big)/\sim, \qquad x \sim g(x) \ \text{ for } x \in \mathbb{S}^{k-1}.$$

The sublevel sets of the torus, up to homotopy, are:

scan pp. 28–29 are in reverse order (circled 16, 15)

- a disk $\simeq$ a point (0-cell);
- the cylinder $\simeq$ a disk with a 1-handle attached (a "U" shape) $\simeq$ a point with a 1-cell attached, i.e. a circle;
- the torus minus a disk $\simeq$ a disk with two 1-handles $\simeq$ a point with two 1-cells attached, i.e. a figure eight $\mathbb{S}^1 \vee \mathbb{S}^1$;
- the torus $\simeq$ the figure eight with a 2-cell attached along the loop $aba^{-1}b^{-1}$.

p. 15: sketches of the handle pictures; one equivalence marked "?"

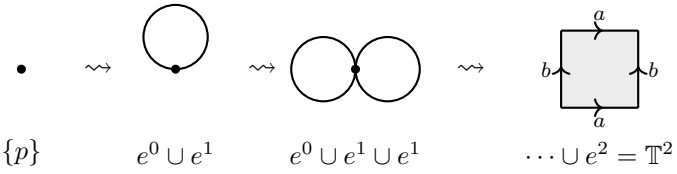


Figure: homotopy types of the sublevel sets M_a as a passes p, q, r, s.

The homotopy type (and the diffeomorphism type, outside of a neighbourhood) changes exactly at the critical points of the function f: those points where the gradient is zero, $\nabla f = 0$.

If we look at the Hessian $\operatorname{Hess} f_x = \big(\partial^2 f/\partial u_i \partial u_j(x)\big)$ in local coordinates u, which here is non-degenerate at each critical point, we see (writing the signature as (#positive, #negative) eigenvalues):

$$\operatorname{sig}(\operatorname{Hess} f_p) = (2,0), \qquad \operatorname{sig}(\operatorname{Hess} f_q) = \operatorname{sig}(\operatorname{Hess} f_r) = (1,1), \qquad \operatorname{sig}(\operatorname{Hess} f_s) = (0,2).$$

We call the negative part the *index*, i.e. the number of decreasing directions at the critical point, and we see that it corresponds to the dimension of the attached cell: index 0 at p (0-cell), index 1 at q and r (1-cells), index 2 at s (2-cell).

This is why we want to start with smooth functions.

Definition 8.15 (Morse function)**.** We will call a smooth function $f\colon M \to \mathbb{R}$ *Morse* if it has no degenerate critical points, i.e. at every critical point x (where $df_x = 0$) the Hessian Hess f_x is a non-singular matrix.

Fix: notes say "no nondegenerate critical pts"

We will formalise this next time.

[Added] For reference, the two basic theorems (Milnor, *Morse Theory*, Thms. 3.1 and 3.2): let $f\colon M \to \mathbb{R}$ be smooth and write $M^a = f^{-1}(-\infty, a]$. (A) If $a < b$, $f^{-1}[a,b]$ is compact and contains no critical points, then M^a is diffeomorphic to M^b, and M^a is a deformation retract of M^b. (B) If x is a non-degenerate critical point of index λ with $f(x) = c$, and $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no other critical point, then for small $\varepsilon > 0$, $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon} \cup e^{\lambda}$.

Chapter 9

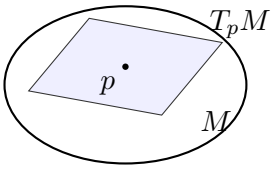
Lecture 9: More on functions and their critical points

Last time we defined a smooth function $f\colon M \to \mathbb{R}$ on a manifold M to be *Morse* if it has no degenerate critical points. That is, every critical point of f is non-degenerate.

Fix: notes say "no non-degenerate critical points"

9.1 What does this mean?

To make sense of this definition we need to understand criticality, so we introduce the tangent space T_pM at a point $p \in M$.



We consider velocities of curves

$$\gamma\colon [-1,1] \to M, \qquad \gamma(0) = p,$$

and, with respect to a chart $\varphi\colon U \to \mathbb{R}^n$ with $p \in U \subset M$, we ask that $\varphi \circ \gamma\colon [-1,1] \to \mathbb{R}^n$ be differentiable at 0. The classes of such curves, where two curves are identified when their tangents $(\varphi \circ \gamma)'(0)$ at 0 agree, define the *tangent space* T_pM at p. (This identification does not depend on the chart, since the transition maps are diffeomorphisms; the tangent vector represented by γ is written $\gamma'(0)$.)

Given a smooth map $f\colon M \to N$, there is an induced *linear* map

$$f_*\colon T_pM \to T_{f(p)}N, \qquad f_*\big(\gamma'(0)\big) = (f \circ \gamma)'(0),$$

called the *differential* or *push-forward* of f.

Fix: notes: target T_pN

Definition 9.1. Let $f\colon M \to \mathbb{R}$ be smooth. A point $p \in M$ is a *critical point* of f if

$$f_*\colon T_pM \to T_{f(p)}\mathbb{R}$$

is the zero map. The value $f(p)$ is then called a *critical value*.

In local coordinates $(x_1, \dots, x_n)$ about p this is written

$$\frac{\partial f}{\partial x_i}\Big|_p = 0 \qquad \text{for all } i = 1, \dots, n.$$

Also, there is a bilinear form on T_pM, well defined at a critical point p, which can be written similarly in coordinates as

$$H_f(p) = \Big(\frac{\partial^2 f}{\partial x_i\, \partial x_j}\Big|_p\Big)_{i,j}, \qquad \text{also written } f_{**}(p).$$

This is the *Hessian* of f at p.

Definition 9.2. A critical point p of f is *non-degenerate* if the Hessian $H_f(p)$ is non-degenerate, i.e. its matrix is non-singular. The *index* of f at p is the number of negative eigenvalues of $H_f(p)$ (the dimension of a maximal subspace of T_pM on which $H_f(p)$ is negative definite).

The notion of index is well defined in this case. Indeed, under a change of coordinates the Hessian at a critical point changes by $H \mapsto J^{\mathsf{T}} H J$ with J the (invertible) Jacobian, and by Sylvester's law of inertia the number of negative eigenvalues is unchanged.

9.2 Why do we care about non-degenerate critical points?

numbered 6.2–6.4 in the notes; read as 9.2–9.4

Lemma 9.3 (Morse Lemma)**.** *If p is a non-degenerate critical point of a smooth function f, there is a local coordinate system $(x_1, \dots, x_n)$ about p with p at the origin such that*

$$f = f(p) \; - \sum_{i=1}^{\lambda} x_i^2 + \sum_{j=\lambda+1}^{n} x_j^2,$$

where λ is the index of f at p.

Proof (sketch). We may shift so that $f(p) = 0$ and $p = 0$ in some chart. By the fundamental theorem of calculus and the integral form of Taylor's theorem,

$$f(x_1, \dots, x_n) = \int_0^1 \frac{d}{dt} f(tx)\, dt = \int_0^1 \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(tx_1, \dots, tx_n)\, x_i\, dt,$$

so $f = \sum_i x_i g_i$ with g_i smooth and $g_i(0) = \frac{\partial f}{\partial x_i}(0) = 0$. Applying the same argument to each g_i, and in fact

$$f(x_1, \dots, x_n) = \sum_{i,j} h_{ij}(x_1, \dots, x_n)\, x_i x_j$$

for some smooth functions h_{ij}. We can symmetrize ($h_{ij} = h_{ji}$, replacing h_{ij} by $\frac{1}{2}(h_{ij} + h_{ji})$) and compute

$$h_{ij}(0) = \frac{1}{2}\, \frac{\partial^2 f}{\partial x_i\, \partial x_j}(0)$$

in some coordinate system for T_pM. This can be inductively diagonalized in a neighbourhood of p, for example via the spectral theorem plus a continuity argument.

Fix: notes: ∂f for $\partial^2 f$

Example in dimension 2. Write

$$f(x,y) = x^2 h_{11} + 2xy\, h_{12} + y^2 h_{22}, \qquad \frac{\partial^2 f}{\partial x^2}\Big|_{(0,0)} = 2h_{11}(0,0), \text{ etc.}$$

By a linear coordinate change we may assume $h_{11}(0,0) \neq 0$; by continuity there is a neighbourhood on which $h_{11} \neq 0$. Then we can define a local change of coordinates

$$\bar{x} = \sqrt{|h_{11}|}\Big(x + \frac{h_{12}}{h_{11}}\, y\Big) \quad \Longrightarrow \quad \bar{x}^2 = |h_{11}| \Big(x^2 + 2\frac{h_{12}}{h_{11}} xy + \frac{h_{12}^2}{h_{11}^2} y^2\Big).$$

In the case $h_{11} > 0$,

$$\bar{x}^2 = h_{11}x^2 + 2h_{12}xy + \frac{h_{12}^2}{h_{11}} y^2,$$

and since

$$f = \underbrace{x^2 h_{11} + 2xy\, h_{12}}_{= \bar{x}^2 - \frac{h_{12}^2}{h_{11}} y^2} + y^2 h_{22}, \qquad \text{we get} \qquad f = \bar{x}^2 + \Big(h_{22} - \frac{h_{12}^2}{h_{11}}\Big) y^2.$$

Non-degeneracy at 0 means $h_{11}h_{22} - h_{12}^2 \neq 0$ there, so the coefficient of y^2 is non-zero near 0, and $\bar{y} = \sqrt{\big|h_{22} - h_{12}^2/h_{11}\big|}\; y$ finishes the job. (The Jacobian of $(x,y) \mapsto (\bar{x}, \bar{y})$ at 0 is invertible, so this is a coordinate change by the inverse function theorem.) The proof proceeds the same way for the other cases and for higher dimensions.

[Added] In general, after a linear change we have $h_{11}(0) \neq 0$; put $\bar{x}_1 = \sqrt{|h_{11}|}\,\big(x_1 + \sum_{j>1} x_j h_{1j}/h_{11}\big)$, so that $f = \pm \bar{x}_1^2 + \sum_{i,j \geq 2} x_i x_j\, h'_{ij}$ with h'_{ij} smooth and symmetric and $(h'_{ij}(0))$ non-singular; induct on n. (See Milnor, *Morse Theory*, Lemma 2.2.)

□

Corollary 9.4. *Non-degenerate critical points are isolated.*

[Added] Indeed, in Morse coordinates $df = (-2x_1, \ldots, -2x_\lambda, 2x_{\lambda+1}, \ldots, 2x_n)$ vanishes only at the origin.

9.3 One-parameter groups of diffeomorphisms

Definition 9.5. A *one-parameter group of diffeomorphisms* of a manifold M is a smooth map

$$\Phi \colon \mathbb{R} \times M \to M$$

such that

1. for each $t \in \mathbb{R}$, the map $\varphi_t \colon M \to M$ defined as $\varphi_t(p) = \Phi(t,p)$ is a diffeomorphism of M onto M, and
2. for all $t, s \in \mathbb{R}$, $\varphi_{t+s} = \varphi_t \circ \varphi_s$.

This defines a vector field X on M: for a smooth $f \colon M \to \mathbb{R}$, X is characterized by

$$X_q(f) = \lim_{h \to 0} \frac{f(\varphi_h(q)) - f(q)}{h}.$$

Φ vs. φ_t distinction added in blue on the page

So this is an assignment of a direction to each point q of M, and X_q takes a directional derivative of f. We say that X *generates* the group φ.

Lemma 9.6. *A smooth vector field on M which vanishes outside of a compact set $K \subset M$ generates a unique one-parameter group of diffeomorphisms of M.*

Proof. Consider a smooth curve $\gamma\colon t \mapsto \gamma(t) \in M$ with velocity vector $\frac{d\gamma}{dt} \in T_{\gamma(t)}M$, characterized by

$$\frac{d\gamma}{dt} f = \lim_{h\to 0} \frac{f(\gamma(t+h)) - f(\gamma(t))}{h}.$$

Let φ be a one-parameter group of diffeomorphisms generated by a vector field X as in the lemma. Then for fixed q, the curve $t \mapsto \varphi_t(q)$ satisfies

$$\frac{d\varphi_t(q)}{dt} = X_{\varphi_t(q)}, \qquad \varphi_0(q) = q, \tag{9.1}$$

since

$$\frac{d\varphi_t(q)}{dt}(f) = \lim_{h\to 0} \frac{f(\varphi_{t+h}(q)) - f(\varphi_t(q))}{h} = \lim_{h\to 0} \frac{f\big(\varphi_h(\varphi_t(q))\big) - f(\varphi_t(q))}{h} = X_{\varphi_t(q)}(f).$$

This already gives uniqueness: any group generated by X consists of solutions of (9.1), which are unique.

Existence. By existence and uniqueness of solutions of ODEs (Picard–Lindelöf, applied in local coordinates), the system (9.1) has a unique solution that depends smoothly on the initial conditions. Hence for every $q \in M$ there is a neighbourhood U of q and an $\varepsilon > 0$ such that (9.1) has a unique smooth solution for initial points in U and $|t| < \varepsilon$.

Since K is compact, finitely many such neighbourhoods cover K, so there is a minimal $\varepsilon > 0$ for such a cover of K. For $q \notin K$, let $\varphi_t(q) = q$ (this solves (9.1) since X vanishes there). Hence there is a unique smooth solution $\varphi_t(q)$, $|t| < \varepsilon$, for all $q \in M$. Note that

$$\varphi_{t+s} = \varphi_t \circ \varphi_s \qquad \text{for } |t|, |s|, |t+s| < \varepsilon$$

(both sides solve (9.1) in t with the same initial value $\varphi_s(q)$). So φ_t is a diffeomorphism for $|t| < \varepsilon$ with inverse φ_{-t}.

? "by LNP..."; step size ε vs. $\varepsilon/2$ unclear

But we can iterate this process for $t \geq \varepsilon$: writing $t = k\,\frac{\varepsilon}{2} + r$ with $k = \lfloor 2t/\varepsilon \rfloor$ and $0 \leq r < \frac{\varepsilon}{2}$, define $\varphi_t = \varphi_{\varepsilon/2} \circ \cdots \circ \varphi_{\varepsilon/2} \circ \varphi_r$ (k factors), and $\varphi_{-t} = (\varphi_t)^{-1}$. One checks that this is well defined, smooth, and satisfies the group law. □

9.4 The diffeomorphism theorem

title "Homotopy Theorem" crossed out, replaced by "Diffeomorphism"

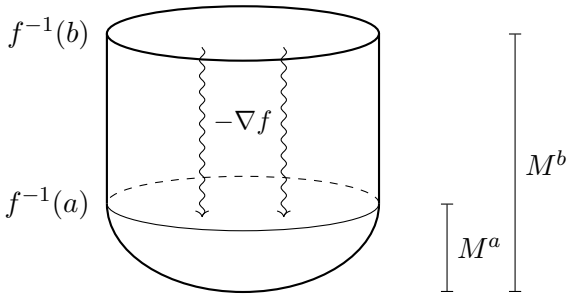
For $f\colon M \to \mathbb{R}$ we have the *sublevel sets*

$$M^a := f^{-1}(-\infty, a] = \{p \in M : f(p) \leq a\}.$$

notes mix M_a and M^a

Theorem 9.7. *Let $f\colon M \to \mathbb{R}$ be a smooth function, let $a < b$, and suppose $f^{-1}[a,b]$ is compact and contains no critical points of f. Then M^a is diffeomorphic to M^b. Moreover, M^a is a deformation retract of M^b.*

Idea. Push $f^{-1}(b)$ down to $f^{-1}(a)$ along $-\nabla f$.



Proof. M admits a Riemannian metric; we need it to define $\langle X, Y\rangle$, the inner product of tangent vectors. So we can define the gradient ∇f, which is the vector field characterized by

$$\langle X, \nabla f\rangle = Xf$$

for any vector field X; that is, $\langle X, \nabla f\rangle$ is the directional derivative of f along X. Note that ∇f vanishes exactly at the critical points of f. If $\gamma\colon \mathbb{R} \to M$ is a curve with velocity vector $\frac{d\gamma}{dt}$, then

$$\left\langle \frac{d\gamma}{dt}, \nabla f \right\rangle = \frac{d(f\circ\gamma)}{dt}.$$

Fix: notes: $\gamma\colon \mathbb{R} \to \mathbb{R}$

Define a smooth function $\rho\colon M \to \mathbb{R}$ that is

$$\rho = \frac{1}{\langle \nabla f, \nabla f\rangle} \quad \text{on } f^{-1}[a,b],$$

and 0 outside of a compact neighbourhood of $f^{-1}[a,b]$ (chosen to contain no critical points, which is possible since the critical set is closed and $f^{-1}[a,b]$ is compact). Then the vector field X defined as

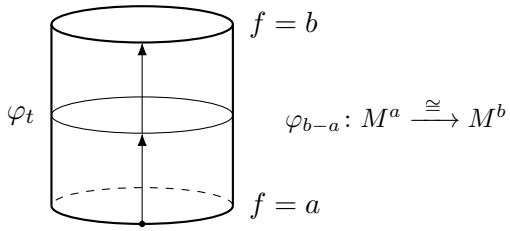
$$X_q = \rho(q)\,(\nabla f)_q$$

satisfies the hypotheses of Lemma 9.6 (it is smooth and vanishes outside a compact set), so there is a one-parameter group of diffeomorphisms $\varphi_t\colon M \to M$ generated by X.

For fixed q, consider $t \mapsto f(\varphi_t(q))$. If $\varphi_t(q) \in f^{-1}[a,b]$, then

$$\frac{df(\varphi_t(q))}{dt} = \left\langle \frac{d\varphi_t(q)}{dt}, \nabla f \right\rangle = \langle X, \nabla f\rangle = 1.$$

So $t \mapsto f(\varphi_t(q))$ is linear with derivative 1 while $f(\varphi_t(q))$ lies in $[a,b]$. Therefore $\varphi_{b-a}\colon M \to M$ is a diffeomorphism which carries M^a onto M^b.



Furthermore, define $r_t\colon M^b \to M^b$, $0 \le t \le 1$, by

p. 15: sketch of this flow

Fix: notes: $r_t\colon M_b \to M_a$

$$r_t(q) = \begin{cases} q & \text{if } f(q) \le a, \\ \varphi_{t(a-f(q))}(q) & \text{if } a \le f(q) \le b. \end{cases}$$

Then $r_0 = \mathrm{id}$, and r_1 is a retraction from M^b to M^a; r_t is continuous since the two formulas agree on $f^{-1}(a)$. Hence r_t is a deformation retraction of M^b onto M^a. (Cf. Milnor, *Morse Theory*, Theorem 3.1.) □

Chapter 10

Lecture 10: Configuration spaces and Morse-like results

10.1 Recap and the handle attachment theorem

p. 16, boxed: "need to be rewritten, out of order / missing pieces"

For smooth functions we have the diffeomorphism result (Theorem 9.7) that says: if $f\colon M \to \mathbb{R}$, $a < b$, $f^{-1}[a,b]$ is compact, and there are no critical values in $[a,b]$, then M^a is diffeomorphic to M^b.

There is also a handle attachment theorem (not proved here) for smooth $f\colon M \to \mathbb{R}$:

Theorem 10.1 (Handle attachment)**.** *Let p be a non-degenerate critical point of f with index λ and $f(p) = c$. Suppose there is $\varepsilon > 0$ such that $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no critical points other than p. Then $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon} \cup_g e^\lambda$, that is, $M^{c-\varepsilon}$ with a λ-cell attached along a map $g\colon \partial e^\lambda \to M^{c-\varepsilon}$.*

(Milnor, *Morse Theory*, Theorem 3.2.)

Morse-type inequalities

Let c_λ be the number of critical points of index λ of a Morse function on a compact manifold M. The homotopy type changes by attaching various cells of index λ as we "climb" M via the Morse function. Consequently

$$\sum_\lambda (-1)^\lambda c_\lambda = \chi(M),$$

and in general we have the constraints

$$c_\lambda \;\geq\; b_\lambda(M),$$

where $b_\lambda(M)$ is the λ-th Betti number of M (the rank of the λ-th homology group). (These are the weak Morse inequalities.) The key point for us is that the Betti numbers can be computed for many spaces we like.

10.2 Betti numbers for the configuration space of points on $\mathbb{S}^2$

Let $B\,\mathrm{Conf}(N)$ denote the configuration space of N points on $\mathbb{S}^2$ modulo rotations, $B\,\mathrm{Conf}(N) = \mathrm{Conf}(N, \mathbb{S}^2)/\mathrm{SO}(3)$. Via stereographic projection from the first point, for $N \geq 3$ we have a

? meaning of "B" not on page; quotient by SO(3) inferred

diffeomorphism

$$B\operatorname{Conf}(N) \xrightarrow{\cong} \operatorname{Conf}(N-1,\mathbb{R}^2)/\mathrm{SO}(2).$$

[Added] Indeed, $\mathrm{SO}(3)$ acts transitively on $\mathbb{S}^2$, so we may rotate u_1 to the north pole; the remaining freedom is the stabilizer $\mathrm{SO}(2)$ of the pole, and stereographic projection from the pole identifies $\mathbb{S}^2 \setminus \{u_1\}$ with $\mathbb{R}^2$, $\mathrm{SO}(2)$-equivariantly (rotations about the origin). For $N \geq 3$ the actions are free.

That allows us to compute Betti numbers from known Betti numbers via the fibration $\mathrm{SO}(2) \to \operatorname{Conf}(N-1,\mathbb{R}^2) \to \operatorname{Conf}(N-1,\mathbb{R}^2)/\mathrm{SO}(2)$: the Poincaré polynomial is

$$P\big(B\operatorname{Conf}(N)\big) = \frac{P\big(\operatorname{Conf}(N-1,\mathbb{R}^2)\big)}{P(\mathrm{SO}(2))},$$

where

$$P\big(\operatorname{Conf}(N-1,\mathbb{R}^2)\big) = (1+t)(1+2t)\cdots\big(1+(N-2)t\big), \qquad P(\mathrm{SO}(2)) = 1+t.$$

[Added] Here $P(X)(t) = \sum_i b_i(X)\, t^i$. The formula for $\operatorname{Conf}(n,\mathbb{R}^2)$ is due to Arnold (1969). The division is justified because the $\mathrm{SO}(2)$-bundle is trivial: the map $u \mapsto (u_2-u_1)/|u_2-u_1| \in \mathbb{S}^1$ is $\mathrm{SO}(2)$-equivariant, so $\operatorname{Conf}(N-1,\mathbb{R}^2) \cong \mathrm{SO}(2) \times \big(\operatorname{Conf}(N-1,\mathbb{R}^2)/\mathrm{SO}(2)\big)$, and the Künneth formula applies. Hence

$$P\big(B\operatorname{Conf}(N)\big) = (1+2t)(1+3t)\cdots\big(1+(N-2)t\big).$$

The picture behind the formula for $\operatorname{Conf}(n,\mathbb{R}^2)$ is that of punctured planes, and the induction up in n: forgetting the last point gives a fibration $\operatorname{Conf}(n,\mathbb{R}^2) \to \operatorname{Conf}(n-1,\mathbb{R}^2)$ whose fibre is the plane with $n-1$ punctures, with Poincaré polynomial $1+(n-1)t$ (Fadell–Neuwirth).

So the Betti numbers b_λ of $B\operatorname{Conf}(N)$ are:

	$\lambda=0$	$\lambda=1$	$\lambda=2$	$\lambda=3$
$N=3$	1	0	0	0
$N=4$	1	2	0	0
$N=5$	1	5	6	0
$N=6$	1	9	26	24
$\vdots$	1			

In general $b_0 = 1$, and the top Betti number is $b_{N-3} = (N-2)!$.

One can use this idea to describe extensions and induction arguments for configuration spaces in general; for example the $K(\pi,1)$ type of configuration spaces comes from the fact that such spaces really change nicely (the Fadell–Neuwirth fibrations: inductively, $\operatorname{Conf}(n,\mathbb{R}^2)$ is an iterated fibration with aspherical fibres, so it is aspherical, with fundamental group the pure braid group P_n).

some crossed-out words here

10.3 Min-type functions

numbered 10.2 again in the notes

Definition 10.2. A *min-type function* is given by a parametrized family of functions

$$f(x,p)\colon X \times M \to \mathbb{R},$$

where X is a parameter space, compact…; for our purposes, finite and discrete. We require $f(x,\cdot)$ to be smooth (or smooth enough) for each parameter x. The min-type function is then

$$F(p) = \min_{x\in X} f(x,p).$$

Our function, the injectivity radius

$$\rho\colon \mathrm{Conf}(N)\to\mathbb{R}, \qquad \rho(u) = \min_{\substack{i\neq j\\ u_i,u_j\in u}} d(u_i,u_j),$$

is of this type (with X the set of pairs $\{i,j\}$ and $f(\{i,j\},u) = d(u_i,u_j)$, the geodesic distance on $\mathbb{S}^2$). (Here "injectivity radius" means the minimal distance; the largest radius of disjoint spherical caps centred at the u_i is $\rho(u)/2$.)

In general, topological regularity is characterized by the non-emptiness of an intersection of gradient cones of this type.

[Added] Precisely: for $p\in M$ let $X(p) = \{x\in X : f(x,p) = F(p)\}$ be the active parameters, and let

$$C_p = \{v\in T_pM :\ df_x|_p(v) > 0 \text{ for all } x\in X(p)\}$$

be the cone of directions that increase every active function, hence increase F to first order. Where $C_p\neq\emptyset$, a gradient-like vector field can be built locally, and the flow argument of Theorem 9.7 goes through for superlevel sets $\{F\geq c\}$ (cf. Baryshnikov–Bubenik–Kahle, *Min-type Morse theory for configuration spaces of hard spheres*, IMRN 2014).

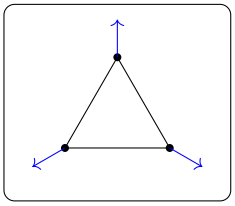
Corollary 10.3. *The topology of the superlevel sets of F only changes at critical points x, which only occur when the gradient cone is empty, i.e. when*

$$\sum_x w_x\, df_x\big|_u = 0 \qquad \textit{for some } w_x\geq 0 \textit{ not all zero, } x\in X(u).$$

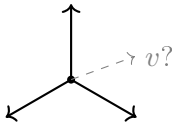
The equivalence of "$C_u = \emptyset$" and the existence of such weights is Gordan's theorem of the alternative (a finite-dimensional Farkas/separation lemma).

From our picture for the injectivity radius, this is when increasing some collection of active distances simultaneously is not possible.

? p. 21 right column only partly legible



regular:
all can move apart



critical: no v in
the positive (dual) cone

10.3.1 The contact graph and balanced configurations

Let $u\in\mathrm{Conf}(N)$ with $\rho(u)\geq\theta$ (we will take $\rho(u) = \theta$). There is a *contact graph*:

p. 22 left column crossed out

- vertices: the points $u_i\in u$;
- edges: u_iu_j whenever $d(u_i,u_j) = \theta$.

Edges may be assigned positive weights w_{ij}, and if the resulting weighted graph is *force balanced*, this is equivalent to the cone being empty.

[Added] Write $e_{ij} \in T_{u_i}\mathbb{S}^2$ for the unit tangent vector at u_i pointing along the geodesic towards u_j. The differential of $d(u_i, u_j)$ has component $-e_{ij}$ at u_i and $-e_{ji}$ at u_j. So the condition of Corollary 10.3 reads: there are weights $w_{ij} \geq 0$, not all zero, on the edges of the contact graph with

$$\sum_{j \sim i} w_{ij}\, e_{ij} = 0 \qquad \text{at every vertex } u_i,$$

i.e. the "forces" along the edges balance at each point. The edges with $w_{ij} > 0$ form the *balanced (stressed) subgraph.*

So the only obstacles to topological regularity are exactly the force balanced configurations of points.

Examples...

left blank on p. 23

Definition 10.4. $\mathrm{Conf}(N, \theta) = \{u \in \mathrm{Conf}(N, \mathbb{S}^2) : \rho(u) \geq \theta\}$, and $\mathrm{Conf}(N, 0) := \mathrm{Conf}(N, \mathbb{S}^2)$.

? "$N \geq 3$" or "$N > 3$"

Theorem 10.5. *Suppose $N \geq 3$. Then for $0 < \theta < \pi/N$, $\mathrm{Conf}(N, \theta)$ is a strong deformation retract of, and hence homotopy equivalent to, $\mathrm{Conf}(N, 0)$.*

Proof (sketch). First, there are no balanced configurations with $\rho(u) = \theta'$ for $0 < \theta' < \pi/N$; then push each point apart using the flow of a gradient-like vector field for ρ...

[Added] *No balanced configurations.* (i) A balanced subgraph is not contained in any open hemisphere: if all its vertices lie in $\{y : \langle y, e\rangle > 0\}$, take a vertex v of the subgraph minimizing $\langle v, e\rangle$. For each neighbour u, the vector e_{vu} is a positive multiple of $u - \langle u, v\rangle v$, whose e-component is $\langle u, e\rangle - \langle u, v\rangle\langle v, e\rangle \geq \langle v, e\rangle(1 - \langle u, v\rangle) > 0$ (using $\langle u, e\rangle \geq \langle v, e\rangle$ and $\langle u, v\rangle < 1$). So every force at v has a positive e-component, and they cannot sum to zero. (ii) Each connected component of the balanced subgraph is itself balanced. If a component has k vertices, a spanning tree has $k-1$ edges of length θ', total length $L = (k-1)\theta'$. A tree has a centre c within distance $L/2$ (along the tree) of all its points, so all vertices lie in the spherical ball of radius $L/2$ about c. By (i), $L/2 \geq \pi/2$, so $(k-1)\theta' \geq \pi$, so $\theta' \geq \pi/(k-1) \geq \pi/(N-1) > \pi/N$. (So this crude argument even gives the theorem for $\theta < \pi/(N-1)$. The sharp bound is $\theta < 2\pi/N$: a balanced graph has every valence ≥ 2, hence at least as many edges as vertices, and Kusner–Kusner–Lagarias–Shlosman (Thm. 4.14, cited below) show its total length is at least 2π, with equality only for the N points equally spaced on a great circle. Compare the statement at the start of Lecture 11.)

Deformation. So ρ has no critical values in $(0, \theta]$, and the min-type version of Theorem 9.7 lets us push $\{\rho \geq \eta\}$ onto $\{\rho \geq \theta\}$ for every $0 < \eta < \theta$; some care is needed near $\rho = 0$, where $\mathrm{Conf}(N, 0)$ is not compact. (Cf. Kusner–Kusner–Lagarias–Shlosman, *Configuration spaces of equal spheres touching a given sphere: the twelve spheres problem*, arXiv:1611.10297 (2016); in *New Trends in Intuitive Geometry*, Bolyai Soc. Math. Stud. 27, Springer, 2018.)

□

? "$N = 4$" or "$N > 4$"

For $N = 4 \ldots$, $\mathrm{Conf}(N, \theta)$ for small θ is therefore path connected. (Indeed $\mathrm{Conf}(N, \mathbb{S}^2)$ is connected, since the diagonals removed have real codimension 2.)

left blank in the notes

Example 10.6 ($N = 4$)**.** Balanced subgraphs need every vertex of valence ≥ 2, and a vertex of valence 2 must have its two edges in opposite directions, so it lies on a great circle through its two neighbours. One finds: the square on a great circle ($\theta = \pi/2$), and the regular tetrahedron ($\theta = \arccos(-1/3) \approx 109.47°$, the maximum of ρ). An equilateral triangle on a great circle ($\theta = 2\pi/3$) plus a free fourth point is *not* realizable, since every point of $\mathbb{S}^2$ is within $\pi/2$ of one of the three. This is the editor's check; compare the question marks on p. 29.

10.3.2 Outlook

pp. 24–25: headings only

- Strata for Tammes maxima: table... (The Tammes problem asks for $\max \rho$ over $\mathrm{Conf}(N, \mathbb{S}^2)$; it is solved for $N \leq 14$ and $N = 24$, with $N = 13, 14$ due to Musin–Tarasov, 2012/2015.)
- An algorithm for searching for critical configurations by evolution of force configurations...
- Local maxima; rings...
- Rigidity?, critical conditions and weights...
- Candidate graph sorting...

10.4 Topological regularity revisited

pp. 26–29 seem to re-explain §10.3; possibly a later lecture

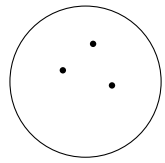
? p. 26 has only these two lines

What is this? This is a force balanced configuration.

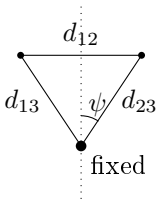
That is, for the injectivity radius, the gradient cone is the intersection of the convex cones of motions that increase the distance between each active pair of points.

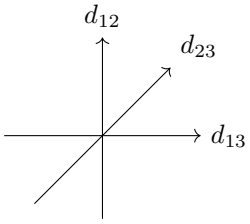
Picture: three points. The configuration space of 3 points on $\mathbb{S}^2$ is 6-dimensional. Fixing one point leaves a 4-dimensional space; fixing also the rotation angle ψ about it leaves a 3-dimensional space, which is locally parametrized by the distances d_{12}, d_{13}, d_{23}. The cone for d_{12} is the upper half space $\{d_{12}$ increasing$\}$, etc.

angle called θ in the notes



3 points on $\mathbb{S}^2$





coordinates near u

When the three active differentials are linearly independent, the three half-spaces meet in an open octant, so the cone is non-empty and the value is regular.

So:

θ is a topologically regular value $\iff$ neighbourhoods of the value have superlevel sets that deformation retract to each other.

(For $\theta' < \theta$ near θ, $\{\rho \geq \theta\} \hookrightarrow \{\rho \geq \theta'\}$ is a deformation retract.) So critical values are where the topology changes...

For topologically regular values, this deformation retraction is defined by the flow of some vector field defined by f, the gradient flow as in the last lecture (Theorem 9.7). So the topology can only change at an obstruction to gradient flow. For a min-type function, this obstruction is given by a convex cone, namely the intersection of all variational cones that increase f at the parameters $x \in X$.

Exercise 10.7. Classify all distinct realizable balanced configurations of N points on a sphere.

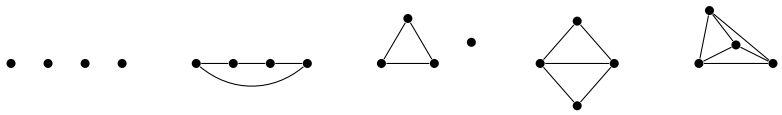
First, list the graphs. For 3 points, the candidate contact graphs are:



For example, to be force balanced every vertex needs valence ≥ 2 (and a vertex of valence 2 needs its two edges opposite, i.e. along a great circle). For $N = 3$ only the triangle survives, realized as three points at mutual distance $2\pi/3$ on a great circle.

? reads "valence = 2"

Some graphs on 4 points:



"?" by the triangle + point and the rhombus; "opposite", "great circles". One more sketch, apparently K_4, drawn at right.

p. 29 ends with a sketch of two great circles

For the rhombus with a diagonal, the two valence-2 vertices need opposite edges, which forces great circles; whether such configurations are realizable is a question of geometry... (It is not: the two valence-3 vertices would have to be at distance θ and at distance 2θ along a great circle, forcing $\theta = 2\pi/3$, and then the two valence-2 vertices would coincide.)

Chapter 11

Lecture 11: Critical points, jamming and entropy

11.1 Critical points and jamming

Throughout, a configuration is an ordered N-tuple $X = (x_1, \dots, x_N)$ of points on the sphere $\mathbb{S}^2$, and d denotes the geodesic (angular) distance. Let $f(X) = \min_{i<j} d(x_i, x_j)$ and write $\mathrm{Conf}(N;\theta) = \{X : f(X) \geq \theta\}$. A configuration in $\mathrm{Conf}(N;\theta)$ is the same thing as a packing of N spherical caps of angular radius $\rho = \theta/2$.

Last time we stated the following theorem, which characterizes the regular values of f in terms of the geometry of the contact graph.

Definition 11.1 (Contact graph)**.** Let $X \in \mathrm{Conf}(N;\theta)$ with $f(X) = \theta$, where $\theta = 2\rho$. The *contact graph* of X is the set of edges on $\mathbb{S}^2$ (geodesic arcs) joining those points x_i, x_j that are at distance exactly θ.

[Added] For an edge $\{i,j\}$ of the contact graph let $u_{ij} \in T_{x_i}\mathbb{S}^2$ be the unit tangent vector at x_i pointing along the geodesic towards x_j (well defined since $0 < \theta < \pi$). A tangent vector $v = (v_1, \dots, v_N)$, $v_i \in T_{x_i}\mathbb{S}^2$, changes the length of the edge at rate

crossed out: "geodesic skeleton of", "struts"

$$\left.\frac{d}{dt}\right|_{t=0} d(x_i + tv_i,\, x_j + tv_j) \;=\; -\langle u_{ij}, v_i\rangle - \langle u_{ji}, v_j\rangle.$$

We say v lies in the *tangent cone* (of increasing directions) at X if this rate is strictly positive for every contact edge, and that θ is a *regular value* of f if every X with $f(X) = \theta$ admits such a v; otherwise θ is a *critical value* and the offending X are *critical configurations*.

Theorem 11.2. *A configuration X with $f(X) = \theta$ has no element in its tangent cone if and only if there exist weights $w_{ij} = w_{ji} \geq 0$ on the contact edges, not all zero, such that the contact graph is* force balanced*:*

$$\sum_{j:\,\{i,j\}\ \text{contact}} w_{ij}\, u_{ij} \;=\; 0 \qquad \text{for every vertex } i.$$

In particular, θ is a regular value if and only if no configuration at level θ admits such a balanced (non-negative, not all zero) system of weights.

[Added] *Proof sketch.* Let A be the linear map $v \mapsto \big(-\langle u_{ij}, v_i\rangle - \langle u_{ji}, v_j\rangle\big)_{\{i,j\}}$ from $\bigoplus_i T_{x_i}\mathbb{S}^2$ to $\mathbb{R}^E$, E the set of contact edges. By Gordan's alternative (a form of Farkas' lemma), either there is v with $Av > 0$ componentwise, or there is $w \in \mathbb{R}^E$, $w \geq 0$, $w \neq 0$ with $A^T w = 0$. Computing $A^T w$ vertex by vertex gives exactly $-\sum_j w_{ij} u_{ij} = 0$. Together with the deformation theorem from last lecture (if $[a, b]$ contains no critical values, then $\mathrm{Conf}(N; b)$ is a deformation retract of $\mathrm{Conf}(N; a)$), this is the basic tool for computing the topology of $\mathrm{Conf}(N; \theta)$ as θ increases.

"(non-negative, not all 0)" replaces crossed-out "(positive)"

The simplest application is to points on a circle.

Theorem 11.3. *For N points on the circle $\mathbb{S}^1$,* $\mathrm{Conf}(N; 0) \simeq \mathrm{Conf}(N; 2\pi/N)$.

? Blotted symbol before $2\pi/N$; read here as the circle case. Possibly "<": the $\mathbb{S}^2$ theorem, see remark.

[Added] *Alternative reading.* The blot may be "$\theta <$", i.e. the statement for $\mathbb{S}^2$: $\mathrm{Conf}(N; \theta)$ is a strong deformation retract of $\mathrm{Conf}(N; 0)$ for $0 \leq \theta < 2\pi/N$. This is Theorem 4.14 of Kusner–Kusner–Lagarias–Shlosman (cited in §11.2). It sharpens the bound π/N of Lecture 10, since a balanced contact graph has total length at least 2π, and the smallest critical value is attained only by N equally spaced points on a great circle. The proof sketch below covers only the circle.

[Added] *Proof sketch.* Here $\mathrm{Conf}(N; 0)$ means the usual ordered configuration space (distinct points). On $\mathbb{S}^1$ the vectors u_{ij} at x_i are $\pm$ the unit tangent, so force balance at a vertex forces it to have one contact on each side with equal weights. Following the weights along a chain of contacts, a chain with an endpoint is unbalanced at that endpoint; hence the contact graph of a critical configuration is the whole N-cycle, i.e. X is a regular N-gon and $\theta = 2\pi/N$. Thus there are no critical values in $(0, 2\pi/N)$, so $\mathrm{Conf}(N; \theta) \simeq \mathrm{Conf}(N; 0)$ for $\theta < 2\pi/N$. At $\theta = 2\pi/N$ itself the space consists of the labelled regular N-gons, a disjoint union of $(N-1)!$ circles (one for each cyclic order of the labels); and $\mathrm{Conf}(N; 0)$ has the same homotopy type, each of its $(N-1)!$ components being a circle times an open simplex.

This also gives an algorithm for exploring critical and maximal configurations: *ρ-evolution.* Starting from any configuration, increase ρ by moving the points in a direction of the tangent cone; update the contact graph whenever new contacts form, and continue until no increasing direction exists, i.e. until the contact graph becomes force balanced. The configuration reached is then a critical configuration, generically a local maximum of f (a *jammed* configuration).

? Page ends with "update..." and "conflicting..." (or "conf-flip"?); not expanded.

11.2 Examples with pictures

p. 2: "$N = 3$, $N = 4$: full picture" – pictures were presumably drawn on the board.

For $N = 3$ and $N = 4$ one can give the full picture of the critical points.

[Added] For $N = 3$: a vertex with a single contact cannot be balanced, so every vertex of a critical contact graph has degree 2, and the two tangent vectors at it must be opposite; thus the three points lie on a great circle, equally spaced. The only critical value is the maximum $\theta = 2\pi/3$, and $\mathrm{Conf}(3; \theta) \simeq \mathrm{Conf}(3; 0)$ for $\theta < 2\pi/3$. For $N = 4$: the maximum is the regular tetrahedron, $\theta = \arccos(-1/3) \approx 109.47°$. The square inscribed in a great circle ($\theta = \pi/2$) is also critical (each vertex has two opposite contacts), but it is not a local maximum: pushing alternate vertices slightly north and south increases all four contact distances to second order, deforming the square towards the tetrahedron.

For larger N:

- For various N the maximum is the solution of the *Tammes problem.* (The Tammes problem is solved for $N \leq 14$ and $N = 24$; e.g. $N = 13$ and $N = 14$ by Musin–Tarasov, $N = 24$ by Robinson.)

? "$N = 10$ – local max?": whether there are non-global local maxima, presumably.

- $N = 10$: local maxima?
- $N = 12$: local maxima? Higher critical points. The jitterbug?

[Added] For $N = 12$ the global maximum is the icosahedron ($\theta = \arctan 2 \approx 63.43°$). The cuboctahedron (the kissing configuration of the FCC lattice, $\theta = 60°$) is a critical configuration that is not a local maximum; Buckminster Fuller's "jitterbug" motion deforms the cuboctahedron into the icosahedron. (See R. Kusner, W. Kusner, J. Lagarias, S. Shlosman, *Configuration spaces of equal spheres touching a given sphere: the twelve spheres problem*, arXiv:1611.10297 (2016); in *New Trends in Intuitive Geometry*, Bolyai Soc. Math. Stud. 27, Springer, 2018. That paper shows the cuboctahedral (FCC) configuration is critical. That it is not a local maximum follows from the jitterbug: along it the triangle edges lengthen from 60° to $\arctan 2$, while the square diagonals shrink from 90° to $\arctan 2$.)

11.3 Local maxima and Edwards entropy

Definition 11.4 (Edwards' ansatz)**.** All jammed states of equal volume have equal probability.

"Edw." crossed out and replaced by "Ansatz"

[Added] This is the hypothesis of S. F. Edwards and R. B. S. Oakeshott (*Theory of powders*, Physica A, 1989). The *Edwards entropy* at volume V is the logarithm of the number (or measure) of jammed states of volume V; it plays the role that the Boltzmann entropy plays for the microcanonical ensemble.

This is related to the evolution of the ρ-flow: if configurations are chosen at random and then evolved by ρ-evolution, the probability of a given jammed state is the measure of its basin of attraction, and Edwards' ansatz says these basins are (for equal volumes) equally large. It is also related to the model … It is not always true ? Numerical studies of small disc packings have found that jammed states are in general *not* equally likely, and that the frequencies depend on the preparation protocol. *Update:* later work (Martiniani, Schrenk, Ramola, Chakraborty, Frenkel, *Nature Physics*, 2017) gave numerical evidence that the ansatz does hold for soft-sphere packings exactly at the jamming threshold.

? "Also the model…"; crossed out: "ex. annealed"

11.3.1 Boltzmann measure: the discrete setting

Consider N particles, each in one of k states. Each state has an energy E_i, $i \in \{1, \dots, k\}$. Let n_i be the number of particles in state i. The system is closed, so

$$E = \sum_{i=1}^{k} E_i \, n_i, \qquad N = \sum_{i=1}^{k} n_i$$

are fixed. The number of ways to place the particles with occupation numbers $\{n_1, \dots, n_k\}$ is the multinomial coefficient

he writes both n_i and N_i; unified to n_i

$$W = \frac{N!}{n_1! \, n_2! \cdots n_k!}.$$

By the law of large numbers, the optimal partition (the one maximizing W) dominates for N large. (More precisely: W, as a function of the frequencies n_i/N, is $e^{NH+o(N)}$ with $H = -\sum (n_i/N) \ln(n_i/N)$, so frequencies whose entropy is smaller than the maximum by a fixed

amount are exponentially rare.) So we determine the partition that maximizes W; since log is monotone increasing we may replace W by $\log W$. By Stirling's formula,

$$\log W = (N\ln N - N) - \sum_{i=1}^{k}(n_i \ln n_i - n_i) \;+\; \text{error} \quad (\text{error } = O(k\ln N)).$$

We maximize subject to the two constraints with Lagrange multipliers α, β:

$$f = \ln W + \alpha\Big(N - \sum_{i=1}^{k} n_i\Big) + \beta\Big(E - \sum_{i=1}^{k} E_i n_i\Big).$$

Since $\frac{\partial}{\partial n}(n\ln n - n) = \ln n$,

$$0 = \frac{\partial f}{\partial n_i} = -\ln n_i - (\alpha + \beta E_i), \qquad \text{so} \qquad n_i = e^{-\alpha - \beta E_i}.$$

Summing over i gives $N = e^{-\alpha} Z$, which eliminates α:

$$\frac{n_i}{N} = \frac{e^{-\beta E_i}}{\sum_{j=1}^{k} e^{-\beta E_j}}, \qquad Z = \sum_{j=1}^{k} e^{-\beta E_j} \quad (\text{the } \textit{partition function}).$$

Definition 11.5. Here $\beta = \dfrac{1}{T}$ (or $\dfrac{1}{k_B T}$ with Boltzmann's constant k_B) is the inverse temperature, and

$$F = -T \ln Z$$

is the *free energy*. The *Boltzmann measure* is

$$\Pr(\text{state } i) = \frac{e^{-\beta E_i}}{Z}.$$

The mean energy per particle is

$$\bar{E} = \frac{E}{N} = \sum_{i=1}^{k} E_i \frac{n_i}{N} = \sum_{i=1}^{k} \frac{E_i e^{-\beta E_i}}{Z} = \frac{-\frac{\partial}{\partial \beta} Z}{Z} = -\frac{\partial}{\partial \beta} \ln Z.$$

Fix: page has $e^{\beta E_i}$ and final "$-\partial_\beta Z$"; signs and ln corrected

Lemma 11.6. *$-\ln Z$ is concave as a function of β. Equivalently, $\bar{E} = -\frac{\partial}{\partial \beta}\ln Z$ is monotone (non-increasing in β, i.e. non-decreasing in T).*

Proof idea. We must show $\dfrac{\partial^2}{\partial \beta^2} \ln Z \geq 0$. Why? Write it as the variance of E, which is ≥ 0.

[Added] Indeed, with $\langle g \rangle = \sum_i g(E_i) e^{-\beta E_i}/Z$,

$$\frac{\partial^2}{\partial \beta^2} \ln Z = \frac{Z''}{Z} - \left(\frac{Z'}{Z}\right)^2 = \langle E^2 \rangle - \langle E \rangle^2 = \mathrm{Var}(E) \geq 0,$$

with equality only if all E_i are equal.

□

Fix: page has "> 0"; it is ≥ 0

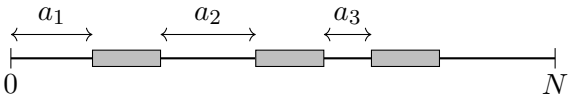
Fix: page: "$\frac{1}{\beta}\ln Z$"; sign restored as in $F = -T\ln Z$

Thus $-\dfrac{1}{\beta}\ln Z$ is the free energy. As $T \to \infty$ (i.e. $\beta \to 0$), $\Pr(i) \to \dfrac{1}{k}$: all states become equally likely. As $T \to 0$ the measure concentrates on the states of minimal energy.

Remark 11.7. For hard particles the energy of a configuration is 0 if the particles do not overlap and $+\infty$ if they do. The Boltzmann weight $e^{-\beta E}$ is then the indicator of the admissible configurations, so the Boltzmann measure is normalized volume on the configuration space, at every temperature. This is why volumes of configuration spaces appear as partition functions below.

11.3.2 The hard disc model in an interval

Place hard discs of diameter 1 (that is, hard rods of length 1) in the interval $[0, N]$, and let $a_1, a_2, a_3, \dots$ be the gaps in front of the first, second, third, ... disc.



With three discs the configuration space is

$$\Omega_{N,3} = \{(a_1, a_2, a_3) : \ 0 \le a_1, a_2, a_3, \quad a_1 + a_2 + a_3 \le N - 3\},$$

a simplex with

$$\operatorname{vol} \Omega_{N,3} = \frac{(N-3)^3}{3!}.$$

p. 7: sketch of the simplex with axes a_1, a_2, a_3

In general, for $k \le N$ discs,

$$\operatorname{vol}(\Omega_{N,k}) = \frac{(N-k)^k}{k!}.$$

(The gaps describe the discs up to relabelling; for labelled discs multiply by $k!$.)

Now give each particle an energy E_0 and define the *fugacity*

$$z = e^{-\beta E_0}.$$

(In the usual notation $z = e^{\beta\mu}$ with chemical potential μ; here $\mu = -E_0$.) The grand canonical partition function is

$$Z_N(z) = \sum_{k=0}^{N} z^k \operatorname{vol}(\Omega_{N,k}) = \sum_{k=0}^{N} \frac{(N-k)^k}{k!} z^k.$$

This is the classical *Tonks gas* (L. Tonks, 1936), one of the few exactly solvable models of hard particles.

Exercise 11.8. Using Stirling's formula on the largest term, show that $\frac{1}{N} \ln Z_N(z) \to P$, where $P > 0$ solves $z = Pe^P$, and that the dominant number of discs satisfies $k/N \to P/(1+P)$. (P is β times the pressure; note that the density k/N increases from 0 to 1 as z goes from 0 to ∞.)

11.4 A quasi-one-dimensional model with open ends

We study discs in a narrow channel, where a combinatorial analysis is possible. Take discs of radius $r = 1$ in a channel whose width w satisfies

crossed out: "Model"

$$2 < w < 2 + \sqrt{3}.$$

[Added] The centres then lie in a strip of width $h = w - 2 \in (0, \sqrt{3})$. Discs cannot pass each other ($h < 2$). If two consecutive discs touch while lying against opposite walls, their centres are horizontally $L = \sqrt{4 - h^2}$ apart, and $1 < L < 2$; the condition $h < \sqrt{3}$ is exactly $L > 1$, which guarantees that each disc can touch only its two neighbours in the row (discs i and $i+2$ are always more than 2 apart). Consecutive discs touching along the same wall are horizontally 2 apart.

Let the length of the configuration be t. (Discs are confined only by the walls; the ends are open, so "jammed" means that t is locally minimal for the given number of discs.)

p. 9: three sketches; third shows a narrow channel with discs in a staircase

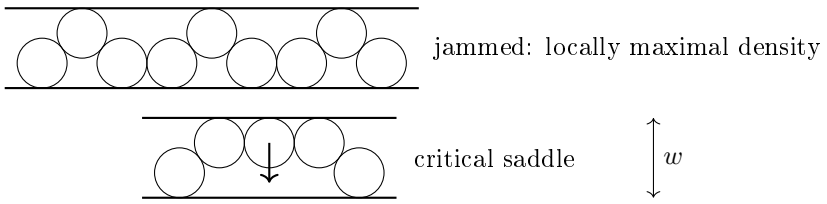
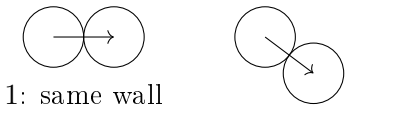


Figure 11.1: Discs in a narrow channel ($h = 1.2$, $L = 1.6$). Top: a jammed configuration, contact word 00100100. Bottom: three consecutive discs on the same wall (word 011 0); the middle one is force balanced but can be pushed into the gap, so this is a critical point but not a local maximum.

11.4.1 Combinatorics: how many maxima for given t?

The length is determined by the number of adjacent pairs of discs at opposite walls versus at the same wall. Record each contact between consecutive discs by a letter:



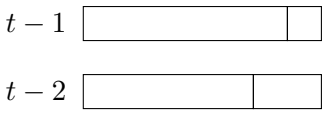
Two consecutive 1's are impossible in a jammed configuration: the middle disc of three discs along one wall can move away from the wall (Figure 11.1, bottom). So counting jammed configurations amounts to counting binary words with no two adjacent 1's. This is equivalent to a domino tiling problem, or a "stars and bars" count: such words are sequences formed of the blocks

$$0 = \square, \qquad 01 = \square\square \text{ (a domino).}$$

(Precisely: prefixing a 0, words of length ℓ with no 11 correspond bijectively to tilings of a $1 \times (\ell + 1)$ strip by squares and dominoes.)

This satisfies the Fibonacci recursion (we count configurations up to the up/down reflection of the channel): a tiling of length t is a tiling of length $t-1$ followed by a square, or a tiling of length $t-2$ followed by a domino; there are $1, 2, \ldots$ tilings of lengths $1, 2, \ldots$ Hence the number of tilings of length m is the Fibonacci number F_{m+1}, and the number of binary words of length ℓ with no 11 is $F_{\ell+2}$ ($F_1 = F_2 = 1$).

? "(not up/down symmetric)" – reading as: words count configurations modulo reflection.
crossed out: "# words ending in 0 of length ℓ"



Given exactly k ones in a word of length ℓ, we are inserting k ones into a sequence of $\ell - k$ zeros,

$$\wedge 0 \wedge 0 \wedge 0 \wedge \cdots \wedge 0 \wedge,$$

at most one into each of the $\ell - k + 1$ slots. So for each k there are $\binom{\ell - k + 1}{k}$ choices of order. (Summing over k gives $\sum_k \binom{\ell-k+1}{k} = F_{\ell+2}$.)

So if the length of a "1" is 2 and that of a "0" is L, the configuration with word of length ℓ (i.e. $\ell+1$ discs) containing k ones has length

$$t = 2k + (\ell - k)L + 2,$$

the $+2$ coming from the two ends.

Remark 11.9. With truly open ends, an end disc lying against the same wall as its neighbour can roll around that neighbour towards the opposite wall, shortening t. If such end configurations are excluded, the count becomes $\binom{\ell-k-1}{k}$ instead of $\binom{\ell-k+1}{k}$; the asymptotics below are unchanged.

? Check: with open ends, a 1 in the first or last slot seems not locally length-minimizing.

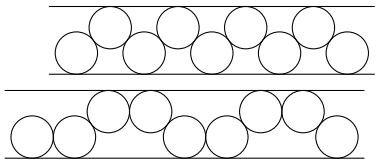
We can also compute the maximum of $\binom{n-k+1}{k}$ over k (with $n=\ell$): as $n\to\infty$ the maximizing k satisfies

$$\frac{k}{n} \longrightarrow \frac{5}{10} - \frac{\sqrt{5}}{10} = \frac{1}{2} - \frac{\sqrt{5}}{10} \approx 0.276.$$

[Added] Indeed, with $x = k/n$, Stirling gives $\frac{1}{n}\ln\binom{n-k+1}{k} \to g(x) = (1-x)\ln(1-x) - x\ln x - (1-2x)\ln(1-2x)$, and $g'(x)=0$ becomes $(1-2x)^2 = x(1-x)$, i.e. $5x^2-5x+1=0$, whose root in $(0,\frac12)$ is $x = \frac{5-\sqrt5}{10}$. At this point $g(x) = \ln\varphi$, $\varphi = \frac{1+\sqrt5}{2}$, which matches the growth $F_{n+2}\sim\varphi^n$: the maximal term carries the whole exponential growth.

And there is a "unique" densest packing: the zigzag with all contacts at opposite walls (the word $00\cdots0$, length $\ell L+2$), unique up to reflection. At the other extreme the least dense jammed configuration alternates pairs on the two walls (word $0101\cdots$, or $1010\cdots$ with ends allowed).

? Second sketch annotated "← ... last ... as well": read as the least dense case.



Exercise 11.10. Using the law of large numbers and Stirling's approximation, show that if all jammed configurations of $n+1$ discs are equally likely (Edwards' ansatz without the volume constraint), a typical configuration has a fraction

$$\frac{1}{2} - \frac{\sqrt{5}}{10}$$

of 1's. Deduce that its length per disc is close to $2x + (1-x)L$ with $x = \frac12 - \frac{\sqrt5}{10}$. Compare to coin flips, where the fraction would be $\frac12$.

"Ex. law of large ... & approx. ..."; partly illegible

11.4.2 Outlook

- Probabilities of critical points. Flow to jammed configurations... The resulting distribution is probably concentrated on the maxima, since critical saddles should have probability near 0. (A saddle attracts only a lower-dimensional set of starting configurations under the flow.)

? First word unclear ("prune"? "prior"?).

- $\Longrightarrow$ Question: compute these values!
- Also... for large N this is hard, so we study low-order cases.

? last line read as "so study low order cases"

Chapter D

Interlude D: Min-max, configuration spaces and braids

[Added] This interlude is new: it is not in the handwritten notes. It takes three threads from Lectures 7–11 (the deformation theorem of Lecture 9, the max-min distance function and its balanced contact graphs in Lectures 10–11, and the braid groups of Lecture 7) and follows each one into a neighbouring field. To keep it readable it is set in black; all of it should be read as added material.

Lectures 8–11 used Morse theory in two ways. For smooth functions, the Morse inequalities count critical points of *non-degenerate* functions by Betti numbers. For the min-type function

$$\rho(u) = \min_{i<j} d(u_i, u_j) \qquad \text{on } \operatorname{Conf}(N, \mathbb{S}^2),$$

we only had a deformation theorem: the topology of $\{\rho \geq \theta\}$ changes only at balanced configurations. This leaves some natural questions open.

- The deformation theorem alone already forces critical points to exist, without any non-degeneracy. How many, and why?
- What do the Betti numbers of $\operatorname{Conf}(N, \mathbb{S}^2)/\mathrm{SO}(3)$ computed in Lecture 10 say about energies on that space, and why are they products like $(1+2t)(1+3t)\cdots$?
- Lecture 7 defined braid groups as fundamental groups and Lecture 8 found the Weyl chambers of A_{N-1} inside $\operatorname{Conf}(N, \mathbb{R})$. Are these two facts related?

The answers lead to *min-max theory* (Section D.1), to Morse theory on *configuration spaces of hard particles* (Sections D.2–D.3), and to *braid and Artin groups* (Section D.4). Section D.5 collects theorems from these areas.

Throughout, M is a connected smooth manifold with a Riemannian metric and $f\colon M \to \mathbb{R}$ is smooth. As in Lecture 9, $M^a = f^{-1}(-\infty, a]$. We say that f has *compact sublevel sets* if every M^a is compact. This holds when M is compact, and also for energies such as $\sum_{i<j} |u_i - u_j|^{-1}$ on $\operatorname{Conf}(N, \mathbb{S}^2)$, which blow up at collisions.

D.1 From Morse theory to min-max

D.1.1 The deformation lemma and the minimax principle

The proof of Theorem 9.7 gives slightly more than its statement. We record the version we need.

Lemma D.1 (Deformation lemma)**.** *Let f have compact sublevel sets, let $a < b$ and $\delta > 0$, and suppose that $f^{-1}[a-\delta, b+\delta]$ contains no critical point. Then there are diffeomorphisms $\eta_t \colon M \to M$, $0 \le t \le 1$, depending smoothly on t, with $\eta_0 = \mathrm{id}$, $\eta_1(M^b) \subseteq M^a$, and $\eta_t(x) = x$ for all t whenever $f(x) \le a - \delta$ or $f(x) \ge b + \delta$.*

Proof. Choose a smooth $\chi \colon \mathbb{R} \to [0,1]$ with $\chi = 1$ on $[a,b]$ and support in $(a-\delta, b+\delta)$, and put

$$X = \chi(f)\,\frac{\nabla f}{|\nabla f|^2}.$$

This is smooth, because $\nabla f \neq 0$ on the support of $\chi(f)$, and it vanishes outside the compact set $f^{-1}[a-\delta, b+\delta]$. By Lemma 9.6 it generates a one-parameter group φ_t. Along a flow line, $\frac{d}{dt} f(\varphi_t(x)) = df(X) = \chi(f) \in [0,1]$, with value 1 while $f \in [a,b]$. Set $\eta_t = \varphi_{-t(b-a)}$. Running the flow backwards for time $b-a$, a point with $f(x) \le b$ either starts below a and stays there (f is non-increasing), or loses height at unit rate until it reaches a. So $\eta_1(M^b) \subseteq M^a$. Points where $X = 0$ are fixed by the flow, which gives the last claim. □

The lemma says that *if there is no critical value near c, then everything up to level $c+\varepsilon$ can be pushed below $c-\varepsilon$.* Turning this around produces critical points.

Theorem D.2 (Minimax principle)**.** *Let f have compact sublevel sets. Let $Z \subseteq M$ be a set (possibly empty) and let $\mathcal{F}$ be a family of non-empty subsets of M such that $\eta_1(A) \in \mathcal{F}$ whenever $A \in \mathcal{F}$ and (η_t) is an isotopy of diffeomorphisms of M fixing Z pointwise. Put*

$$c = \inf_{A \in \mathcal{F}} \sup_{x \in A} f(x).$$

If $c < \infty$ and $\sup_Z f < c$, then c is a critical value of f.

Proof. Since f attains its minimum, $c > -\infty$. The critical points in $f^{-1}[c-1, c+1]$ form a compact set, so their values form a compact set. If c is not critical, there is therefore an $\varepsilon > 0$ such that $[c-3\varepsilon, c+3\varepsilon]$ contains no critical value; shrinking ε, we may also assume $\sup_Z f < c - 2\varepsilon$. Apply Lemma D.1 with $a = c-\varepsilon$, $b = c+\varepsilon$, $\delta = \varepsilon$. The isotopy fixes $\{f \le c - 2\varepsilon\} \supseteq Z$. Pick $A \in \mathcal{F}$ with $\sup_A f \le c + \varepsilon$. Then $\eta_1(A) \in \mathcal{F}$ and $\eta_1(A) \subseteq M^{c-\varepsilon}$, so $\sup_{\eta_1(A)} f \le c - \varepsilon < c$, contradicting the definition of c. □

Example D.3.

1. $\mathcal{F} = \{\{x\} : x \in M\}$ gives $c = \min f$. If M is compact, $\mathcal{F} = \{M\}$ gives $c = \max f$.
2. *Homology classes.* For $0 \neq \alpha \in H_k(M)$ let $\mathcal{F}_\alpha$ be the family of compact $A \subseteq M$ such that α lies in the image of $H_k(A) \to H_k(M)$. A diffeomorphism isotopic to the identity acts trivially on homology, so $\mathcal{F}_\alpha$ is invariant. The critical value c_α is "the lowest level at which α can be realized". For the height function on the torus of Example 8.13, the classes in H_1 give the two saddle values $f(q)$ and $f(r)$.
3. *Mountain pass.* For $p, q \in M$, take $Z = \{p, q\}$ and $\mathcal{F} = \{\gamma([0,1]) : \gamma \text{ a path from } p \text{ to } q\}$.

Corollary D.4 (Mountain pass)**.** *Let f have compact sublevel sets and let $p \neq q$ be strict local minima of f. Then f has a third critical point, at the level*

$$c = \inf_{\gamma\colon p \rightsquigarrow q} \max_{t \in [0,1]} f(\gamma(t)) \;>\; \max\big(f(p), f(q)\big).$$

Proof. Say $f(p) \le f(q)$. Since q is a strict local minimum, a small geodesic sphere S around q (not containing p inside) satisfies $\min_S f > f(q)$. Every path from p to q crosses S, so $c \ge \min_S f > f(q) \ge f(p)$. Also $c < \infty$ because M is connected. Theorem D.2 with Example D.3(3) applies. □

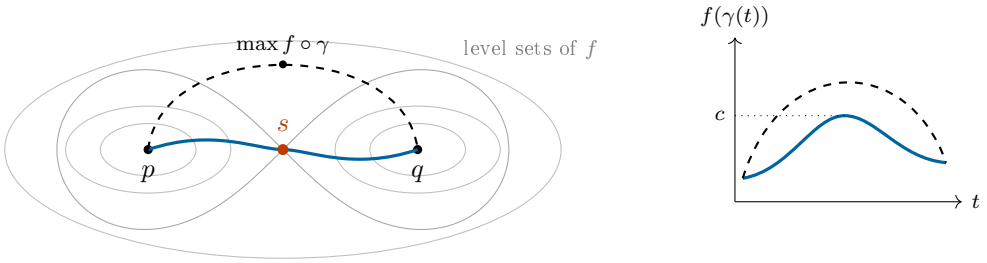


Figure D.1: The mountain pass. Every path from p to q has a highest point; the dashed path climbs higher than necessary. The minimax level c is attained by paths through the saddle s, where the level set through s is a figure eight.

D.1.2 Lusternik–Schnirelmann category

The minimax principle gives one critical value per invariant family. To get many critical points we need many families, and a way to tell them apart. Lusternik and Schnirelmann (around 1930) measured the families by how hard it is to contract them.

Definition D.5. For $A \subseteq M$, the *category of A in M*, $\operatorname{cat}_M(A)$, is the least m such that $A \subseteq U_1 \cup \cdots \cup U_m$ with each U_i open in M and each inclusion $U_i \hookrightarrow M$ homotopic to a constant map. The *Lusternik–Schnirelmann category* of M is $\operatorname{cat}(M) = \operatorname{cat}_M(M)$.

So $\operatorname{cat}(\text{point}) = 1$ and $\operatorname{cat}(\mathbb{S}^n) = 2$ (two hemispheres). This is the unnormalized convention; many papers subtract 1. The following properties are immediate. Category is monotone in A and subadditive, $\operatorname{cat}_M(A \cup B) \le \operatorname{cat}_M(A) + \operatorname{cat}_M(B)$. A homeomorphism of M homotopic to the identity preserves it. A finite set has category 1: a union of small disjoint balls can be contracted inside the connected manifold M.

Theorem D.6 (Cup-length bound)**.** *If there are classes $u_1, \dots, u_k \in H^*(M)$ of positive degree (any coefficients) with $u_1 \smile \cdots \smile u_k \neq 0$, then $\operatorname{cat}(M) \ge k + 1$.*

Proof. Suppose $M = U_1 \cup \cdots \cup U_k$ with each $U_i \hookrightarrow M$ null-homotopic. Then the restriction of u_i to U_i factors through the cohomology of a point, which vanishes in positive degree, so $u_i|_{U_i} = 0$. By the long exact sequence of the pair (M, U_i), u_i is the image of some $\tilde{u}_i \in H^*(M, U_i)$. The relative cup product (defined for open sets) gives

$$\tilde{u}_1 \smile \cdots \smile \tilde{u}_k \in H^*(M, U_1 \cup \cdots \cup U_k) = H^*(M, M) = 0,$$

and it maps to $u_1 \smile \cdots \smile u_k$. Hence the product is 0, a contradiction. □

To count critical points we also need a version of the deformation lemma that is allowed to stop near the critical points at level c.

Lemma D.7 (Deformation with a hole)**.** *Let f have compact sublevel sets, let $c \in \mathbb{R}$, let K_c be the (compact) set of critical points with value c, and let $U \supseteq K_c$ be open. Then there are $\varepsilon > 0$ and a diffeomorphism η of M, isotopic to the identity, with $\eta\big(M^{c+\varepsilon} \setminus U\big) \subseteq M^{c-\varepsilon}$.*

Proof. Choose open sets $K_c \subseteq V \subseteq \overline{V} \subseteq U_1 \subseteq \overline{U}_1 \subseteq U$ with $\overline{U}_1$ compact.

Step 1: the gradient is bounded below away from V near level c. There are $\varepsilon_0, \sigma > 0$ with $|\nabla f| \geq \sigma$ on $f^{-1}[c-\varepsilon_0, c+\varepsilon_0] \setminus V$. Otherwise there are $x_n \notin V$ with $f(x_n) \to c$ and $\nabla f(x_n) \to 0$. They lie in the compact set $f^{-1}[c-1, c+1]$, so a subsequence converges to some $x \notin V$ with $\nabla f(x) = 0$ and $f(x) = c$. Then $x \in K_c \subseteq V$, a contradiction.

Step 2: the vector field. Let $\psi \colon M \to [0,1]$ be smooth with $\psi = 0$ on V and $\psi = 1$ outside U_1, and let χ be a cutoff with $\chi = 1$ on $[c-\varepsilon, c+\varepsilon]$ and support in $(c-\varepsilon_0, c+\varepsilon_0)$, for an $\varepsilon \leq \varepsilon_0/2$ fixed below. Put $X = \chi(f)\psi \nabla f / |\nabla f|^2$. By Step 1 it is smooth, compactly supported, and $|X| \leq 1/\sigma$. Along its backward flow f is non-increasing, and $\frac{d}{dt} f = -\chi(f)\psi$.

Step 3: flow lines from outside U do not reach U_1 in time 2ε. Let $d = \operatorname{dist}(M \setminus U, \overline{U}_1) > 0$. A flow line has speed at most $1/\sigma$, so in time T it travels at most T/σ. Choose ε with $2\varepsilon < d\sigma$. A flow line starting at $x \notin U$ then stays outside U_1, where $\psi = 1$, for $0 \leq t \leq 2\varepsilon$. If moreover $f(x) \leq c + \varepsilon$, then f drops at unit rate while it lies in $[c-\varepsilon, c+\varepsilon]$, so after time 2ε we have $f \leq c - \varepsilon$. Take η to be the backward flow for time 2ε. □

Theorem D.8 (Lusternik–Schnirelmann)**.** *Every smooth function on a compact connected manifold M has at least $\operatorname{cat}(M)$ critical points.*

Proof. Let $m = \operatorname{cat}(M)$ and, for $1 \leq k \leq m$, let $\mathcal{F}_k$ be the family of closed $A \subseteq M$ with $\operatorname{cat}_M(A) \geq k$. These families are invariant under diffeomorphisms isotopic to the identity and nested, $\mathcal{F}_1 \supseteq \cdots \supseteq \mathcal{F}_m \ni M$. So the values $c_k = \inf_{A \in \mathcal{F}_k} \max_A f$ satisfy $c_1 \leq c_2 \leq \cdots \leq c_m \leq \max f$, and each is critical by Theorem D.2. If they are distinct we have m critical points. Suppose instead that $c_k = c_{k+1} = c$. We claim that K_c is infinite. If not, let U be a union of small disjoint balls around the points of K_c, so that $\operatorname{cat}_M(\overline{U}) = 1$. Take ε, η from Lemma D.7 and $A \in \mathcal{F}_{k+1}$ with $\max_A f \leq c + \varepsilon$. The closed set $B = A \setminus U$ has $\operatorname{cat}_M(B) \geq \operatorname{cat}_M(A) - \operatorname{cat}_M(\overline{U}) \geq k$, so $\eta(B) \in \mathcal{F}_k$. But $\eta(B) \subseteq M^{c-\varepsilon}$, which gives $c_k \leq c - \varepsilon$, a contradiction. □

The argument also shows that if two of the minimax values coincide, then the critical set at that level is not just infinite but has category at least 2 in M.

Example D.9 (Three critical points on the torus)**.** On $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ the de Rham classes $a = [dx]$ and $b = [dy]$ in $H^1(\mathbb{T}^2; \mathbb{R})$ have $a \smile b = [dx \wedge dy] \neq 0$. By Theorem D.6, $\operatorname{cat}(\mathbb{T}^2) \geq 3$, so *every* smooth function on $\mathbb{T}^2$ has at least three critical points. A *Morse* function needs at least $b_0 + b_1 + b_2 = 4$ (Lecture 10), as for the height function of Lecture 8. The bound 3 is sharp. Consider

$$f(x, y) = \sin(\pi x)\, \sin(\pi y)\, \sin(\pi(x + y)),$$

which is well defined on $\mathbb{R}^2/\mathbb{Z}^2$ because each shift $x \mapsto x + 1$ or $y \mapsto y + 1$ changes the sign of exactly two factors. Using $\sin u \cos v + \cos u \sin v = \sin(u + v)$,

$$\partial_x f = \pi \sin(\pi y) \sin(\pi(2x + y)), \qquad \partial_y f = \pi \sin(\pi x) \sin(\pi(x + 2y)).$$

If $y \equiv 0$, then $\partial_y f = \pi \sin^2(\pi x)$ vanishes only for $x \equiv 0$, and likewise if $x \equiv 0$. Otherwise $2x + y \equiv x + 2y \equiv 0 \pmod 1$, so $x \equiv y$ and $3x \equiv 0$. The critical points are therefore exactly

$$(0,0)\ (f=0), \qquad \left(\tfrac{1}{3},\tfrac{1}{3}\right)\ \left(f = \tfrac{3\sqrt{3}}{8},\ \max\right), \qquad \left(\tfrac{2}{3},\tfrac{2}{3}\right)\ \left(f = -\tfrac{3\sqrt{3}}{8},\ \min\right).$$

Near the origin $f \approx \pi^3 xy(x+y)$, a *monkey saddle*: a degenerate critical point at which the zero set consists of three lines through the point (Figure D.2).

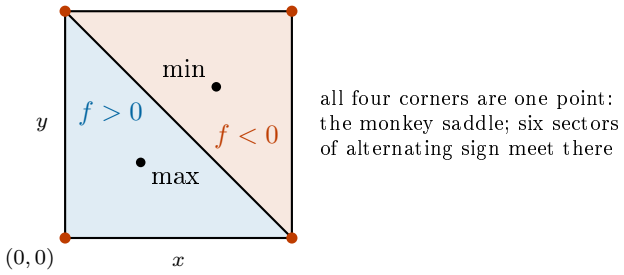


Figure D.2: The function $\sin(\pi x)\sin(\pi y)\sin(\pi(x+y))$ on the torus $\mathbb{R}^2/\mathbb{Z}^2$. Its zero set is the three closed curves $x \equiv 0$, $y \equiv 0$, $x+y \equiv 0$, which cut the torus into two triangles. Each triangle contains one extremum, at its centroid.

Remark D.10 (Back to Interlude A). Up to the constant $(2i)^3$, f is the *Weyl denominator* $\prod_{j<k}\left(e^{i(\theta_j-\theta_k)/2} - e^{-i(\theta_j-\theta_k)/2}\right)$ of SU(3), written in the coordinates $x = (\theta_1-\theta_2)/2\pi$, $y = (\theta_2-\theta_3)/2\pi$ on the maximal torus of Section A.8 (modulo its centre). Its zero set is the set of non-regular elements, where two eigenvalues collide, that is, the walls of $\mathrm{Conf}(3,\mathbb{R})$ wrapped around the circle. Its two extrema are the regular elements whose eigenvalues are *equally spaced* on the circle, in the two cyclic orders. So maximizing $|f|$ is the circle version of maximizing the injectivity radius ρ in Lecture 7.

D.1.3 Closed geodesics and minimal surfaces

Min-max was born in geometry, with the space of closed curves in place of M and length in place of f. Birkhoff (Trans. AMS, 1917) swept a Riemannian 2-sphere by a one-parameter family of closed curves, like the circles of latitude, and took

$$W = \inf_{\text{sweepouts}} \max_{t}\ \mathrm{length}(\gamma_t).$$

Lemma D.1 has an infinite-dimensional analogue, a curve-shortening process applied to the whole family. If no closed geodesic had length W, this process would push every sweepout below level W. So *every Riemannian metric on* $\mathbb{S}^2$ *has a closed geodesic*, of length W. Lusternik and Schnirelmann (1929) used three linked families to obtain *three simple closed geodesics*. Their deformation step had a gap, which was closed only much later; the proof now accepted uses Grayson's curve-shortening flow (Ann. of Math., 1989). The number three is the category of $\mathbb{R}P^2$. On the round sphere the simple closed geodesics are the great circles, parametrized by $\mathbb{R}P^2$, and the $\mathbb{Z}/2$ cup-length of $\mathbb{R}P^2$ is 2.

Almgren and Pitts replaced curves by hypersurfaces and length by area. Their min-max theory (Pitts' book, 1981), with the regularity theory of Schoen and Simon (1981), produces a smooth closed embedded minimal hypersurface in every closed Riemannian manifold of dimension

between 3 and 7. In the last fifteen years this machinery has solved two famous problems; see Theorems D.26 and D.27 in the gallery. The Willmore conjecture is a statement about tori, and Marques and Neves prove it by min-max over a *five*-parameter family. They show that the minimal surface it produces has Morse index at most 5 and is not a great sphere. By a theorem of Urbano (Proc. AMS, 1990), such a minimal surface in $\mathbb{S}^3$ is a Clifford torus, of area $2\pi^2$. This is Lusternik–Schnirelmann in spirit: a topologically non-trivial family forces a critical point, and an index count identifies it.

D.2 Min-type functions and Morse inequalities for configurations

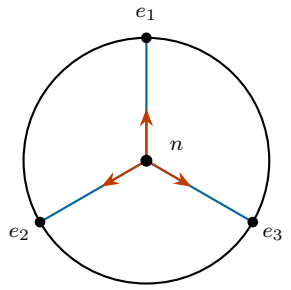
D.2.1 Minimax for the injectivity radius

Now return to ρ on $\mathrm{Conf}(N, \mathbb{S}^2)$, or on $B\,\mathrm{Conf}(N) = \mathrm{Conf}(N, \mathbb{S}^2)/\mathrm{SO}(3)$ as in Lecture 10. The roles of sub- and superlevel sets are reversed: we study $\mathrm{Conf}(N, \theta) = \{\rho \geq \theta\}$, which shrinks as θ grows. By Theorem 11.2, the critical configurations are those whose contact graph carries a balanced system of non-negative weights. The deformation theorem quoted in the proof sketch of Lecture 11 states that if $[\theta_1, \theta_2]$ contains no critical value, then $\mathrm{Conf}(N, \theta_2)$ is a deformation retract of $\mathrm{Conf}(N, \theta_1)$. (See Baryshnikov–Bubenik–Kahle for the general min-type framework, and Kusner–Kusner–Lagarias–Shlosman for the sphere.) Given this, the proof of Theorem D.2 applies to ρ with all inequalities reversed:

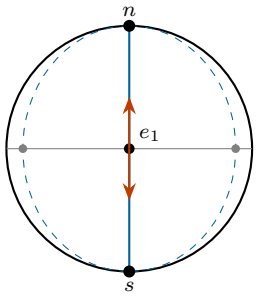
if $\mathcal{F}$ is a deformation-invariant family and $\theta_{\mathcal{F}} = \sup_{A \in \mathcal{F}} \min_A \rho > 0$, then $\theta_{\mathcal{F}}$ is a critical value, realized by a balanced configuration.

One ingredient should be checked: the critical values of ρ in $[\theta_0, \pi]$, $\theta_0 > 0$, form a closed set. In a limit of balanced configurations, contact edges can only be gained, and the normalized weights have a convergent subsequence, so the limit is balanced.

The extreme case is $\mathcal{F} = \{\text{points}\}$. Then $\theta_{\mathcal{F}} = \max \rho$ is the solution of the *Tammes problem*, the top critical value of ρ. Beyond it $\mathrm{Conf}(N, \theta)$ is empty.



from above: the forces at the pole n cancel (120° apart)



from the side: the forces at an equatorial point cancel

Figure D.3: A balanced configuration for $N = 5$: the two poles and three points on the equator. The contact graph (blue) has the six pole-to-equator edges, all of length $\pi/2$, with equal weights. This is a Tammes configuration, $\max \rho = \pi/2$. It is *not* isolated: the equatorial points may slide freely as long as their gaps stay at least $\pi/2$. Critical sets of min-type functions can be large.

Example D.11 (A min-max value computed exactly)**.** Take $N = 4$, labelled. The regular tetrahedra (the maxima of ρ, $\rho = \arccos(-\frac{1}{3})$) come in two classes modulo $\mathrm{SO}(3)$, mirror images of each other. The signed volume $\det(u_2 - u_1, u_3 - u_1, u_4 - u_1)$ is rotation invariant and has opposite signs on the two classes, so every path from one to the other passes through a configuration whose four points lie in a plane, hence on a circle. On a circle, the closest pair of four points is at angular distance at most $\pi/2$, with equality only for the square on a great circle (Exercise D.38). On the other hand, the path

$$u(s) = \big((\cos s, 0, \sin s),\ (-\cos s, 0, \sin s),\ (0, \cos s, -\sin s),\ (0, -\cos s, -\sin s)\big), \qquad |s| \le \arcsin \tfrac{1}{\sqrt{3}},$$

runs from one tetrahedron ($s < 0$) through the square ($s = 0$) to the other ($s > 0$). Along it, $\rho = \min\big(\arccos(-\sin^2 s),\ \pi - 2\,|s|\big) \ge \pi/2$. Hence the mountain-pass value between the two tetrahedra is exactly $\pi/2$, and the pass is the balanced square of Lecture 11. This is the "pushing alternate vertices" deformation of Lecture 11 read as a min-max statement.

D.2.2 What the Betti numbers of Lecture 10 force

Lecture 10 computed the Poincaré polynomial

$$P\big(B\,\mathrm{Conf}(N)\big)(t) = \prod_{k=2}^{N-2} (1 + kt).$$

Setting $t = 1$ and $t = -1$:

$$\sum_{\lambda} b_\lambda\big(B\,\mathrm{Conf}(N)\big) = \prod_{k=2}^{N-2}(k+1) = \frac{(N-1)!}{2}, \quad \chi\big(B\,\mathrm{Conf}(N)\big) = \prod_{k=2}^{N-2}(1-k) = (-1)^{N-3}(N-3)!.$$

For $N = 6$ this is $1 + 9 + 26 + 24 = 60 = 5!/2$, matching the table in Lecture 10, and $\chi = 1 - 9 + 26 - 24 = -6$.

For $N \ge 3$, $\mathrm{SO}(3)$ acts freely on $\mathrm{Conf}(N, \mathbb{S}^2)$ (a non-trivial rotation fixes only two points of the sphere), so $B\,\mathrm{Conf}(N)$ is a smooth manifold of dimension $2N - 3$. Let E be a smooth $\mathrm{SO}(3)$-invariant function on $\mathrm{Conf}(N, \mathbb{S}^2)$ that tends to $+\infty$ at collisions, for example the Riesz energy $E_s = \sum_{i<j} |u_i - u_j|^{-s}$ ($s > 0$). Its sublevel sets are compact, and the Morse theory of Lectures 8–10 applies to the induced function on $B\,\mathrm{Conf}(N)$ (Milnor, *Morse Theory*, Thm. 3.5).

Proposition D.12. *If E induces a Morse function on $B\,\mathrm{Conf}(N)$, then E has at least $(N-1)!/2$ critical $\mathrm{SO}(3)$-orbits of labelled configurations, at least b_λ of them of index λ. If there are finitely many, then $\sum_\lambda (-1)^\lambda c_\lambda = (-1)^{N-3}(N-3)!$.*

Example D.13 ($N = 4$)**.** Here $b_0 = 1$, $b_1 = 2$ and $\chi = -1$. For the Coulomb energy E_1, the regular tetrahedron and the square on a great circle are critical by symmetry. In $B\,\mathrm{Conf}(4)$ they give $4!/12 = 2$ tetrahedral classes and $4!/8 = 3$ square classes (the orders 12 and 8 are their rotation groups). A numerical computation of the Hessian (editor's check) shows that both are non-degenerate modulo rotations, with index 0 and 1. The count $2 \cdot (+1) + 3 \cdot (-1) = -1 = \chi$ is consistent, and $c_0 = 2 \ge 1$, $c_1 = 3 \ge 2$. The squares are mountain passes between the tetrahedra, exactly as for ρ in Example D.11.

Remark D.14 (Labelled versus unlabelled). These bounds count *labelled* configurations. The symmetric group Σ_N acts on $B\,\mathrm{Conf}(N)$, and an unlabelled critical configuration whose rotation group is G accounts for $N!/\,|G|$ labelled classes. So the bound of Proposition D.12 only says $\sum 1/\,|G| \geq 1/(2N)$, summed over the unlabelled critical configurations. Topology forces essentially *nothing* about how many genuinely different equilibria an energy has. The plane shows the same effect in cohomology. $\mathrm{Conf}(N,\mathbb{C})$ has total Betti number $N!$, but by a theorem of Arnold (1970) the rational cohomology of the *unordered* space $\mathrm{Conf}(N,\mathbb{C})/\Sigma_N$ is that of a circle. Forgetting the labels kills almost all of the rational topology. Numerical experiments, by contrast, suggest that the number of distinct local minima of energies such as E_1 grows quickly with N. That growth is a fact about geometry, not about topology.

Remark D.15 (Without non-degeneracy). The proof of Theorem D.8 goes through for any function with compact sublevel sets. If such a function has finitely many critical points, then a large sublevel set is a deformation retract of the whole space, and that set has full category. Over $\mathbb{Q}$ the cohomology of $\mathrm{Conf}(N-1,\mathbb{C})$ is generated in degree 1 and has top degree $N-2$ (Section D.3), so its cup-length is $N-2$. Since $\mathrm{Conf}(N-1,\mathbb{C})/\mathrm{SO}(2) \cong B\,\mathrm{Conf}(N)$ and the circle bundle is trivial (Lecture 10), Künneth gives cup-length $N-3$ for $B\,\mathrm{Conf}(N)$. Hence *every* smooth proper $\mathrm{SO}(3)$-invariant energy, degenerate or not, has at least $N-2$ critical orbits. This is far weaker than $(N-1)!/2$: degenerate critical points can absorb many non-degenerate ones, as the monkey saddle of Example D.9 absorbs two saddles.

D.3 Configuration spaces of the plane, and of thick particles

D.3.1 Arnold's computation

Lecture 10 quoted the Poincaré polynomial of $\mathrm{Conf}(n,\mathbb{R}^2) = \mathrm{Conf}(n,\mathbb{C})$. Here is the proof. For $i \neq j$ let

$$\omega_{ij} = \frac{1}{2\pi}\, d\arg(z_i - z_j),$$

a closed 1-form on $\mathrm{Conf}(n,\mathbb{C})$ with $\omega_{ij} = \omega_{ji}$. Its integral over a loop is the number of times z_i winds around z_j.

Theorem D.16 (Fadell–Neuwirth 1962; Arnold 1969). *1. Forgetting the last point, the map* $\pi\colon \mathrm{Conf}(n,\mathbb{C}) \to \mathrm{Conf}(n-1,\mathbb{C})$ *is a locally trivial fibration with fibre* $\mathbb{C} \setminus \{n-1 \text{ points}\}$, *and it has a section.*

2. $\mathrm{Conf}(n,\mathbb{C})$ *is aspherical, i.e.* $\pi_k = 0$ *for* $k \geq 2$. *Its fundamental group, the pure braid group* P_n, *is an iterated semidirect product* $P_n \cong F_{n-1} \rtimes P_{n-1}$ *of free groups.*
3. $H^*(\mathrm{Conf}(n,\mathbb{C});\mathbb{Z})$ *is torsion-free with Poincaré polynomial* $\prod_{k=1}^{n-1}(1+kt)$. *It is generated by the classes* $[\omega_{ij}]$, *and all relations follow from* $\omega_{ij}^2 = 0$ *and the* Arnold relations $\omega_{ij}\omega_{jk} + \omega_{jk}\omega_{ki} + \omega_{ki}\omega_{ij} = 0$.

Proof of (2) and of the Poincaré polynomial, given (1). (2) The fibre F is homotopy equivalent to a wedge of $n-1$ circles, so it is aspherical. Inductively the base is aspherical, and the base case $\mathrm{Conf}(1,\mathbb{C}) = \mathbb{C}$ is contractible. The long exact sequence $\pi_k(F) \to \pi_k(\mathrm{Conf}(n,\mathbb{C})) \to \pi_k(\mathrm{Conf}(n-1,\mathbb{C}))$ then gives $\pi_k(\mathrm{Conf}(n,\mathbb{C})) = 0$ for $k \geq 2$, and in degree one it gives $1 \to F_{n-1} \to P_n \to P_{n-1} \to 1$. The sequence splits by the section $(z_1,\dots,z_{n-1}) \mapsto (z_1,\dots,z_{n-1},1+\sum |z_i|)$.

(3) The fibre over $(z_1,\dots,z_{n-1})$ is the plane punctured at the z_i. Its H^1 is free with basis dual to small loops around the punctures. The global classes $[\omega_{in}]$, $1 \leq i \leq n-1$, restrict to exactly this

dual basis in every fibre: going once around z_i changes $\arg(z_n - z_i)$ by 2π and leaves $\arg(z_n - z_j)$ unchanged. So the Leray–Hirsch theorem applies, $H^*(\mathrm{Conf}(n,\mathbb{C})) \cong H^*(\mathrm{Conf}(n-1,\mathbb{C})) \otimes H^*(F)$ as modules, and $P_n(t) = P_{n-1}(t)\,(1+(n-1)t)$. □

For $n = 3$ the change of variables $(z_1, z_2, z_3) \mapsto \left(z_1,\ z_2 - z_1,\ \frac{z_3 - z_1}{z_2 - z_1}\right)$ gives $\mathrm{Conf}(3,\mathbb{C}) \cong \mathbb{C} \times \mathbb{C}^* \times (\mathbb{C} \setminus \{0,1\})$, with Poincaré polynomial $(1+t)(1+2t)$. Dividing by the affine group $z \mapsto az + b$ in this way is the plane version of Lecture 10's reduction of $\mathrm{Conf}(N, \mathbb{S}^2)$ by $\mathrm{SO}(3)$.

D.3.2 Thick particles

For points, $\mathrm{Conf}(n,\mathbb{C})$ has the clean topology above. The physical systems of Lecture 11 are made of *hard discs*, and there the topology depends on the radius. Fix a region R and let $\mathrm{Conf}(n, r; R)$ be the space of n non-overlapping discs of radius r in R. This is a superlevel set of a min-type "tautological" function: the minimum of half of all pairwise distances and of all distances from centres to ∂R. Its critical points are again balanced configurations, now with forces from the walls as well. This is the min-type Morse theory of Baryshnikov, Bubenik and Kahle (Theorem D.28). Some landmarks:

- For small r nothing happens: $\mathrm{Conf}(n, r; R) \simeq \mathrm{Conf}(n, R)$. In a box with shortest side L, the first critical radius is $L/2n$, at the configuration of n discs in a row across the box.
- Carlsson, Gorham, Kahle and Mason (Phys. Rev. E 85, 2012) computed the homology of small hard-disc systems numerically and found topology changing already for very few discs, at the balanced configurations.
- In an infinite strip, Alpert, Kahle and MacPherson (J. Appl. Comput. Topology, 2021) found three "homological phases", named after gas, liquid and solid (Theorem D.29). In their units (discs of diameter 1), the channel of Lecture 11 has width between 1 and $1 + \sqrt{3}/2$, just above the trivial case of width 1.
- The thick-disc spaces are not just the point spaces with holes filled in. Ramsey's *thick braids* (Theorem D.30) are loops of discs that are non-trivial even though the corresponding loops of points can be contracted.

D.3.3 Robots, motion planning and algorithms

Configuration spaces are also the state spaces of machines. Ghrist (in a 2001 volume dedicated to J. Birman) studied n robots moving on a network of tracks, the configuration space of n points on a graph. He showed that it is aspherical, with homological dimension bounded by the number of *essential* vertices (valence at least 3); Abrams' thesis (2000) gave a cubical model.

Farber (Discrete Comput. Geom., 2003) asked how discontinuous a motion planner must be. A *motion planner* on $U \subseteq X \times X$ is a continuous map s assigning to each $(x, y) \in U$ a path from x to y. The *topological complexity* $\mathrm{TC}(X)$ is the least number of open sets covering $X \times X$ that carry motion planners. This is a Lusternik–Schnirelmann-type invariant, with the same unnormalized convention as in Definition D.5. If $\mathrm{TC}(X) = 1$, then $s(x_0, \cdot)$ is a contraction of X. So a non-contractible space, such as $\mathrm{Conf}(n, \mathbb{R}^2)$ for $n \geq 2$, needs at least two planning rules (Exercise D.42). For n robots in $\mathbb{R}^m$ the answer is known exactly (Theorem D.31).

Smale (J. Complexity, 1987) turned the same idea on numerical analysis. A monic polynomial of degree d with distinct roots is the same thing as a point of $\mathrm{Conf}(d,\mathbb{C})/\Sigma_d$, the unordered set

of its roots. An algorithm that outputs approximations to the roots must branch ("if ... then ...") at least as often as the *Schwarz genus* of the covering $\mathrm{Conf}(d,\mathbb{C}) \to \mathrm{Conf}(d,\mathbb{C})/\Sigma_d$ requires. Using the cohomology of braid groups, Smale showed that for sufficiently accurate approximations the number of branchings exceeds $(\log_2 d)^{2/3}$. Vassiliev (Funct. Anal. Appl., 1988; see also his book *Complements of Discriminants of Smooth Maps*, AMS, 1992) computed the genus exactly when d is a prime power: it equals d. So for these degrees the lower bound becomes linear in d. The genus is always at most d, and Arone (Contemp. Math. 399, 2006) showed that it is strictly smaller than d when d is neither a prime power nor twice a prime power.

D.4 Braids

D.4.1 Artin's presentation

Recall from Lecture 7 that $B_N = \pi_1\big(\mathrm{Conf}(N,\mathbb{C})/\Sigma_N\big)$ and $P_N = \pi_1(\mathrm{Conf}(N,\mathbb{C}))$. Put the base point at $1, 2, \dots, N$ on the real line. Let σ_i be the loop in which the points at i and $i{+}1$ exchange places by a half turn, counterclockwise, with the others fixed. In a braid diagram (time running downwards, seen from a fixed side) we draw this as strand i crossing over strand $i+1$.

Theorem D.17 (Artin 1925, 1947)**.** *B_N is generated by $\sigma_1, \dots, \sigma_{N-1}$ with the relations*

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1} \quad (1 \le i \le N-2), \qquad \sigma_i\sigma_j = \sigma_j\sigma_i \quad (|i-j| \ge 2).$$

The map $B_N \to \Sigma_N$ sending σ_i to the transposition $(i\ i{+}1)$ has kernel P_N.

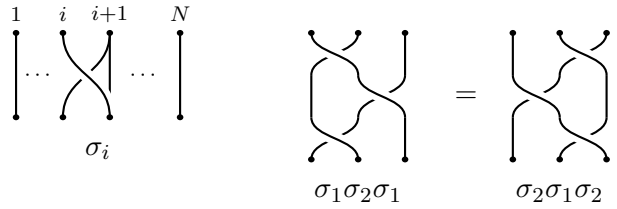


Figure D.4: Left: the Artin generator σ_i. Right: the braid relation. In both braids the strand starting on the left passes over the other two, and the strands starting in the middle and on the right cross once. The two diagrams differ by sliding the middle crossing past the strand that lies over it.

The far commutation relation says that exchanges in disjoint regions of the plane can be performed in either order. This uses the fact that *points take up no room*, a point we return to in Theorem D.30. A first gem: the braid relation forces all generators to become equal in the abelianization, and a winding number shows that nothing more collapses, so B_N is infinite.

Proposition D.18. *For $N \ge 2$ the abelianization of B_N is $\mathbb{Z}$, generated by the image of any σ_i. The abelianization map is induced by the discriminant $\Delta(z) = \prod_{i<j}(z_i - z_j)^2 \colon \mathrm{Conf}(N,\mathbb{C})/\Sigma_N \to \mathbb{C}^*$, and it sends every σ_i to 1.*

Proof. In an abelian quotient, $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ becomes $\sigma_i = \sigma_{i+1}$, so the abelianization is cyclic, generated by σ_1. Conversely, Δ is symmetric, so it is defined on unordered configurations, and it takes values in $\mathbb{C}^*$. Along σ_i, the argument of $z_i - z_{i+1}$ changes by π while every other

factor of Δ returns to itself (the factors $(z_i - z_k)$ and $(z_{i+1} - z_k)$ are exchanged), so $\Delta \circ \sigma_i$ winds once around 0. Hence $\Delta_*\colon B_N \to \pi_1(\mathbb{C}^*) = \mathbb{Z}$ is onto with $\sigma_i \mapsto 1$, and the abelianization is $\mathbb{Z}$. □

Example D.19 ($N = 2$)**.** $\mathrm{Conf}(2,\mathbb{C}) \cong \mathbb{C} \times \mathbb{C}^*$ via $(z_1, z_2) \mapsto \big(\frac{z_1+z_2}{2}, w = z_1 - z_2\big)$, and Σ_2 acts by $w \mapsto -w$. Squaring identifies $\mathbb{C}^*/\pm 1$ with $\mathbb{C}^*$, so $\mathrm{Conf}(2,\mathbb{C})/\Sigma_2 \cong \mathbb{C} \times \mathbb{C}^*$, $B_2 \cong \mathbb{Z}$ generated by σ_1, and $P_2 = 2\mathbb{Z}$ generated by the full twist σ_1^2 (Figure D.5).

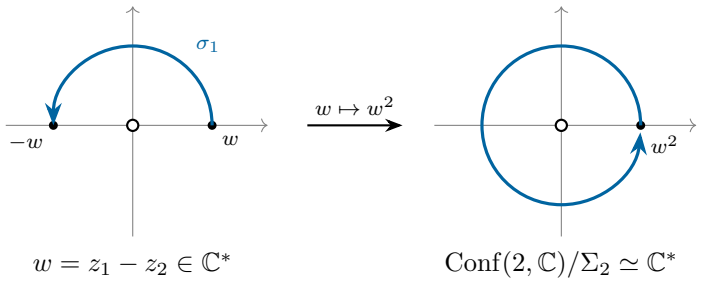


Figure D.5: Two points in the plane. The half turn exchanging z_1 and z_2 is not a loop in $\mathrm{Conf}(2,\mathbb{C})$, but it becomes one after the points are unlabelled. Under the identification $w \mapsto w^2$ it is a generator of $\pi_1(\mathbb{C}^*)$. This is σ_1, and the discriminant of $(z - z_1)(z - z_2)$ is w^2.

D.4.2 Braids from reflection groups

Now combine Lecture 7 with Lecture 8 and Interlude A. The space $\mathrm{Conf}(N,\mathbb{C})$ is the complement in $\mathbb{C}^N$ of the hyperplanes $z_i = z_j$. These are the *complexified* reflecting hyperplanes of the Weyl group $W(A_{N-1}) = \Sigma_N$. Their real parts cut $\mathbb{R}^N$ into the $N!$ contractible Weyl chambers of Lecture 8. Over $\mathbb{C}$ a hyperplane has real codimension 2, so the complement is connected, but now loops can wind around the walls. The Weyl group acts freely on the complement, and

$$\pi_1\Big(\big(\mathbb{C}^N \setminus \textstyle\bigcup_{i<j}\{z_i = z_j\}\big)/W\Big) = B_N.$$

The generator σ_i is a half turn around the wall $z_i = z_{i+1}$ of the fundamental chamber $x_1 < \cdots < x_N$, that is, around the wall of the simple reflection s_i. Compare the two presentations:

$$W:\quad s_i^2 = 1,\ \ (s_i s_j)^{m_{ij}} = 1; \qquad\qquad B_N:\quad \underbrace{\sigma_i\sigma_j\sigma_i\cdots}_{m_{ij}} = \underbrace{\sigma_j\sigma_i\sigma_j\cdots}_{m_{ij}},$$

with $m_{ij} = 3$ for adjacent nodes of the Dynkin diagram A_{N-1} and $m_{ij} = 2$ otherwise. The braid group is the Coxeter group with the relations $s_i^2 = 1$ deleted. This works for every diagram of Lectures 4–5.

Definition D.20. For a Coxeter matrix (m_{ij}), the *Artin group* (or Artin–Tits group) is $A = \langle \sigma_1, \dots, \sigma_r \mid \underbrace{\sigma_i\sigma_j\sigma_i\cdots}_{m_{ij}} = \underbrace{\sigma_j\sigma_i\sigma_j\cdots}_{m_{ij}}\ (i \neq j)\rangle$. Adding $\sigma_i^2 = 1$ recovers the Coxeter group.

Theorem D.21 (Brieskorn 1971; Deligne 1972)**.** *Let W be a finite real reflection group acting on $V = \mathbb{R}^r$, and let Y_W be the complement in $V \otimes \mathbb{C}$ of the complexified reflecting hyperplanes. Then W acts freely on Y_W and $\pi_1(Y_W/W)$ is the Artin group of the Coxeter diagram of W (Brieskorn). Moreover Y_W, and hence Y_W/W, is aspherical (Deligne).*

So every finite Coxeter diagram, including the E_8 of the densest packing in dimension 8, has a braid group. For $W(B_n)$ the quotient Y_W/W is the space of n unordered points in $\mathbb{C}^*$, and its Artin group is the braid group of an annulus (Exercise D.41). For *infinite* Coxeter groups the analogous statement is the $K(\pi,1)$ *conjecture*, still open in general. For the affine groups, whose fundamental alcoves appeared in Section A.8 as configurations of points on a circle, it was proved by Paolini and Salvetti (Theorem D.32).

Remark D.22 (Further directions).

- *Linearity.* Braid groups have faithful finite-dimensional representations, by Bigelow and by Krammer (Theorem D.33).
- *Anyons.* For identical quantum particles, exchanges are loops in the *unlabelled* configuration space. In $\mathbb{R}^3$ this space has fundamental group Σ_N, which allows only bosons and fermions. In the plane the group is B_N, and a one-dimensional representation $\sigma_i \mapsto e^{i\vartheta}$ gives particles with any exchange phase. This was observed by Leinaas and Myrheim (Nuovo Cimento, 1977), and Wilczek named the particles *anyons* (1982). Non-abelian representations of B_N are the basis of proposals for topological quantum computation (Kitaev, 2003; Freedman–Kitaev–Larsen–Wang, Bull. AMS, 2003). The difference between labelled and unlabelled configurations that we met in Section D.2 here has physical consequences.

D.5 Gallery of theorems

Each statement is followed by a sentence or two on why it belongs in this course.

Min-max and critical points

Theorem D.23 (Lusternik–Fet 1951; Birkhoff 1917 for $\mathbb{S}^2$)**.** *Every closed Riemannian manifold carries a non-constant closed geodesic.*

The proof is min-max over a non-trivial homotopy class of maps into the free loop space, which is the minimax principle of Theorem D.2 in infinite dimensions.

Theorem D.24 (Lusternik–Schnirelmann 1929; Grayson, Ann. of Math. 1989)**.** *Every Riemannian metric on $\mathbb{S}^2$ has at least three simple closed geodesics.*

This is the theorem behind Theorem D.8, with $\operatorname{cat}(\mathbb{R}P^2) = 3$. Franks (1992) and Bangert (1993) showed that there are always infinitely many closed geodesics, not necessarily simple.

Theorem D.25 (Conley–Zehnder, Invent. Math. 1983)**.** *Every Hamiltonian diffeomorphism of the standard symplectic torus $\mathbb{T}^{2n}$ has at least $2n+1$ fixed points, and at least 2^{2n} if all fixed points are non-degenerate.*

This is Arnold's conjecture for tori. The two numbers are exactly $\operatorname{cat}(\mathbb{T}^{2n})$ and $\sum_\lambda b_\lambda(\mathbb{T}^{2n})$, the degenerate and the Morse counts of Example D.9.

Theorem D.26 (Marques–Neves, Ann. of Math. 179 (2014))**.** *Every immersed torus Σ in $\mathbb{R}^3$ satisfies*

$$\mathcal{W}(\Sigma) = \int_\Sigma H^2 \, dA \;\geq\; 2\pi^2, \qquad H = \tfrac{1}{2}(\kappa_1 + \kappa_2).$$

More precisely, every embedded closed surface of genus $g \geq 1$ in $\mathbb{S}^3$ has $\int_\Sigma (1+H^2)\, dA \geq 2\pi^2$, with equality if and only if Σ is the Clifford torus up to conformal transformations of $\mathbb{S}^3$.

This is the Willmore conjecture (1965). In $\mathbb{R}^3$ the extremal tori are, up to conformal maps, the torus of revolution with radii in ratio $\sqrt{2} : 1$. The proof is a min-max argument over a five-parameter family (Section D.1.3).

Theorem D.27 (Marques–Neves, Invent. Math. 2017; Song, Ann. of Math. 2023)**.** *Every closed Riemannian manifold of dimension $3 \le n+1 \le 7$ contains infinitely many smooth, closed, embedded minimal hypersurfaces. Marques and Neves proved this under the assumption of positive Ricci curvature; Song proved it in general.*

This was conjectured by Yau (1982). The proofs use min-max over families of arbitrarily many parameters, the analogue of the classes $c_1 \le c_2 \le \cdots$ in the proof of Theorem D.8.

Configuration spaces of hard particles

Theorem D.28 (Baryshnikov–Bubenik–Kahle, IMRN 2014)**.** *For hard balls of radius r in a rectangular box in $\mathbb{R}^d$ whose shortest side has length L, the critical points of the tautological function are the mechanically balanced configurations. There are no critical values in $(0, L/2n)$, so $\mathrm{Conf}(n, r) \simeq \mathrm{Conf}(n)$ for $r < L/2n$, and the topology changes just above $L/2n$.*

This is the box version of Theorem 10.5 and of the threshold $2\pi/N$ on the sphere quoted in Lecture 11. The critical configuration is a row of balls spanning the box, the analogue of N points equally spaced on a great circle.

Theorem D.29 (Alpert–Kahle–MacPherson, J. Appl. Comput. Topology 5 (2021))**.** *Let $C(n, w)$ be the space of n non-overlapping discs of diameter 1 in an infinite strip of integer width $w \ge 2$, and fix j. If $j \le w-2$, then $H_j(C(n,w)) \cong H_j(\mathrm{Conf}(n, \mathbb{R}^2))$ and b_j grows like n^{2j}. If $j \ge w-1$, then b_j grows exponentially in n. More precisely, writing $j = q(w-1) + r$ with $0 \le r < w-1$, $b_j \asymp (q+1)^n \, n^{qw+2r}$.*

Exponential growth of Betti numbers is a topological counterpart of the exponential count of jammed states in the channel of Lecture 11, where Fibonacci numbers appeared.

Theorem D.30 (P. Ramsey, arXiv:2311.16776, 2023)**.** *For four discs of radius r in the unit disc with $\frac{1}{4} < r \le \frac{1}{3}$, there is a non-trivial element of π_1 of the configuration space that becomes trivial when each disc is replaced by its centre. It is represented by the commutator $\sigma_1^{-1}\sigma_3^{-1}\sigma_1\sigma_3$. For $n \ge 5$ discs and $\frac{1}{n} < r \le \frac{1}{n-1}$, the configuration space contains a non-contractible $(n-3)$-sphere.*

In B_4 the relation $\sigma_1\sigma_3 = \sigma_3\sigma_1$ makes this commutator trivial. Thick discs have no room to perform the two exchanges independently, so far commutation *fails* for them. This is a preprint (P. Ramsey, *Thick braids and other non-trivial homotopy in configuration spaces of hard discs*).

Theorem D.31 (Farber–Grant, Proc. AMS 2009)**.** *For $n, m \ge 2$, $\mathrm{TC}(\mathrm{Conf}(n, \mathbb{R}^m)) = 2n-1$ if m is odd and $2n-2$ if m is even.*

Farber and Yuzvinsky had proved the cases $m = 2$ and m odd. The lower bound is a cup-length bound in the spirit of Theorem D.6, using products of "zero-divisors" $u \otimes 1 - 1 \otimes u$ in $H^*(\mathrm{Conf}(n, \mathbb{R}^m) \times \mathrm{Conf}(n, \mathbb{R}^m))$.

Braid and Artin groups

Theorem D.32 (Paolini–Salvetti, Invent. Math. 224 (2021)). *The $K(\pi,1)$ conjecture holds for affine Artin groups: for every affine Coxeter group, the quotient of the complexified complement of the reflection arrangement by the group is a classifying space for the affine Artin group.*

This extends Theorem D.21 from the finite Weyl groups of Lectures 4–6 to their affine versions, whose alcoves are the simplices of Interlude A.

Theorem D.33 (Bigelow, J. AMS 2001; Krammer, Ann. of Math. 2002). *The Lawrence–Krammer representation $B_n \to \mathrm{GL}_{n(n-1)/2}\big(\mathbb{Z}[q^{\pm1},t^{\pm1}]\big)$ is faithful for all n. In particular braid groups are linear.*

Bigelow's proof is topological: it uses the configuration space of *two* points in the punctured disc.

Rigidity and optimality of packings

Theorem D.34 (Schütte–van der Waerden, Math. Ann. 1953; Musin, Ann. of Math. 2008). *The kissing number is $\tau_3 = 12$ and $\tau_4 = 24$.*

The first settles the Newton–Gregory problem. In the language of these notes it says that $\max\rho$ on $\mathrm{Conf}(13,\mathbb{S}^2)$ is less than $\pi/3$. Musin's proof modifies Delsarte's linear programming method.

Theorem D.35 (Tammes problem: Musin–Tarasov 2012 and 2015; Robinson 1961). *The Tammes problem is solved for $N = 13$ (Musin–Tarasov, Discrete Comput. Geom. 48 (2012); optimal angle $\approx 57.1367^\circ$, unique up to isometry), $N = 14$ (Musin–Tarasov, Experiment. Math. 24 (2015); $\approx 55.6706^\circ$) and $N = 24$ (Robinson, Math. Ann. 1961; the snub cube).*

The proofs for 13 and 14 enumerate the possible *irreducible contact graphs* by computer. These are the balanced graphs of Lecture 11, so each proof is the statement that one particular critical configuration of ρ is the top one. Together with the classical cases ($N \le 12$, by Fejes Tóth, Schütte–van der Waerden, Danzer and Böröczky), this covers $N \le 14$ and $N = 24$.

Exercises

Exercise D.36. For $f = \sin(\pi x)\sin(\pi y)\sin(\pi(x+y))$ on $\mathbb{R}^2/\mathbb{Z}^2$, show that the index of ∇f at the origin is -2 (expand f to third order) and check the Poincaré–Hopf formula $1+1-2 = \chi(\mathbb{T}^2)$. Why can a function on $\mathbb{T}^2$ with three critical points never be Morse?

Exercise D.37. Using $H^*(\mathbb{R}P^n;\mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^{n+1})$, show that $\mathrm{cat}(\mathbb{R}P^n) \ge n+1$. Deduce that every smooth function on $\mathrm{SO}(3) \cong \mathbb{R}P^3$, for example a smooth function of the orientation of a rigid polytope (Lecture 7), has at least four critical points. How many does a Morse function need?

Exercise D.38. Complete Example D.11. (a) Show that if four distinct points lie on a circle in $\mathbb{S}^2$, then two of them are at angular distance at most $\pi/2$, with equality only for a square on a great circle. (b) Verify the formula for $\rho(u(s))$ and that it is at least $\pi/2$.

Exercise D.39. Compute $P(B\,\mathrm{Conf}(7))$ and check that the Betti numbers sum to 360 and that $\chi = 24$. If a Morse energy on $B\,\mathrm{Conf}(7)$ has c_0 local minima, show that it has at least $c_0 + 13$ critical points of index 1. (Use the strong Morse inequality $c_1 - c_0 \ge b_1 - b_0$.)

Exercise D.40. Show that $\Delta = (\sigma_1\sigma_2\sigma_1)^2$ commutes with σ_1 and σ_2 in B_3, and that it is the full twist of all three strands. Show that $(\sigma_1\sigma_2)^3 = \Delta$. Under the abelianization of Proposition D.18, what is the image of Δ?

Exercise D.41. The Weyl group $W(B_n) = (\mathbb{Z}/2)^n \rtimes \Sigma_n$ has reflecting hyperplanes $x_i = \pm x_j$ and $x_i = 0$. Show that the map $(z_1, \dots, z_n) \mapsto \{z_1^2, \dots, z_n^2\}$ identifies $Y_{W(B_n)}/W(B_n)$ with the space of n unordered distinct points in $\mathbb{C}^*$. Write down the Artin relations for the diagram B_n, with $m = 4$ on the double edge, and interpret the extra generator as a strand going around the puncture.

Exercise D.42. (a) Show that $\mathrm{TC}(X) = 1$ if and only if X is contractible. (b) Show that $\mathrm{TC}(\mathbb{S}^1) = 2$ by writing down two explicit motion planners. (c) Two robots move without collision on a circular track. Identify the configuration space and compute its TC.

Exercise D.43. Show that $C(n, 1)$, the space of n discs of diameter 1 in a strip of width 1, has exactly $n!$ connected components and that each is contractible. Compare with $\mathrm{Conf}(n, \mathbb{R})$ in Lecture 7. (This is the case $w = 1$, excluded from Theorem D.29; Alpert, Kahle and MacPherson record $b_0 = n!$ and $b_j = 0$ for $j \geq 1$.)

Further reading

- J. Milnor, *Morse Theory*, Annals of Math. Studies 51, 1963. Part I is the basis of Lectures 8–10 and of Lemma D.1.
- O. Cornea, G. Lupton, J. Oprea, D. Tanré, *Lusternik–Schnirelmann Category*, AMS Math. Surveys and Monographs 103, 2003. Category, cup-length, and critical point theory.
- M. Struwe, *Variational Methods*, Springer. Min-max and the mountain pass in infinite dimensions.
- F. C. Marques, A. Neves, *The Willmore conjecture*, arXiv:1409.7664. A survey of the proof.
- Y. Baryshnikov, P. Bubenik, M. Kahle, *Min-type Morse theory for configuration spaces of hard spheres*, Int. Math. Res. Not. 2014(9), 2577–2592.
- C. Kassel, V. Turaev, *Braid Groups*, Springer GTM 247, 2008.
- M. Farber, *Invitation to Topological Robotics*, EMS Zurich Lectures in Advanced Mathematics, 2008.
- R. Ghrist, *Elementary Applied Topology*, 2014. Configuration spaces, robots, and Morse theory for applications.

Chapter E

Interlude E: Random packings, entropy and information

[Added] This interlude is new: it is not in the handwritten notes. It picks up three threads that the course left loose: the coin flips and the gas in a box of Lecture 7, the Edwards entropy and the Fibonacci words of Lecture 11, and the project topics "error correcting codes and information theory" and "entropy of jammed systems" listed at the end of Lecture 6. To keep it readable it is set in black; all of it should be read as added material.

Most of this course was about the *best* configurations: the densest packing, the largest kissing configuration, the maximum of the injectivity radius. Lecture 7 already suggested a second point of view. For large N one does not ask for the best configuration but for the *typical* one, and one answers with Stirling's formula. Lecture 11 then did exactly this for jammed discs in a channel, and found the golden ratio.

In this interlude we follow that second point of view: hard spheres as statistical mechanics with no energy at all (Section E.1); random and jammed packings (Section E.2); why dense liquids do not simply crystallize (Sections E.3–E.5); Edwards' statistical mechanics of the Lecture 11 channel, solved completely (Section E.6); and information theory (Section E.7), where the channel's maximal Edwards entropy turns out to be a Shannon capacity and E_8 and Leech come from perfect codes.

Throughout, ln is the natural logarithm and $\log_2$ the logarithm to base 2. Boltzmann's constant is set to 1.

E.1 Hard spheres as statistical mechanics

Recall the remark at the end of §11.3.1: for hard particles the energy of a configuration is 0 if no two particles overlap and $+\infty$ if two do, so the Boltzmann weight $e^{-\beta E}$ is the indicator function of the admissible configurations, at every temperature. For N balls of diameter 1 with centres in a region $\Omega \subset \mathbb{R}^d$ of volume V, the admissible configurations form the *hard-sphere configuration space*

$$\mathrm{Conf}_{\geq 1}(N,\Omega) = \{(x_1,\dots,x_N) \in \Omega^N : \ |x_i - x_j| \geq 1 \text{ for all } i \neq j\},$$

a closed subset of the configuration space $\mathrm{Conf}(N,\Omega)$ of Lecture 7 (Definition 7.1). On the sphere it is the space $\mathrm{Conf}(N,\theta)$ of Definition 10.4, whose topology we computed with min-type Morse

theory. The (configurational) *partition function* is its volume,

$$Z_N(\Omega) = \frac{1}{N!}\operatorname{vol}\operatorname{Conf}_{\geq 1}(N,\Omega),$$

the factor $1/N!$ accounting for the indistinguishability of the particles. The *entropy* is $S = \ln Z_N$, and since the energy is identically zero the free energy is $F = -TS$. Temperature only sets a scale. The *pressure* is

$$\beta P = \frac{\partial S}{\partial V},$$

the rate at which the configuration space grows when the box grows. (Change Ω by a homothety to make this a derivative in one variable.) So for hard spheres *everything is entropy*. This is worth remembering in Section E.6, where Edwards runs the same dictionary in the opposite direction and treats volume as if it were energy.

The Kepler problem is about where the free energy is finite

Let δ_d be the maximal packing density of balls in $\mathbb{R}^d$, and let

$$\rho_{\mathrm{cp}} = \frac{\delta_d}{\operatorname{vol} B^d(\frac{1}{2})}$$

be the corresponding maximal *number density* of centres for balls of diameter 1. In dimension 3, Hales' theorem gives $\delta_3 = \pi/\sqrt{18}$ and so $\rho_{\mathrm{cp}} = \sqrt{2}$, the number density of the face-centred cubic lattice with nearest-neighbour distance 1.

Proposition E.1. *Let $Q_\ell = [0,\ell]^d$, and let $N = N(\ell)$ with $N/\ell^d \to \rho$ as $\ell \to \infty$.*

1. *If $N > \rho_{\mathrm{cp}}(\ell+1)^d$, then $Z_N(Q_\ell) = 0$. In particular $Z_N(Q_\ell) = 0$ for all large ℓ when $\rho > \rho_{\mathrm{cp}}$.*
2. *If $\rho < \rho_{\mathrm{cp}}$, then $\liminf_{\ell\to\infty} \frac{1}{N}\ln Z_N(Q_\ell) > -\infty$.*

Proof. (1) Given N admissible centres in Q_ℓ, the balls of radius $\frac{1}{2}$ around them lie in the cube $[-\frac{1}{2}, \ell+\frac{1}{2}]^d$ of side $\ell+1$. Repeat this cube periodically. The result is a periodic packing of density $N\operatorname{vol}B^d(\frac{1}{2})/(\ell+1)^d$, which is at most δ_d.

(2) Choose $\rho < \rho'' < \rho_{\mathrm{cp}}$. There is a periodic packing of balls of diameter 1 with some number density $\rho' > \rho''$, since periodic packings come arbitrarily close to the optimal density (Exercise 1.7 of Lecture 1). Scale it by $\lambda = (\rho'/\rho'')^{1/d} > 1$. The centres now have number density ρ'' and are at mutual distance at least λ, so each centre may be moved independently anywhere in a ball of radius $r = (\lambda-1)/2$ without creating an overlap. The cube Q_ℓ contains $\rho''\ell^d - O(\ell^{d-1})$ of these wiggle balls, which is more than N for large ℓ; keep N of them. Assigning the N labelled particles to the N wiggle balls in any of the $N!$ ways gives disjoint pieces of $\operatorname{Conf}_{\geq 1}(N, Q_\ell)$, so $Z_N \geq \frac{1}{N!}N!\,(\operatorname{vol}B^d(r))^N$ and $\frac{1}{N}\ln Z_N \geq \ln\operatorname{vol}B^d(r)$. □

The limit $s(\rho) = \lim \frac{1}{N}\ln Z_N$ exists for $\rho < \rho_{\mathrm{cp}}$ (subadditivity; see Ruelle's book), so the free energy is finite at every density below ρ_{cp} and infinite at every density above it. At ρ_{cp} itself one expects $s = -\infty$ and a divergent pressure, as for the Tonks gas below, but the proposition does not decide this. Thus the theorems of Hales in $\mathbb{R}^3$, and of Viazovska and Cohn–Kumar–Miller–Radchenko–Viazovska in $\mathbb{R}^8$ and $\mathbb{R}^{24}$, determine the density at which the free energy of the hard-sphere gas becomes infinite.

One dimension: the Tonks gas

On a line everything can be computed. Lecture 11 (§11.3.2) found that the gaps between n rods of length 1 in an interval of length ℓ form a simplex of volume $(\ell-n)^n/n!$. Ordering the labels gives a factor $n!$, so

$$Z_n(\ell) = \frac{1}{n!}\cdot n!\cdot\frac{(\ell-n)^n}{n!} = \frac{(\ell-n)^n}{n!}.$$

Proposition E.2 (Tonks 1936). *The hard-rod gas has equation of state*

$$\beta P = \frac{\partial}{\partial \ell}\ln Z_n(\ell) = \frac{n}{\ell-n} = \frac{\rho}{1-\rho}, \qquad \rho = \frac{n}{\ell},$$

and entropy per particle $\frac{1}{n}\ln Z_n \to 1 + \ln\frac{1-\rho}{\rho}$ *as* $n,\ell\to\infty$ *with* $n/\ell\to\rho\in(0,1)$.

The first formula is a one-line computation, and the second is Stirling's formula. Compare Exercise 11.8, where the same system was treated in the grand canonical ensemble: there the density was $\rho = P/(1+P)$, which is the same equation of state solved for P. The pressure is a real-analytic function of $\rho\in[0,1)$ and blows up at $\rho = 1 = \rho_{\mathrm{cp}}$, as Proposition E.1 predicts. There is no phase transition. That is not an accident of the model.

Theorem E.3 (van Hove 1950). *Consider particles on a line interacting by a pair potential that has a hard core ($\phi(r) = +\infty$ for $r < a$) and finite range ($\phi(r) = 0$ for $r > R$), and is bounded on $[a, R]$. Then the thermodynamic limit of the pressure is a real-analytic function of the temperature and the density: one-dimensional systems of this kind have no phase transitions.*

The proof uses a *transfer operator*: by the hard core and the finite range, the partition function is an iterated integral $\langle f, \mathcal{T}^n g\rangle$ of a positive integral operator, whose simple top eigenvalue (Perron–Frobenius–Jentzsch) depends analytically on the parameters. Section E.6 has the same mechanism with a 2×2 matrix. (Cuesta and Sánchez, J. Stat. Phys. 2004, discuss which hypotheses are needed, and give one-dimensional models with transitions when they fail.)

Two and three dimensions: transitions seen only on the computer

In $\mathbb{R}^3$ the hard-sphere gas is believed to *freeze*. In 1957 Alder and Wainwright (molecular dynamics) and Wood and Jacobson (Monte Carlo) published, back to back in J. Chem. Phys. 27, the first computer evidence of a fluid–solid transition. In the accepted picture (Hoover and Ree 1968) the fluid is stable up to packing fraction about 0.494, the close-packed crystal from about 0.545, and the two coexist in between. All of this is *numerical*: that hard spheres crystallize, or have any phase transition at all, is open in every dimension $d\geq 2$. In $\mathbb{R}^2$ long-range positional order is impossible (Mermin 1968 for smooth potentials; Richthammer 2016 for hard discs, see the gallery), and what can survive is *orientational* bond order, the hexatic phase. Bernard and Krauth (Phys. Rev. Lett. 107, 2011) found numerically that hard discs melt in two steps: a first-order liquid–hexatic transition and a continuous hexatic–solid transition. Rigorous results are mostly at zero temperature, where the problem becomes an optimization problem of the kind studied in this course (Theil's theorem in the gallery). The first continuum model with finite-range interactions and a *proved* phase transition was the two-species Widom–Rowlinson model (Ruelle, Phys. Rev. Lett. 27, 1971), with hard cores only between unlike particles.

E.2 Random packings

E.2.1 Rényi's parking problem

Lecture 1 (Proposition 1.10) gave the easiest lower bound for packing densities: a *saturated* packing, one to which no ball can be added, has density at least 2^{-d}. The most natural random way of producing a saturated packing is to throw balls at random, one at a time, and keep each one that does not overlap those already placed, until there is no room left. This is *random sequential adsorption* (RSA). In dimension one it is Rényi's *parking problem*: cars of length 1 park at uniformly random free positions along a kerb of length x until no gap of length ≥ 1 remains.

Let $M(x)$ be the expected number of cars that park. Clearly $M(x) = 0$ for $x < 1$, and $M(x) = 1$ for $1 \leq x < 2$. For $x \geq 1$, condition on the position $t \in [0, x-1]$ of the first car, which is uniform. It leaves two independent kerbs, of lengths t and $x-1-t$:

$$M(x) = 1 + \frac{1}{x-1}\int_0^{x-1} \big(M(t) + M(x-1-t)\big)\,dt = 1 + \frac{2}{x-1}\int_0^{x-1} M(t)\,dt. \tag{E.1}$$

Theorem E.4 (Rényi 1958). *$M(x)/x \to m$ as $x \to \infty$, where*

$$m = \int_0^\infty \exp\Big(-2\int_0^t \frac{1-e^{-u}}{u}\,du\Big)\,dt = 0.7475979202\ldots$$

Proof. Since the cars are disjoint subintervals of $[0,x]$, $0 \leq M(x) \leq x$, so all Laplace transforms below converge for $s > 0$. Put $N(y) = M(y+1)$ for $y \geq 0$ and multiply (E.1) by $y = x-1$:

$$yN(y) = y + 2\int_0^y M(t)\,dt.$$

Take Laplace transforms $\hat f(s) = \int_0^\infty e^{-sy} f(y)\,dy$. The transform of $yN(y)$ is $-\hat N'(s)$, that of $\int_0^y M$ is $\hat M(s)/s$, and $\hat M(s) = e^{-s}\hat N(s)$ because M vanishes on $[0,1)$. So

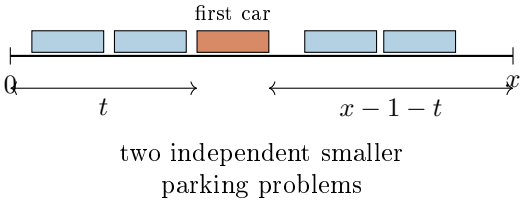
$$\hat N'(s) + \frac{2e^{-s}}{s}\hat N(s) = -\frac{1}{s^2}.$$

Solutions of the homogeneous equation are constant multiples of $\exp\big(2\int_s^\infty e^{-u}u^{-1}du\big)$, which tends to a nonzero constant as $s \to \infty$, whereas $\hat N(s) \leq \int_0^\infty e^{-sy}(y+1)\,dy \to 0$. Hence $\hat N$ is the unique decaying solution,

$$\hat N(s) = \int_s^\infty \frac{1}{t^2}\exp\Big(2\int_s^t \frac{e^{-u}}{u}\,du\Big)\,dt = \frac{1}{s^2}\int_s^\infty \exp\Big(-2\int_s^t \frac{1-e^{-u}}{u}\,du\Big)\,dt,$$

where we wrote $e^{-u} = 1-(1-e^{-u})$ and used $\exp(2\int_s^t du/u) = t^2/s^2$. As $s \to 0^+$ the last integral tends to m by dominated convergence (for $t \geq 1$ the integrand is at most $\exp(-2\int_1^t \frac{1-e^{-u}}{u}du) = O(t^{-2})$). Hence $\hat M(s) = e^{-s}\hat N(s) \sim m/s^2$.

Now $U(x) = \int_0^x M$ is nondecreasing and $\int_0^\infty e^{-sx}dU(x) = \hat M(s) \sim m\,s^{-2}$, so Karamata's Tauberian theorem (Feller, *An Introduction to Probability Theory*, Vol. II, §XIII.5) gives $U(x) \sim mx^2/2$. Finally (E.1) itself says $M(x) = 1 + 2U(x-1)/(x-1)$, so $M(x) \sim mx$. □



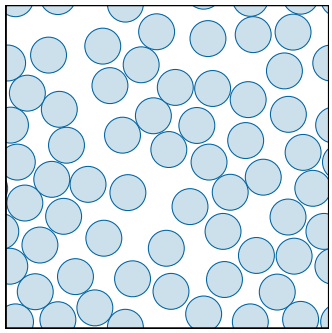


Figure E.1: Left: the recursion behind Rényi's integral equation (E.1). Right: a saturated random sequential adsorption of 57 discs of diameter 1 in a periodic 9×9 square (computed for this figure). Every point of the square is within distance 1 of a centre, yet no two discs touch.

Numerical check. Quadrature gives $m = 0.74759792025$ (Rényi, Publ. Math. Inst. Hung. Acad. Sci. 3, 1958). Twenty simulated kerbs of length 10^5 gave 0.74762 ± 0.00012. Solving (E.1) on a grid shows that $M(x) - mx$ settles down to $m - 1 \approx -0.2524$ already for $x \approx 10$. The density $m \approx 0.748$ lies well above the saturation bound $\frac{1}{2}$ and well below 1.

In the plane, RSA of discs saturates at packing fraction about 0.547 (Feder 1980). This is a numerical value; Zhang and Torquato (Phys. Rev. E 88, 2013) give 0.5470735(28), and in $\mathbb{R}^3$ 0.3841307(21). (A quick simulation for this interlude, ten saturated periodic configurations of about 1740 discs each, gave 0.5468 ± 0.0009.) No closed form is known in any dimension $d \geq 2$.

Notice what RSA does *not* produce. In a saturated RSA configuration, with probability one no two discs touch: the contact graph of Lectures 10–11 is empty. RSA packings are saturated but not jammed; a small shake would let every disc move. To get mechanically stable random packings one has to compress.

E.2.2 Random close packing and the maximally random jammed state

Pour ball bearings into a container and shake it: the packing fraction settles near 0.64. This *random close packing* (RCP) value goes back to the experiments of Bernal and of Scott (both in *Nature*, 1960); Scott and Kilgour (1969) measured about 0.637. It is well below the crystal value 0.7405, and remarkably reproducible.

Torquato, Truskett and Debenedetti (*Is random close packing of spheres well defined?*, Phys. Rev. Lett. 84, 2000) argued that it is not: by compressing faster or slower one trades a little disorder for a little density, with no canonical place to stop. They proposed instead the *maximally random jammed* (MRJ) state, the jammed packing minimizing a chosen order parameter, again at about 0.64 (numerical). This needs a definition of "jammed"; Torquato and Stillinger (J. Phys. Chem. B 105, 2001) give a hierarchy. For a packing in a container with rigid walls:

- *locally jammed*: no single ball can be moved while all the others are held fixed;
- *collectively jammed*: locally jammed, and no subset of balls can be moved collectively, the walls being fixed;
- *strictly jammed*: collectively jammed, also under small volume-nonincreasing deformations of the walls (for periodic packings: of the lattice).

For instance, with suitable walls, the simple cubic packing is collectively but not strictly jammed (shearing the box lets it slide), FCC is strictly jammed, and the honeycomb disc packing is locally

but not collectively jammed. (In a torus there are no walls to stop a row, and the square and simple cubic packings are not even collectively jammed; see Figure E.2.) To first order, a ball with the others held fixed is stuck exactly when the directions to its contacts do not all lie in a closed hemisphere. In particular a locally jammed ball has at least $d+1$ contacts.

E.2.3 Jamming, force balance and Maxwell counting

Collective jamming is the version of Lectures 10–11 that sees all balls at once. Take N balls with centres x_i and radii r_i in a flat torus $\mathbb{R}^d/\Lambda$ (periodic boundary conditions), with contact set $E = \{\{i,j\} : |x_i - x_j| = r_i + r_j\}$, and let $u_{ij} = (x_j - x_i)/|x_j - x_i|$. A velocity $v = (v_1, \dots, v_N) \in (\mathbb{R}^d)^N$ changes the contact distances at the rates

$$(Av)_{ij} = \langle u_{ij}, v_j - v_i \rangle, \qquad \{i,j\} \in E.$$

The linear map $A\colon (\mathbb{R}^d)^N \to \mathbb{R}^E$ is the *rigidity matrix* of the contact graph. To first order a motion keeps the balls disjoint if and only if $Av \geq 0$ componentwise. Translations $v_i = c$ always satisfy $Av = 0$; call them *trivial*.

Definition E.5. The packing is *first-order jammed* if every v with $Av \geq 0$ is trivial.

For sphere packings, first-order jamming is equivalent to collective jamming (with the lattice fixed): a sphere packing is rigid if and only if it is infinitesimally rigid. This is a theorem of Connelly (1988) for packings in a container, extended to periodic packings by Donev, Torquato, Stillinger and Connelly (J. Comput. Phys. 197, 2004). It is special to spheres; it fails for ellipsoids.

Proposition E.6. *A packing is first-order jammed if and only if*

1. $\ker A$ *consists of the trivial motions (the contact graph is infinitesimally rigid as a bar framework), and*
2. *there are* strictly positive *weights* $w_{ij} > 0$ *on all contacts with* $\sum_j w_{ij} u_{ij} = 0$ *at every ball* i *(a positive self-stress: the forces balance at every ball).*

Proof. If (1) and (2) hold and $Av \geq 0$, then

$$0 = \sum_i \Big\langle v_i, \sum_j w_{ij} u_{ij} \Big\rangle = - \sum_{\{i,j\} \in E} w_{ij} \langle u_{ij}, v_j - v_i \rangle = -\langle w, Av \rangle,$$

a sum of non-positive terms, so $Av = 0$, and v is trivial by (1). Conversely, (1) is necessary because $Av = 0$ implies $Av \geq 0$. For (2), Stiemke's alternative (a relative of Gordan's alternative used in the proof of Theorem 11.2) says that either there is v with $Av \geq 0$ and $Av \neq 0$, or there is $w > 0$ with $A^T w = 0$. In a first-order jammed packing the first option is excluded, and $A^T w = 0$ is exactly the balance condition. □

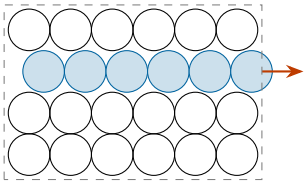
So the jammed packings are the force-balanced contact graphs of Lecture 10 with two upgrades: the weights are positive on *every* contact, and the graph is rigid. (Compare Corollary 10.3, where critical configurations of the min-distance function only needed weights ≥ 0, not all zero.) The rigidity condition already forces a lot of contacts.

Corollary E.7 (Maxwell counting)**.** *A first-order jammed packing of N balls in a d-dimensional torus has at least $dN - d + 1$ contacts, so the mean number of contacts per ball satisfies*

$$z = \frac{2\,|E|}{N} \;\geq\; 2d - \frac{2(d-1)}{N}.$$

Proof. By (1), $\ker A$ has dimension d, so $\operatorname{rank} A = dN - d$. By (2), A^T has a nonzero kernel, so $|E| \geq \operatorname{rank} A + 1 = dN - d + 1$. □

This is Maxwell's 1864 count for frames. Conversely, for generic radii each contact is one equation $|x_i - x_j| = r_i + r_j$ on $dN - d$ free coordinates plus one free parameter, the common scale of the radii at which the packing jams, and generic equations should not be overdetermined. Together the counts give $z = 2d$: jammed frictionless packings should be *isostatic*. The upper bound is a theorem in the plane (Connelly–Gortler–Solomonides–Yampolskaya, gallery) and open for $d \geq 3$, but it holds very precisely in simulations. O'Hern, Silbert, Liu and Nagel (Phys. Rev. E 68, 2003), compressing soft repulsive spheres to the threshold "point J" where they first jam, found $z = 3.98$ in 2d and 5.98 in 3d (both ± 0.02, rattlers removed) and a 3d monodisperse threshold $\phi^* = 0.639 \pm 0.001$, essentially the RCP value.



square lattice in a torus: $z = 4 = 2d$,
but the shaded row slides; to first order
$Av = 0$ for this motion, so it is not jammed

Figure E.2: Maxwell's count is necessary, not sufficient. In the square packing every disc has four contacts, yet a whole row can slide along itself: its vertical contacts open only to second order.

E.3 Geometric frustration

Why does a dense liquid not simply crystallize? In 1952 F. C. Frank (*Supercooling of liquids*, Proc. R. Soc. Lond. A 215) pointed out that the question has a *local* answer. Put twelve atoms around a thirteenth. There are three natural candidates: the cuboctahedron (the FCC neighbourhood), its twisted relative from HCP, and the icosahedron. Lecture 11 met the first and the last, which the jitterbug connects. For Lennard-Jones atoms Frank computed that the icosahedral cluster has the lowest energy of the three: it has 30 near-neighbour pairs among the outer twelve, against 24. But icosahedral symmetry is incompatible with a lattice, so a liquid of locally icosahedral clusters must rearrange substantially before it can crystallize, and it can be supercooled. This is the first instance of *geometric frustration*: the locally preferred structure cannot be extended to fill space.

For hard spheres the same tension appears as a statement about densities. The densest *local* arrangement around one ball beats the densest global one. Fejes Tóth conjectured, and Hales and McLaughlin proved (J. Amer. Math. Soc. 23, 2010), that in any packing of unit balls in $\mathbb{R}^3$ every Voronoi cell (Lecture 1, Definition 1.15) has volume at least that of the regular dodecahedron of inradius 1. That cell would give density

$$\frac{\operatorname{vol} B^3}{\operatorname{vol}(\text{dodecahedron of inradius 1})} \approx 0.7547 \;>\; \frac{\pi}{\sqrt{18}} \approx 0.7405.$$

The dodecahedral cell is the Voronoi cell of a ball surrounded by an icosahedron of twelve touching balls, and regular dodecahedra do not tile space. The local optimum is not the global one.

Frank's icosahedron is often described as twenty regular tetrahedra, the densest arrangement of four balls, glued around the centre. That description is slightly wrong, and the error is the heart of the matter.

E.4 Twenty tetrahedra do not make an icosahedron

Frank's icosahedral cluster (and the jitterbug of Lecture 11) rests on a small geometric fact: a regular icosahedron is *not* made of 20 regular tetrahedra. Cone the icosahedron from its centre and you get 20 congruent tetrahedra, but they are slightly flattened. The edges from the centre are a little *shorter* than the edges of the icosahedron:

$$\frac{\text{circumradius}}{\text{edge}} = \sin\frac{2\pi}{5} \approx 0.951 < 1.$$

In terms of balls: twelve unit balls placed at the vertices of an icosahedron around a central unit ball can all touch the central ball, but then they do not touch each other. This slack is why the twelve outer balls can roll around (the kissing configuration in $\mathbb{R}^3$ is not rigid), and it is the root of the Newton–Gregory dispute over whether a thirteenth ball fits.

The usual proof is a computation, for instance with the dihedral angle $\arccos\frac{1}{3}$ of the regular tetrahedron. The following argument needs no computation. It uses only the plane fact that *six* equal circles fit exactly around a seventh, together with the triangle inequality on the sphere.

Proposition E.8. *A regular icosahedron is not the union of* 20 *regular tetrahedra sharing a vertex at its centre.*

The argument. Suppose it were, with edge length a. Put balls of radius $a/2$ at the centre and at the 12 vertices. Then the central ball touches all twelve, and each vertex ball touches its five neighbours. Project the twelve contact points radially onto a sphere about the centre, and measure distances by angle. Neighbouring vertices are then exactly 60° apart, because the centre and two neighbours form an equilateral triangle.

Stand the icosahedron on a vertex: a north pole N, the antipodal vertex S, and two *tropical rings* of five vertices each. The upper ring consists of the neighbours of N, the lower ring of the neighbours of S, and the rings are staggered by 36° of longitude. The equatorial band between the rings is made of ten triangles (Figure E.3, left). Every vertex of the upper ring is 60° from N, so the upper ring lies at latitude 30° N. Likewise the lower ring lies at latitude 30° S.

Now *rotate the lower ring* by 36° about the axis NS, keeping its latitude. Each lower vertex ℓ now sits directly below an upper vertex u, and the ten equatorial triangles have become five equatorial quadrilaterals, the "squares" (Figure E.3, right). Each of the five meridians through N and S now carries four of our points,

$$N,\ u,\ \ell,\ S \qquad \text{at angles} \qquad 0^\circ,\ 60^\circ,\ 120^\circ,\ 180^\circ$$

along the meridian, since u is 60° from N, ℓ is 60° from S, and N and S are 180° apart. The meridian is a great circle, and 60° steps are exactly the spacing of a *ring of six* balls around the central one, as in the hexagonal packing (Figure E.4). So after the rotation, u and ℓ touch: the five meridians have become five half-rings of six, one great circle each.

But before the rotation u and ℓ also touched, since they were neighbours in the icosahedron, and in between we only slid ℓ sideways along its latitude circle. The closest point of a latitude

circle to u is the one on u's meridian, so sliding ℓ off the meridian moves it strictly *farther* from u. In other words, the deformation from ten equatorial triangles to five equatorial squares, made without changing any latitude, shortened the distance from u to ℓ. Yet it is $60°$ at both ends. That is a contradiction. □

In one line, the same argument is the triangle inequality on the sphere:

$$d(u,\ell) \;\geq\; d(N,\ell) - d(N,u) \;=\; (180° - 60°) - 60° \;=\; 60°,$$

with equality only if u lies on the arc from N to ℓ, that is, on ℓ's meridian. Each upper vertex u has two lower neighbours, and they cannot both lie on u's meridian, so for at least one of them $d(u,\ell) > 60°$. (In the icosahedron both are off the meridian, by $\pm 36°$.) The ring of six is the picture of the equality case.

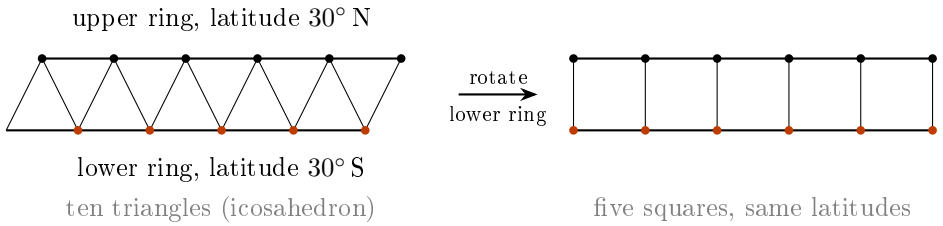


Figure E.3: The equatorial band, unrolled (the left and right ends are identified; longitudes in units of $72°$). Rotating the lower ring by $36°$ turns the ten equatorial triangles into five quadrilaterals without changing any latitude.

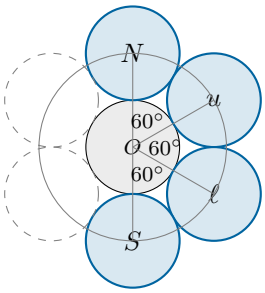


Figure E.4: After the rotation, each meridian is half of a ring of six balls around the central ball: N, u, ℓ, S are consecutive at $60°$ steps. The dashed balls complete the ring.

Remark E.9 (What the argument shows, quantitatively). In the hypothetical configuration with rings at latitudes $\pm 30°$, an upper and a lower vertex whose longitudes differ by $36°$ are at angular distance $\arccos\left(\frac{3}{4}\cos 36° - \frac{1}{4}\right) \approx 69.1°$, not $60°$. In the true icosahedron the rings sit lower, at latitudes $\pm\arctan\frac{1}{2} \approx \pm 26.6°$, every edge subtends $\arctan 2 \approx 63.4°$ at the centre, and the gaps between neighbouring outer balls are what let the twelve balls roll.

Remark E.10 (Two related classical facts). (1) Regular tetrahedra do not even fill the space around an *edge*: the dihedral angle is $\arccos\frac{1}{3} \approx 70.53°$, and five such angles leave a gap of about $7.36°$. (Aristotle claimed that regular tetrahedra tile space; see Lagarias–Zong, *Mysteries in packing regular tetrahedra*, Notices AMS 59 (2012).) The same angle appears here: five points at $60°$ from N and $60°$ from each other would need angles of $\arccos\frac{1}{3}$ at N, and five of these

do not close up. (2) In curved space the frustration disappears. In the 3-sphere, 600 regular tetrahedra (with the right curvature) fit together exactly, five around each edge, forming the regular polytope $\{3,3,5\}$ with 120 vertices, each surrounded by a regular icosahedron. This is the "ideal" icosahedral order of Frank's liquids; the flat-space packing problem has to compromise.

? Wording of the attribution is a placeholder for the author to edit.

Remark E.11 (Source). This argument came out of a conversation between the author and T. C. Hales.

E.5 Ideal order in curved space

The previous section measured the frustration: five regular tetrahedra around an edge leave a gap of about 7.36°. As noted there, the gap closes in positive curvature. The dihedral angle of a regular spherical tetrahedron grows with its size from the flat value $\arccos\frac{1}{3}\approx 70.53°$, and for edge 36° on the unit $\mathbb{S}^3$ it is exactly $72° = 360°/5$. Six hundred such tetrahedra are the cells of the regular polytope $\{3,3,5\}$ (Coxeter, *Regular Polytopes*), whose 120 vertices are at angular distance at least 36°, each with 12 nearest neighbours forming a regular icosahedron. Balls of angular radius 18° at these points realize Frank's local structure *everywhere*. By a theorem of Cohn and Kumar (gallery) these 120 points even minimize every completely monotonic pair potential on $\mathbb{S}^3$: the ideal order is a zero-temperature ground state, though in curved space.

This is the starting point of the curved-space approach to liquids and glasses of Sadoc, Mosseri and others, and of Nelson (*Order, frustration, and defects in liquids and glasses*, Phys. Rev. B 28, 1983): a dense flat liquid is the ideal $\{3,3,5\}$ structure "flattened", with the curvature concentrated on line defects (disclinations) along which six rather than five tetrahedra share an edge. See Sadoc and Mosseri, *Geometrical Frustration* (Cambridge, 1999).

E.6 Edwards entropy

E.6.1 The ansatz and its tests

A pile of sand is not at thermal equilibrium; k_BT is negligible next to the energy needed to lift a grain. Yet tapped or sheared granular packings settle into reproducible states, and Edwards and Oakeshott (*Theory of powders*, Physica A 157, 1989) proposed a statistical mechanics for them in which the *volume* plays the role of energy. This is Definition 11.4: all jammed (mechanically stable) states of the same volume are equally probable. Writing $\Omega(V)$ for the number (or a suitable measure) of jammed states of volume V, the dictionary is

thermal statistical mechanics	Edwards' granular ensemble
energy E	volume V
all states of energy E	jammed states of volume V
entropy $S(E)=\ln\Omega(E)$	Edwards entropy $S(V)=\ln\Omega(V)$
temperature $T=\partial E/\partial S$	compactivity $X=\partial V/\partial S$
Boltzmann weight $e^{-E/T}$	weight $e^{-V/X}$

The compactivity measures how much more compact the packing could be. As $X\to 0^+$ the weight concentrates on the densest jammed states, and as $X\to\infty$ on the most numerous

ones. Compare Section E.1: for the thermal hard-sphere gas $\partial S/\partial V = \beta P$, while for Edwards $\partial S/\partial V = 1/X$.

In the continuum, jammed packings are isolated points of configuration space, so "equal probability" needs a measure, and a real protocol samples a jammed state with probability equal to the volume of its *basin of attraction*, as in the ρ-evolution of Lecture 11. Edwards' ansatz then says that basins of equal-volume states are equally large; as noted in Lecture 11, small disc packings violate this. The most careful test so far (Martiniani, Schrenk, Ramola, Chakraborty and Frenkel, *Nature Physics* 13, 2017) computed basin volumes by thermodynamic integration for $N = 64$ polydisperse discs in 2d (hard cores with soft repulsive shells), with $N = 32$ to 128 for finite-size scaling. Above jamming the basin volumes are broadly distributed and the ansatz fails; the distribution narrows towards the unjamming point, extrapolated to $\phi^* = 0.841(3)$, where all packings become equally probable (numerically). Baule, Morone, Herrmann and Makse (Rev. Mod. Phys. 90, 2018) review the theory and its tests.

Where Edwards' ensemble can be written down *exactly*, its predictions can be checked against the ansatz-free dynamics. The channel of Lecture 11 is the simplest such model, and we now solve it.

E.6.2 The channel model solved

Recall the setting of §11.4. Discs of radius 1 lie in a channel of width $w = 2+h$ with $0 < h < \sqrt{3}$, so that each disc touches at most its two neighbours. Two consecutive discs in contact at opposite walls are horizontally $L = \sqrt{4-h^2} \in (1,2)$ apart, and two at the same wall are 2 apart. A jammed configuration of $n = \ell + 1$ discs is recorded by its contact word $\sigma \in \{0,1\}^\ell$ with no two adjacent 1's (1 for same wall, 0 for opposite walls). A word with k ones has length

$$t(\sigma) = 2k + (\ell - k)L + 2 = \ell L + (2 - L)k + 2,$$

and the number of such words is

$$\Omega(\ell, k) = \binom{\ell - k + 1}{k},$$

or $\binom{\ell-k-1}{k}$ if the end discs are restricted, as flagged in Lecture 11. We take the volume to be the channel length, $V = t$ (multiply by w for the area). Since t is an affine function of k, fixing the volume is the same as fixing the number k of defects. Write $x = k/\ell \in [0, \tfrac{1}{2}]$ for the defect density, and use η for the packing fraction (the letter φ is reserved for the golden ratio).

Proposition E.12 (Edwards entropy of the channel). *As $\ell \to \infty$ with $k/\ell \to x \in [0, \tfrac{1}{2}]$,*

$$\frac{1}{\ell} \ln \Omega(\ell, k) \;\longrightarrow\; s(x) = (1-x)\ln(1-x) - x \ln x - (1-2x)\ln(1-2x),$$

for both counts. The function s is strictly concave on $[0, \tfrac{1}{2}]$, vanishes at both ends, and has its maximum $s(x^*) = \ln \varphi$ at $x^* = \frac{5-\sqrt{5}}{10}$. The volume per contact is $v(x) = L + (2-L)x$ and the packing fraction is

$$\eta(x) = \frac{\pi}{w\, v(x)} = \frac{\pi}{w\,(L + (2-L)x)}.$$

Proof. Stirling's formula in the form $\ln \binom{a}{b} = a\, H\!\left(\frac{b}{a}\right) + O(\ln a)$, with $H(p) = -p \ln p - (1-p)\ln(1-p)$, applied with $a = \ell(1-x) + O(1)$ and $b = \ell x$, gives the limit $(1-x)H\!\left(\frac{x}{1-x}\right)$.

Expanding, with $1-\frac{x}{1-x}=\frac{1-2x}{1-x}$, this is $s(x)$. The shift $O(1)$ in a does not change the limit, so the two counts agree to leading order. Next,

$$s'(x)=\ln\frac{(1-2x)^2}{x(1-x)}, \qquad s''(x)=\frac{1}{1-x}-\frac{1}{x}-\frac{4}{1-2x}<0,$$

since $\frac{1}{1-x}\le 2<\frac{4}{1-2x}$ on $[0,\frac12)$. The critical point solves $(1-2x)^2=x(1-x)$, that is $5x^2-5x+1=0$, as in Lecture 11. At x^* one has $1-x^*=\frac{5+\sqrt5}{10}=\frac{\varphi}{\sqrt5}$, $x^*=\frac{1}{\sqrt5\,\varphi}$ and $1-2x^*=\frac{1}{\sqrt5}$, so that $s(x^*)=\frac{\varphi}{\sqrt5}\ln\frac{\varphi}{\sqrt5}-\frac{1}{\sqrt5\varphi}\ln\frac{1}{\sqrt5\varphi}-\frac{1}{\sqrt5}\ln\frac{1}{\sqrt5}$. Since $\frac{\varphi}{\sqrt5}-\frac{1}{\sqrt5\varphi}-\frac{1}{\sqrt5}=0$, the terms in $\ln\sqrt5$ cancel and what is left is $\big(\frac{\varphi}{\sqrt5}+\frac{1}{\sqrt5\varphi}\big)\ln\varphi=\ln\varphi$, using $\varphi+\varphi^{-1}=\sqrt5$. The formula for v is t/ℓ up to $O(1/\ell)$, and η is the area πn of the discs divided by the area wt of the channel, in the limit. □

The densest jammed state is the zigzag, $x=0$, with $\eta_{\max}=\pi/(wL)$; the loosest is $x=\frac12$, with $\eta_{\min}=2\pi/(w(L+2))$. The zigzag word $00\cdots0$ is unique, the loosest word $1010\cdots1$ is unique for odd ℓ (for even ℓ there are $\frac{\ell}{2}+1$ words with $\frac{\ell}{2}$ ones), and both have Edwards entropy zero. For $h=1.2$, the channel of Figure 11.1, one has $L=1.6$, $w=3.2$ and

$$\eta_{\max}\approx 0.6136, \qquad \eta(x^*)\approx 0.5739, \qquad \eta_{\min}\approx 0.5454.$$

Compactivity and the equation of state. Changing k by one changes V by $2-L$ and $\ln\Omega$ by $s'(x)+o(1)$, so

$$\frac{1}{X}=\frac{\partial S}{\partial V}=\frac{s'(x)}{2-L}=\frac{1}{2-L}\ln\frac{(1-2x)^2}{x(1-x)}. \tag{E.2}$$

Proposition E.13 (Equation of state). *For $X\in(-\infty,0)\cup(0,\infty]$, the defect density at compactivity X is*

$$x(X)=\frac12\Big(1-\frac{1}{\sqrt{1+4e^{-(2-L)/X}}}\Big),$$

and $\eta(X)=\pi/\big(w(L+(2-L)x(X))\big)$. In particular:

- *as $X\to0^+$, $x\sim e^{-(2-L)/X}\to0$ and $\eta\to\eta_{\max}$ (densest);*
- *$X=\pm\infty$ corresponds to $x=x^*$, the most numerous states;*
- *the states looser than typical, $x^*<x<\frac12$, have* negative *compactivity, and as $X\to0^-$, $x\to\frac12$ and $\eta\to\eta_{\min}$ (loosest).*

Proof. Put $y=e^{(2-L)/X}$. By (E.2), $(1-2x)^2=y\,x(1-x)$, that is $(4+y)x^2-(4+y)x+1=0$, whose root in $[0,\frac12]$ is $x=\frac12\big(1-\sqrt{y/(y+4)}\big)$, which is the formula. The limits follow by inspection: $y\to\infty$, $y=1$, and $y\to0$ respectively. The sign of X is the sign of $s'(x)$, which changes at x^*. □

The small-X behaviour $x\approx e^{-(2-L)/X}$ is an Arrhenius law: a defect costs "volume" $2-L$, just as a thermal excitation costs energy. The negative branch is the analogue of negative temperatures for spin systems, whose energy is bounded above. Here the jammed volume is bounded above, and past the entropy maximum adding volume *decreases* the number of states. For $h=1.2$, for example, $X=0.25,1,-1,-0.25$ give $x\approx0.128,0.239,0.311,0.390$ and $\eta\approx0.595,0.579,0.569,0.559$.

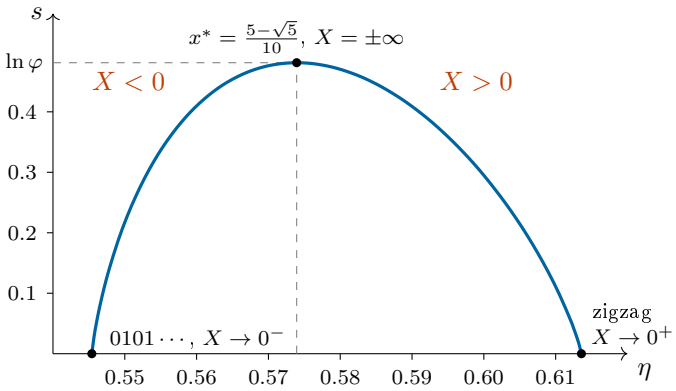


Figure E.5: Edwards entropy per contact of the channel model as a function of the packing fraction, for $h = 1.2$ ($L = 1.6$, $w = 3.2$); computed from Proposition E.12. Since V decreases as η increases, the compactivity $X = \partial V/\partial S$ is positive on the dense side of the maximum and negative on the loose side.

The canonical ensemble is a transfer matrix. Instead of fixing V, weight every jammed state by $e^{-V/X}$:

$$Z_\ell(X) = \sum_{\sigma} e^{-t(\sigma)/X} = e^{-2/X} \sum_{\sigma} a^{\#0(\sigma)}\, b^{\#1(\sigma)}, \qquad a = e^{-L/X},\ b = e^{-2/X},$$

the sum running over words of length ℓ with no 11. Keep track of the last letter. If $(c_\ell(0), c_\ell(1))$ are the weighted sums over words ending in 0 and in 1, then $(c_1(0), c_1(1)) = (a, b)$ and

$$\big(c_{\ell+1}(0), c_{\ell+1}(1)\big) = \big(c_\ell(0), c_\ell(1)\big)\, T, \qquad T = \begin{pmatrix} a & b \\ a & 0 \end{pmatrix},$$

since after a 0 we may write either letter and after a 1 only a 0. So we have $Z_\ell(X) = e^{-2/X}(a, b)\, T^{\ell-1}(1, 1)^T$. Since $T \geq 0$ and $T^2 > 0$, Perron–Frobenius gives

$$\frac{1}{\ell} \ln Z_\ell(X) \longrightarrow \ln \lambda(X), \qquad \lambda(X) = \frac{a + \sqrt{a^2 + 4ab}}{2},$$

the larger root of $\lambda^2 - a\lambda - ab = 0$. This is analytic in $X \neq 0$: as in van Hove's theorem, the one-dimensional channel has no Edwards phase transition.

Proposition E.14 (Equivalence of ensembles)**.** *The mean defect density in the canonical ensemble at compactivity X, $\lim_\ell \langle k\rangle/\ell = b\,\partial_b \ln\lambda$ (with a held fixed), equals the microcanonical $x(X)$ of Proposition E.13. Moreover*

$$\ln \lambda(X) = \max_{0 \leq x \leq 1/2} \Big(s(x) - \frac{v(x)}{X} \Big),$$

so $-X \ln\lambda$ is the Legendre transform of the entropy (an "Edwards free energy" per contact).

Proof. Write $\lambda = a\, g(u)$ with $u = b/a = e^{-(2-L)/X}$ and $g(u) = \frac{1}{2}(1 + r)$, $r = \sqrt{1 + 4u}$. Then

$$b\,\partial_b \ln\lambda = u\,\frac{g'(u)}{g(u)} = \frac{u \cdot 2/r}{1 + r} = \frac{r^2 - 1}{2r(1 + r)} = \frac{1}{2}\Big(1 - \frac{1}{r}\Big),$$

and $1/r = \left(1 + 4e^{-(2-L)/X}\right)^{-1/2}$, as in Proposition E.13. For the second statement, group the words by k: $Z_\ell = \sum_k \Omega(\ell, k) e^{-(\ell L + (2-L)k + 2)/X}$ has at most $\ell + 1$ terms, so $\frac{1}{\ell} \ln Z_\ell$ converges to the maximum over x of $s(x) - v(x)/X$ (Laplace's method). Since s is strictly concave, the maximizer is unique and is characterized by $s'(x) = (2 - L)/X$, which is again (E.2). □

At $X = \infty$ we have $a = b = 1$, and T becomes the Fibonacci matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with $\lambda = \varphi$. The identity $\ln \lambda(\infty) = \max s = \ln \varphi$ is Lecture 11's observation that "the maximal term carries the whole exponential growth". Exercise 11.10 asked for the typical defect fraction x^*; we now see it as the $X = \infty$ point of a whole equation of state. Section E.7 identifies the measure behind it.

E.6.3 Comparison with Bowles and Ashwin

The same model was analysed independently, and in the same spirit, by Bowles and Ashwin (*Edwards entropy and compactivity in a model of granular matter*, Phys. Rev. E 83, 2011). They use discs of diameter σ between walls a distance H apart, with $H/\sigma < 1 + \sqrt{3/4}$, so that each disc touches only its neighbours and one wall. With $\sigma = 2$ and $H = w$ this is exactly our condition $h < \sqrt{3}$. Their bond volumes $v_0 = H\sqrt{(2 - H)H}$ (cross-channel, $\sigma = 1$) and $v_1 = H$ (same wall) are our L and 2, rescaled by $\frac{1}{2}$ and multiplied by the width. They count $\Omega_J = \binom{N-M}{M}$ jammed states with M defects among N bonds, which differs from Lecture 11's count only at the ends, and their entropy per particle and compactivity are exactly our $s(x)$ and (E.2), in their units. They note that $X \to \infty$ at $\theta = \frac{1}{2} - \frac{\sqrt{5}}{10}$ and they restrict attention to $X > 0$, remarking that denser-defect packings would need negative compactivities. They also show that all the jammed packings are isostatic, and that two subsystems exchanging defects equilibrate at equal compactivity. Ashwin and Bowles had earlier (*Complete jamming landscape of confined hard discs*, Phys. Rev. Lett. 102, 2009) treated the wider channel $1 + \sqrt{3/4} < H/\sigma < 2$ with a transfer matrix on 32 local tiles.

The open-ends caveat of Lecture 11 is precisely the boundary term on which the two counts differ; it affects none of the limits above.

E.7 Information theory

E.7.1 Entropy and maximum entropy

Lecture 7 counted coin flips and Lecture 11 maximized a multinomial coefficient. In both, the number of sequences of length N with letter frequencies p is $e^{NH(p) + O(\ln N)}$, $H(p) = -\sum_i p_i \ln p_i$. Shannon (1948) took H as the measure of *information*: such a source can be encoded with $H(p)/\ln 2$ bits per letter and no fewer. Jaynes (*Information theory and statistical mechanics*, Phys. Rev. 106, 1957) turned the Lagrange multiplier computation of §11.3.1 into a principle: *the least biased distribution compatible with what one knows has maximal entropy.*

Proposition E.15 (Maximum entropy gives Boltzmann)**.** *Let $E_1, \dots, E_k$ be energies and $q_i = e^{-\beta E_i}/Z$ a Boltzmann distribution with mean energy $\bar{E} = \sum q_i E_i$. Every probability vector p with $\sum p_i E_i = \bar{E}$ satisfies $H(p) \le H(q)$, with equality only for $p = q$.*

Proof. The *relative entropy* $D(p\|q) = \sum p_i \ln(p_i/q_i)$ is non-negative and vanishes only for $p = q$, by Jensen's inequality for ln. Since $-\ln q_i = \beta E_i + \ln Z$,

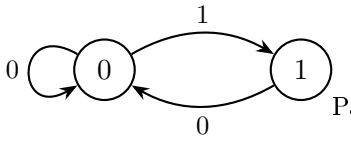
$$0 \le D(p\|q) = -H(p) + \beta \sum_i p_i E_i + \ln Z = -H(p) + \beta \bar{E} + \ln Z = -H(p) + H(q).$$

□

Edwards' ansatz is the same principle with volume for energy: the flat measure on jammed states of volume V is the maximum entropy measure given V, and the canonical weight $e^{-V/X}$ is the maximum entropy measure given the mean volume.

E.7.2 The golden-mean channel

Shannon's 1948 paper also treated *constrained* channels, where not every sequence of symbols may be sent. Its first example is the telegraph, whose symbols have different durations and may not follow each other arbitrarily. The capacity of such a channel is the exponential growth rate of the number of admissible messages. The simplest constraint is "no two consecutive 1's", the $(1,\infty)$ run-length-limited constraint of magnetic and optical recording. In symbolic dynamics the admissible bi-infinite sequences form the *golden-mean shift*. These are exactly the contact words of the channel model.



$$T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \varphi$$

Parry: $P(0 \to 1) = \varphi^{-2}$, $P(0 \to 0) = \varphi^{-1}$

Figure E.6: The golden-mean shift. Admissible words are paths in this graph, where the state is the last letter written. Its adjacency matrix is the Fibonacci matrix. With the weights $a = e^{-L/X}$, $b = e^{-2/X}$ on the edges labelled 0 and 1, it becomes the Edwards transfer matrix of the channel.

Theorem E.16 (Shannon 1948; Parry 1964). *1. The number of binary words of length n with no two consecutive 1's is F_{n+2}, and the capacity of the constraint is*

$$C = \lim_{n\to\infty} \frac{1}{n} \log_2 F_{n+2} = \log_2 \varphi \approx 0.6942 \text{ bits per symbol.}$$

2. Among stationary Markov chains supported on admissible sequences, the entropy rate is maximized uniquely by the chain with $P(0 \to 1) = \varphi^{-2}$, $P(0 \to 0) = \varphi^{-1}$, $P(1 \to 0) = 1$ (the Parry measure*). Its entropy rate is $\ln \varphi$ nats $= C$, and its frequency of 1's is $\frac{5-\sqrt{5}}{10}$.*

Proof. (1) The count is Lecture 11's Fibonacci argument. Equivalently it is $(1,1)T^{n-1}(1,1)^T$ with T the Fibonacci matrix, whose eigenvalues are φ and $-\varphi^{-1}$, so the count is $c\varphi^n + O(\varphi^{-n})$.

(2) Such a chain is determined by $p = P(0 \to 1)$, since after a 1 a 0 is forced. Its stationary law is $\pi_0 = \frac{1}{1+p}$, $\pi_1 = \frac{p}{1+p}$, and its entropy rate is $f(p) = \pi_0 H(p) = H(p)/(1+p)$. Setting $f'(p) = 0$ gives $(1+p)\ln\frac{1-p}{p} = H(p)$, which simplifies to $2\ln(1-p) = \ln p$, that is $(1-p)^2 = p$, so $p = \frac{3-\sqrt{5}}{2} = \varphi^{-2}$ and $1-p = \varphi^{-1}$. Then $H(p) = 2p\ln\varphi + (1-p)\ln\varphi = (1+p)\ln\varphi$, so $f(p) = \ln\varphi$, and $\pi_1 = \frac{\varphi^{-2}}{1+\varphi^{-2}} = \frac{1}{\varphi^2+1} = \frac{5-\sqrt{5}}{10}$. Uniqueness: f is continuous on $[0,1]$, vanishes at $p = 0$ and at $p = 1$, and has only one critical point in $(0,1)$. □

(The variational principle of symbolic dynamics upgrades (2) to all shift-invariant measures: the Parry measure is the unique measure of maximal entropy, and its entropy is the topological entropy $\ln\varphi$. See Lind and Marcus in the Further reading.)

Corollary E.17. *The maximal Edwards entropy per contact of the channel model equals the Shannon capacity of the golden-mean constraint,* $\ln\varphi$ *nats* $= \log_2\varphi$ *bits. Edwards' flat measure on jammed channel packings (the case* $X=\infty$*) is, in the limit, the Parry measure. More generally, the canonical Edwards ensemble at compactivity* X *is the Markov measure that maximizes entropy minus mean volume divided by* X*, for the weights* a, b *of Figure E.6: an equilibrium (Gibbs) measure in the sense of dynamics (Exercise E.32).*

So the typical jammed state of Lecture 11, with its fraction $\frac{5-\sqrt{5}}{10}$ of same-wall contacts, is the typical message of an optimally used golden-mean channel.

Remark E.18 (Capacity per unit length). Shannon's telegraph asks for capacity per unit *time* when symbols have durations t_i; for free concatenation it is $\ln X_0$ with $\sum_i X_0^{-t_i}=1$. A no-11 word is a free concatenation of the blocks 0 and 10, of channel lengths L and $2+L$, so jammed configurations (any number of discs) of length at most t number $e^{C't+o(t)}$ with $e^{-C'L}+e^{-C'(L+2)}=1$. This is $\lambda(X)=1$ with $X=1/C'$, and by Proposition E.14 $C'=\max_x s(x)/v(x)$. For $h=1.2$: $C'\approx 0.2817$ at $x\approx 0.266$, so $X\approx 3.55$.

E.7.3 Noise spheres: Shannon's packing argument

Codes enter sphere packing through noise. A sender transmits a vector $x\in\mathbb{R}^n$ (one real number per use of the channel) subject to a power constraint $|x|^2\le nP$. The receiver sees $y=x+z$, where z has independent $N(0,N)$ coordinates. The capacity of this *additive white Gaussian noise* channel is the largest rate R (bits per coordinate) such that 2^{nR} codewords can be chosen and decoded with error probability tending to 0 as $n\to\infty$.

Theorem E.19 (Shannon 1948, 1949)**.** *The capacity of the AWGN channel with signal power* P *and noise power* N *is*

$$C=\tfrac{1}{2}\log_2\left(1+\frac{P}{N}\right)\ \textit{bits per coordinate.}$$

Why packing? The heuristic is the gas in a box of Lecture 7 again. By the law of large numbers $|z|^2=nN(1+O(n^{-1/2}))$: in high dimension the noise is concentrated on a thin shell of radius $\sqrt{nN}$ around the transmitted codeword, just as the particle counts were concentrated near their mean. It is also nearly orthogonal to x, so $|y|^2\approx|x|^2+|z|^2\le n(P+N)$. Reliable decoding requires the noise balls of radius $\sqrt{nN}$ around different codewords to be essentially disjoint, and they all lie in a ball of radius about $\sqrt{n(P+N)}$. Comparing volumes gives at most $\big(\sqrt{n(P+N)}/\sqrt{nN}\big)^n=(1+P/N)^{n/2}$ codewords, the bound in the theorem. Shannon's 1949 paper (*Communication in the presence of noise*, Proc. IRE 37) turned this picture into a proof; achievability uses codewords chosen *at random*, the analogue of the non-constructive lower bounds of Lecture 1.

Demanding *zero* error against every noise vector of length less than ρ gives a genuine packing problem. For codewords of equal power, on the sphere of radius $\sqrt{nP}$, this means angular separation θ with $\sin(\theta/2)\ge\rho/\sqrt{nP}$, and the best such codes are counted by $M(n,\theta)$ of Definition 3.3. So the Kabatiansky–Levenshtein bound of §3.4 is a coding theorem, published as one (Problems of Information Transmission, 1978), and through Corollary 3.6 it is also the best known exponential upper bound on packing density.

E.7.4 Perfect codes, Golay and Leech

For binary codes (Definition 6.8) the Hamming distance replaces the Euclidean one, and the noise ball is a Hamming ball.

Proposition E.20 (Hamming's sphere packing bound)**.** *A binary code $C \subseteq \{0,1\}^n$ with minimal distance $d(C) \geq 2e+1$ satisfies*

$$|C| \sum_{i=0}^{e} \binom{n}{i} \leq 2^n.$$

Proof. The Hamming balls of radius e around the codewords are disjoint, by the triangle inequality, and each contains $\sum_{i \leq e} \binom{n}{i}$ words. □

A code attaining equality is *perfect*: its Hamming balls tile the cube, so every received word lies within distance e of exactly one codeword. The trivial perfect codes are a single word, the whole cube, and the binary repetition codes of odd length. The nontrivial examples are rare and beautiful.

- The *Hamming codes* (1950): $n = 2^r - 1$, $e = 1$, $|C| = 2^{n-r}$, and indeed $2^{n-r}(1+n) = 2^n$.
- The binary *Golay code* (1949): $n = 23$, $e = 3$, $|C| = 2^{12}$, and

$$2^{12}\left(1 + 23 + \binom{23}{2} + \binom{23}{3}\right) = 2^{12}(1 + 23 + 253 + 1771) = 2^{12} \cdot 2^{11} = 2^{23}.$$

- The ternary Golay code: $n = 11$, $e = 2$, 3^6 words over $\mathbb{F}_3$, and $3^6(1 + 11 \cdot 2 + \binom{11}{2} \cdot 4) = 3^6 \cdot 243 = 3^{11}$.

In fact these are all of them (Theorem E.28 in the gallery).

Adding a parity bit turns the Hamming code of length 7 into the extended Hamming code $[8,4,4]$, and Construction A (Definition 6.9) turns that into E_8, as we saw in Lecture 6. Adding a parity bit to the Golay code gives the extended Golay code $\mathcal{G}_{24}$, a $[24,12,8]$ code whose 759 words of weight 8 (the *octads*) form a Steiner system $S(5,8,24)$. Leech (1964, 1967) built the Leech lattice from it. In the form given by Conway and Sloane (Ch. 4, §11), $\sqrt{8}\,\Lambda_{24}$ is the set of $x \in \mathbb{Z}^{24}$ such that, for some $m \in \{0,1\}$, all $x_i \equiv m \pmod 2$, each set $\{i : x_i \equiv \nu \pmod 4\}$ ($\nu \in \mathbb{Z}/4$) is a word of $\mathcal{G}_{24}$, and $\sum_i x_i \equiv 4m \pmod 8$. The minimal vectors have norm 32 in these coordinates, and there are

$$\underbrace{\tbinom{24}{2} \cdot 4}_{(\pm 4, \pm 4, 0^{22})} + \underbrace{759 \cdot 2^7}_{(\pm 2^8, 0^{16}) \text{ on an octad}} + \underbrace{24 \cdot 2^{12}}_{(\mp 3, \pm 1^{23})} = 1104 + 97152 + 98304 = 196560$$

of them, the kissing number in dimension 24 quoted in Lecture 3. So E_8 and the Leech lattice, the two lattices now known to be optimal packings beyond dimension 3, come from the perfect Hamming code of length 7 and the perfect Golay code of length 23 by adding a parity bit.

E.7.5 Lattice codes

Construction A also runs the other way: a lattice $\Lambda \subset \mathbb{R}^n$ is a code for the Gaussian channel, with infinitely many codewords. To respect the power constraint one keeps only the points of Λ in a bounded region. A good choice of region, which keeps the code a group, is the Voronoi cell of a coarser lattice $\Lambda' \subset \Lambda$, which gives the *nested lattice codes* Λ/Λ'. Erez and Zamir (IEEE Trans. Inform. Theory 50, 2004) proved that, with a random dither and a scaling at the receiver, nested lattice codes with lattice decoding achieve the capacity $\frac{1}{2}\log_2(1+P/N)$. Decoding is then the nearest-point (Voronoi cell) problem of Lecture 1.

E.8 Gallery of theorems

Results stated in the text (Rényi, Tonks and van Hove, Shannon, Kabatiansky–Levenshtein) are not repeated here.

Theorem E.21 (Hales, Ann. of Math. 162, 2005; Viazovska, Ann. of Math. 185, 2017; Cohn–Kumar–Miller–Radchenko–Viazovska, Ann. of Math. 185, 2017)**.** *The optimal sphere packing densities in dimensions* 3, 8, 24 *are* $\pi/\sqrt{18}$, $\pi^4/384$ *and* $\pi^{12}/12!$. *Equivalently, the hard-sphere gas of unit diameter in* $\mathbb{R}^3$, $\mathbb{R}^8$, $\mathbb{R}^{24}$ *has finite free energy at number densities below, and infinite free energy above,* $\sqrt{2}$, 16 *and* 2^{24} *respectively.*

By Proposition E.1, ρ_{cp} is the optimal density divided by $\operatorname{vol} B^d(\frac{1}{2})$; it is the number density of FCC, E_8 and Leech, scaled to minimal distance 1.

Theorem E.22 (Hales–McLaughlin, J. Amer. Math. Soc. 23, 2010)**.** *In a packing of unit balls in* $\mathbb{R}^3$, *every Voronoi cell has volume at least that of a regular dodecahedron of inradius* 1.

Fejes Tóth's dodecahedral conjecture: the locally best density ≈ 0.7547 exceeds the global optimum 0.7405. This is geometric frustration as a theorem.

Theorem E.23 (Theil, Comm. Math. Phys. 262, 2006)**.** *For a class of short-range, Lennard-Jones-like pair potentials in the plane, the ground-state energy per particle of* N *particles converges as* $N \to \infty$ *to the energy per particle of the triangular lattice: pieces of the triangular lattice are asymptotically optimal.*

Crystallization at zero temperature, the problem whose finite-temperature version for hard spheres is open (§E.1). Heitmann–Radin (1980) had done "sticky discs".

Theorem E.24 (Richthammer, Comm. Math. Phys. 345, 2016)**.** *In the hard disc model in a* $2n \times 2n$ *box with arbitrary boundary condition, the mean square displacement of a particle near the centre is at least* $c \log n$.

Hard discs have no long-range positional order: the most a two-dimensional "crystal" can have is the hexatic orientational order seen numerically by Bernard and Krauth.

Theorem E.25 (Connelly–Gortler–Solomonides–Yampolskaya, Discrete Comput. Geom. 2020)**.** *A collectively jammed packing of* n *discs in a flat torus* $\mathbb{R}^2/\Lambda$ *whose radii and lattice are generic has exactly* $2n-1$ *contacts, and its contact graph is isostatic.*

The isostatic conjecture of Section E.2 as a theorem in the plane (the lattice is held fixed, and must itself be generic). Corollary E.7 gives the lower bound $2n-1$; the content is the matching upper bound, $2n-2$ positions modulo translation plus the common scale of the radii. The authors leave dimension ≥ 3 open.

Theorem E.26 (Cohn–Kumar, J. Amer. Math. Soc. 20, 2007)**.** *A configuration on* $\mathbb{S}^{n-1}$ *that has* m *distinct distances and is a spherical* $(2m-1)$*-design ("sharp") minimizes the energy* $\sum_{i \neq j} f(|x_i - x_j|^2)$ *among configurations of the same size, for every completely monotonic* f. *The same holds for the* 120 *vertices of the* 600*-cell.*

Examples: the icosahedron, the 240 roots of E_8, the 196560 minimal vectors of the Leech lattice. These are zero-temperature ground states for a whole family of interactions at once, including Frank's ideal $\{3,3,5\}$ order.

Theorem E.27 (Cohn–Kumar–Miller–Radchenko–Viazovska, Ann. of Math. 196, 2022)**.** *The E_8 and Leech lattices are universally optimal among point configurations in $\mathbb{R}^8$ and $\mathbb{R}^{24}$: they minimize energy for every potential that is a completely monotonic function of squared distance, for example Gaussians $e^{-\alpha r^2}$ and inverse powers.*

So the zero-temperature Gaussian core model is solved in dimensions 8 and 24 at every density. The sphere packing theorems are recovered in the limit $\alpha \to \infty$.

Theorem E.28 (Tietäväinen, SIAM J. Appl. Math. 24, 1973; van Lint 1971; Zinov'ev–Leont'ev 1973)**.** *Every nontrivial perfect code over a finite field has the same length, number of words and minimal distance as a Hamming code or one of the two Golay codes.*

Tilings of the Hamming cube by Hamming balls are classified by their parameters (codes with the Hamming parameters need not be linear). The Golay code's tiling is the discrete shadow of the Leech lattice.

Theorem E.29 (Erez–Zamir, IEEE Trans. Inform. Theory 50, 2004)**.** *Nested lattice codes with dithering, MMSE scaling and lattice decoding achieve the capacity $\frac{1}{2}\log_2(1+P/N)$ of the Gaussian channel.*

Structured codes, decoded by finding the nearest lattice point, reach Shannon's capacity.

Exercises

Exercise E.30. (a) Solve (E.1) on $[2,3]$: show $M(x) = 1 + \frac{2(x-2)}{x-1}$ there. (b) Show that every saturated packing of unit intervals in $[0,x]$ has at least $(x-1)/2$ intervals, and deduce $m \geq \frac{1}{2}$ without computing anything.

Exercise E.31. (a) Show that the honeycomb disc packing is locally jammed but not collectively jammed: find a collective motion. (b) Verify the claim of Figure E.2.

Exercise E.32. Let $r = (r_0, r_1)^T$ be a positive right eigenvector of $T = \left(\begin{smallmatrix} a & b \\ a & 0 \end{smallmatrix}\right)$ for $\lambda = \lambda(X)$. Show that $P(i \to j) = T_{ij} r_j / (\lambda r_i)$ is a stochastic matrix, that its stationary chain has frequency of 1's equal to $x(X)$, and that for $X = \infty$ it is the Parry measure of Theorem E.16.

Exercise E.33. Compute the variance of the number of defects k in the canonical ensemble to leading order in ℓ, by differentiating $\ln \lambda$ twice in $\ln b$. Show that it is $\ell / \big(-s''(x)\big)$, $x = x(X)$ (compare Lemma 11.6).

Exercise E.34. (a) Check that $1 + 90 + \binom{90}{2} = 2^{12}$, so the sphere packing bound allows a perfect binary code of length 90 with 2^{78} words and $e = 2$. (b) Show that no such code exists. *Hint:* assume $0 \in C$, so every word of weight 3 lies at distance ≤ 2 from a unique codeword, necessarily of weight 5. Count the weight-3 words that have 1's in the first two positions and are covered by codewords of weight 5 with 1's in the first two positions: each such codeword covers exactly three of them.

Exercise E.35. (a) Let $z \in \mathbb{R}^n$ have independent $N(0,N)$ coordinates. Show that $|z|^2/n \to N$ in probability, and compare with the gas in a box of §7.3.2. (b) Show that a fixed x with $|x|^2 = nP$ satisfies $\langle x, z\rangle / n \to 0$ in probability, and deduce $|x+z|^2/n \to P+N$. (c) If disjoint balls of radius $\sqrt{n(N-\varepsilon)}$ lie in a ball of radius $\sqrt{n(P+N+\varepsilon)}$, bound their number by $2^{n(C+\delta(\varepsilon))}$.

Further reading

- D. Ruelle, *Statistical Mechanics: Rigorous Results*, Benjamin, 1969.
- S. Torquato and F. H. Stillinger, *Jammed hard-particle packings: From Kepler to Bernal and beyond*, Rev. Mod. Phys. 82 (2010).
- A. Baule, F. Morone, H. J. Herrmann and H. A. Makse, *Edwards statistical mechanics for jammed granular matter*, Rev. Mod. Phys. 90 (2018).
- T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley.
- D. Lind and B. Marcus, *An Introduction to Symbolic Dynamics and Coding*, Cambridge, 1995.
- J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Springer, Chs. 1–5.

Chapter F

Interlude F: Computation and certified proof

[Added] This interlude is new: it is not in the handwritten notes. It collects the computational ideas that sit behind several results quoted in the course (the Kepler conjecture, the Tammes problem for 13 and 14 points, the kissing numbers in dimensions 4, 8 and 24) and tries them out on small problems that the notes already contain. To keep it readable it is set in black; all of it should be read as added material. The short Python programs mentioned below are in the directory `code/` next to these notes.

Look again at the proof of Thue's theorem in Lecture 1. It begins with an infinite object, a saturated packing of the plane, and it ends with a statement about a single triangle (Lemma 1.19, in the form its proof actually uses): a triangle with sides at least 2 and circumradius less than 2 has area at least $\sqrt{3}$. Everything in between is bookkeeping: a Delaunay triangulation cuts the packing into pieces, a weighted average reduces the density of the whole to the density of the pieces, and the boundary terms disappear in the limit. The Fejes Tóth inequality of Lecture 2 has the same shape. A triangulation of the sphere is cut into $2n-4$ spherical triangles, Euler's formula counts them, and one inequality about spherical triangles finishes the job.

This is the shape of almost every sharp result in the subject.

(1) *Reduce* a continuum of configurations to finitely many combinatorial *cases*, indexed by a geometric object: a triangulation, a contact graph, a Delaunay star, a perfect form.

(2) *Prune* the cases by cheap necessary conditions: angle sums, areas, Euler's formula, planarity.

(3) *Settle* each surviving case by an inequality (it cannot occur, or it is worse than the candidate) or by solving equations (it is the candidate).

(4) *Certify* every step, so that the proof can be checked by somebody who does not trust the person, or the program, that found it.

In Lecture 1 there was one case and the inequality took two lines. In Hales's proof of the Kepler conjecture there are thousands of cases and tens of thousands of inequalities. This interlude is about what happens in between: how cases are indexed and enumerated (Section F.2), how searches are organized (Section F.3), which pieces of linear algebra and optimization do the work (Sections F.4–F.5), what algebra can and cannot decide (Section F.6), and how a computer can

produce a *proof* rather than a number (Section F.7). In Section F.8 we put the pieces together and prove Thue's triangle lemma by a certified computer search.

F.1 The shape of a computer-assisted proof

Here are the problems from the course, arranged by the four steps above.

problem	cases indexed by	each case settled by	reference
disks in $\mathbb{R}^2$	Delaunay triangles	area $\geq \sqrt{3}$	Lecture 1
N caps on $\mathbb{S}^2$	triangulations, Euler	spherical area inequality	Lecture 2
critical points of ρ	balanced contact graphs	force balance (an LP)	Lectures 10–11
Tammes, N=13, 14	irreducible graphs	LP and inequalities	Musin–Tarasov
kissing, $n = 8, 24$	— (one case)	one polynomial	Section F.4
Kepler, $\mathbb{R}^3$	tame plane graphs	nonlinear inequalities, LPs	Hales
lattices in $\mathbb{R}^n$	perfect forms	Voronoi's algorithm	Lect. 6, Int. B

Two remarks before we start. First, the line between "by hand" and "by computer" is not where one might think. The kissing number 240 in dimension 8 is proved by a single polynomial, which a computer found but which anyone can check with pencil and paper (Example F.7). On the other hand, the enumeration of all tame graphs for the Kepler conjecture cannot be checked by hand at all. Second, what matters in the end is not whether a computer was used but whether every step produced a *certificate*: a finite object (a list of graphs, a polynomial with rational coefficients, a collection of boxes, a vector of weights) together with a simple procedure that checks it. Most of this interlude is about which certificates are available and how to produce them.

F.2 Indexing cases by geometric objects

F.2.1 Contact graphs

Lectures 10 and 11 attached to a configuration $X = (x_1, \dots, x_N)$ on $\mathbb{S}^2$ with minimal distance $\theta = \rho(X)$ its *contact graph* (Definition 11.1), with edges the geodesic arcs of length θ. Two simple lemmas turn this graph into a good index for case analysis.

Lemma F.1 (Contact graphs are planar)**.** *Let $0 < \theta < \pi$ and let $X \subset \mathbb{S}^2$ have all pairwise distances $\geq \theta$. Then two contact edges (arcs of length θ between points of X) can meet only at a common endpoint.*

Proof. Two minor arcs from a common endpoint a lie on great circles through a, which meet again only at $-a$, so they meet only at a. Suppose the arcs $[a, b]$ and $[c, d]$ have four distinct endpoints and cross at a point p. By the triangle inequality,

$$d(a,c) + d(b,d) \leq d(a,p) + d(p,c) + d(b,p) + d(p,d) = d(a,b) + d(c,d) = 2\theta.$$

Both terms on the left are $\geq \theta$, so both equal θ and both triangle inequalities are equalities. Since $d(a, c) = \theta < \pi$, equality in $d(a, c) \leq d(a, p) + d(p, c)$ means that p lies on the minor arc from a to c. That arc and $[a, b]$ leave a in the same direction (towards p) and both have length θ, so $c = b$, a contradiction. □

Lemma F.2 (Angle lemma). *Let $u, v, w \in \mathbb{S}^2$ with $d(u,v) = d(v,w) = \theta \in (0, \pi)$ and $d(u,w) \geq \theta$. Then the angle φ at v between the arcs vu and vw satisfies $\varphi \geq \alpha(\theta)$, where*

$$\cos \alpha(\theta) = \frac{\cos \theta}{1 + \cos \theta}$$

is the angle of the equilateral spherical triangle of side θ. Consequently a point of X with $k \geq 2$ contacts has $k\,\alpha(\theta) \leq 2\pi$.

Proof. By the spherical law of cosines, $\cos d(u,w) = \cos^2\theta + \sin^2\theta \cos\varphi$. The hypothesis $\cos d(u,w) \leq \cos\theta$ gives $\cos\varphi \leq (\cos\theta - \cos^2\theta)/\sin^2\theta = \cos\theta/(1+\cos\theta)$. The k angles between consecutive contact arcs at a point add up to 2π. □

For $\theta = \pi/3$ this gives $\cos\alpha = \frac{1}{3}$: *at most five contacts*, because $5 \arccos\frac{1}{3} < 2\pi < 6\arccos\frac{1}{3}$. This is the same number $\arccos\frac{1}{3} \approx 70.53°$ as the dihedral angle of the regular tetrahedron in Section E.4, and for the same reason. For $\theta = \pi/2$ we get $\alpha = \pi/2$ and at most four contacts.

F.2.2 A worked example: the critical configurations of five points

Exercise 10.7 of Lecture 10 asked for the classification of balanced configurations, and the notes did $N = 3$ and $N = 4$ by hand. Let us do $N = 5$ the way a computer would.

Let X be a critical configuration of ρ on $\mathrm{Conf}(5, \mathbb{S}^2)$, with balancing weights $w \geq 0$, not all zero, as in Theorem 11.2. Let S be the *stressed graph*: the edges with $w_e > 0$, together with all five points (points without stressed edges are isolated vertices of S; they feel no force). We look for all graphs that can occur as S, using four rules.

R1 S is planar. (Lemma F.1, since $\theta \leq \max\rho < \pi$.)

R2 No vertex has degree 1. (A single positive force cannot balance.)

R3 No vertex of degree 2 lies in a triangle of S. (If v has exactly two stressed edges, the two forces are opposite, so the neighbours u, w lie on a great circle on either side of v and $d(u,w) = \min(2\theta, 2\pi - 2\theta)$. If also $d(u,w) = \theta$, then $\theta = 2\pi/3$. But four unit vectors with pairwise inner products $\leq -\frac{1}{2}$ would have $|x_1 + \cdots + x_4|^2 \leq 4 - 12 \cdot \frac{1}{2} < 0$.)

R4 If two vertices a, b have three common neighbours, then a, b are not adjacent, and their common neighbours span no triangle. (The circles of radius θ about a and b meet in at most two points unless they coincide, which forces $b = -a$ and $\theta = \pi/2$. Then $d(a,b) = \pi \neq \theta$, and the common neighbours lie on the equator between a and b, where three points cannot be pairwise $\pi/2$ apart.)

The program `code/contact5.py` runs through the 34 graphs on five vertices (from the *Atlas of Graphs* built into the `networkx` library). One is empty; R1 removes K_5; R2 removes 18; R3 removes 7 and R4 removes one more. Six graphs survive (Figure F.1).

Now the last, geometric, step, which here is easy enough to do by hand.

- K_4 plus a point: four pairwise equidistant points form a regular tetrahedron, so $x_1 + x_2 + x_3 + x_4 = 0$. For any fifth point y the four numbers $\langle y, x_i\rangle$ add up to 0, so one of them is ≥ 0 and $d(y, x_i) \leq \pi/2 < \theta$.

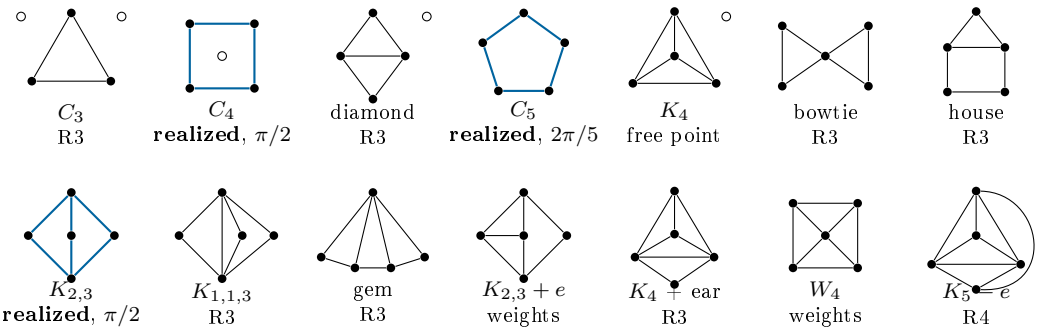


Figure F.1: The 14 graphs on five vertices that pass R1 and R2 (hollow: a point without stressed contacts). R3 and R4 remove eight of them. Of the six that remain, three are removed by a one-line geometric argument and three (blue) are realized by critical configurations.

- W_4: by R4 applied to the two pairs of opposite rim vertices, $\theta = \pi/2$, the rim is a square on a great circle and the hub is its pole. At a rim vertex the two rim forces are opposite and the force towards the hub is perpendicular to both, so its weight must vanish.
- $K_{2,3} + e$: the same argument at an endpoint of the extra edge (north and south forces opposite, the extra force along the equator).
- The survivors are realized: C_5 by the regular pentagon on a great circle ($\theta = 2\pi/5$); C_4 by a square on a great circle with the fifth point at a pole ($\theta = \pi/2$); $K_{2,3}$ by the two poles and three points on the equator with all gaps in $[\pi/2, \pi)$, as in Figure D.3 of Interlude D. For C_5 and C_4 equal weights balance. For $K_{2,3}$ put the same weight w_k on both edges at the k-th equatorial point (so the equatorial points balance); since all gaps are less than π, the three directions at a pole are not in a closed half-plane, so there are $w_1, w_2, w_3 > 0$ balancing the poles. (Equal weights work only for gaps $2\pi/3$.)

Proposition F.3. *The critical values of ρ on $\mathrm{Conf}(5, \mathbb{S}^2)$ are $2\pi/5$ and $\pi/2$. At $2\pi/5$ the critical configurations are the regular pentagons on great circles; at $\pi/2$ they are the configurations with two antipodal points and three points on the equator between them, with all gaps $\geq \pi/2$.*

Together with Lecture 11 this is the complete list for $N \leq 5$. The lowest critical value is $2\pi/N$, as predicted by the Kusner–Kusner–Lagarias–Shlosman theorem quoted there. For $N = 6$ the same rules leave many more graphs, and the geometric step needs actual computation: one has to solve, for each graph, the equations "all edges have length θ, all non-edges are longer, the forces balance" (Section F.6). That is the general pattern.

F.2.3 The same idea at full scale

Tammes for $N = 13, 14$. The Tammes problem asks for $\max \rho$. Musin and Tarasov call a contact graph *irreducible* (or jammed) if no single point can be shifted so as to increase its distance to all its neighbours and no "Danzer flip" is possible, and they show that for $N > 6$ the contact graph of an optimal configuration is irreducible. The contact graph of an irreducible configuration is planar, every vertex has degree 0, 3, 4 or 5 (the upper bound is Lemma F.2), every face is a convex spherical polygon with sides θ and at most $\lfloor 2\pi/\theta \rfloor$ vertices, and isolated vertices lie in faces with at least six vertices. These are the analogues of R1–R4. They enumerate

the planar graphs with these properties by computer, discard most of them by area and angle inequalities (in the spirit of the proof of the Fejes Tóth inequality: the face areas add up to 4π, and each face has a minimal area depending on its shape), and treat the few survivors by optimization. This gives the optimal 13- and 14-point configurations (Section F.9). Generating planar graphs with prescribed degrees and face sizes is itself a well-developed subject; the standard tool is `plantri` (Brinkmann and McKay).

Kepler. In Hales's proof the cases are indexed by *tame plane graphs*. Around each ball one draws, on a sphere about its centre, the planar graph whose vertices are (the radial projections of) the neighbouring centres within a fixed distance (2.51 in the original proof, 2.52 in the revised and formal one) and whose edges join neighbours that are close to each other. The local density (the *score* of a *decomposition star*, later replaced by *Marchal cells*) is then bounded case by case. "Tame" is a list of combinatorial conditions of the R1–R4 type (degree bounds, face sizes, a weight inequality), designed so that a non-tame graph is excluded by an easy argument. Hales's original archive had 5128 tame graphs. When Nipkow, Bauer and Schultz re-did the enumeration inside the proof assistant Isabelle (2006), they confirmed that the archive was complete and found it redundant, reducing it to 2771 graphs. The revised proof uses a larger archive of 18 762 graphs.

Lattices. For lattice packings the index is a *perfect form* (Theorem B.27 of Interlude B). Voronoi's algorithm walks from one perfect form to its neighbours across the facets of a polyhedral cone, so the enumeration is a graph search. Its cost is dominated by symmetry: one must decide whether a new form is equivalent to one already found. In dimension 8 there are 10 916 perfect forms (Dutour Sikirić, Schürmann and Vallentin, 2007).

F.3 Branch and bound, and meet in the middle

F.3.1 Branch and bound

Suppose we want to show that a function F on a compact set K satisfies $F \geq 0$, or to find $\max_K F$. *Branch and bound* keeps a list of pieces of K (boxes, cells, partial assignments). For each piece B it computes a cheap *bound*: a number $\underline{F}(B) \leq \inf_B F$ (for the inequality) or $\overline{F}(B) \geq \sup_B F$ (for the maximum). If the bound settles the piece, the piece is discarded; otherwise it is *branched*, that is, split into smaller pieces. Two properties make it work: the bounds must be *valid* (never wrong), and they must become *sharp* as the pieces shrink.

Max-min problems. The min-type function ρ of Lecture 10 is well suited to this. Let $B_1, \dots, B_N \subset \mathbb{S}^2$ be small cells and suppose $x_i \in B_i$. Then

$$\rho(x_1, \dots, x_N) \;\leq\; \overline{\rho}(B_1, \dots, B_N) := \min_{i<j} \; \max_{y \in B_i,\, z \in B_j} d(y, z),$$

and the right side is a minimum of $\binom{N}{2}$ numbers that are easy to compute. If $\overline{\rho} < \rho(X_0)$ for a known configuration X_0, no configuration with $x_i \in B_i$ can be optimal. The difficulty is the number of pieces: with cells of size ε there are about $(4\pi/\varepsilon^2)^N$ choices, and symmetry multiplies the work by $|\mathrm{SO}(3) \times \Sigma_N|$-many copies of the same configuration. For the analogous problem of n equal circles in a square, Markót and Csendes made this approach work with interval arithmetic and a clever elimination of regions (the "method of active areas"), proving the optimal packings for

$n = 28, 29, 30$ (about 50, 50 and 20 hours of computing in 2005); Markót later did $n = 31, 32, 33$ (2021). For the Tammes problem with 13 points, the graph enumeration of Section F.2 is used instead: fixing the combinatorics first removes both the symmetry and most of the dimension.

Branch and bound on integer points. The same idea finds short vectors in a lattice. Let G be a positive definite Gram matrix and $C > 0$; we want all $x \in \mathbb{Z}^n$ with $x^T G x \le C$. Completing squares (the LDL^T decomposition, in exact rational arithmetic) writes

$$x^T G x = \sum_{i=1}^{n} q_{ii}\Big(x_i + \sum_{j>i} q_{ij} x_j\Big)^2, \qquad q_{ii} > 0.$$

Choose x_n first: the $i = n$ term alone must be $\le C$, which leaves finitely many integers. Given x_n, the remaining budget bounds x_{n-1}, and so on. This is the algorithm of Fincke and Pohst (1985), a branch and bound whose pieces are the partial vectors $(x_k, \dots, x_n)$. Run on the Cartan matrix of E_8 (`code/fincke_pohst.py`), it visits 739 nodes and finds the 240 roots; with $C = 4$ it visits 5477 nodes and finds the 2400 nonzero vectors of norm at most 4. The size of the tree depends heavily on the basis: a basis with long, nearly parallel vectors gives a very unbalanced tree. The LLL algorithm of Lenstra, Lenstra and Lovász (1982) computes, in polynomial time, a reduced basis whose first vector is within a factor $2^{(n-1)/2}$ of the shortest, and in practice LLL followed by Fincke–Pohst enumeration is how minimal vectors are found. Voronoi's algorithm needs exactly this at every step: the minimal vectors of each new form.

F.3.2 Meet in the middle

The idea of *meet in the middle* is to split a search over pairs (u, v) into two half-searches and match the halves through a table. The classic example is *subset sum*: given n integers, is there a subset with sum T? Split the integers into two halves, list the $2^{n/2}$ subset sums of each half, sort one list, and look up $T - s$ for every s in the other. This takes about $2^{n/2}$ steps instead of 2^n (Horowitz and Sahni, 1974), and Schroeppel and Shamir (1981) reduced the memory to about $2^{n/4}$.

Here is the same idea counting vectors of E_8, as defined in Lecture 6 (Definition 6.4). Write $x = (u, v)$ with $u, v \in \mathbb{R}^4$. Then $x \in E_8$ exactly when u, v are both integral or both in $(\mathbb{Z} + \frac{1}{2})^4$, and the coordinate sum is even. So it suffices to tabulate the half-vectors u by the pair $\big(|u|^2,\ \text{sum of } u \text{ modulo } 2\big)$, separately for the two cosets, and to multiply the counts of matching classes. The program `code/e8_mitm.py` lists only 625 half-vectors and recovers

$$240, \quad 2160, \quad 6720, \quad 17520 \qquad \text{vectors of norm } 2, 4, 6, 8,$$

which is $240\,\sigma_3(m)$ as predicted by the theta series (equation (B.5) in Interlude B), whereas the full list has 26 641 vectors. In lattice language: E_8 is a union of four cosets of $D_4 \oplus D_4$, and the theta series of an orthogonal sum is the product of the theta series. This "gluing" is how Conway and Sloane describe many lattices.

To be honest about where meet in the middle is used: as an *algorithm* it is central in coding theory and cryptography. Stern's method (1989) for finding low-weight codewords splits an information set in two halves and looks for collisions, and Howgrave-Graham (2007) combined lattice reduction with a meet-in-the-middle search to attack the NTRU cryptosystem. In the proofs about packings quoted in this course it appears mainly in the gluing form above, as a way to count or list vectors without listing the whole space.

F.4 Matrix techniques

F.4.1 Gram matrices

A configuration of N unit vectors is determined, up to an orthogonal transformation, by its Gram matrix $G_{ij} = \langle x_i, x_j \rangle$.

Proposition F.4. *A real symmetric $N \times N$ matrix G is the Gram matrix of N vectors in $\mathbb{R}^d$ if and only if G is positive semidefinite of rank at most d. The vectors are then unique up to $\mathrm{O}(d)$.*

Proof. If $G = V^T V$ with V a $d \times N$ matrix, then $y^T G y = |Vy|^2 \geq 0$ and $\operatorname{rank} G \leq d$. Conversely, diagonalize $G = Q \Lambda Q^T$ with $\Lambda \geq 0$ having at most d nonzero entries, and take V to be the nonzero rows of $\Lambda^{1/2} Q^T$. If $V^T V = W^T W$, the map $Vy \mapsto Wy$ is a well defined isometry between the column spaces, and it extends to an element of $\mathrm{O}(d)$. □

So "is there a configuration of N points on $\mathbb{S}^{d-1}$ with all angles at least θ?" becomes "is there a positive semidefinite matrix with ones on the diagonal, off-diagonal entries $\leq \cos\theta$, and rank at most d?" Without the rank condition this is a *semidefinite program*. The rank condition is what makes the problem hard, and dropping it is what makes bounds computable. Sometimes, however, the sign pattern alone settles everything.

Proposition F.5 (Rankin 1955)**.** *Let $x_1, \dots, x_N \in \mathbb{R}^d$ be nonzero.*
(a) If $\langle x_i, x_j \rangle < 0$ for all $i \neq j$, then $N \leq d + 1$.
(b) If $\langle x_i, x_j \rangle \leq 0$ for all $i \neq j$, then $N \leq 2d$.

Proof. Claim: vectors with pairwise non-positive inner products that lie in an open half-space $\{\langle y, e \rangle > 0\}$ are linearly independent. Indeed, a linear relation can be written $u = \sum_{i \in P} a_i x_i = \sum_{j \in Q} b_j x_j$ with $a, b \geq 0$ and $P \cap Q = \emptyset$. Then $|u|^2 = \sum a_i b_j \langle x_i, x_j \rangle \leq 0$, so $u = 0$, and $0 = \langle u, e \rangle = \sum a_i \langle x_i, e \rangle$ forces $a = 0$; similarly $b = 0$. For (a), apply the claim to $x_1, \dots, x_{N-1}$ and $e = -x_N$. For (b), choose e with $\langle x_i, e \rangle \neq 0$ for all i and apply the claim to the vectors on each side. □

Since $d(x, y) \geq \pi/2$ on $\mathbb{S}^2$ means $\langle x, y \rangle \leq 0$, this settles two cases of the Tammes problem at once and bounds a third: five or six points cannot all be more than $\pi/2$ apart (by (a), $N \leq 4$), and the triangular bipyramid and the octahedron show that $\pi/2$ is attained for $N = 5, 6$; seven points cannot all be $\pi/2$ apart (by (b), $N \leq 6$), so the Tammes distance for $N = 7$ is less than $\pi/2$ (its actual value, $\approx 77.87°$, needs much more work).

F.4.2 Positive definite kernels and the linear programming bound

The proof of Proposition F.5 used one positive semidefinite matrix. The linear programming bound uses infinitely many at once. For $S^{n-1} \subset \mathbb{R}^n$ let G_k be the Gegenbauer polynomial $C_k^{(n-2)/2}$, normalized by $G_k(1) = 1$. By the addition theorem for spherical harmonics, $G_k(\langle x, y \rangle)$ is a positive multiple of $\sum_m Y_m(x) Y_m(y)$, summed over an orthonormal basis of the harmonic polynomials of degree k. Hence for every finite $C \subset S^{n-1}$

$$\sum_{x,y \in C} G_k(\langle x, y \rangle) \;\geq\; 0, \tag{F.1}$$

that is, the matrix $\big(G_k(\langle x, y \rangle)\big)_{x,y \in C}$ is positive semidefinite (Schoenberg, 1942).

Theorem F.6 (Delsarte–Goethals–Seidel 1977; Kabatiansky–Levenshtein 1978)**.** *Let* $f(t) = \sum_{k=0}^{D} c_k G_k(t)$ *with* $c_0 > 0$ *and* $c_k \geq 0$ *for all* k*, and suppose* $f(t) \leq 0$ *for* $-1 \leq t \leq \cos\theta$*. Then every* $C \subset S^{n-1}$ *with all angles at least* θ *has* $|C| \leq f(1)/c_0$*.*

Proof. Sum $f(\langle x, y\rangle)$ over all pairs. The off-diagonal terms are ≤ 0, so the sum is at most $|C|\, f(1)$. By (F.1) every $k \geq 1$ contributes ≥ 0 and $k = 0$ contributes $c_0 |C|^2$. Hence $c_0 |C|^2 \leq |C|\, f(1)$. □

This is the third time in these notes that a bound comes from *throwing away squares*: Blichfeldt dropped $|\sum x_i|^2$ (Lemma 2.13), Cohn and Elkies dropped one square for each frequency (Remark B.22), and here each G_k hides a sum of squares.

Example F.7 (The kissing number in dimension 8)**.** Take $n = 8$, $\theta = \pi/3$, and the polynomial of Odlyzko–Sloane and Levenshtein,

$$f(t) = (t+1)\Big(t + \frac{1}{2}\Big)^2 t^2 \Big(t - \frac{1}{2}\Big).$$

Clearly $f \leq 0$ on $[-1, \frac{1}{2}]$. Expanding in the G_k for $n = 8$ (exact rational arithmetic; a few lines of `sympy`) gives

$$(c_0, \ldots, c_6) = \Big(\frac{3}{320},\ \frac{3}{40},\ \frac{15}{64},\ \frac{39}{80},\ \frac{399}{640},\ \frac{9}{16},\ \frac{33}{128}\Big),$$

all positive, and $f(1)/c_0 = \frac{9}{4} \cdot \frac{320}{3} = 240$. So $\tau_8 \leq 240$, and E_8 gives equality. The same computation with $f(t) = (t+1)(t+\frac{1}{2})^2(t+\frac{1}{4})^2 t^2 (t-\frac{1}{4})^2(t-\frac{1}{2})$ (degree 10) in dimension 24 gives 196 560, with all eleven coefficients $c_0, \ldots, c_{10}$ positive. Note the roots: they are the inner products $-1, -\frac{1}{2}, 0, \frac{1}{2}$ occurring in the E_8 kissing configuration, and $-1, \pm\frac{1}{2}, \pm\frac{1}{4}, 0$ in the Leech one. Equality in the proof forces $f(\langle x, y\rangle) = 0$ for $x \neq y$, which is how one guesses f, and how Bannai and Sloane (1981) proved that both kissing configurations are unique.

The polynomial was found by solving a linear program (maximize c_0 subject to $f(1) = 1$ and to sampled constraints $f(t_\ell) \leq 0$), but the *certificate* is the polynomial: rational coefficients, checked exactly. In dimensions 3 and 4 the LP bound is not sharp (it gives 13 and 25). Musin proved $\tau_4 = 24$ (2008) by a modification of the method. Replacing pairs by triples of points gives *semidefinite* programming bounds (Bachoc and Vallentin, 2008), which improved the upper bounds in dimensions $5, 6, 7, 9, 10$; Leijenhorst and de Laat (2024) improved them in dimensions 11 to 23. For packings in $\mathbb{R}^n$ the continuous analogue is the Cohn–Elkies bound (Theorem B.21), and its three-point refinement (Cohn, de Laat and Salmon, 2022) improves the density bounds in dimensions 4–7 and 9–16.

F.4.3 Rigidity matrices and stresses

The other matrix in this subject is the *rigidity matrix* A of a contact graph: the linear map that sends a motion $v = (v_1, \ldots, v_N)$ to the rates of change of the contact distances (Lecture 11, and Interlude E, Proposition E.6). Its kernel is the space of infinitesimal motions, and the kernel of A^T is the space of *stresses*, systems of edge weights that balance at every vertex. Theorem 11.2 says that a configuration is critical when $\ker A^T$ meets the non-negative orthant. Two computational remarks. First, if the configuration has algebraic coordinates, ranks and kernels can be computed exactly over a number field, and a positive stress, once found numerically, can be checked exactly. Second, deciding whether $\ker A^T$ contains a non-negative nonzero vector is a linear program: maximize nothing subject to $A^T w = 0$, $w \geq 0$, $\sum w_e = 1$. Its feasibility is exactly Gordan's alternative from Lecture 10. The next section explains why this is no accident.

F.5 KKT: forces are Lagrange multipliers

The Tammes problem $\max \rho$ is a nonsmooth optimization problem, since ρ is a minimum of smooth functions. The standard trick is to add a variable. Write s for the cosine of the minimal distance and consider

$$\text{minimize } s \quad \text{subject to} \quad \langle x_i, x_j\rangle \le s \ \ (i<j), \qquad |x_i|^2 = 1 \ \ (1\le i\le N). \tag{F.2}$$

Now everything is smooth (indeed polynomial), and a configuration X is a local maximum of ρ if and only if $(X, \cos\rho(X))$ is a local minimum of (F.2). The basic necessary condition for smooth constrained problems is due to F. John (1948) for inequality constraints, and to Mangasarian and Fromovitz (1967) with equality constraints.

Theorem F.8 (Fritz John conditions)**.** *Let p be a local minimum of f subject to $g_k \le 0$ and $h_l = 0$, all C^1. Then there are $\lambda_0 \ge 0$, $\lambda_k \ge 0$ and $\mu_l \in \mathbb{R}$, not all zero, with*

$$\lambda_0 \nabla f(p) + \sum_k \lambda_k \nabla g_k(p) + \sum_l \mu_l \nabla h_l(p) = 0, \qquad \lambda_k g_k(p) = 0 \text{ for all } k.$$

If $\lambda_0 > 0$ one may take $\lambda_0 = 1$; these are the Karush–Kuhn–Tucker (KKT) *conditions.*

Usually one needs a *constraint qualification* (for instance, linear independence of the gradients of the active constraints) to guarantee $\lambda_0 > 0$. For the Tammes problem it comes for free, and the multipliers turn out to be the forces of Lecture 11.

Proposition F.9. *A configuration X with $\rho(X) = \theta \in (0,\pi)$ satisfies the KKT conditions for (F.2) (with $s = \cos\theta$) if and only if it is critical in the sense of Theorem 11.2. The multipliers of the constraints $\langle x_i, x_j\rangle \le s$ are balancing weights on the contact graph. In particular every local maximum of ρ is critical.*

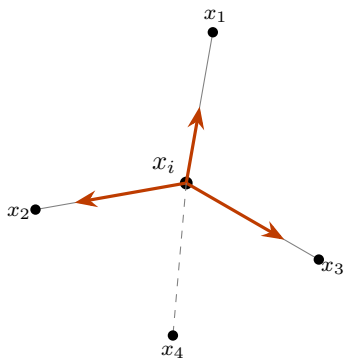
Proof. With $g_{ij} = \langle x_i, x_j\rangle - s$ and $h_i = |x_i|^2 - 1$, the derivative of the Lagrangian in s and in x_i gives

$$\lambda_0 = \sum_{i<j} \lambda_{ij}, \qquad \sum_{j\ne i} \lambda_{ij} x_j + 2\mu_i x_i = 0 \quad (1\le i\le N),$$

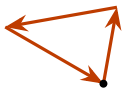
with $\lambda_{ij} = \lambda_{ji}$, and $\lambda_{ij} = 0$ unless $\langle x_i, x_j\rangle = s$, that is, unless ij is a contact. For a contact, $x_j = \cos\theta\, x_i + \sin\theta\, u_{ij}$, with u_{ij} the unit tangent vector at x_i pointing to x_j. The tangential part of the second equation is therefore $\sin\theta \sum_j \lambda_{ij} u_{ij} = 0$: the λ_{ij} balance the contact graph. If all λ_{ij} vanished, then $\lambda_0 = 0$ and $2\mu_i x_i = 0$, so all multipliers would vanish; hence the weights are not all zero and $\lambda_0 = \sum \lambda_{ij} > 0$. Conversely, balancing weights w_{ij}, scaled so that $\sum w_{ij} = 1$, give KKT multipliers $\lambda_{ij} = w_{ij}$, $\lambda_0 = 1$ and $\mu_i = -\frac{1}{2}\cos\theta \sum_j w_{ij}$ (the normal part of the second equation). The last sentence follows from the Fritz John theorem. □

So the dictionary is: *multipliers are forces*, *complementary slackness says that only contacts carry force*, and *Gordan's alternative is LP duality*. The weights are found by a linear program, and the forces at each vertex close up into a polygon (Figure F.2); that polygon is the first step of the Maxwell–Cremona correspondence between stresses and reciprocal diagrams.

Remark F.10 (Necessary, not sufficient). KKT points need not be maxima: the square of Lecture 11 is critical but not a local maximum of ρ. A sufficient condition comes from Interlude E. If X is *first-order jammed* (every nontrivial infinitesimal motion strictly shortens some contact,



weights 0.68, 1, 1 on the solid contacts; the dashed contact carries no force



the three forces, head to tail, close up: $\sum_j w_{ij} u_{ij} = 0$

Figure F.2: Force balance at one vertex of a critical configuration, seen in the tangent plane at x_i. The multipliers of Proposition F.9 are the lengths of the arrows. A fourth contact with zero weight is allowed: the weights are ≥ 0, not necessarily > 0.

which by Proposition E.6 means a positive stress on every contact plus infinitesimal rigidity modulo rotations), then a compactness argument on the unit sphere of motions shows that X is a strict local maximum of ρ up to rotation. The ρ-evolution algorithm of Lecture 11 is, in this language, an *active set method*: it moves in a direction that increases all active constraints until a new constraint becomes active, and it stops at a KKT point.

F.6 Polynomial methods

F.6.1 Equations for balanced configurations

Once a contact graph is fixed, a critical configuration is a solution of polynomial equations: $|x_i|^2 = 1$, $\langle x_i, x_j \rangle = c$ on the edges, and the balance equations $\sum_j w_{ij}(x_j - c\,x_i) = 0$, modulo rotations, together with the inequalities $\langle x_i, x_j \rangle < c$ off the edges and $w \geq 0$. Polynomial algebra solves the equations exactly.

Example F.11 (The square antiprism, $N = 8$)**.** The Tammes configuration for 8 points (Schütte and van der Waerden, 1951) is a square antiprism with all 16 edges equal. Put the two squares at heights $\pm h$ on circles of radius r, rotated by 45° against each other. A side of a square has squared length $2r^2$ and a slanted edge has $(2 - \sqrt{2})r^2 + 4h^2$, and $\cos\theta = 1 - r^2$. With $s = \sqrt{2}$ and $r_2 = r^2$, the equations are

$$r_2 + h^2 - 1 = 0, \qquad s^2 - 2 = 0, \qquad s\,r_2 - 4h^2 = 0, \qquad c - 1 + r_2 = 0.$$

A Gröbner basis in the lexicographic order $r_2 > h > s > c$ (Buchberger's algorithm, one line in `sympy`) is

$$\{\, r_2 + c - 1, \;\; h^2 - c, \;\; 2s - 7c - 1, \;\; 7c^2 + 2c - 1 \,\}.$$

It is triangular, like the result of Gaussian elimination, and its last element is a polynomial in c alone. So $c = \cos\theta$ satisfies $7c^2 + 2c - 1 = 0$, that is $\cos\theta = (2\sqrt{2} - 1)/7$ and $\theta \approx 74.8585^\circ$.

For larger systems one needs to decide how many real roots a univariate polynomial has in an interval. *Sturm's theorem* (1829) does it exactly: with $p_0 = p$, $p_1 = p'$ and $p_{k+1} =$

$-\operatorname{rem}(p_{k-1}, p_k)$, the number of roots in $(a, b]$ is the drop in the number of sign changes of $(p_0, p_1, \dots)$ from a to b. For $p = 7c^2 + 2c - 1$ the sequence is $(7c^2 + 2c - 1,\ 14c + 2,\ \frac{8}{7})$, with signs $(-, +, +)$ at 0 and $(+, +, +)$ at 1: exactly one root in $(0, 1]$. Descartes' rule of signs and its refinements (Vincent; Collins and Akritas, 1976) give faster *real root isolation*. *Resultants* eliminate one variable at a time, and *homotopy continuation* (as in the programs Bertini and PHCpack) follows the solutions of an easy system along a path to all isolated complex solutions of the given one, numerically. Numerical solutions can be certified afterwards: by Smale's *α-theory*, if $\alpha(f, x) = \beta(f, x)\gamma(f, x) < (13 - 3\sqrt{17})/4 \approx 0.1577$, where $\beta = \left\|Df(x)^{-1} f(x)\right\|$ is the Newton step and $\gamma = \sup_{k \geq 2} \left\|Df(x)^{-1} D^k f(x)/k!\right\|^{1/(k-1)}$, then Newton's method started at x converges quadratically to a true zero. The program `alphaCertified` (Hauenstein and Sottile, 2012) checks these conditions in exact or interval arithmetic.

F.6.2 Tarski–Seidenberg: everything is decidable

A *first-order formula over* $\mathbb{R}$ is built from polynomial equations and inequalities with rational coefficients, using $\wedge, \vee, \neg$ and quantifiers $\exists x, \forall x$ over real numbers. For example, "there are 13 points on $\mathbb{S}^2$ with pairwise distances at least $\pi/3$" is

$$\exists x_1, \dots, x_{13} \in \mathbb{R}^3 : \ \bigwedge_i |x_i|^2 = 1 \ \wedge \ \bigwedge_{i<j} \langle x_i, x_j \rangle \leq \tfrac{1}{2}.$$

Theorem F.12 (Tarski 1948/1951; Seidenberg 1954)**.** *Every first-order formula over* $\mathbb{R}$ *is equivalent to a quantifier-free one, and the equivalent formula can be computed. In particular the truth of every first-order sentence over* $\mathbb{R}$ *is decidable.*

So $\tau_3 \leq 12$ could, *in principle*, be decided by running an algorithm. There is also a nice theoretical consequence. Apply quantifier elimination to the formula in one free variable c: "there are N unit vectors with $\langle x_i, x_j \rangle \leq c$". The result defines a finite union of points and intervals with real algebraic endpoints, and its least element is the cosine of the Tammes distance. *Hence every Tammes value* $\cos d_N$ *is an algebraic number*, as are the optimal values of all the finite configuration problems of the course, without anyone having to find them.

Why not just do it? Collins's *cylindrical algebraic decomposition* (1975), the first practical algorithm, has running time roughly $(sd)^{2^{O(k)}}$ for s polynomials of degree d in k variables, *doubly* exponential in k, and Davenport and Heintz (1988) showed that quantifier elimination can be inherently doubly exponential. For a single block of existential quantifiers there are better algorithms (Grigoriev–Vorobjov, Renegar, Basu–Pollack–Roy) with running time of the order $s^{k+1} d^{O(k)}$, singly exponential. For the 13-sphere sentence above, $k = 39$ and $s = 91$; that is far beyond any computer. In practice, cylindrical algebraic decomposition handles problems with a handful of variables. Tarski's theorem decides everything in principle and nothing big in practice. The art of the subject is to reduce a big problem to many small ones, each with few variables, and then to treat those by the methods of the next section.

F.7 Verification

F.7.1 Interval arithmetic

Interval arithmetic (R. E. Moore, 1966) computes with closed intervals $X = [\underline{x}, \overline{x}]$ instead of numbers:

$$X + Y = [\underline{x} + \underline{y},\ \overline{x} + \overline{y}], \qquad X \cdot Y = [\min P, \max P], \quad P = \{\underline{x}\underline{y}, \underline{x}\overline{y}, \overline{x}\underline{y}, \overline{x}\overline{y}\},$$

and so on. On a computer each lower endpoint is rounded down and each upper endpoint up (the IEEE 754 standard provides these *directed rounding* modes), or one uses exact rational endpoints, as we will. By induction on the length of an expression one gets the *fundamental theorem*: if an expression F is evaluated on intervals, the result contains $F(x)$ for every choice of x in the intervals. So a single evaluation on a box gives a *proved* enclosure of the range of F on the box.

The price is overestimation, the *dependency problem*: each occurrence of a variable is treated as independent. For $f(x) = x - x^2$ on $X = [0, 1]$, the natural evaluation $X - X \cdot X = [0, 1] - [0, 1] = [-1, 1]$, while the true range is $[0, \frac{1}{4}]$. Rewriting $f = \frac{1}{4} - (x - \frac{1}{2})^2$ gives the exact range. The *centred form* $f(m) + f'(X)(X - m)$, here $\frac{1}{4} + [-1, 1] \cdot [-\frac{1}{2}, \frac{1}{2}] = [-\frac{1}{4}, \frac{3}{4}]$, is still not exact, but its overestimation is $O(w^2)$ in the width w of the box, against $O(w)$ for the natural evaluation. This is why bounds become sharp as boxes shrink, which is what branch and bound needs.

Certified zeros. Interval arithmetic also proves that solutions exist.

Proposition F.13 (Interval Newton)**.** *Let $f \in C^1[a, b]$, $X = [a, b]$, $m \in X$, and suppose that the interval $f'(X)$ (any enclosure of f' on X) does not contain 0. Put $N(X) = m - f(m)/f'(X)$. If $N(X) \subseteq X$, then f has exactly one zero in X, and it lies in $N(X)$.*

Proof. Since $f' \neq 0$ on X, f is strictly monotone, so there is at most one zero. Say $f' > 0$ and $f(m) \geq 0$ (the other cases are symmetric). By the mean value theorem $f(a) = f(m) - f'(\xi)(m - a)$ for some $\xi \in X$, and $m - f(m)/f'(\xi) \in N(X) \subseteq X$ gives $f(m)/f'(\xi) \leq m - a$, so $f(a) \leq 0$. By the intermediate value theorem there is a zero x^* in $[a, m]$. Finally $x^* = m - f(m)/f'(\eta)$ for some η between x^* and m, so $x^* \in N(X)$. □

For the antiprism, take $f(c) = 7c^2 + 2c - 1$, $X = [0.26, 0.27]$ and $m = 0.265$. Then $f(m) = 0.021575$, $f'(X) = [5.64, 5.78]$, and $N(X) = [0.26117\ldots, 0.26126\ldots] \subset X$, so the Tammes cosine for 8 points lies in $N(X)$. One more step shrinks the interval to width $4 \cdot 10^{-9}$: Newton's quadratic convergence, now with a proof attached. In several variables the same role is played by the *Krawczyk operator* $K(X) = m - Yf(m) + (I - Yf'(X))(X - m)$, with Y an approximate inverse of $f'(m)$: if $K(X)$ lies in the interior of X, then f has a unique zero in X (Krawczyk 1969; Moore 1977). This is how a numerically computed balanced configuration is turned into a theorem.

F.7.2 Dual numbers and jets

To use centred forms or interval Newton one needs enclosures of derivatives on boxes. They come almost for free from *dual numbers*: compute in $\mathbb{R}[\varepsilon]/(\varepsilon^2)$. For a polynomial,

$$f(a + b\varepsilon) = f(a) + f'(a)\, b\, \varepsilon,$$

because $\varepsilon^2 = 0$ kills everything else. So evaluating f at $x+\varepsilon$ with the ordinary rules $(a+a'\varepsilon)(b+b'\varepsilon) = ab + (ab' + a'b)\varepsilon$ produces $f(x)$ and $f'(x)$ together. This is *forward-mode automatic differentiation*. With k variables one carries a gradient $(a, a'_1, \dots, a'_k)$ instead; with $\mathbb{R}[\varepsilon]/(\varepsilon^{r+1})$ one carries the Taylor coefficients up to order r, a *jet*. Now let the entries be intervals. Evaluating on a box B then gives, in one pass, enclosures of $f(B)$ and of $\nabla f(B)$, by the fundamental theorem. We use exactly this in Section F.8.

Going one step further, a *Taylor model* (Berz and Makino, 1998) represents a function on a box by a polynomial of degree r together with an interval bounding the remainder. Arithmetic on Taylor models keeps the dependency in the polynomial part, so the overestimation is $O(w^{r+1})$. The nonlinear inequalities of the Kepler conjecture were proved with second-order versions of this idea ("Taylor intervals"), and Solovyev and Hales (2013) implemented them inside the proof assistant HOL Light.

F.7.3 Sums of squares

A polynomial p that is a sum of squares of polynomials is visibly nonnegative, and the squares are a certificate. Write $p = m^T Q m$ with m the vector of monomials of degree at most $\frac{1}{2}\deg p$. Then p is a sum of squares if and only if this holds for some positive semidefinite Q, since $Q = L^T L$ gives $p = |Lm|^2$. The conditions on Q are linear equations (matching coefficients) plus $Q \succeq 0$: *finding a sum-of-squares certificate is a semidefinite program* (Parrilo, 2000/2003; Lasserre, 2001).

Not every nonnegative polynomial is a sum of squares (Hilbert, 1888). Motzkin's $x^4y^2 + x^2y^4 - 3x^2y^2 + 1$ is nonnegative by the inequality between arithmetic and geometric means applied to $x^4y^2, x^2y^4, 1$, but it is not a sum of squares. To prove $p \geq 0$ only on a set $K = \{g_1 \geq 0, \dots, g_r \geq 0\}$ one looks for certificates of the form

$$p = \sigma_0 + \sum_i \sigma_i g_i \qquad \text{(Putinar, 1993)}, \qquad \text{or} \qquad p = \sum_{I \subseteq \{1,\dots,r\}} \sigma_I \prod_{i \in I} g_i \qquad \text{(Schmüdgen, 1991)},$$

with sums of squares σ. Schmüdgen showed that a polynomial strictly positive on a compact K always has the second kind of certificate, and Putinar that, under an extra hypothesis on K (Archimedean), it has the first. The Positivstellensatz of Krivine (1964) and Stengle (1974) gives certificates for the *emptiness* of a semialgebraic set, and so for nonnegativity in general, at the price of denominators. Compare: the σ_i are polynomial versions of the KKT multipliers of Section F.5, and the multiplier $\lambda(z)$ of Section F.8 is an example.

The semidefinite program is solved numerically, so the matrix Q it returns is only approximately positive semidefinite and only approximately satisfies the linear equations. Peyrl and Parrilo (2008) showed how to round it to an *exact* rational certificate when p is strictly inside the cone: project the rounded Q onto the affine space of exact solutions and check $Q \succeq 0$ exactly, for example by an exact Cholesky factorization. The same rounding idea turns the numerical solutions of the LP and SDP bounds of Section F.4 into rigorous bounds.

F.7.4 Formal proof

A certificate still has to be checked by *some* program, and programs have bugs. A *proof assistant* (HOL Light, Isabelle, Coq, Lean) reduces this risk to a small trusted kernel that checks every logical step.

- *Kepler.* Hales's proof was published in 2005 (with a joint paper of Hales and Ferguson). The refereeing could not certify all of the computer code, and Hales started the Flyspeck project to produce a formal proof. It was completed on 10 August 2014, in HOL Light and Isabelle. The formal proof checks more than 23 000 nonlinear inequalities (after subdividing domains), which took about 5000 processor hours on the Microsoft Azure cloud, together with 43 078 linear programs and the tame graph enumeration of Nipkow, Bauer and Schultz.

- *Dimension* 8. Viazovska's proof (Interlude B) relies on two inequalities between modular forms. In her paper they were checked with computer help: the functions were replaced by truncated expansions at the two ends of the interval, the truncation errors were bounded rigorously, and the inequalities for the truncations were checked by a finite calculation. Romik (2023) later gave a proof without computer calculations. A Lean formalization of the whole dimension-8 theorem, begun by Hariharan and Viazovska in 2024, was completed in February 2026; its final stages were produced by the autoformalization system Gauss of Math, Inc. (Hariharan, Birkbeck, Lee, Ma, Mehta, Poiroux and Viazovska, arXiv:2604.23468). According to EPFL's announcement (March 2026) the dimension-24 case followed.

- *Outside this course.* The four colour theorem (Appel and Haken, 1976–77; a simpler proof by Robertson, Sanders, Seymour and Thomas, 1997) was formalized in Coq by Gonthier (2005, published 2008). Tucker's proof (1999, 2002) that the Lorenz equations have a strange attractor, Smale's 14th problem, is a classic of interval analysis: it combines rigorous integration of the flow with a normal form near the origin.

F.8 A certified proof of Thue's triangle lemma

We now prove the inequality of Lemma 1.19 by computer, using branch and bound, exact interval arithmetic, dual numbers and a KKT multiplier. Of course Lecture 1 proves it in two lines. The point is to watch every piece of the machinery work on an example where we know the answer.

The polynomial form. Let a triangle have sides $a, b, c \geq 2$ and circumradius $R \leq 2$, and let c be the longest side. Put $x = a^2$, $y = b^2$, $z = c^2$. Heron's formula says

$$16\,\mathrm{Area}^2 = H(x,y,z) := 2xy + 2yz + 2zx - x^2 - y^2 - z^2,$$

and $R = abc/(4\,\mathrm{Area})$ gives $R^2 = xyz/H$, so $R \leq 2$ means $g := 4H - xyz \geq 0$. Since $c \leq 2R \leq 4$, the domain is

$$D = \{4 \leq x \leq z,\ 4 \leq y \leq z,\ z \leq 16,\ g \geq 0\},$$

and the inequality of Lemma 1.19 (area $\geq \sqrt{3}$, i.e. density $\leq \pi/\sqrt{12}$) follows from $H \geq 48$ on D. (We allow $R = 2$; this only makes the claim stronger.) All transcendental functions are gone: this is the first rule of thumb.

Where it is tight. $H = 48$ at the equilateral triangle $(4,4,4)$, a corner of D. It is also 48 at $(4,4,12)$, the triangle with sides $2, 2, 2\sqrt{3}$ and largest angle $2\pi/3$, which has $R = 2$ exactly. (It is excluded by the strict inequality $R < 2$ in Lecture 1, but it lies in the closure, so the inequality on D is sharp at two points.) Tight points are the enemy of branch and bound: near them no bound of the form "$H \geq 48$ on this box" can be proved with any slack, and a naive search never ends.

The KKT multiplier. At $(4,4,12)$ the constraint $g \geq 0$ is active. Solving the KKT equations $\nabla H = \mu_x e_x + \mu_y e_y + \lambda \nabla g$ there gives $\nabla H = (24, 24, -8)$, $\nabla g = (48, 48, -48)$, hence $\lambda = \frac{1}{6}$ and $\mu_x = \mu_y = 16$. At $(4,4,4)$ the constraint g is not active ($g = 128$) and its multiplier is 0. On the edge $x = y = 4$ that joins the two tight points,

$$H - 48 = (z-4)(12-z), \qquad g = 4z(12-z), \qquad \text{so} \qquad H - 48 = \frac{z-4}{4z}\, g.$$

This suggests the variable multiplier $\lambda(z) = \frac{z-4}{4z}$, which is 0 at $z = 4$ and $\frac{1}{6}$ at $z = 12$. A two-line computation (checked symbolically in the script) gives the identity

$$H - 48 = \frac{P}{z} + \frac{z-4}{4z}\, g, \qquad P = \frac{xy-16}{4}\, z^2 + \big(16 - (x-8)(y-8)\big) z - 4(x-y)^2. \tag{F.3}$$

We write $P = Az^2 + Bz + C$ with $A = \frac{xy-16}{4}$, $B = 16 - (x-8)(y-8)$, $C = -4(x-y)^2$. Since $\lambda(z) \geq 0$ and $g \geq 0$ on D, it suffices to prove

$$P(x,y,z) \geq 0 \qquad \text{on the polyhedron } \{4 \leq x \leq z,\ 4 \leq y \leq z \leq 16\}. \tag{F.4}$$

The curved constraint has disappeared. The price: P vanishes on the whole edge $x = y = 4$.

The search. We prove (F.4) by branch and bound over boxes, starting from $[4,16]^3$ and bisecting the longest side. A box is *settled* if one of three tests succeeds:

(i) it misses the region ($z < x$ or $z < y$ everywhere on the box);
(ii) *value test:* the interval evaluation of P on the box has lower end ≥ 0;
(iii) *derivative test:* interval enclosures of $\partial P/\partial x$ and $\partial P/\partial y$ (from dual numbers) have lower ends ≥ 0, so P is smallest on the face $x = \underline{x}$, $y = \underline{y}$, and the interval evaluation of P on that face has lower end ≥ 0.

Test (iii) is what handles the tight edge: on it, $A = B = C = 0$ exactly, so the evaluation on the face gives the interval $[0,0]$. All arithmetic is in exact rationals (Python's `fractions.Fraction`), so there is no rounding at all. Here is the heart of `code/thue_bb.py`, slightly abridged:

```
from fractions import Fraction as Fr
class I:                                  # closed interval, rational endpoints
    def __init__(s, lo, hi=None): s.lo, s.hi = Fr(lo), Fr(lo if hi is None else hi)
    def __add__(s, t): t = lift(t); return I(s.lo + t.lo, s.hi + t.hi)
    def __neg__(s): return I(-s.hi, -s.lo)
    def __mul__(s, t):
        t = lift(t); p = [s.lo*t.lo, s.lo*t.hi, s.hi*t.lo, s.hi*t.hi]
        return I(min(p), max(p))
    def sq(s):                            # a square is >= 0
        lo, hi = sorted([abs(s.lo), abs(s.hi)])
        return I(0 if s.lo <= 0 <= s.hi else lo*lo, hi*hi)
class D:                                  # dual number: value and gradient, all intervals
    def __init__(s, v, g): s.v, s.g = v, g
    def __add__(s, t): t = liftD(t); return D(s.v + t.v, [a + b for a, b in zip(s.g, t.g)])
    def __mul__(s, t):                    # Leibniz rule
        t = liftD(t); return D(s.v*t.v, [s.v*b + t.v*a for a, b in zip(s.g, t.g)])
    def sq(s): return s*s
def P(x, y, z):
    A = Fr(1, 4)*(x*y - 16); B = 16 - (x - 8)*(y - 8); C = -4*(x - y).sq()
    return A*z.sq() + B*z + C
def grad(X, Y, Z):
    e = lambda k: [I(1) if j == k else I(0) for j in range(3)]
```

```
    return P(D(X, e(0)), D(Y, e(1)), D(Z, e(2))).g
def check(X, Y, Z):
    if Z.hi < X.lo or Z.hi < Y.lo: return True             # (i)   outside the region
    if P(X, Y, Z).lo >= 0: return True                      # (ii)  value test
    gx, gy, _ = grad(X, Y, Z)                               # (iii) derivative test
    return gx.lo >= 0 and gy.lo >= 0 and P(I(X.lo), I(Y.lo), Z).lo >= 0
def bb(box):                      # returns only if P >= 0 is proved on the box
    if check(*box): return
    k = max(range(3), key=lambda j: box[j].hi - box[j].lo)
    m = (box[k].lo + box[k].hi) / 2
    for half in (I(box[k].lo, m), I(m, box[k].hi)):
        b = list(box); b[k] = half; bb(tuple(b))
bb((I(4, 16), I(4, 16), I(4, 16)))
```

(Here `lift` and `liftD` turn numbers into point intervals and constant dual numbers; subtraction and the reversed operations are defined in the same way.)

The result. The program terminates in under a second. It examines 53 boxes; the 27 leaves are settled by test (i) in 9 cases, by test (ii) in 15 and by test (iii) in 3; the deepest box is at depth 8. Since every leaf passed a valid test and the leaves cover $[4,16]^3$, this proves (F.4), hence $H \geq 48$ on D, hence the inequality of Lemma 1.19. (The equality case is not addressed by the search.)

Without test (iii) the search never ends. With the depth capped at $3k$, the number of unsettled boxes is $8, 16, 32, 64, 128, 256$ for $k = 3, \ldots, 8$, all of them strung along the edge $x = y = 4$: each halving of the box size doubles the work. Figure F.3 shows both runs on the slice $z = 6$.

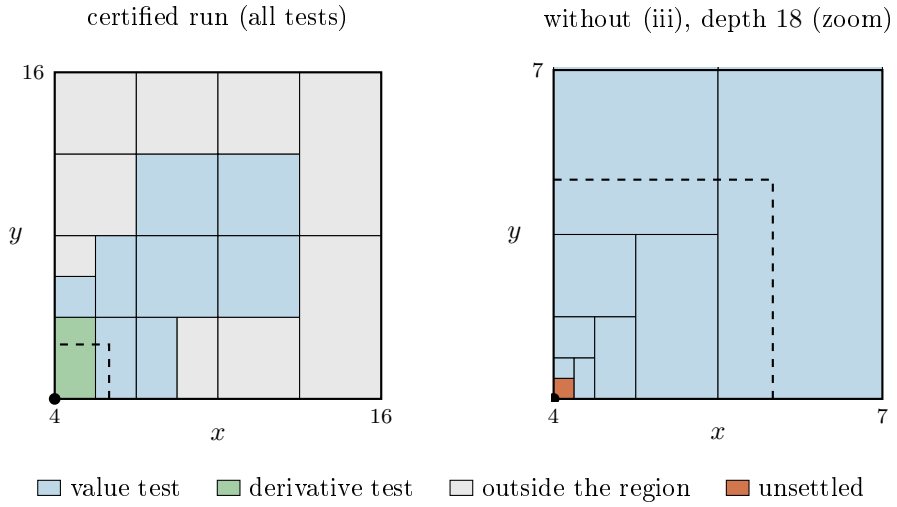


Figure F.3: Branch and bound for (F.4), shown on the slice $z = 6$ (the boxes are three-dimensional; we draw their intersections with this plane). The dashed square is the region $x, y \leq z = 6$, and the dot is the tight edge $x = y = 4$. Left: the certified run, with 18 boxes on this slice. Right: without the derivative test the boxes pile up geometrically towards the tight edge, and the corner box is never settled.

Remark F.14 (What was learned). (1) The tight points dictated the method: a strict inequality would have been proved by test (ii) alone, and the multiplier would not have been needed. Every

sharp inequality in Hales's proof needed a local argument near its equality case of exactly this kind. (2) The multiplier $\lambda(z)$ is a Positivstellensatz-type certificate in the sense of Section F.7: (F.3) reads $z(H-48) = P + \frac{z-4}{4}\,g$, and only the nonnegativity of P on a polyhedron is left to the computer. A semidefinite program could look for a sum-of-squares certificate for P too (Exercise F.32). (3) The derivative test is the monotonicity argument of Lecture 1 in disguise: there the area $\frac{1}{2}ac\sin B$ is increased by lengthening a and c.

F.9 Gallery of theorems

The theorems below were proved with essential help from computers, or are certified by finite objects that were found by computer. Each comes with a sentence on how it fits the course.

Theorem F.15 (Hales, Ann. of Math. 162 (2005); Hales–Ferguson, Discrete Comput. Geom. 36 (2006))**.** *No packing of congruent balls in* $\mathbb{R}^3$ *has density greater than* $\pi/\sqrt{18}$.

The formal proof is T. Hales et al., *A formal proof of the Kepler conjecture*, Forum Math. Pi 5 (2017), e2. Cases are indexed by tame plane graphs (Section F.2) and settled by linear programs and interval arithmetic; the whole of this interlude is a miniature of it.

Theorem F.16 (Hales–McLaughlin, J. Amer. Math. Soc. 23 (2010))**.** *Every Voronoi cell of a packing of unit balls in* $\mathbb{R}^3$ *has volume at least that of the regular dodecahedron of inradius* 1.

Fejes Tóth's dodecahedral conjecture (Interlude E, Section E.3), proved by the same computer-assisted methodology as the Kepler conjecture.

Theorem F.17 (Musin–Tarasov, Discrete Comput. Geom. 48 (2012); Experiment. Math. 24 (2015))**.** *The optimal Tammes configurations for* $N = 13$ *and* $N = 14$ *are unique up to isometry, with minimal angular distances* $\approx 57.1367^\circ$ *and* $\approx 55.6706^\circ$.

The proofs enumerate irreducible contact graphs: the full-scale version of the five-point example of Section F.2. The case $N = 13$ is the "strong thirteen spheres problem": how large can 13 equal balls touching a unit ball be? Since $57.1367^\circ < 60^\circ$, it contains a new proof of $\tau_3 = 12$.

Theorem F.18 (Odlyzko–Sloane, J. Combin. Theory Ser. A 26 (1979); Levenshtein, Soviet Math. Dokl. 20 (1979))**.** $\tau_8 = 240$ *and* $\tau_{24} = 196\,560$.

Example F.7. The computer found the polynomials; the proof is checkable by hand. Bannai and Sloane (Canad. J. Math. 33, 1981) proved uniqueness of the configurations.

Theorem F.19 (Musin, Ann. of Math. 168 (2008))**.** $\tau_4 = 24$.

The plain LP bound gives only 25. Musin modifies Delsarte's method so that the auxiliary polynomial may be positive on part of the range, and controls that part by a separate geometric argument.

Theorem F.20 (Bachoc–Vallentin, J. Amer. Math. Soc. 21 (2008))**.** $\tau_5 \le 45$, $\tau_6 \le 78$, $\tau_7 \le 135$, $\tau_9 \le 366$, $\tau_{10} \le 567$.

Three-point semidefinite bounds, a matrix version of Theorem F.6. N. Leijenhorst and D. de Laat (Math. Program. Comput., 2024) improved the bounds in dimensions 11–23, and H. Cohn, D. de Laat and A. Salmon (*Three-point bounds for sphere packing*, arXiv:2206.15373) gave the analogous improvement of the Cohn–Elkies bound for packings in dimensions 4–7 and 9–16.

Theorem F.21 (Markót–Csendes, SIAM J. Optim. 16 (2005); Markót, J. Global Optim. 81 (2021)). *The best known packings of n equal circles in a square are optimal for $n = 28, \dots, 33$.*

Interval branch and bound in configuration space, the method of Section F.3 with a great deal of engineering.

Theorem F.22 (Brakensiek–Heule–Mackey–Narváez, IJCAR 2020; J. Automat. Reason. 66 (2022)). *Every tiling of $\mathbb{R}^7$ by unit cubes contains two cubes sharing a full face (Keller's conjecture holds in dimension 7).*

Keller's conjecture was known for $n \le 6$ (Perron) and false for $n \ge 8$ (Lagarias–Shor for $n \ge 10$, Mackey for $n = 8$). The case $n = 7$ was reduced to the non-existence of a clique of size 128 in certain graphs and settled by a SAT solver; the proofs of unsatisfiability (about 200 gigabytes) were checked by an independent, formally verified checker. A packing problem for cubes, settled by a certificate of non-existence.

Theorem F.23 (Appel–Haken 1977; Robertson–Sanders–Seymour–Thomas, J. Combin. Theory Ser. B 70 (1997); Gonthier, Notices AMS 55 (2008)). *Every planar graph is 4-colourable.*

The model case of "cases indexed by planar graphs": an unavoidable set of configurations (633 in the 1997 proof), each shown reducible by computer. Gonthier's Coq proof removed the doubts about the code.

Theorem F.24 (Tucker, C. R. Acad. Sci. Paris 328 (1999); Found. Comput. Math. 2 (2002)). *The Lorenz equations, with the classical parameters, have a robust strange attractor.*

Smale's 14th problem, solved by interval arithmetic: an example far from geometry of the methods of Section F.7.

Theorem F.25 (Lower bounds by construction, 2025). $\tau_{11} \ge 593$ *(AlphaEvolve, Google DeepMind, arXiv:2506.13131);* $\tau_{10} \ge 510$, $\tau_{14} \ge 1932$ *(M. Ganzhinov,* Highly symmetric lines*, Linear Algebra Appl. 722 (2025); arXiv:2207.08266).*

A lower bound is certified by a list of vectors and an exact check of all pairwise inner products. Finding the list is the hard part; checking it is easy. This is the opposite situation to the upper bounds above.

Exercises

Exercise F.26. Using Lemma F.2, show that in a configuration with $\rho \ge \pi/3$ every point has at most five contacts, and that the bound is still five for the Tammes configuration of 13 points ($\theta \approx 57.14^\circ$). Show that the bound drops to four exactly when $\theta > \arctan 2$, the edge of the regular icosahedron.

Exercise F.27. Modify the program `code/contact5.py` to run the rules R1–R4 on the 156 graphs on six vertices. Which graphs survive? Find the critical configurations whose stressed graphs are the octahedron, the hexagon C_6, and C_5 plus an isolated point. Are there others?

Exercise F.28. Prove the equality case of Proposition F.5(b): if $2d$ nonzero vectors in $\mathbb{R}^d$ have pairwise non-positive inner products, they are $\pm$ an orthogonal basis up to positive scaling. Deduce that the octahedron is the unique optimal configuration for $N = 6$, while for $N = 5$ the optimum is not unique.

Exercise F.29. Verify Example F.7: expand f in Gegenbauer polynomials for $n = 8$ and compute $f(1)/c_0$. Then use the same polynomial in dimension 3. Show that its coefficients are still non-negative, and that it only proves $\tau_3 \le 1890/127 \approx 14.9$. Why should one expect a polynomial with roots at $-\frac{1}{2}, 0, \frac{1}{2}$ to be good in dimension 8 but not in dimension 3?

Exercise F.30. For $N = 4$, compute the KKT multipliers of Proposition F.9 at the regular tetrahedron and at the square on a great circle. Show that the square is not first-order jammed: find a nontrivial infinitesimal motion v with $Av = 0$, and check that along the corresponding curve all four contact distances increase to second order. Relate this to Example D.11 of Interlude D.

Exercise F.31. (a) Compute the natural interval extension of $f(x) = x^2 - 2x$ on $[0, 2]$, and of the equivalent forms $x(x-2)$ and $(x-1)^2 - 1$. (b) Use Proposition F.13 with exact rational arithmetic to enclose $\sqrt{2}$ in an interval of width less than 10^{-20}, starting from $[1, 2]$.

Exercise F.32. (a) Check that the three coefficients A, B, C of P vanish on the edge $x = y = 4$. Compute $\partial P/\partial x$ and $\partial P/\partial y$, and show by interval arithmetic that both are ≥ 0 on the box $[4, 5.5] \times [4, 7] \times [4, 7]$, one of the three boxes settled by test (iii). (b) Using an SDP solver, look for a certificate $P = \sigma_0 + \sum_i \sigma_i g_i$ on the polyhedron of (F.4) with $g \in \{x-4,\ y-4,\ z-x,\ z-y,\ 16-z\}$, and round it to exact rationals.

Exercise F.33. Write the statement "there are 5 points on $\mathbb{S}^2$ with pairwise distances greater than $\pi/2$" as a first-order sentence. How many variables and polynomials does it have? Using Proposition F.5, decide it by hand. Why does the existence of a decision procedure imply that $\cos d_N$ is algebraic for every N?

Exercise F.34. Modify `code/e8_mitm.py` to count the vectors of norm 2 and 4 in D_4 (split $2 + 2$) and in E_7 (take E_7 to be the vectors of E_8 orthogonal to the root $e_1 - e_2$, and keep the first two coordinates in the same half). You should find $24, 24$ and $126, 756$.

Further reading

- R. E. Moore, R. B. Kearfott, M. J. Cloud, *Introduction to Interval Analysis*, SIAM, 2009. Interval arithmetic, interval Newton and Krawczyk, with code.
- W. Tucker, *Validated Numerics: A Short Introduction to Rigorous Computations*, Princeton University Press, 2011. Includes automatic differentiation and the Lorenz attractor.
- G. Blekherman, P. A. Parrilo, R. R. Thomas (eds.), *Semidefinite Optimization and Convex Algebraic Geometry*, MOS–SIAM Series on Optimization, 2013. Sums of squares, Positivstellensätze, SDP.
- S. Basu, R. Pollack, M.-F. Roy, *Algorithms in Real Algebraic Geometry*, 2nd ed., Springer, 2006. Sturm sequences, cylindrical algebraic decomposition, complexity.
- D. Cox, J. Little, D. O'Shea, *Ideals, Varieties, and Algorithms*, Springer UTM. Gröbner bases.
- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Springer; Ch. 9 (bounds for codes and sphere packings), Ch. 13 (Odlyzko–Sloane, bounds on kissing numbers) and Ch. 14 (Bannai–Sloane, uniqueness).
- T. C. Hales et al., *A formal proof of the Kepler conjecture*, Forum Math. Pi 5 (2017), e2; and T. C. Hales, *Dense Sphere Packings: A Blueprint for Formal Proofs*, LMS Lecture Note Series 400, Cambridge, 2012.

- J. C. Lagarias (ed.), *The Kepler Conjecture: The Hales–Ferguson Proof*, Springer, 2011.
- H. Cohn, *Packing, coding, and ground states*, PCMI lecture notes, arXiv:1603.05202. LP and SDP bounds from the point of view of these notes.

Chapter G

Interlude G: Packings in the laboratory

[Added] This interlude is new: it is not in the handwritten notes. It follows the aside at the end of Lecture 6 ("ellipsoid packings; polygon packings; honeycomb and bubble problems") and the physical remarks scattered through Lectures 7–11 out of the mathematics and into the laboratory: squashed peas, foams, colloids, opals, lasers made of sand, sticky spheres and dice. To keep it readable it is set in black; all of it should be read as added material.

Everything in these notes so far has been about ideal balls: exactly round, exactly hard, exactly equal. Physicists have been squeezing, shaking and photographing real ones for three centuries, and many of the objects of the course were met on a laboratory bench before they were met in a proof. The Voronoi cell was a squashed pea, the face-centred cubic lattice was an opal, and the contact graph was a pattern of paint marks on ball bearings.

In physics one has to be careful about the status of a statement, more careful than in the rest of the course. Throughout, claims are tagged [theorem] (proved), [simulation] (seen on a computer), [experiment] (measured) or [model] (a calculation in an idealized model). Most of the famous numbers of the subject (the random close packing fraction 0.64, the freezing point 0.494 of hard spheres, the jamming exponents) are of the second and third kind; Figure G.4 sorts them. Interlude E discussed the statistical mechanics behind several of them (hard spheres, random close packing, Maxwell counting, frustration, Edwards' entropy), and we refer to it rather than repeat it.

G.1 Squashed peas, lead shot and soap

The first Hales

Lecture 1 introduced the Voronoi cell (Definition 1.15) as a bookkeeping device: it assigns to each ball the part of space that is closer to it than to any other ball. There is a much more physical way to see Voronoi cells. Take soft balls, and press them together until there is no space left. If every ball keeps its centre and grows at the same rate, each one ends up filling exactly its Voronoi cell.

The first famous squashing experiment was made by a Hales, although not the one who proved the Kepler conjecture. Stephen Hales, a clergyman and physiologist, reported in his *Vegetable Staticks* (1727):

> I compressed several fresh parcels of Pease in the same Pot, with a force equal to 1600, 800, and 400 pounds; ... being thereby formed into pretty regular Dodecahedrons.

(The peas swelled with water inside an iron pot, under a weighted lid.) What could "pretty regular dodecahedrons" have been? Not regular pentagonal dodecahedra, as a short argument shows.

Proposition G.1. *Congruent regular dodecahedra do not tile* $\mathbb{R}^3$.

Proof. The dihedral angle of the regular dodecahedron is $\delta = \arccos(-1/\sqrt{5}) \approx 116.57°$. Suppose the dodecahedra tile, and let e be an edge of one tile T. The edges and vertices of the other tiles meet the open segment e in finitely many points or in subsegments of e, so we may pick a point $p \in e$ that is not a vertex of any tile and at which no edge of another tile crosses e transversally. Let $T' \neq T$ be a tile containing p. If p lies in the relative interior of a face of T', that face must contain the direction of e: otherwise the face plane would cross e at p and the half-space bounded by it would overlap the interior of T near p. If p lies on an edge of T', then that edge runs along e. So in a small disc perpendicular to e at p, the tiles cut out sectors of angle δ (wedges along e) and π (half-planes), with disjoint interiors, filling the disc. Hence $k\delta + m\pi = 2\pi$ for integers $k \geq 1$, $m \geq 0$. Since $\pi/2 < \delta < \pi$, the case $m = 1$ is impossible, and so is $m = 2$. For $m = 0$ we would need $k\delta = 2\pi$, but $3\delta \approx 349.7°$ and $4\delta \approx 466.3°$. □

So three regular dodecahedra around an edge leave a gap of about 10.3°. This is the dodecahedral counterpart of the 7.36° gap left by five regular tetrahedra around an edge (Section E.4), and the two facts are dual to each other: a regular dodecahedron is the Voronoi cell of a ball surrounded by a regular icosahedron of twelve neighbours, whose Delaunay cells would be the twenty slightly flattened tetrahedra of Interlude E. By the theorem of Hales and McLaughlin quoted there, no Voronoi cell of a unit ball packing is smaller than this dodecahedron, and by Proposition G.1 not all of them can be equal to it. The peas must have been something else.

Kepler's pomegranate

A century earlier, Kepler had already given the right answer. In *Strena seu de nive sexangula* (1611), the little book on snowflakes that also contains his packing conjecture, he describes the seeds of a pomegranate, squeezed together as the fruit grows, and the rhombic shape they take: the rhombic dodecahedron. Here is the computation behind it.

Proposition G.2 (The Voronoi cell of FCC)**.** *Let* $D_3 = \{x \in \mathbb{Z}^3 : x_1 + x_2 + x_3 \text{ even}\}$ *be the FCC lattice, with minimal distance* $\sqrt{2}$*. Its Voronoi cell at* 0 *is*

$$V = \{x \in \mathbb{R}^3 : \ |x_i| + |x_j| \leq 1 \text{ for all } i \neq j\}.$$

This is the rhombic dodecahedron: the cube $[-\frac{1}{2}, \frac{1}{2}]^3$ *with a square pyramid of height* $\frac{1}{2}$ *erected on each face. It has* 12 *rhombic faces, one for each minimal vector* $\pm e_i \pm e_j$*,* 6 *vertices of degree four at* $(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$*, and* 8 *vertices of degree three at* $(\pm\frac{1}{2}, \pm\frac{1}{2}, \pm\frac{1}{2})$*. Its volume is* $2 = \operatorname{covol}(D_3)$.

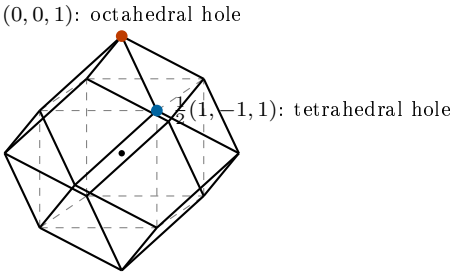
Proof. The Voronoi cell is $\{x : \langle x, v\rangle \leq \frac{1}{2}|v|^2 \ \ \forall v \in D_3 \setminus 0\}$. For the 12 minimal vectors $v = \pm e_i \pm e_j$ the condition reads $\pm x_i \pm x_j \leq 1$, so V contains the Voronoi cell. For the converse we show that every other $v \in D_3$ gives a redundant condition. Let $x \in V$ and put $a_i = |x_i|$; then $a_i + a_j \leq 1$, so each $a_i \leq 1$ and at most one a_i exceeds $\frac{1}{2}$. With $m_i = |v_i| \in \mathbb{Z}_{\geq 0}$,

$$\tfrac{1}{2}|v|^2 - \langle x, v\rangle \ \geq \ \sum_i \Big(\tfrac{1}{2}m_i^2 - a_i m_i\Big).$$

A term with $m_i = 0$ is 0, and a term with $m_i \geq 2$ is at least $\frac{1}{2}m_i(m_i-2) \geq 0$. A term with $m_i = 1$ is $\frac{1}{2} - a_i$, which is negative for at most one index i_0. If that happens, v_{i_0} is odd, and since $v_1+v_2+v_3$ is even, some other coordinate v_j is odd too. If $m_j = 1$, the two terms add up to $1 - a_{i_0} - a_j \geq 0$; if $m_j \geq 3$, the j-term is at least $\frac{3}{2}$ and the i_0-term at least $-\frac{1}{2}$. Either way the sum is ≥ 0. So V is the Voronoi cell. (For every simply laced root lattice the same holds, with the roots in place of the $\pm e_i \pm e_j$; see Lemma H.6 of Interlude H.)

On the cube $\max|x_i| \leq \frac{1}{2}$ all the inequalities hold. Over the face $x_3 = \frac{1}{2}$ of the cube, the inequalities $|x_1| + x_3 \leq 1$ and $|x_2| + x_3 \leq 1$ cut out the pyramid with apex $(0,0,1)$, and similarly for the other faces. The volume is $1 + 6 \cdot \frac{1}{3} \cdot 1 \cdot \frac{1}{2} = 2$. Each face $\{\pm x_i \pm x_j = 1\} \cap V$ is a rhombus with two degree-three and two degree-four vertices, for instance $(1,0,0), (\frac{1}{2}, \frac{1}{2}, \pm\frac{1}{2}), (0,1,0)$ on the face $x_1 + x_2 = 1$. □

The two kinds of vertices are the two kinds of holes of FCC (Interlude H, Section H.3): the degree-four vertices, at distance 1 from the centre, are octahedral holes, and the degree-three ones, at distance $\frac{\sqrt{3}}{2}$, are tetrahedral holes. The rhombic faces have diagonals $\sqrt{2}$ and 1. In the hexagonal close packing (HCP) the layers above and below each ball are mirror images of each other, and the cell becomes the *trapezo-rhombic dodecahedron*: the rhombic dodecahedron cut across a three-fold axis, with one half turned by 60° (six rhombi and six trapezoids, the same volume). Both cells have twelve faces, one for each of the twelve kissing neighbours. Whatever Stephen Hales saw, the experiment below suggests that most of his peas were less regular than either.



cell	faces	$A/V^{2/3}$
ball (does not tile)	–	4.836
rhombic dodecahedron (FCC)	12	5.345
truncated octahedron (BCC)	14	5.315
Kelvin (curved faces)	14	5.306
Weaire–Phelan (average)	13.5	5.288
cube	6	6

Figure G.1: Left: the Voronoi cell of FCC is a cube (dashed) with a pyramid on each face. Right: surface area per cell for cells of unit volume. The flat polyhedra were computed for this table; the Kelvin and Weaire–Phelan values are numerical minimizations (Surface Evolver), as quoted by Weaire and Phelan.

Lead shot: what disorder does

Hales' peas presumably went into the pot anyhow. In 1939 the botanist J. W. Marvin repeated the experiment with lead shot, compressed in a cylinder [experiment]. Shot that had been carefully stacked in a close-packed arrangement became rhombic dodecahedra. Shot that had simply been poured in became irregular polyhedra with on average 14.17 faces, not twelve. So disorder costs faces: a randomly packed ball has more Voronoi neighbours than a crystalline one, even though it has fewer *touching* neighbours.

Bernal turned this into a programme around 1960. To model the structure of liquids he poured steel ball bearings into rubber balloons, kneaded them, ran black paint into the pile and let it set, and then took the pile apart ball by ball, recording which pairs had been in contact

by the paint marks (Bernal, *Nature* 183, 1959; Bernal and Mason, *Nature* 188, 1960). Bernal and Mason counted about 6.4 contacts per ball [experiment], close to the isostatic value $2d = 6$ of Corollary E.7. Bernal also classified the holes of a random packing: besides the tetrahedra and octahedra of FCC there are three larger "canonical holes" bounded by equilateral triangles (*Proc. R. Soc. A* 280, 1964). The density of these packings, about 0.64, is the random close packing value of Interlude E (Section E.2).

Soap: the cell that minimizes its area

Squeeze the peas harder, or replace them by bubbles, and the cells choose their own shapes: a dry foam minimizes its total film area subject to fixed cell volumes. Plateau (*Statique expérimentale et théorique des liquides*, 1873) found the rules empirically: films meet in threes, at 120°, along curves (the Plateau borders), and the borders meet in fours at points, at the tetrahedral angle $\arccos(-\frac{1}{3}) \approx 109.47°$. This is the angle between two lines from the centre of a regular tetrahedron to its vertices, the Tammes angle for $N = 4$ points of Lecture 11 (Section 11.2). Jean Taylor proved a century later that these are the only singularities a soap film can have (gallery, Theorem G.10).

Which cell has the least area? Kelvin (1887) proposed a slightly curved truncated octahedron, the Voronoi cell of the body-centred cubic lattice, with 8 hexagonal and 6 square faces. For a century it was the best known. In 1994 Weaire and Phelan found a better foam by computer [simulation]: a periodic structure with two kinds of cells of equal volume, pentagonal dodecahedra and 14-hedra with 12 pentagons and 2 hexagons, in the ratio 1 : 3. Its area is about 0.3% smaller (Figure G.1). The Weaire–Phelan foam is the Voronoi diagram, relaxed, of the A15 (β-tungsten) structure, one of the *Frank–Kasper* phases of metallurgy, whose Delaunay cells are all nearly regular tetrahedra. So the best known foam is Frank's frustrated tetrahedral packing of Interlude E seen from the dual side. It took until 2012 to make it in the laboratory: monodisperse bubbles tend to crystallize into Kelvin's structure next to flat walls, and Gabbrielli, Meagher, Weaire, Brakke and Hutzler (*Phil. Mag. Lett.* 92, 2012) obtained Weaire–Phelan foam in a container whose walls were shaped to fit it [experiment]. Whether Weaire–Phelan is optimal is open, and so is Kelvin's problem itself.

Matzke (*Amer. J. Botany* 33, 1946) blew a foam of about a thousand equal bubbles one at a time and examined the 600 in its interior. He found on average 13.70 faces per bubble and not a single Kelvin cell [experiment]. Euler's formula gives a feeling for where such numbers come from.

Proposition G.3 (Coxeter's statistical honeycomb)**.** *Let P be a convex polyhedron with exactly three faces at each vertex, and let $\bar{n}$ be the average number of sides of its faces. Then the number of faces is*

$$F = \frac{12}{6 - \bar{n}}.$$

If every face of a foam had $\bar{n} = 2\pi/\arccos\frac{1}{3} \approx 5.104$ sides, the number of regular tetrahedra that fit around an edge, every cell would have $F \approx 13.40$ faces.

Proof. With V vertices and E edges, $3V = 2E$ (three edges at each vertex, two vertices on each edge) and $\bar{n}F = 2E$. Euler's formula $V - E + F = 2$ becomes $F - \frac{E}{3} = 2$, that is $F(1 - \frac{\bar{n}}{6}) = 2$. For the second statement: by Plateau's rules every vertex of a dry foam is simple, and each face of a cell is dual to an edge of the Delaunay complex of the cell centres. The number of sides of the face is the number of Delaunay cells around that edge. If those cells were regular tetrahedra, the dihedral angle $\arccos\frac{1}{3}$ would have to go $\bar{n}$ times around the edge. □

Coxeter (*Illinois J. Math.* 2, 1958) called such idealized averages "statistical honeycombs". The formula is a dictionary: $\bar{n} = 5$ gives the dodecahedron ($F = 12$), Kelvin's cell has $\bar{n} = \frac{8\cdot 6+6\cdot 4}{14} = \frac{36}{7}$ ($F = 14$), Weaire–Phelan has $F = 13.5$, Matzke's bubbles 13.70, and Marvin's shot 14.17. (Voronoi cells of completely random points have on average $2 + \frac{48\pi^2}{35} \approx 15.54$ faces, a theorem of Meijering, *Philips Res. Rep.* 8, 1953.) Five tetrahedra around every edge would give $\bar{n} = 5$ exactly, and then all the cells would be the regular dodecahedra of Proposition G.1. This is impossible in flat space, but it is exactly what happens in the curved space of Section E.5. The Voronoi cells of the 120 vertices of the 600-cell in $\mathbb{S}^3$ are the 120 regular dodecahedra of the 120-cell, spherical dodecahedra whose dihedral angle is exactly 120°, three around each edge. Stephen Hales' "regular dodecahedrons" are what the peas would have become in a pot shaped like the 3-sphere.

In the plane everything is settled: the hexagonal honeycomb is the least-perimeter way to divide the plane into equal areas (Hales, gallery, Theorem G.11). Bubble rafts on water were used by Bragg and Nye (*Proc. R. Soc. A* 190, 1947) as models of two-dimensional crystals, complete with dislocations and grain boundaries.

G.2 Colloids: atoms you can watch

The hard-sphere gas, made visible

Interlude E (Section E.1) treated hard spheres as a system with no energy at all, in which everything is entropy. Atoms are too soft for this, but colloids, particles between about 100 nanometres and a few micrometres across, suspended in a liquid, come remarkably close. The standard hard-sphere colloid is a sphere of poly(methyl methacrylate) carrying a thin brush of polymer that stops the spheres from sticking, suspended in a liquid of nearly the same refractive index, which removes most of the van der Waals attraction and lets light through.

Colloids are big enough to see in an optical microscope, and slow. By the Stokes–Einstein relation a sphere of radius a in a liquid of viscosity η diffuses with $D = k_B T/(6\pi\eta a)$. For $a = 1\,\mu$m in water at room temperature, $D \approx 2 \times 10^{-13}\,\mathrm{m}^2/\mathrm{s}$, and the time to diffuse its own radius is $a^2/D \approx 5$ seconds. An atom in a liquid does the same in picoseconds. So colloids are atoms that can be filmed, and since the only energy scale is $k_B T$, the phase behaviour depends only on the packing fraction φ.

Pusey and van Megen (*Nature* 320, 1986) prepared hard-sphere suspensions at a range of concentrations and watched them for days [experiment]. At low φ the samples stayed fluid; in a window of concentrations they separated into a fluid and a crystal, with the crystal fraction growing across the window; above it they crystallized throughout; and above about $\varphi \approx 0.58$ they did not crystallize at all, but stayed amorphous, a colloidal *glass* (Pusey and van Megen, *Phys. Rev. Lett.* 59, 1987). This is the sequence predicted by the computer for hard spheres, with coexistence between 0.494 and 0.545 (Interlude E) [simulation]. (The volume fractions were in fact calibrated by identifying the onset of freezing with 0.494, so the experiment tests the shape of the phase diagram rather than that number; on this scale melting came out near 0.535. Polydispersity, a few percent in real colloids, also shifts the phase boundaries, and in simulations a size spread above about 7% suppresses crystallization altogether.) Whether hard spheres crystallize at all is still not a theorem in any dimension $d \geq 2$, as Interlude E explained. In microgravity, on the Space Shuttle, dense samples that had stayed glassy on earth for more than a year crystallized completely within two weeks (Zhu et al., *Nature* 387, 1997), so part of the

colloidal "glass" is gravity's doing.

Why order wins: a free-volume calculation

That a system with no energy should *order* when compressed seems paradoxical, since ordering sounds like a loss of entropy. The resolution is that the relevant entropy is the room each ball has to rattle. The following idealized model, the *cell model*, makes this quantitative.

Proposition G.4 (Cell model [model])**.** *Suppose each of N hard balls of diameter 1 in $\mathbb{R}^d$ moves independently in a cage formed by its neighbours, frozen at the sites of a lattice packing scaled from its close-packed density ρ_{cp} to density $\rho = N/V$, and suppose the volume of each cage is $c(a-1)^d$, where $a = (\rho_{\mathrm{cp}}/\rho)^{1/d}$ is the nearest-neighbour distance. Then the entropy per ball is $s = d\ln(a-1) + \ln c$ and the equation of state is*

$$\frac{\beta P}{\rho} = \frac{1}{1-(\rho/\rho_{\mathrm{cp}})^{1/d}}.$$

Proof. The partition function is $Z = (c(a-1)^d)^N$, so $S = \ln Z = Nd\ln(a-1) + N\ln c$. The volume is $V = Na^d/\rho_{\mathrm{cp}}$, so $dV/da = Nda^{d-1}/\rho_{\mathrm{cp}}$. With $\beta P = \partial S/\partial V$ as in Section E.1,

$$\beta P = \frac{\partial S/\partial a}{\partial V/\partial a} = \frac{Nd/(a-1)}{Nda^{d-1}/\rho_{\mathrm{cp}}} = \frac{\rho_{\mathrm{cp}}}{a^{d-1}(a-1)},$$

and dividing by $\rho = \rho_{\mathrm{cp}}/a^d$ gives $\beta P/\rho = a/(a-1) = 1/(1-a^{-1})$. □

In dimension one this is *exactly* the Tonks gas of Proposition E.2: with $\rho_{\mathrm{cp}} = 1$ it reads $\beta P/\rho = 1/(1-\rho)$. In one dimension the cell picture is not a model at all, since the gaps between consecutive rods really are free lengths. In three dimensions the cell model is crude, but not bad: for FCC at the melting density $\varphi = 0.545$ it gives $\beta P/\rho \approx 10.3$, against about 11 in simulations.

Now compare a crystal with an amorphous packing at the same density. The amorphous packing jams at $\varphi_J \approx 0.64$ instead of $\pi/\sqrt{18} \approx 0.7405$, so in the same model its cages have linear size $(\varphi_J/\varphi)^{1/3} - 1$ instead of $(0.7405/\varphi)^{1/3} - 1$. At $\varphi = 0.55$ the ratio of the two cage sizes is 2.01: each crystalline ball has twice the room to move in every direction. The vibrational entropy gained by crystallizing is therefore about $3\ln 2 \approx 2.1\,k_B$ per ball. Against this the disordered state has its *configurational* entropy, the logarithm of the number of distinct disordered arrangements, which wins at low density; roughly speaking, freezing is where the two balance. Order is the high-entropy state of a crowded system.

FCC or HCP?

Every stacking of hexagonal layers (each layer in the hollows of the one below, the Barlow packings) has density $\pi/\sqrt{18}$, so Hales' theorem does not choose between them. Nor does counting: with L layers there are $3\cdot 2^{L-1}$ stacking sequences, but each layer has m balls, so the stacking entropy per ball is $(\ln 3 + (L-1)\ln 2)/(Lm) \to 0$ as $m \to \infty$. The choice is made by the vibrational entropy, which sees the difference between FCC and HCP only at the third shell of neighbours. Bolhuis, Frenkel, Mau and Huse (*Nature* 388, 1997) computed that FCC is favoured, by about $10^{-3}k_BT$ per particle near melting [simulation]; Mau and Huse (*Phys. Rev. E* 59, 1999) found $0.00115(4)\,k_B$ per sphere at close packing, and that FCC has the highest entropy of all stackings. The difference is so small that colloidal crystals grown in the laboratory often stack *randomly* (Pusey, van Megen, Bartlett, Ackerson, Rarity and Underwood, *Phys. Rev. Lett.* 63, 1989) [experiment].

Looking inside. Confocal microscopy locates every particle of a three-dimensional sample, in glasses and in growing crystal nuclei (Gasser et al., *Science* 292, 2001). In two dimensions, colloidal hard spheres confined to a plane show the melting scenario that Bernard and Krauth found in simulations (Section E.1): a first-order liquid–hexatic transition followed by a continuous hexatic–solid transition (Thorneywork, Abbott, Aarts and Dullens, *Phys. Rev. Lett.* 118, 2017) [experiment].

G.3 Opals and photonic crystals

A gemstone made of lattice planes

Precious opal flashes pure colours that change as the stone is turned. In 1964 J. V. Sanders and his colleagues at CSIRO looked at opal in the electron microscope and found a regular array of silica spheres, a few hundred nanometres across, stacked in hexagonal layers (Jones, Sanders and Segnit, *Nature* 204, 1964; Sanders, *Nature* 204, 1964). The colours are Bragg diffraction of light by the layers [experiment]. To see which layers and which colours, we need the planes of a lattice, and those are counted by the dual lattice $\Lambda^* = \{y : \langle x, y\rangle \in \mathbb{Z}\ \forall x \in \Lambda\}$ of Interlude B (Theorem B.15).

Proposition G.5 (Lattice planes)**.** *Let $\Lambda \subset \mathbb{R}^3$ be a lattice and $w \in \Lambda^*$ a primitive vector (not a multiple kw' of some $w' \in \Lambda^*$ with $k \geq 2$). Then Λ is the disjoint union of its intersections with the parallel planes $\{x : \langle x, w\rangle = k\}$, $k \in \mathbb{Z}$, each of which is nonempty, and these planes are $1/|w|$ apart. Conversely, every equally spaced family of parallel planes that covers Λ, each plane containing a lattice point, arises this way, so the largest spacing between lattice planes is $1/\min\{|w| : w \in \Lambda^* \setminus 0\}$.*

Proof. The values $\langle x, w\rangle$, $x \in \Lambda$, form a subgroup $g\mathbb{Z}$ of $\mathbb{Z}$ with $g \geq 1$. Then $\langle x, w/g\rangle \in \mathbb{Z}$ for all $x \in \Lambda$, so $w/g \in \Lambda^*$, and primitivity forces $g = 1$. So every integer k is a value, every lattice point lies on one of the planes, and the planes $\langle x, w\rangle = k$ are $1/|w|$ apart. Conversely, suppose Λ lies on the planes $\langle x, u\rangle \in s\mathbb{Z}$, for a unit vector u and some $s > 0$, and each of these planes contains a lattice point. Then $w = u/s$ has $\langle x, w\rangle \in \mathbb{Z}$ for all $x \in \Lambda$, so $w \in \Lambda^*$, and every integer is a value, so w is not a multiple kw' with $k \geq 2$. □

Example G.6 (The colour of an opal)**.** For $\Lambda = D_3$ the dual is $D_3^* = \mathbb{Z}^3 \cup (\mathbb{Z}^3 + \frac{1}{2}(1,1,1))$, the body-centred cubic lattice, whose shortest vectors are the eight vectors $\frac{1}{2}(\pm1, \pm1, \pm1)$ of length $\frac{\sqrt{3}}{2}$. So the widest-spaced planes of FCC are perpendicular to the body diagonals, at distance $2/\sqrt{3}$: these are the hexagonal close-packed layers. Rescaling from minimal distance $\sqrt{2}$ to sphere diameter D, the layer spacing is

$$d = \frac{2}{\sqrt{3}} \cdot \frac{D}{\sqrt{2}} = D\sqrt{\tfrac{2}{3}},$$

which is also the height of a regular tetrahedron of edge D. The tetrahedron shows that the spacing is the same for every Barlow stacking, so the colour does not care about stacking faults. Bragg's law for light inside a medium of effective refractive index n_{eff} reflects the wavelength

$$\lambda = 2n_{\text{eff}}\, d \sin\theta,$$

where θ is the glancing angle. At normal incidence, with $n_{\text{eff}} \approx 1.45$ for silica, $\lambda \approx 2.37\, D$. The visible range 400–700 nm corresponds to D between about 170 and 300 nm. Turning the stone changes θ, and the colour slides towards the blue.

Photonic band gaps

In 1987 Yablonovitch (*Phys. Rev. Lett.* 58, 2059), who wanted to switch off spontaneous emission, and John (*Phys. Rev. Lett.* 58, 2486), who wanted to localize light, independently proposed to do with photons what a crystal does with electrons. In a medium whose refractive index is periodic with lattice Λ, Maxwell's equations separate into Bloch waves, and the frequencies form bands over the Brillouin zone, the Voronoi cell of Λ^* (up to the factor 2π). A *complete photonic band gap* is an interval of frequencies in which light cannot propagate in any direction.

The Brillouin zone of an opal is the Voronoi cell of BCC: the truncated octahedron, that is, Kelvin's cell from Section G.1. A gap opens on each face of the zone, at the frequency where the Bragg condition for the corresponding planes holds, and for a complete gap the gaps on all faces must overlap. With D_3^* as above, the centres of the hexagonal faces (the points the physicists call L) are at distance $\frac{\sqrt{3}}{4} \approx 0.433$, the centres of the square faces (X) at $\frac{1}{2}$, and the vertices (W) at $\frac{\sqrt{5}}{4} \approx 0.559$. The free-photon frequencies at W and at L differ by the factor $\sqrt{5/3} \approx 1.29$, which is hard to bridge. In fact Ho, Chan and Soukoulis (*Phys. Rev. Lett.* 65, 1990) found that dielectric spheres on an FCC lattice have no complete gap in the lowest bands, because of a band degeneracy at W, while spheres on the *diamond* structure have one, already at an index contrast of about 2 [simulation].

The diamond structure is $D_3^+ = D_3 \cup (D_3 + \frac{1}{2}(1,1,1))$: FCC together with one of its two classes of tetrahedral holes (Interlude H, Section H.3). This is the $n = 3$ case of the construction D_n^+ of Lecture 6 (Section 6.6), which is a lattice only for even n (Exercise B.44). As a packing of balls it is very loose: the nearest distance drops to $\frac{\sqrt{3}}{2}$, there is one ball per unit volume, and the density is $\frac{4}{3}\pi(\frac{\sqrt{3}}{4})^3 = \frac{\pi\sqrt{3}}{16} \approx 0.340$, less than half that of FCC. The best crystal for light is among the worst for packing. Weevils found it first: the iridescent scales of the beetle *Lamprocyphus augustus* contain a diamond-structured photonic crystal of chitin (Galusha et al., *Phys. Rev. E* 77, 2008) [experiment].

Opals can be made to work after all by inverting them. Let colloidal spheres crystallize into a synthetic opal, fill the gaps with a material of high refractive index, and dissolve the spheres. The result, an *inverse opal* of air spheres, has a complete gap between its eighth and ninth bands once the index contrast exceeds about 2.8 (Busch and John, *Phys. Rev. E* 58, 1998) [simulation]. Silicon, with index about 3.5, qualifies, and silicon inverse opals with a complete gap near 1.5 μm, the wavelength of optical fibres, were made by Blanco et al. (*Nature* 405, 2000; a gap of roughly 5% of its centre frequency) and Vlasov, Bo, Sturm and Norris (*Nature* 414, 2001) [experiment]. For a theorem guaranteeing gaps in a model of high-contrast media, see the gallery (Theorem G.18).

G.4 From crystal to glass: random lasers and granular lasers

Take the same colloidal spheres and let them settle quickly, before they can crystallize, or add salt so that they stick on contact. The result is a *photonic glass* (García, Sapienza, Blanco and López, *Adv. Mater.* 19, 2007): a random packing of monodisperse spheres, the optical counterpart of Bernal's ball bearings. Light no longer diffracts; it scatters many times and performs a random walk, with steps of the order of the *transport mean free path* ℓ. Over distances much larger than ℓ, the energy density W of the light diffuses, $\partial_t W = D\Delta W$ with $D = v\ell/3$ (v the speed of light in the medium), the continuum limit of the coin flips in the gas in a box of Lecture 7 (Section 7.3.2).

Now soak the packing in a laser dye and pump it. Letokhov (*Sov. Phys. JETP* 26, 1968)

predicted that a scattering medium with gain will lase without any mirrors, once it is large enough: the scattering keeps the light inside long enough to be amplified. The threshold is a pretty piece of spectral theory.

Proposition G.7 (Letokhov's critical radius [model]). *Let $W(r,t)$ solve $\partial_t W = D\Delta W + \frac{v}{\ell_g} W$ in a ball of radius R in $\mathbb{R}^3$, with $W = 0$ on the boundary, where $D = v\ell/3$ and ℓ_g is the gain length (the distance over which the intensity grows by a factor e). Then every solution decays to 0 if $R < R_{\mathrm{cr}}$ and generic solutions grow exponentially if $R > R_{\mathrm{cr}}$, where*

$$R_{\mathrm{cr}} = \pi \sqrt{\frac{\ell\, \ell_g}{3}}.$$

Proof. Expand in the Dirichlet eigenfunctions $-\Delta\phi_k = \lambda_k \phi_k$ of the ball:

$$W = \sum_k c_k\, e^{(v/\ell_g - D\lambda_k)t} \phi_k.$$

Everything decays if and only if $v/\ell_g < D\lambda_1$. The first eigenvalue is simple, with a positive eigenfunction, which is therefore invariant under rotations. For radial ϕ, the function $u = r\phi$ satisfies $u'' = -\lambda u$ with $u(0) = u(R) = 0$, so $u = \sin(\pi r/R)$ and $\lambda_1 = \pi^2/R^2$. The threshold $v/\ell_g = \frac{v\ell}{3} \cdot \frac{\pi^2}{R^2}$ gives the formula. □

The threshold volume scales like $(\ell\ell_g)^{3/2}$: stronger scattering means a smaller random laser. And ℓ depends on the packing. For independent scatterers, $\ell \approx 1/(n\sigma_{\mathrm{tr}})$, with number density $n = 6\varphi/(\pi D^3)$ and transport cross-section σ_{tr} of order D^2 for grains much larger than the wavelength, so ℓ is of order D/φ and $R_{\mathrm{cr}} \propto \varphi^{-1/2}$: compacting a granular gain medium lowers its threshold. (Near jamming, positional correlations between the grains reduce scattering, and the independent-scatterer estimate has to be corrected by the structure factor of the packing.)

Lawandy, Balachandran, Gomes and Sauvain (*Nature* 368, 1994) saw laser action in a dye solution with titania particles [experiment], and Cao and coworkers (*Phys. Rev. Lett.* 82, 1999) saw sharp lasing modes in zinc oxide powder, where the feedback comes from closed loops of multiply scattered light. In a photonic glass the Mie resonances of the equal spheres can be used to select the lasing wavelength (Gottardo et al., *Nature Photonics* 2, 2008). Wiersma and Cavalieri (*Nature* 414, 2001) built a random laser out of compressed glass powder whose pores were filled with a dye-doped liquid crystal. The alignment of the liquid crystal, and with it the strength of the scattering, depends on temperature, so the laser can be tuned by heating it, and it switches off at the phase transition of the liquid crystal near 42°C [experiment]. This was already a granular laser, tuned through the scattering of its grains.

Granular lasers. Then the packing itself was made to move. Folli, Puglisi, Leuzzi and Conti (*Shaken granular lasers*, *Phys. Rev. Lett.* 108, 2012) immersed about 1500 millimetre-sized spheres in a dye solution, put the container on a vibrating plate and pumped it with a pulsed laser [experiment]. The random laser emission depends on whether and how the grains move: at strong shaking a second emission peak appeared, attributed to transient cavities formed by a few grains. In a sequel (Folli, Ghofraniha, Puglisi, Leuzzi and Conti, *Sci. Rep.* 3, 2013) with 1 mm glass beads in a rhodamine solution, the laser light distinguished a solid-like regime, in which the grains keep their neighbours and the spectra are steady, from a fluidized regime in which the density fluctuations switch the system above and below threshold, so that the spectra

alternate between fluorescence and lasing. The authors call the result a "granular" laser with a tunable emission spectrum: the configuration of the beads, and whether the packing is solid-like or fluidized, is part of the optical cavity.

G.5 Sticky spheres

From hard to sticky

Add small polymer coils to a colloidal suspension and the spheres attract each other: when two spheres come closer than the size of a coil, the coils are excluded from the gap between them, and the osmotic pressure of the coils pushes the spheres together (Asakura and Oosawa, *J. Chem. Phys.* 22, 1954). The range of this *depletion* attraction is the size of the coil, which can be a few percent of the sphere diameter. In the limit of infinitely deep and narrow attraction, Baxter's *sticky hard spheres* (*J. Chem. Phys.* 49, 1968), the energy of a configuration is minus the number of contacts: it is read off the contact graph of Lectures 10–11 (Definition 11.1), now for balls in $\mathbb{R}^3$.

So the ground states of n sticky spheres are the packings of n unit balls with the most contacts. This *contact number problem* is a cousin of the kissing problem, and it is hard. In the plane it is solved: n unit discs have at most $\lfloor 3n - \sqrt{12n-3} \rfloor$ contacts (Harborth, gallery), attained by hexagonal patches. In $\mathbb{R}^3$ only bounds are known (Bezdek and Reid, gallery). For small n the answers come from enumeration.

Rigidity and counting

A cluster that is held together only by its contacts should be rigid. The rigidity matrix A of Interlude E (Section E.2; see also Interlude F, Section F.4) has a row $(\dots, -u_{ij}, \dots, u_{ij}, \dots)$ for each contact. Now there is no box: a free cluster can be translated and rotated.

Proposition G.8 (Maxwell count for clusters [theorem])**.** *Let $n \geq 3$ unit balls in $\mathbb{R}^3$ have centres not all on a line. If the contact framework is infinitesimally rigid (every infinitesimal motion preserving all contact distances to first order is a rigid motion), then it has at least $3n-6$ contacts, and if moreover $n \geq 4$, every ball has at least three contacts.*

Proof. The infinitesimal rigid motions $v_i = b + \omega \times x_i$ form a 6-dimensional subspace of $\ker A$ when the centres are not collinear (a motion with $b + \omega \times x_i = 0$ for three non-collinear x_i is zero). Infinitesimal rigidity says this is all of $\ker A$, so $\operatorname{rank} A = 3n-6$, and A needs at least that many rows. For the second claim, suppose ball i has at most two contacts. Pick $w \neq 0$ orthogonal to its contact directions, and put $v_i = w$ and $v_l = 0$ for $l \neq i$; this lies in $\ker A$. A rigid motion vanishing at three non-collinear centres is zero, so if this v is a rigid motion, the other $n-1 \geq 3$ centres lie on a line L, and $x_i \notin L$. Then the other balls form a straight chain along L, and an end ball of the chain touches only its neighbour in the chain and possibly ball i. The centres other than this end ball include x_i and two points of L, so they are not collinear, and the same construction at the end ball gives an element of $\ker A$ that is not a rigid motion. □

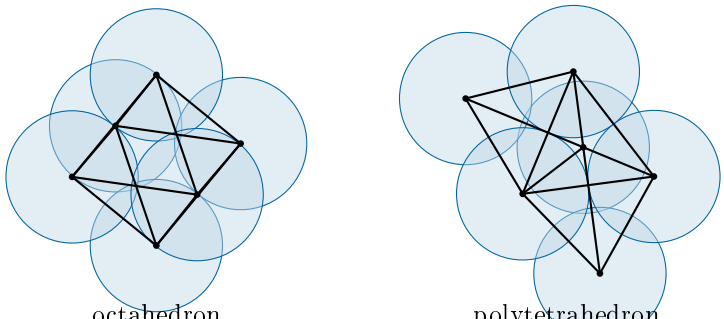
For clusters that are rigid but not infinitesimally rigid the second claim can fail: a ball squeezed exactly between two balls at distance 2 of an otherwise rigid cluster is pinned by its two contacts.

Clusters with exactly $3n-6$ contacts that are rigid are called *isostatic* (or minimally rigid), those with more *hyperstatic*, and rigid ones with fewer *hypostatic*. A rigid cluster that is not infinitesimally rigid is *singular*. Arkus, Manoharan and Brenner (*Phys. Rev. Lett.* 103, 2009; *SIAM*

J. Discrete Math. 25, 2011) enumerated the candidate contact graphs (adjacency matrices with at least $3n-6$ edges and minimum degree 3, the two conditions suggested by Proposition G.8), and solved for the distances with geometric rules. Hoy, Harwayne-Gidansky and O'Hern (*Phys. Rev. E* 85, 2012) went to $n = 11$, and Holmes-Cerfon (*SIAM Review* 58, 2016) to $n = 14$ by a different method, walking from one rigid cluster to the next along the one-parameter families obtained by breaking a contact. Her counts:

n	6	7	8	9	10	11	12	13	14
rigid clusters	2	5	13	52	263	1659	11980	98529	895478
of which hypostatic	0	0	0	0	1	20	159	1308	11245
of which hyperstatic	0	0	0	0	3	21	183	1411	11241
most contacts	12	15	18	21	25	29	33	36	40

These are numerical results [simulation], believed complete apart possibly from a few highly singular clusters, but not proved complete. The first singular cluster appears at $n = 9$, and the first clusters with more than $3n-6$ contacts at $n = 10$: there the ground states stop being isostatic and become fragments of the hexagonal close packing. Even the statement that $3n-6$ is the maximal number of contacts for $6 \le n \le 9$ is, strictly speaking, a conjecture, conditional on the completeness of these lists (Bezdek and Khan, arXiv:1601.00145).



octahedron
12 contacts, $\sigma = 24$

polytetrahedron
12 contacts, $\sigma = 2$

Figure G.2: The two rigid clusters of six sticky spheres, with their contact graphs. The octahedron is a piece of FCC (six balls around an octahedral hole). The polytetrahedron is three regular tetrahedra glued face to face; the view is along the normal of one face of the middle tetrahedron. Both have $3n-6 = 12$ contacts, so they have the same energy, but in the sticky-limit model the polytetrahedron is 24 times as likely (Example G.9).

Six spheres: entropy chooses the less symmetric cluster

For $n = 6$ there are two ground states, both with 12 contacts (Figure G.2). They have the same energy, so which one forms is decided by entropy, and the entropy has three parts. Holmes-Cerfon, Gortler and Brenner (*Proc. Natl. Acad. Sci. USA* 110, 2013) showed that in the sticky limit the partition function of a rigid, infinitesimally rigid cluster with m contacts is, up to factors common to all clusters of the same size,

$$z = \kappa^m \cdot \frac{\sqrt{\det I}}{\sigma \sqrt{\det(AA^T)}} \qquad (m = 3n-6), \tag{G.1}$$

where κ is a "stickiness" parameter that depends on the depth and range of the attraction, I is the moment of inertia tensor of the centres (unit masses), σ is the order of the rotation group of the cluster, and A is the rigidity matrix. The three factors are rotational entropy (a cluster with more symmetry has fewer distinguishable orientations, and a more spread-out cluster has more), and vibrational entropy: $\det(AA^T)$ is the product of the nonzero eigenvalues of the stiffness matrix A^TA, and softer clusters wiggle more.

Example G.9 ($n = 6$ [model]). For the octahedron, with centres $\pm\frac{1}{\sqrt{2}}e_i$, the rotation group is that of the cube, $\sigma = 24$, and $I = 2 \cdot \mathrm{Id}$, so $\det I = 8$. The 12×12 matrix AA^T has entries $\langle u_e, u_{e'}\rangle$ (with the signs of the shared endpoints) and $\det(AA^T) = 864$. For the polytetrahedron, with centres

$$(0,0,0),\ (1,0,0),\ (\tfrac{1}{2},\tfrac{\sqrt{3}}{2},0),\ (\tfrac{1}{2},\tfrac{\sqrt{3}}{6},\tfrac{\sqrt{6}}{3}),\ (\tfrac{1}{2},-\tfrac{7\sqrt{3}}{18},\tfrac{2\sqrt{6}}{9}),\ (\tfrac{4}{3},\tfrac{4\sqrt{3}}{9},\tfrac{2\sqrt{6}}{9}),$$

only a half-turn survives, $\sigma = 2$, and an exact computation gives $\det I = \frac{261569}{26244} \approx 9.967$ and $\det(AA^T) = \frac{261569}{972} \approx 269.1$. Hence

$$\frac{z_{\mathrm{poly}}}{z_{\mathrm{oct}}} = \frac{24}{2} \cdot \sqrt{\frac{\det I_{\mathrm{poly}}}{\det I_{\mathrm{oct}}} \cdot \frac{\det(AA^T)_{\mathrm{oct}}}{\det(AA^T)_{\mathrm{poly}}}} = 12 \cdot \sqrt{\frac{972}{26244} \cdot \frac{864}{8}} = 12 \cdot \sqrt{\frac{108}{27}} = 24.$$

Symmetry contributes the factor 12, and the moments of inertia and the vibrations together exactly 2 (about 1.12 and 1.79). So the model predicts that the octahedron forms with probability $\frac{1}{25} = 4\%$.

Meng, Arkus, Brenner and Manoharan (*Science* 327, 2010) assembled clusters of colloidal spheres with a short-range attraction in microscopic wells and counted them [experiment]: the polytetrahedron was observed more than 20 times as often as the octahedron, so the octahedron made up only about 4–5% of the six-particle clusters, in good agreement with the 4% (a ratio of 24) of Example G.9. For clusters of up to ten particles, highly symmetric clusters were strongly suppressed by rotational entropy. So the cluster that is a piece of the crystal loses, and the polytetrahedral one wins, as in Frank's picture of liquids (Section E.3), though here the cause is rotational entropy rather than energy. In sticky systems the crystal comes back at $n \geq 10$, through the hyperstatic clusters, which gain a factor κ for every extra contact. Kallus and Holmes-Cerfon (*Phys. Rev. E* 95, 2017) extended (G.1) to singular clusters, whose zero-frequency modes make the harmonic formula diverge, computed the partition functions of all known rigid clusters with $n \leq 19$, and found that the hyperstatic ones, which look like fragments of a close-packed lattice, dominate the landscape: this is how crystalline order emerges from sticky spheres, already at $n = 10$.

The configuration space is stratified. The space of n sticky spheres modulo rigid motions is a union of strata, one for each contact graph G, of dimension $3n - 6 - |E(G)|$ when G is regular. Rigid clusters are the 0-dimensional strata, and breaking one contact of an isostatic cluster gives a 1-dimensional stratum, a path to other clusters. This is the configuration space of Lectures 7–10 (and Interlude D, Section D.3) cut up by contact graphs instead of by levels of the min-distance function; see Holmes-Cerfon's review (*Annu. Rev. Condens. Matter Phys.* 8, 2017).

G.6 Minimal rigidity and random jamming

Near the jamming point

Interlude E showed that a first-order jammed packing of frictionless spheres has at least $2d$ contacts per ball on average, and that jammed packings found in simulations have exactly this many: they are isostatic at the jamming threshold $\varphi_J \approx 0.64$ (point J of O'Hern, Silbert, Liu and Nagel). What happens just above φ_J? Make the spheres slightly soft, with a repulsion proportional to the overlap (harmonic spheres), and compress. Then (Figure G.3) [simulation]

$$z - 2d \;\propto\; (\varphi - \varphi_J)^{1/2}, \qquad P \;\propto\; \varphi - \varphi_J,$$

found for a bubble model of foams by Durian (*Phys. Rev. Lett.* 75, 1995) and in detail by O'Hern et al. (*Phys. Rev. E* 68, 2003). In the laboratory, Majmudar, Sperl, Luding and Behringer (*Phys. Rev. Lett.* 98, 2007) measured the contact numbers of photoelastic discs, which light up under stress, and found a jump at φ_J followed by growth with exponent about 0.5, and pressure growing with exponent about 1.1 [experiment]. The square root, rather than the linear growth one might expect, says that new contacts are cheap: the isostatic packing is marginal.

There is a pretty counting argument behind the marginality, due to Wyart, Nagel and Witten (*Europhys. Lett.* 72, 2005) [model]. Cut a cube of side L (in units of the ball diameter) out of a packing with $z = 2d + \Delta z$ and free its boundary. The contacts crossing the boundary, about L^{d-1} of them, are lost, and in an isostatic packing each lost contact releases one floppy mode. The interior has about $\frac{1}{2}\Delta z\, L^d$ redundant contacts to block them. So floppy modes exist in boxes of size up to

$$\ell^* \sim \frac{1}{\Delta z} \sim (\varphi - \varphi_J)^{-1/2},$$

a length that diverges at the jamming point. Extended soft modes of this size explain the excess of low-frequency vibrations (the "boson peak") of jammed packings.

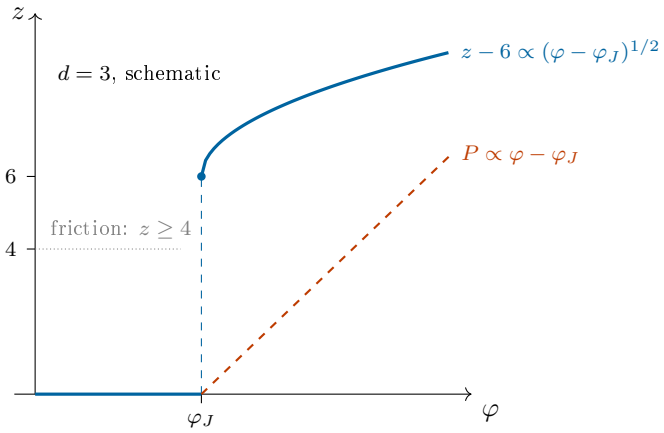


Figure G.3: Schematic behaviour of soft frictionless spheres near the jamming point in $d = 3$: below φ_J there are no contacts (after rattlers are removed); at φ_J the mean contact number jumps to the isostatic value $2d$; above it the excess grows like a square root, while the pressure of harmonic spheres grows linearly. Frictional packings can be stable with fewer contacts.

Friction, and the loosest random packing

Real grains have friction, and friction lowers the number of contacts needed. The counting is the same as in Corollary E.7, but with forces instead of motions [model]. A frictional contact carries a force *vector* with d components (normal and tangential), so a packing with mean coordination z has $\frac{zN}{2}\cdot d$ unknown force components. Each ball must satisfy force balance (d equations) and torque balance ($\binom{d}{2}$ equations), $N\frac{d(d+1)}{2}$ in all. For generic packings to carry a load we need at least as many unknowns as equations, $z \geq d+1$. Without friction a contact carries only a normal force and there is no torque on a sphere, so $\frac{zN}{2} \geq dN$, which is $z \geq 2d$ again. On the other hand each contact is one equation on the positions, whether or not there is friction, so for generic sizes $\frac{zN}{2} \leq dN$, as in Interlude E. Hence

$$d+1 \;\leq\; z \;\leq\; 2d,$$

with $z = d+1$ approached for very rough grains and $z = 2d$ for frictionless ones: $4 \leq z \leq 6$ in three dimensions.

Fewer contacts allow looser packings. Onoda and Liniger (*Phys. Rev. Lett.* 64, 1990) let glass spheres settle slowly through liquids of nearly the same density and extrapolated to zero effective gravity, finding a *random loose packing* fraction 0.555 ± 0.005 [experiment]. Song, Wang and Makse (*Nature* 453, 2008) proposed a mean-field theory of jammed packings in which the packing fraction is tied to the mean geometric coordination by $\varphi = z/(z+2\sqrt{3})$ [model]. It gives $\varphi = 6/(6+2\sqrt{3}) \approx 0.634$ for $z = 6$ (frictionless, random close packing) and $4/(4+2\sqrt{3}) \approx 0.536$ for $z = 4$ (infinite friction, random loose packing), and a phase diagram of jammed matter between these two lines. With adhesion the constraint count drops further, and simulations of small sticky grains give much looser stable packings; Liu, Li, Baule and Makse (*Soft Matter* 11, 2015) conjecture a limiting loosest packing at $z = 2$ with $\varphi = 1/2^3$, which their simulations approach but do not reach [model].

How packings are made

Slow sedimentation gives loose packings; vibration, shearing and tapping compact them. Knight, Fandrich, Lau, Jaeger and Nagel (*Phys. Rev. E* 51, 1995) tapped glass beads in a tall narrow tube and found that the density approaches its final value only logarithmically in the number of taps t [experiment],

$$\varphi(t) = \varphi_\infty - \frac{\Delta\varphi_\infty}{1 + B\ln(1 + t/\tau)},$$

with φ_∞ between about 0.637 and 0.647 in the fits of Nowak, Knight, Ben-Naim, Jaeger and Nagel (*Phys. Rev. E* 57, 1998). So tapping creeps towards random close packing and essentially stops there. Once the beads are annealed, cycling the tapping intensity up and down retraces a reversible curve of density against intensity (Nowak et al.), a standard testing ground for Edwards' statistical mechanics of powders (Section E.6).

G.7 Dice, candies and Ulam's conjecture

Counting for other shapes

Lecture 6 ended with an aside on packing ellipsoids and polygons (Section 6.5). In the laboratory, non-spherical particles are the rule. The counting of Section G.6 goes through for frictionless

convex particles with f degrees of freedom each: an isostatic packing has $2f$ constraints per particle on average. Spheres have $f = 3$; spheroids (ellipsoids of revolution) $f = 5$, since rotation about the axis does nothing; general ellipsoids and tetrahedra $f = 6$.

Donev, Cisse, Sachs, Variano, Stillinger, Connelly, Torquato and Chaikin (*Science* 303, 2004) poured M&M candies, which are oblate spheroids, into a flask and measured a random packing fraction of 0.685 ± 0.01, well above the 0.64 of spheres [experiment]. Their simulations gave about 0.71 for spheroids and up to 0.735 for general ellipsoids with aspect ratio about 1.3 [simulation]. The packing fraction has a cusp-like minimum at the sphere: any small deformation helps. The M&Ms had 9.82 contacts on average, close to $2f = 10$, but jammed general ellipsoids in the simulations had at most about 11.4, short of $2f = 12$. Such *hypostatic* packings are possible because the equivalence of jamming and first-order jamming, which Interlude E noted is special to spheres, fails for ellipsoids: the curvature of the surfaces can block motions at second order, so a jammed packing of ellipsoids need not satisfy the first-order count.

Tetrahedral dice

Aristotle believed that regular tetrahedra fill space, and Section E.4 explained why they do not even fill the space around an edge. Jaoshvili, Esakia, Porrati and Chaikin (*Phys. Rev. Lett.* 104, 2010) poured tetrahedral dice, of the kind used in role-playing games, into containers and measured, by volume and by magnetic resonance imaging of the interior, a random packing fraction of 0.76 ± 0.02 [experiment]: denser than random spheres (by far) and even denser than the densest sphere packing. A die touches only about 6.3 neighbours on average, far below $2f = 12$. But contacts between polyhedra come in kinds: a face-to-face contact blocks one translation and two rotations, so counts as three constraints, an edge-to-face contact as two, and a vertex-to-face or edge-to-edge contact as one. With these weights the dice had about 12 constraints per particle, isostatic again.

The densest packings of regular tetrahedra were a race between 2006 and 2010 [simulation]. The densest lattice packing, of density $\frac{18}{49} \approx 0.367$ (Hoylman 1970), is far from optimal; Conway and Torquato (2006) found 0.72, Chen (2008) 0.7786, and then Haji-Akbari, Engel, Keys, Zheng, Petschek, Palffy-Muhoray and Glotzer (*Nature* 462, 2009), compressing tetrahedra on the computer, found that they spontaneously form a *dodecagonal quasicrystal*, a cousin of the cut-and-project sets of Interlude C (Section C.6), whose periodic approximant reaches 0.8503. Within months the record passed to explicit periodic packings built from pairs of tetrahedra glued along a face: $\frac{100}{117} \approx 0.8547$ (Kallus, Elser and Gravel), $\frac{12250}{14319} \approx 0.8555$ (Torquato and Jiao) and $\frac{4000}{4671} \approx 0.8563$ (Chen, Engel and Glotzer, gallery), the densest known. These constructions are exact, but nobody knows whether 0.8563 is optimal; the best upper bound is still astronomically close to 1 (Gravel, Elser and Kallus, gallery). Lagarias and Zong (*Notices AMS* 59, 2012) tell the story.

Is the ball the worst?

Every one of these shapes packs more densely than the ball. Ulam suspected that this is always so: among convex bodies in $\mathbb{R}^3$, the ball has the smallest optimal packing density (reported by Martin Gardner, 1995). This is *Ulam's packing conjecture*, and it is open. Numerical work has ruled out many candidates, including the regular tetrahedron. The best result is local: Kallus (gallery, Theorem G.17) proved that every origin-symmetric convex body sufficiently close to the ball, but not a ball, packs more densely than the ball, so the ball is a *local pessimum*. In the

plane the disc is not the worst: the regular heptagon is conjectured to be, and among centrally symmetric convex bodies Reinhardt (1934) conjectured that the worst is a smoothed octagon, of density $\frac{8-4\sqrt{2}-\ln 2}{2\sqrt{2}-1} \approx 0.9024 < \frac{\pi}{\sqrt{12}} \approx 0.9069$. For centrally symmetric bodies in the plane the densest packing is a lattice packing (the aside of Lecture 6), so Reinhardt's problem is a problem about lattices; it is open too.

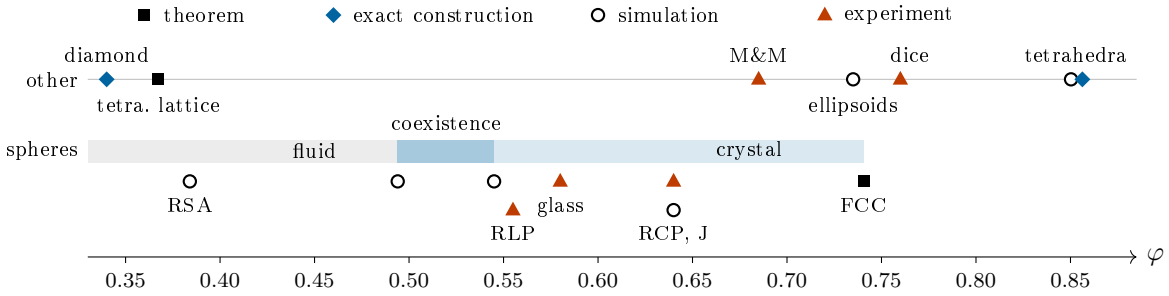


Figure G.4: Packing fractions of this interlude and of Interlude E, by status. Spheres: random sequential adsorption (0.384), the fluid–crystal coexistence region (0.494–0.545), random loose packing (0.555), the colloidal glass (about 0.58), random close packing and point J (about 0.64), and the Kepler bound $\pi/\sqrt{18}$. Other shapes: the diamond packing of balls ($\pi\sqrt{3}/16$), the densest lattice packing of regular tetrahedra (18/49), M&M candies, simulated random ellipsoids, tetrahedral dice, the tetrahedral quasicrystal approximant (0.8503) and the densest known tetrahedron packing (4000/4671).

G.8 Gallery of theorems

Results quoted in Interludes E and F (Hales, Hales–McLaughlin, Connelly et al., Richthammer, Theil) are not repeated.

Theorem G.10 (J. E. Taylor, Ann. of Math. 103 (1976))**.** *Every soap-film-like surface in* $\mathbb{R}^3$*, in the sense of Almgren's* $(\mathbf{M}, \varepsilon, \delta)$*-minimal sets, is locally either a smooth minimal surface, or three smooth sheets meeting at* $120°$ *along a curve, or four such curves meeting at a point at angles* $\arccos(-\frac{1}{3})$*, with six sheets, in the shape of the cone over the edges of a regular tetrahedron.*

Plateau's experimental laws (Section G.1) as a theorem. The tetrahedral angle is the Tammes angle for $N = 4$.

Theorem G.11 (T. C. Hales, Discrete Comput. Geom. 25 (2001))**.** *Any partition of the plane into regions of equal area has perimeter at least that of the regular hexagonal honeycomb (in the sense of an averaged limit over large discs).*

The two-dimensional Kelvin problem is solved; the three-dimensional one is open.

Question G.12 (Kelvin's problem, 1887)**.** Which partition of $\mathbb{R}^3$ into cells of unit volume has the least area per cell? Is it the Weaire–Phelan foam?

Kelvin's own candidate was beaten by 0.3% in 1994, numerically but with a margin far beyond the numerical error. That Weaire–Phelan is optimal is open.

Theorem G.13 (H. Harborth, Elem. Math. 29 (1974)). *The maximal number of touching pairs among n non-overlapping unit discs in the plane is $\lfloor 3n - \sqrt{12n-3} \rfloor$.*

The ground states of n sticky discs. For $n = 7$ (a hexagon around a disc) the formula gives $21 - 9 = 12$.

Theorem G.14 (K. Bezdek and S. Reid, J. Geom. 104 (2013)). *Every packing of $n > 2$ unit balls in $\mathbb{R}^3$ has fewer than $6n - 0.926\, n^{2/3}$ touching pairs.*

A lower bound for the sticky-sphere energy, with a surface term of the right order $n^{2/3}$. Exact values are known only for $n \leq 5$ (Section G.5).

Theorem G.15 (E. R. Chen, M. Engel and S. C. Glotzer, Discrete Comput. Geom. 44 (2010)). *There is a periodic packing of regular tetrahedra of density $\frac{4000}{4671} \approx 0.856347$, with four tetrahedra (two face-sharing dimers) per fundamental domain. It is optimal within a three-parameter family of dimer packings.*

Found by numerical compression, then made exact; its global optimality is open. (Lattice packings, whose tetrahedra all point the same way, reach only $\frac{18}{49}$: Hoylman, *Bull. Amer. Math. Soc.* 76, 1970.)

Theorem G.16 (S. Gravel, V. Elser and Y. Kallus, Discrete Comput. Geom. 46 (2011)). *The packing density of regular tetrahedra is at most $1 - 2.6\cdots \times 10^{-25}$.*

A quantitative version of Section E.4; the gap between 0.856 and $1 - 10^{-25}$ shows how hard upper bounds are.

Theorem G.17 (Y. Kallus, Adv. Math. 264 (2014)). *There is $\varepsilon > 0$ such that every origin-symmetric convex body $K \subset \mathbb{R}^3$ that is not a ball, and whose asphericity is less than ε, has packing density greater than $\pi/\sqrt{18}$. In dimensions 4 to 8 and 24 there are origin-symmetric bodies of arbitrarily small asphericity whose lattice packing density is smaller than that of the ball.*

The best evidence for Ulam's conjecture. In dimensions 4–8 and 24 the best lattices for balls ($D_4, \ldots, E_8$ of Lectures 4–6, and Leech) are so good that deforming the ball can make things worse.

Theorem G.18 (A. Figotin and P. Kuchment, SIAM J. Appl. Math. 56 (1996)). *Consider the periodic medium in $\mathbb{R}^d$, $d = 2, 3$, made of cubes of air, $[\delta, 1]^d + \mathbb{Z}^d$, separated by walls of width δ of a material with dielectric constant $\varepsilon > 1$, and the operator $\nabla \cdot \varepsilon(x)^{-1}\nabla$. For any N, if δ is small enough and ε large enough (suitably coupled), the spectrum in $[0, N]$ lies in small intervals around the Neumann eigenvalues of the cube. In particular the spectrum has gaps.*

Band gaps for high contrast, as a theorem (for acoustic waves, or 2-dimensional photonics); the Maxwell case of inverse opals rests on computation (Section G.3).

Exercises

Exercise G.19. (a) Show that the dihedral angle of the regular dodecahedron is $\arccos(-1/\sqrt{5})$. (b) Show that the truncated octahedron is the Voronoi cell of the body-centred cubic lattice $\mathbb{Z}^3 \cup (\mathbb{Z}^3 + \frac{1}{2}(1,1,1))$, and verify the values 5.345 and 5.315 of $A/V^{2/3}$ in Figure G.1 for it and for the rhombic dodecahedron.

Exercise G.20. (a) Use Proposition G.3 to find the average number of sides of a face for Kelvin's cell, for the Weaire–Phelan cells, and for Matzke's bubbles. (b) The rhombic dodecahedron is not simple. Find the analogue of the formula $F = 12/(6 - \bar{n})$ for polyhedra in which every vertex has degree 3 or 4, and check it on the rhombic dodecahedron. (c) Show that a convex polyhedron with three faces at each vertex and only pentagonal and hexagonal faces has exactly 12 pentagons.

Exercise G.21. In the cell model of Proposition G.4, at $\varphi = 0.60$, compare the cage sizes of FCC and of an amorphous packing jamming at 0.64, and estimate the vibrational entropy difference per ball.

Exercise G.22. (a) Show that the number of Barlow stackings of L labelled layers is $3 \cdot 2^{L-1}$. (b) Show that in every Barlow stacking each ball has 12 touching neighbours, and that its Voronoi cell is either the rhombic or the trapezo-rhombic dodecahedron. (c) Using Proposition G.5, find the second largest spacing of lattice planes in FCC, and the corresponding Bragg wavelength of an opal at normal incidence.

Exercise G.23. (a) Redo Proposition G.7 for a cube of side L (Dirichlet conditions on its boundary), and show that the threshold is $L = \pi\sqrt{\ell\ell_g}$. (b) With $\ell = 1/(n\sigma)$ and $n = 6\varphi/(\pi D^3)$, how does the critical number of grains in a ball of radius R_{cr} depend on φ?

Exercise G.24. (a) Check that the six points listed in Example G.9 form three regular tetrahedra glued face to face, and that they have exactly 12 contacts. (b) Verify $\det I = 8$ and $\det(AA^T) = 864$ for the octahedron. (*Hint:* $AA^T = 2\,\mathrm{Id} + B$, where B records the angles between contacts sharing a ball.) (c) For $n = 5$ show that a rigid cluster must be the triangular bipyramid, with 9 contacts.

Exercise G.25. (a) Carry out the frictional counting of Section G.6 for discs in the plane: show that $3 \le z \le 4$. (b) For frictionless tetrahedra, explain why the isostatic count is 12 constraints per particle, and check it against the contact weights of Section G.7.

Exercise G.26. Check Harborth's formula for the hexagonal patches $n = 3k^2 - 3k + 1$: show that such a patch has $3(3k^2 - 5k + 2)$ contacts and that this equals $\lfloor 3n - \sqrt{12n - 3} \rfloor$.

Further reading

- H. S. M. Coxeter, *Close-packing and froth*, Illinois J. Math. 2 (1958). The statistical honeycombs of Proposition G.3.
- D. Weaire (ed.), *The Kelvin Problem: Foam Structures of Minimal Surface Area*, Taylor and Francis, 1996; D. Weaire and S. Hutzler, *The Physics of Foams*, Oxford, 1999.
- P. N. Pusey et al., *Hard spheres: crystallization and glass formation*, Phil. Trans. R. Soc. A 367 (2009), arXiv:0908.1663.
- J. D. Joannopoulos, S. G. Johnson, J. N. Winn and R. D. Meade, *Photonic Crystals: Molding the Flow of Light*, 2nd ed., Princeton, 2008.
- D. S. Wiersma, *The physics and applications of random lasers*, Nature Physics 4 (2008).
- M. Holmes-Cerfon, *Sticky-sphere clusters*, Annu. Rev. Condens. Matter Phys. 8 (2017), arXiv:1709.05138; and *Enumerating rigid sphere packings*, SIAM Review 58 (2016).
- A. J. Liu and S. R. Nagel, *Jamming is not just cool any more*, Nature 396 (1998).
- J. C. Lagarias and C. Zong, *Mysteries in packing regular tetrahedra*, Notices AMS 59 (2012).

Coda

Chapter H

Interlude H: Dynkin diagrams everywhere

[Added] This interlude is new: it is not in the handwritten notes. It collects the places where the Dynkin diagrams of Lectures 4–6 turn up again, in these notes and outside them, and asks in each place what the diagram lets us read off. To keep it readable it is set in black; all of it should be read as added material.

Lecture 5 drew Dynkin diagrams in order to *classify* root systems. Once the list was on the board, the diagrams had done their job. But they kept coming back. Interlude A found them inside matrix algebras, Interlude C folded them into quasicrystals, Interlude D turned them into braid groups. This last interlude takes the diagrams as the main characters and asks a few concrete questions about objects we already know.

- The inequality $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$ that ended the classification in Lecture 5: is it the same inequality that lists the Platonic solids?
- Interlude E proved that the icosahedron is not made of regular tetrahedra. Can the diagram of the icosahedron see that?
- The FCC lattice has octahedral and tetrahedral holes, a cubic and a tetrahedral description, and the cuboctahedron as its kissing configuration. Which of these are visible on the diagram ○–○–○?
- Lecture 6 built E_8 by filling the deep holes of D_8. Can we *see* the holes of E_8 itself?

The answers are all yes, and most of them come from one extra node: the extended diagram of Interlude A (Section A.9, item 7). At the end we tour the other places where the ADE list appears (singularities, quivers, conformal field theory, subfactors, cubic surfaces), and record in each case what the diagram says there.

Throughout, simply laced root systems are normalized with roots of norm $\langle\alpha,\alpha\rangle=2$, so that the Cartan matrix C is the Gram matrix of the simple roots. We use the labelling of E_8 from Lecture 6 (Section 6.2): $\alpha_1,\dots,\alpha_6$ is a chain, $\alpha_7=e_6+e_7$ is attached to α_5, and $\alpha_8=-\frac{1}{2}(1,\dots,1)$ is attached to α_7.

H.1 One inequality, three times

The branch point, as a determinant

Recall the last step of the classification (Section 5.2). A diagram with one branch point has a centre x and three arms of $p-1$, $q-1$, $r-1$ nodes. Call it $T_{p,q,r}$ (Figure H.1); it has $p+q+r-2$ nodes. The lecture projected x onto the span of the arms and found that the diagram is admissible exactly when $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$. The same computation, done with the Cartan matrix, gives an exact formula.

Proposition H.1. *The Cartan matrix of $T_{p,q,r}$ (all edges simple, roots of norm 2) has determinant*

$$\det C(T_{p,q,r}) = pqr\Big(\frac{1}{p}+\frac{1}{q}+\frac{1}{r}-1\Big).$$

In particular $C(T_{p,q,r})$ is positive definite if and only if $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$.

Proof. Order the nodes with the centre first. Then $C=\left(\begin{smallmatrix}2 & b^T\\ b & C'\end{smallmatrix}\right)$, where $C'=C(A_{p-1})\oplus C(A_{q-1})\oplus C(A_{r-1})$ is the Cartan matrix of the three arms and b has an entry -1 at the three arm nodes next to the centre. The Schur complement formula gives

$$\det C = \det C'\cdot\big(2-b^TC'^{-1}b\big).$$

Two facts about the chain A_k finish the job: $\det C(A_k)=k+1$ (expand along the last row: $d_k=2d_{k-1}-d_{k-2}$), and the corner entry of $C(A_k)^{-1}$ is $d_{k-1}/d_k=k/(k+1)$. Hence

$$\det C = pqr\Big(2-\frac{p-1}{p}-\frac{q-1}{q}-\frac{r-1}{r}\Big)=pqr\Big(\frac{1}{p}+\frac{1}{q}+\frac{1}{r}-1\Big).$$

Since C' is positive definite, C is positive definite exactly when the Schur complement is positive. □

The three terms $\frac{p-1}{p}$ are, up to the factor 2 from our normalization, the squared projections $\frac{1}{2}(1-\frac{1}{p})$ of the lecture. So the proof of Lecture 5 *was* a Schur complement. The formula is more than a test of positivity, though. Plugging in:

(p,q,r)	$(1,q,r)$	$(2,2,n)$	$(2,3,3)$	$(2,3,4)$	$(2,3,5)$	$(3,3,3),(2,4,4),(2,3,6)$	$(2,3,7)$
$T_{p,q,r}$	A_{q+r-1}	D_{n+2}	E_6	E_7	E_8	$\tilde{E}_6,\ \tilde{E}_7,\ \tilde{E}_8$	E_{10}
$\det C$	$q+r$	4	3	2	1	0	-1

These are the determinants $\det C$ that Interlude A used to read off the packing densities of the root lattices, now all from one formula. On the E-series $E_n=T_{2,3,n-3}$ the formula is linear:

$$\det C(E_n) = 6(n-3)\Big(\frac{1}{n-3}-\frac{1}{6}\Big)=9-n. \tag{H.1}$$

At $n=9$ the determinant vanishes. $E_9=T_{2,3,6}$ is the extended diagram $\tilde{E}_8$, the diagram of the affine Weyl group that tiles $\mathbb{R}^8$ by alcoves (Section A.8). At $n=10$ the form becomes indefinite, of signature $(9,1)$. The determinant -1 says the E_{10} lattice is unimodular; it is the even unimodular Lorentzian lattice $\mathrm{II}_{9,1}$.

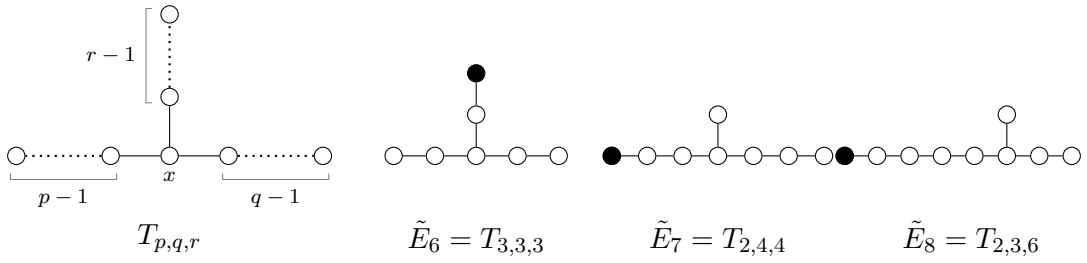


Figure H.1: The diagram $T_{p,q,r}$, with arms of $p-1$, $q-1$, $r-1$ nodes, and the three cases with $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}=1$. These are the extended diagrams $\tilde{E}_6, \tilde{E}_7, \tilde{E}_8$; the black node is the extra (affine) node.

Triangles on the sphere

Now forget Dynkin diagrams and draw a triangle on the unit sphere with angles $\frac{\pi}{p}$, $\frac{\pi}{q}$, $\frac{\pi}{r}$. By Gauss–Bonnet its area is the angle excess,

$$\text{area} = \pi\Big(\frac{1}{p}+\frac{1}{q}+\frac{1}{r}-1\Big),$$

so the triangle exists on the sphere exactly when $\frac{1}{p}+\frac{1}{q}+\frac{1}{r}>1$. It exists in the Euclidean plane when the sum equals 1, and in the hyperbolic plane when it is less than 1. The reflections in its three sides generate a group that tiles the sphere by copies of the triangle. There are $4\pi/\text{area}$ copies, and the rotations (products of two reflections) form a subgroup of index 2:

$$\big|G^+_{p,q,r}\big| = \frac{2}{\frac{1}{p}+\frac{1}{q}+\frac{1}{r}-1}.$$

For $(2,3,3)$, $(2,3,4)$, $(2,3,5)$ this gives 12, 24, 60: the rotation groups of the tetrahedron, the cube (or octahedron), and the icosahedron (or dodecahedron). For $(2,2,n)$ it gives the dihedral group of order $2n$, the symmetries of a regular n-gon viewed as a flat polyhedron (a "dihedron"). The integers p,q,r are the orders of the rotation axes: for the icosahedron, 2 through the edge midpoints, 3 through the face centres and 5 through the vertices. The Euclidean cases $(3,3,3)$, $(2,4,4)$, $(2,3,6)$ are the reflection groups of the equilateral triangle, the isosceles right triangle and the 30–60–90 triangle, the kaleidoscopes of the hexagonal and square lattices.

Comparing with Proposition H.1 we get a curious identity:

$$\det C(T_{p,q,r}) = \frac{2pqr}{\big|G^+_{p,q,r}\big|} \qquad \text{e.g.} \quad \det C(E_8) = \frac{2\cdot 30}{60} = 1, \quad \det C(E_6) = \frac{2\cdot 18}{12} = 3.$$

So *the branch diagrams D_n, E_6, E_7, E_8 are the dihedra and the Platonic solids.* (The chains $A_n = T_{1,q,r}$ do not fit this pattern on the nose; Section H.7 matches them with the cyclic groups.) The arms of E_8 have lengths $1,2,4$ because the icosahedron has rotations of orders $2,3,5$. So far this is a coincidence of numbers. Section H.7 explains it: the resolution of the singularity $\mathbb{C}^2/\Gamma$, for Γ the binary icosahedral group, *is* a $T_{2,3,5}$ of curves.

Coxeter's criterion

There is a third appearance, which puts the first two in one frame. A *Coxeter diagram* has a node for each generator s_i and an edge labelled m_{ij} when $(s_is_j)^{m_{ij}}=1$ (unlabelled means 3, no

edge means 2). Its *Schläfli* (or Gram) matrix is

$$G_{ij} = -\cos\frac{\pi}{m_{ij}}, \qquad G_{ii} = 1,$$

the Gram matrix of unit normals to mirrors meeting at angles π/m_{ij}. For a crystallographic diagram, G is the Cartan matrix rescaled ($G = \frac{1}{2}C$ in the simply laced case).

Theorem H.2 (Coxeter 1934). *Let Γ be a connected Coxeter diagram with Schläfli matrix G. The group $\langle s_i \mid s_i^2, (s_i s_j)^{m_{ij}}\rangle$ is a finite reflection group if and only if G is positive definite. It acts as a discrete group generated by reflections of a Euclidean space, with a simplex as fundamental domain, if and only if G is positive semidefinite with a one-dimensional kernel. The connected positive definite diagrams are $A_n, B_n, D_n, E_6, E_7, E_8, F_4, G_2$, together with the non-crystallographic H_3, H_4 and $I_2(m)$.*

For three mirrors forming a triangle (p,q,r), $\det G = 1 - a^2 - b^2 - c^2 - 2abc$ with $a = \cos\frac{\pi}{p}$, $b = \cos\frac{\pi}{q}$, $c = \cos\frac{\pi}{r}$. We checked numerically that this is positive, zero, negative in the spherical, Euclidean, hyperbolic cases (for example $\det G = 0.0955$ for $(2,3,5)$ and -0.0617 for $(2,3,7)$). A Platonic solid $\{p,q\}$ has the linear diagram $\circ\!\overset{p}{\text{—}}\!\circ\!\overset{q}{\text{—}}\!\circ$, whose determinant is $1-\cos^2\frac{\pi}{p}-\cos^2\frac{\pi}{q}$. It is positive if and only if $\frac{1}{p}+\frac{1}{q} > \frac{1}{2}$, the classical condition for $\{p,q\}$ to exist. This is the triangle $(2,p,q)$ again. So the classification of Platonic solids, of spherical triangle groups and of branched Dynkin diagrams are three readings of one inequality.

H.2 The icosahedron is not made of regular tetrahedra, seen by Coxeter

Interlude E (Section E.4) proved, with a ring of six balls and the triangle inequality on the sphere, that 20 regular tetrahedra glued at a common vertex do not close up into an icosahedron. Around each of the 12 edges from the centre there would be 5 tetrahedra, and 5 dihedral angles $\arccos\frac{1}{3} \approx 70.53°$ leave a gap of $7.36°$. Now look at the same configuration through its Coxeter diagram.

A regular polytope or honeycomb with Schläfli symbol $\{p,q,r\}$ has the linear Coxeter diagram

$$\circ\!\overset{p}{\text{—}}\!\circ\!\overset{q}{\text{—}}\!\circ\!\overset{r}{\text{—}}\!\circ\,.$$

The first three nodes generate the symmetry group of a cell $\{p,q\}$, the last three that of the vertex figure $\{q,r\}$, and the label r says that r cells surround each edge. "Twenty tetrahedra around a vertex, forming an icosahedron" is the local structure of $\{3,3,5\}$: cell a tetrahedron $\{3,3\}$, vertex figure an icosahedron $\{3,5\}$, five cells around an edge.

Proposition H.3. *The Schläfli matrix of $\{p,q,r\}$ has determinant*

$$\det G_{\{p,q,r\}} = \sin^2\frac{\pi}{p}\,\sin^2\frac{\pi}{r} - \cos^2\frac{\pi}{q}.$$

For $\{3,3,r\}$ this is $\frac{1}{2} - \frac{3}{4}\cos^2\frac{\pi}{r}$, and

$$\det G_{\{3,3,r\}} \gtreqless 0 \quad\Longleftrightarrow\quad r\cdot\arccos\tfrac{1}{3} \lesseqgtr 2\pi.$$

The values are $\frac{5}{16}$, $\frac{1}{8}$, $\frac{5-3\tau}{16} \approx 0.0091$, $-\frac{1}{16}$ for $r = 3,4,5,6$, where $\tau = \frac{1+\sqrt{5}}{2}$.

Proof. For a tridiagonal matrix with 1 on the diagonal and off-diagonal entries $-a, -b, -c$, the leading minors satisfy $D_k = D_{k-1} - (\text{entry})^2 D_{k-2}$. So $D_2 = 1 - a^2$, $D_3 = 1 - a^2 - b^2$, and

$$D_4 = D_3 - c^2 D_2 = (1 - a^2)(1 - c^2) - b^2.$$

With $a = b = \frac{1}{2}$ this is $\frac{3}{4}(1 - c^2) - \frac{1}{4} = \frac{1}{2} - \frac{3}{4}c^2$. It vanishes if and only if $c^2 = \cos^2 \frac{\pi}{r} = \frac{2}{3}$, that is, if and only if $\cos \frac{2\pi}{r} = 2c^2 - 1 = \frac{1}{3}$, that is, $\frac{2\pi}{r} = \arccos \frac{1}{3}$. As r grows, c^2 grows, so the sign of $\det G$ is the sign of $2\pi - r \arccos \frac{1}{3}$. For $r = 5$, $c = \cos \frac{\pi}{5} = \frac{\tau}{2}$ and $c^2 = \frac{\tau+1}{4}$, which gives $\frac{1}{2} - \frac{3(\tau+1)}{16} = \frac{5-3\tau}{16}$. □

So the sign of a 4×4 determinant is exactly the sign of the gap left by r regular tetrahedra around an edge. The critical value is $r = 2\pi / \arccos \frac{1}{3} \approx 5.104$, not an integer, so there is no Euclidean honeycomb of regular tetrahedra (Figure H.2):

- $r = 3, 4, 5$: positive definite, spherical. These are the 5-cell $\{3, 3, 3\}$, the 16-cell $\{3, 3, 4\}$ and the 600-cell $\{3, 3, 5\}$, with groups A_4, B_4, H_4.
- $r = 6$: indefinite, hyperbolic. Six tetrahedra around an edge overshoot, and $\{3, 3, 6\}$ is a honeycomb of hyperbolic space with ideal vertices.
- Euclidean: never.

The icosahedral case $r = 5$ is spherical, but only just: $(5 - 3\tau)/16 \approx 0.0091$, against $\frac{5}{16}$ and $\frac{1}{8}$ for $r = 3, 4$. This small number is the 7.36° of Interlude E. A flat icosahedron of regular tetrahedra would need $\det G = 0$. On the 3-sphere the curvature closes the gap, and 600 regular spherical tetrahedra form the 600-cell, the "ideal" icosahedral order of Section E.5. Its Schläfli determinant is positive, which says that H_4 is a finite reflection group and that the tiling closes up on a sphere. In Schläfli's original form the criterion reads $\sin \frac{\pi}{p} \sin \frac{\pi}{r} > \cos \frac{\pi}{q}$, and for $\{3, 3, 5\}$ it is $0.5090 > 0.5$.

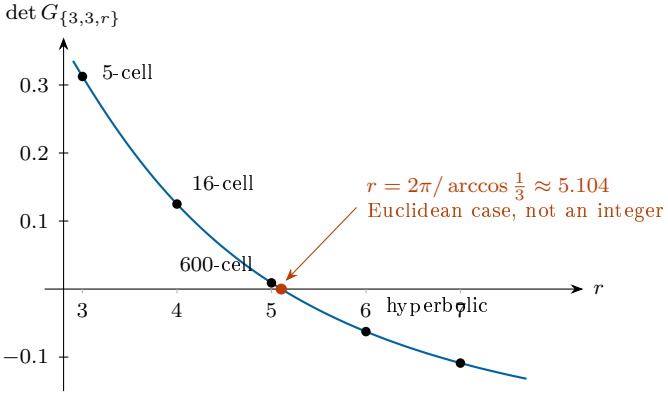


Figure H.2: The Schläfli determinant $\frac{1}{2} - \frac{3}{4}\cos^2 \frac{\pi}{r}$ of $\{3, 3, r\}$, with r treated as a real variable. It crosses zero where r regular tetrahedra exactly fill the space around an edge. The value at $r = 5$ is positive but tiny.

Reading the 600-cell off its diagram. The diagram of H_4 also counts. By Wythoff's construction the k-faces of $\{p, q, r\}$ correspond to cosets of the subgroup generated by all nodes except the $(k + 1)$-st, so their number is $|W|$ divided by the order of that subgroup. With

$|W(H_4)| = 14400$ (Interlude C):

$$\text{vertices: } \frac{14400}{|W(H_3)|} = \frac{14400}{120} = 120, \qquad \text{edges: } \frac{14400}{|W(A_1)|\,|W(I_2(5))|} = \frac{14400}{2\cdot 10} = 720,$$
$$\text{triangles: } \frac{14400}{|W(A_2)|\,|W(A_1)|} = \frac{14400}{6\cdot 2} = 1200, \qquad \text{cells: } \frac{14400}{|W(A_3)|} = \frac{14400}{24} = 600.$$

The vertex stabilizer is H_3, the symmetry group of the icosahedral vertex figure.

Why a lattice cannot do it

The label 5 is also what keeps icosahedral order out of crystals.

Proposition H.4 (Crystallographic restriction)**.** *If a rotation of order m maps a lattice $\Lambda \subset \mathbb{R}^d$ to itself, then its characteristic polynomial has integer coefficients. For $d = 2, 3$ this forces $m \in \{1, 2, 3, 4, 6\}$. More generally, a lattice with a symmetry of order 5 has dimension at least 4. If a group isomorphic to the icosahedral rotation group A_5 preserves a lattice in $\mathbb{R}^d$ and acts on some 3-dimensional subspace as the rotations of an icosahedron, then $d \geq 6$.*

Proof. In a basis of Λ the rotation is an integer matrix. For $d = 2$ its trace $2\cos\frac{2\pi}{m}$ is an integer, and for $d = 3$ the trace $1 + 2\cos\frac{2\pi}{m}$ is; either way $\cos\frac{2\pi}{m} \in \{0, \pm\frac{1}{2}, \pm 1\}$. An integer matrix of order 5 has eigenvalues that are fifth roots of unity, closed under Galois conjugation; a primitive one brings all four primitive ones with it, so $d \geq 4$. For the icosahedral group A_5, the two irreducible real representations of dimension 3 have characters involving $\sqrt{5}$ and are swapped by $\sqrt{5} \mapsto -\sqrt{5}$. The representation on $\Lambda \otimes \mathbb{Q}$ is rational, so if it contains one of them it contains the other equally often, and $d \geq 3 + 3$. (The condition on the 3-dimensional subspace matters: $A_5 \subset \Sigma_5 = W(A_4)$ preserves the 4-dimensional lattice A_4, acting through its rational 4-dimensional representation.) □

For a reflection group the rotation $s_i s_j$ has order m_{ij}, so the proposition is the crystallographic table of Lecture 4 again: $4\cos^2\frac{\pi}{m} \in \{0, 1, 2, 3\}$ allows $m \in \{2, 3, 4, 6\}$, and $4\cos^2\frac{\pi}{5} = \tau + 1$ is irrational. The bounds are attained. The Coxeter element of A_4 has order $h = 5$ (Interlude A, item 3), and Interlude C (Theorem C.20) realizes H_3 inside D_6 and H_4 inside E_8. That is the precise sense in which the E_8 lattice "contains" the icosahedral order that is impossible in a flat crystal: the 240 roots project to two 600-cells. In three dimensions the best one can do is a quasicrystal.

H.3 The FCC lattice, read off its diagrams

The FCC lattice is the root lattice A_3, the densest packing in $\mathbb{R}^3$ (Hales). We now go through its features one by one and find each on a diagram.

Two names: $A_3 = D_3$

Lectures 5 and 6 met FCC twice: as $A_3 = \{x \in \mathbb{Z}^4 : \sum x_i = 0\}$ and, in the remark $D_3 = A_3$ after the classification, as $D_3 = \{x \in \mathbb{Z}^3 : x_1 + x_2 + x_3 \text{ even}\}$. The diagrams agree: D_3 would be a chain of two nodes both joined to a third, which is the chain A_3. That is why Theorem 5.2 starts D_n at $n = 4$. An explicit isometry is

$$f(x) = \tfrac{1}{2}\big(x_1 + x_2 - x_3 - x_4,\ x_1 - x_2 + x_3 - x_4,\ x_1 - x_2 - x_3 + x_4\big).$$

It sends e_1, e_2, e_3, e_4 to the four vertices $\frac{1}{2}(1,1,1)$, $\frac{1}{2}(1,-1,-1)$, $\frac{1}{2}(-1,1,-1)$, $\frac{1}{2}(-1,-1,1)$ of a regular tetrahedron inscribed in the cube, and $e_1 - e_2 \mapsto (0,1,1)$, $e_1 - e_3 \mapsto (1,0,1)$, and so on, so the 12 roots of A_3 go to the 12 roots $\pm e_a \pm e_b$ of D_3. Since the $f(e_i)$ have norm $\frac{3}{4}$ and inner products $-\frac{1}{4}$, one checks $|f(x)|^2 = |x|^2$ whenever $\sum x_i = 0$. In Lie theory this is $\mathfrak{so}_6 \cong \mathfrak{sl}_4$ (Interlude A, end of Section A.5). Geometrically, SL_4 acts on $\Lambda^2\mathbb{C}^4 \cong \mathbb{C}^6$ preserving the quadratic form $\omega \mapsto \omega \wedge \omega \in \Lambda^4\mathbb{C}^4 \cong \mathbb{C}$. Its quadric in $\mathbb{P}^5$ is the Klein quadric, whose points are the lines of $\mathbb{P}^3$.

Keep an eye on the vectors $f(e_i) = \frac{1}{2}(\pm 1, \pm 1, \pm 1)$: the coordinate directions of $\mathbb{R}^4$ point at holes of FCC, as we will see in a moment.

The symmetry group, and the missing inversion

The Weyl group $W(A_3) = \Sigma_4$, of order 24, permutes the four coordinates of $\mathbb{R}^4$; in the cube picture it is the full symmetry group of the inscribed tetrahedron. But the cube has 48 symmetries, and they all preserve D_3. The extra ones come from the diagram. The automorphism group of an irreducible root system is $W \rtimes \mathrm{Aut}(\text{diagram})$ (Humphreys, *Introduction to Lie Algebras*, §12.2), and for a root lattice it is the same as the automorphism group of the lattice, since the roots are the minimal vectors and generate.

Proposition H.5. $\mathrm{Aut}(D_n) = W(B_n)$, *of order* $2^n n!$, *for every* $n \geq 2$ *except* $n = 4$, *where* $\mathrm{Aut}(D_4) = W(F_4)$ *has order* $1152 = 192 \cdot 6$.

Proof. $W(B_n)$ (all signed permutations) preserves D_n, and has order $2^n n!$. The group $W(D_n)$ (signed permutations with an even number of sign changes) has order $2^{n-1} n!$. The diagram of D_n has symmetry group $\mathbb{Z}/2$ for $n \neq 4$ (for $n = 2$, swap the two nodes of $A_1 \times A_1$; for $n = 3$, flip the chain A_3), and Σ_3 for $n = 4$. So $|\mathrm{Aut}\, D_n| = 2 \cdot 2^{n-1} n! = 2^n n!$ for $n \neq 4$, and $\mathrm{Aut}\, D_n \supseteq W(B_n)$ forces equality. For $n = 4$ the order is $6 \cdot 192 = 1152 = |W(F_4)|$, and $W(F_4)$ preserves the D_4 roots, which are its long roots. (A computer search over the possible images of the simple roots confirms the orders 48 for D_3 and 1152 for D_4.) □

For FCC this says: point group $O_h = W(B_3) = W(A_3) \times \{\pm 1\}$. The inversion $x \mapsto -x$ is not in $W(A_3)$; it acts on the diagram A_3 as the flip. (In general $-1 \in W$ exactly when the opposite involution $-w_0$ is trivial, which fails for A_n with $n \geq 2$, for D_n with n odd, and for E_6.) The flip will turn out to swap two kinds of holes.

The holes

Holes are the vertices of the Voronoi cells (Definition 1.15). For the root lattices of this interlude they are exactly the points locally farthest from the lattice (for a general lattice a Voronoi vertex need not be a local maximum of the distance). Around each one, the nearest lattice points span a *Delaunay cell*. In the cube picture FCC has two kinds of holes (with roots $\pm e_a \pm e_b$, so minimal distance $\sqrt{2}$):

- *octahedral holes* at $(1,0,0) + D_3$, at distance 1 from six lattice points 0, $2e_1$, $e_1 \pm e_2$, $e_1 \pm e_3$, which span a regular octahedron of edge $\sqrt{2}$;
- *tetrahedral holes* at $\frac{1}{2}(\pm 1, \pm 1, \pm 1) + D_3$, at distance $\frac{\sqrt{3}}{2}$ from four lattice points, for example $0, (1,1,0), (1,0,1), (0,1,1)$ around $\frac{1}{2}(1,1,1)$, which span a regular tetrahedron of edge $\sqrt{2}$.

The tetrahedral holes come in two classes, $h + D_3$ and $-h + D_3$ with $h = \frac{1}{2}(1,1,1)$, whose tetrahedra point in opposite directions. Together with D_3 itself these are the four cosets of D_3 in the dual lattice $D_3^* = \mathbb{Z}^3 \cup (\mathbb{Z}^3 + h)$:

$$D_3^*/D_3 = \{[0],[1],[2],[3]\} \cong \mathbb{Z}/4, \qquad [0] = D_3, \quad [1] = h + D_3, \quad [2] = e_1 + D_3, \quad [3] = -h + D_3.$$

It is cyclic: $h + h = (1,1,1) \in e_1 + D_3$, so two steps from a lattice point to a tetrahedral hole reach an octahedral hole.

Counting cells. The octahedron has volume $\frac{4}{3}$ and the tetrahedron $\frac{1}{3}$, and the covolume of D_3 is 2. There is one octahedral hole per lattice point (the coset [2] has the same density as D_3) and two tetrahedral holes ([1] and [3]), and indeed

$$1 \cdot \tfrac{4}{3} + 2 \cdot \tfrac{1}{3} = 2.$$

So FCC space is tiled by octahedra and tetrahedra, and the tetrahedra fill exactly one third of it. Now put the FCC balls (radius $\frac{\sqrt{2}}{2}$) at the lattice points and ask what fraction of each Delaunay cell they cover. In a tetrahedron this is $4\Omega_T/(4\pi) \cdot \operatorname{vol} B / \operatorname{vol} T$, with $\Omega_T = 3\arccos\frac{1}{3} - \pi$ the solid angle at a vertex; that is Rogers' $\sigma_3 \approx 0.7796$ (Section 2.7). In the octahedron, with $\Omega_O = 4\arcsin\frac{1}{3}$, it is ≈ 0.7209. The weighted average $\frac{1}{3}(0.7796) + \frac{2}{3}(0.7209) = 0.7405 = \pi/\sqrt{18}$ is the FCC density. So the ratio 0.950 to Rogers' bound in the table of Section 2.7 is the price of the octahedra: they take two thirds of the space, at local density 0.7209 instead of 0.7796.

Around an edge. Each edge of this tiling (a pair of touching balls) lies on two tetrahedra and two octahedra, alternating. The dihedral angle of the octahedron is $\arccos(-\frac{1}{3}) \approx 109.47°$, and since

$$\arccos\tfrac{1}{3} + \arccos\left(-\tfrac{1}{3}\right) = \pi,$$

two of each close up exactly (Figure H.3). So the octahedra fill *precisely* the gap that tetrahedra leave. Five tetrahedra fall 7.36° short of 2π, six overshoot by 63.17°, and FCC settles for two tetrahedra and two octahedra. At a vertex the same bookkeeping gives 8 tetrahedra and 6 octahedra, the 8 triangles and 6 squares of the cuboctahedron, which is the vertex figure.

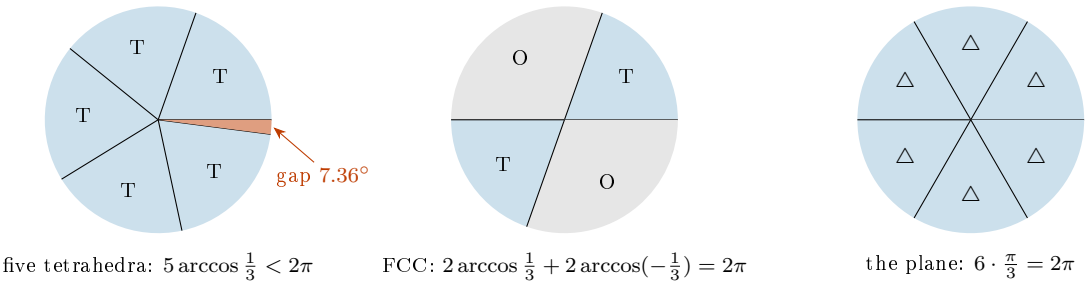


Figure H.3: Cross-sections perpendicular to an edge. Left: five regular tetrahedra leave a gap (the icosahedron problem of Interlude E). Middle: in the FCC Delaunay tiling, octahedra (O) fill the gaps between tetrahedra (T). Right: in the plane, six equilateral triangles close up around a vertex, which is why Rogers' bound is sharp there (Thue).

The holes on the extended diagram

Now the diagram. The extended diagram $\tilde{A}_3$ is a square: the extra node $\alpha_0 = -\theta$ is joined to both ends of the chain. Its four nodes match the four cosets:

node $0 \leftrightarrow [0]$ (lattice points), node $1 \leftrightarrow [1]$, node $2 \leftrightarrow [2]$ (octahedral), node $3 \leftrightarrow [3]$.

The reason will be given in Section H.4: the fundamental alcove of the affine Weyl group has vertices $0, \omega_1, \omega_2, \omega_3$, the ω_i are representatives of the cosets, and under f they become $h = \frac{1}{2}(1,1,1)$, e_1 and $\frac{1}{2}(1,1,-1)$. Two symmetries of the square are visible in the lattice.

- *The flip* fixing nodes 0 and 2 and swapping 1 and 3 is realized by the inversion $x \mapsto -x$. It fixes the lattice and the octahedral holes, and swaps the two classes of tetrahedral holes, "up" and "down".
- *The rotation* by a quarter turn is translation by h: it moves lattice points to tetrahedral holes, those to octahedral holes, and so on around the square. It is a symmetry of the whole pattern of points and holes, though not of the lattice.

The symmetry group of the square, dihedral of order 8, is $(D_3^*/D_3) \rtimes \mathrm{Aut}(A_3) = \mathbb{Z}/4 \rtimes \mathbb{Z}/2$. This is a general fact (Section H.4).

Chemistry. Filling some of the four cosets with atoms gives the classical cubic crystal structures:

$[0] \cup [2] = \mathbb{Z}^3$	rock salt (NaCl): Na on FCC, Cl in the octahedral holes,
$[0] \cup [1]$	zincblende (ZnS), and diamond when both are carbon,
$[0] \cup [1] \cup [3]$	fluorite (CaF_2): F in all tetrahedral holes, two per Ca,
$[0] \cup [1] \cup [2] \cup [3] = D_3^*$	body-centred cubic (BCC).

The last line says that the dual of FCC is BCC, which is the root-lattice statement $A_3^* \cong$ BCC. FCC is the densest packing in $\mathbb{R}^3$, and its dual BCC, the lattice of all its points and holes, is the thinnest *lattice covering* of $\mathbb{R}^3$ (Bambah 1954), with covering density $\frac{5\sqrt{5}\pi}{24} \approx 1.4635$.

Stacking. One dimension down, $\tilde{A}_2$ is a triangle, and the hexagonal lattice has two classes of (triangular) holes swapped by the flip of A_2 (Figure H.4). Lamination (Section 6.3) places the next layer over one class of holes. The choice between the two classes at each step is the choice between FCC ($ABCABC$) and the hexagonal close packing ($ABAB$), and the two choices are exchanged by the symmetry of $\tilde{A}_2$.

The kissing configuration and the jitterbug

The 12 nearest neighbours of a lattice point are the roots of A_3, the vertices of the cuboctahedron. Wythoff's construction reads this off the diagram: the roots form the Weyl group orbit of the highest root $\theta = \omega_1 + \omega_3$, so the cuboctahedron is the A_3 diagram with both end nodes ringed. In the cube picture (B_3) it is the middle node ringed, the "rectified cube". Its 8 triangles face the 8 tetrahedral holes around the ball and its 6 squares face the 6 octahedral holes. The icosahedron, by contrast, is $\{3,5\}$, the H_3 diagram with an end node ringed, and its diagram needs the label 5 that no lattice allows. Lecture 11's jitterbug deforms one into the other. It keeps the 8 triangles over the tetrahedral holes rigid, rotating them, and folds each square over an octahedral hole into two triangles: $8 + 6 \cdot 2 = 20$ faces. (In Coxeter's terminology the icosahedron is the *snub tetrahedron*.) What is lost along the way is the lattice, since the end point carries a forbidden 5.

H.4 Seeing the holes: the extended diagram

Everything in the FCC story came from the four nodes of $\tilde{A}_3$. Here is the general mechanism, which underlies the systematic treatments of Conway–Sloane and Moody–Patera (gallery).

The alcove and its vertices

Let Δ be a simply laced irreducible root system with simple roots $\alpha_1, \dots, \alpha_n$, root lattice L, highest root $\theta = \sum_i m_i \alpha_i$, and fundamental weights ω_i ($\langle \omega_i, \alpha_j \rangle = \delta_{ij}$). The integers $m_i \geq 1$ are the *marks*; put $m_0 = 1$ for the extra node $\alpha_0 = -\theta$ of the extended diagram. The *fundamental alcove* is the simplex

$$\mathcal{A} = \{x : \ \langle x, \alpha_i \rangle \geq 0 \ (1 \leq i \leq n), \ \langle x, \theta \rangle \leq 1\},$$

with one wall for each node of the extended diagram. Its vertices are

$$v_0 = 0, \qquad v_i = \frac{\omega_i}{m_i} \quad (1 \leq i \leq n),$$

since v_i lies on every wall except the wall of α_i. For A_2 this is the triangle in Figure H.4. The affine Weyl group, generated by the reflections in the walls of $\mathcal{A}$, tiles space by copies of $\mathcal{A}$ (Interlude A).

Lemma H.6. *For a simply laced root lattice the Voronoi cell of* 0 *is*

$$V(0) = \{x : \ \langle x, \alpha \rangle \leq 1 \text{ for all roots } \alpha\} = \bigcup_{w \in W} w(\mathcal{A}).$$

Proof for E_8. In general $V(0) = \{x : \langle x, v \rangle \leq \frac{1}{2}|v|^2\}$, where v runs over the *Voronoi-relevant* vectors, and by a theorem of Voronoi, v is relevant if and only if $\pm v$ are the only shortest vectors in its class $v + 2L$ (Conway–Sloane, *Sphere Packings*, Ch. 21). So we sort the short vectors of E_8 into the $2^8 = 256$ classes of $E_8/2E_8$. Two vectors v, w are in the same class when $\frac{1}{2}(v - w) \in E_8$, which needs $\frac{1}{4}|v - w|^2 \in \{0, 2, 4, \dots\}$, since E_8 is even with minimum 2.

- No vector of norm 2 or 4 is in the class of 0, since $\frac{1}{2}v$ would have norm $\frac{1}{2}$ or 1.
- Two roots are congruent only if $\frac{1}{4}(4 - 2\langle \alpha, \beta \rangle) \in \{0, 2\}$, that is, $\beta = \pm\alpha$. So the 240 roots give 120 distinct classes.
- A root α and a vector v of norm 4 are never congruent: we would need both $\frac{1}{4}(6 - 2\langle v, \alpha \rangle)$ and $\frac{1}{4}(6 + 2\langle v, \alpha \rangle)$ (for $v - \alpha$ and $v + \alpha$) in $\{0, 2, 4, \dots\}$, but they add up to 3.
- Two vectors of norm 4 are congruent only if $\frac{1}{4}(8 - 2\langle v, w \rangle) \in \{0, 2, 4\}$, that is, $w = \pm v$ or $w \perp v$. So a class holds at most 8 orthogonal pairs, 16 vectors, and the 2160 vectors of norm 4 (Interlude B, (B.5)) need at least 135 classes.

Since $1 + 120 + 135 = 256$, these are all the classes, each norm-4 class has exactly 16 shortest vectors, and only the root classes have exactly two. So the relevant vectors are the 240 roots, with $\frac{1}{2}|\alpha|^2 = 1$. Finally, on the dominant chamber $\langle x, \alpha \rangle \leq \langle x, \theta \rangle$ for every root (since $\theta - \alpha$ is a nonnegative combination of simple roots), so $V(0)$ meets the chamber in $\mathcal{A}$, and W permutes the chambers. □

Conway and Sloane (*Sphere Packings*, Ch. 21) prove the same for every root lattice: the relevant vectors are the roots. Consequently every vertex of a Voronoi cell, that is, every hole, is a W-image of some v_i. Which v_i are holes, and what do they look like?

Theorem H.7 (Holes from the extended diagram)**.** *Let $i \neq 0$ be a node of the extended diagram $\tilde{\Delta}$ with mark m_i, and let W_i be the (finite) reflection group of the diagram $\tilde{\Delta} \setminus \{i\}$.*

(a) The roots α_j, $j \neq i$ (including α_0), generate a sublattice of L of index m_i.

(b) The lattice points nearest to v_i are the W_i-orbit of 0, at distance $|v_i|$.

(c) If $\tilde{\Delta} \setminus \{i\}$ is connected, then v_i is a vertex of $V(0)$ (a hole), and its Delaunay cell is the Wythoff polytope of the diagram $\tilde{\Delta} \setminus \{i\}$ with node 0 ringed. The holes of this type around 0 number $|W| \,/\, |W(\Delta \setminus \{i\})|$.

(d) If $\tilde{\Delta} \setminus \{i\}$ is disconnected, v_i lies in the relative interior (at the centre) of a face of $V(0)$ of positive dimension, so it is not a hole.

Sketch. (a) The lattice generated by α_j ($j \neq i$, $j \geq 1$) and $\alpha_0 = -\sum_j m_j \alpha_j$ is spanned by the α_j with $j \neq i$ together with $m_i \alpha_i$; its index in L is m_i. (b) The alcoves containing v_i are the images of $\mathcal{A}$ under W_i, the group generated by the walls through v_i. By Lemma H.6, a lattice point λ is nearest to v_i exactly when $v_i \in V(\lambda) = \lambda + \bigcup_w w(\mathcal{A})$, i.e. when v_i lies in one of the alcoves $\lambda + w(\mathcal{A})$. The alcoves form a simplicial tiling on which the affine Weyl group acts preserving the type (node label) of vertices, so v_i can lie in such an alcove only as its vertex of type i, and then λ is the alcove's vertex of type 0. The alcoves with v_i as a vertex are the W_i-images of $\mathcal{A}$, so these λ are the points $W_i \cdot 0$. (c),(d) The group W_i splits as a product over the components of $\tilde{\Delta} \setminus \{i\}$. The components not containing node 0 consist of reflections in linear hyperplanes $\langle x, \alpha_j \rangle = 0$, so they fix 0 (and v_i). The orbit $W_i \cdot 0$ is therefore the Wythoff polytope of the component containing node 0, with node 0 ringed, centred at v_i; it spans v_i plus the span of that component's roots, so it is full-dimensional exactly when that component is everything. A point is a vertex of $V(0)$ exactly when its nearest lattice points (here including 0) affinely span $\mathbb{R}^n$; this gives (c). In case (d), v_i lies in the relative interior of the face of $V(0)$ dual to the lower-dimensional polytope $\mathrm{conv}(W_i \cdot 0)$, and the other components act on that face fixing only v_i. The count is the size of the W-orbit of v_i, whose stabilizer in W is the parabolic subgroup $W(\Delta \setminus \{i\})$ generated by the simple reflections fixing ω_i. □

Two consequences can be read off immediately. First, the *index* in (a) is the mark: deleting a node of mark m produces a root sublattice of index m (this is the lattice shadow of the Borel–de Siebenthal theorem in the gallery). Second, the nodes of mark 1 are special. For them $v_i = \omega_i$, and the ω_i with $m_i = 1$, together with 0, represent the classes of L^*/L. So

$$\#\{\text{nodes of mark 1 in } \tilde{\Delta}\} = |L^*/L| = \det C,$$

and the symmetry group of the extended diagram is $(L^*/L) \rtimes \mathrm{Aut}(\Delta)$: the translations by the ω_i ($m_i = 1$) rotate the extended diagram, as the quarter turn of $\tilde{A}_3$ did. The check: $\tilde{A}_n$ has $n+1$ nodes of mark 1; $\tilde{D}_n$ has 4; $\tilde{E}_6$, $\tilde{E}_7$, $\tilde{E}_8$ have 3, 2, 1; and these are the determinants $n+1, 4, 3, 2, 1$.

Worked examples

A_n. All marks are 1 and $\tilde{A}_n$ is a cycle, so every node is a hole. The hole ω_i (class $[i]$) has

$$|\omega_i|^2 = \frac{i(n+1-i)}{n+1},$$

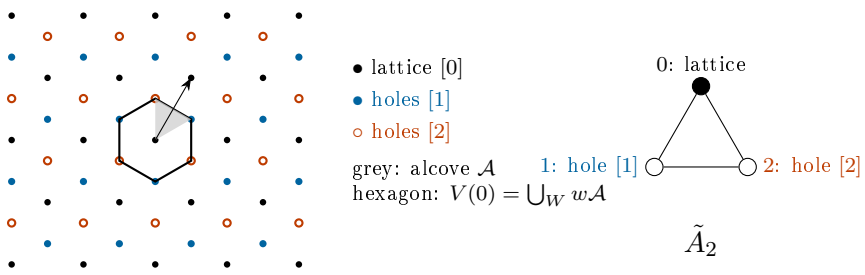


Figure H.4: The hexagonal lattice A_2. The alcove (grey) has vertices $0, \omega_1, \omega_2$; its six W-images make the Voronoi hexagon, whose vertices are the two classes of holes. The nodes of the triangle $\tilde{A}_2$ match the three cosets of A_2 in A_2^*; the flip of A_2 swaps the hole classes (the B/C choice in stacking). The arrow is the root θ.

and its Delaunay cell is the Wythoff polytope of A_n with the i-th node ringed: the *hypersimplex*, the convex hull of the $\binom{n+1}{i}$ vectors with i ones and $n+1-i$ zeros. For $n=3$: $i=1,3$ give tetrahedra at distance $\frac{\sqrt{3}}{2}$, and $i=2$ gives the octahedron at distance 1. For $n=2$ both hole classes are triangles, which is why Rogers' bound is attained by A_2. For $n \geq 3$ the hypersimplex with $i=2$ is not a simplex, so Rogers' bound is never attained by A_n. The deepest holes are in the middle, $i = \lfloor \frac{n+1}{2} \rfloor$.

D_n, and a footnote to Lecture 6. In $\tilde{D}_n$ ($n \geq 4$) the four leaves have mark 1 and all other nodes have mark 2. The inner nodes disconnect the diagram, so the holes are the three leaves other than 0:

- the *vector* hole $\omega_1 = e_1$, at distance 1. Deleting node 1 leaves D_n with node 0 at the end of the long arm, so the Delaunay cell is the cross-polytope, with $2n$ vertices $0, 2e_1, e_1 \pm e_j$;
- the two *spinor* holes $\omega_{n-1}, \omega_n = \frac{1}{2}(1, \ldots, 1, \pm 1)$, at distance $\frac{\sqrt{n}}{2}$. Now node 0 sits at a short leaf, and the Delaunay cell is the half-cube, with 2^{n-1} vertices.

So the deep holes of D_n are the spinor holes $h = \frac{1}{2}(1, \ldots, 1)$ for $n \geq 5$, as Lecture 6 used (Section 6.2). For $n=3$ the roles reverse: the spinor holes are the tetrahedral holes of FCC, at distance $\frac{\sqrt{3}}{2} < 1$, and the deep holes are the octahedral (vector) ones. At $n=4$ all three are at distance 1, and all three Delaunay cells are 16-cells. That tie is triality (Section H.5).

E_8. Here is the full table for $\tilde{E}_8 = T_{2,3,6}$, in the labelling of Lecture 6. The extra node α_0 is attached to α_1, the end of the long arm. For each node we list the mark, the norm of the alcove vertex $v_i = \omega_i / m_i$, the number of nearest lattice points, and what they span (all computed numerically).

node	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
mark m_i	2	3	4	5	6	3	4	2
$\lvert v_i \rvert^2$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$	$\frac{5}{6}$	$\frac{8}{9}$	$\frac{7}{8}$	1
nearest points	2	3	4	5	6	9	8	16
$\tilde{E}_8 \setminus \{i\}$	$A_1 \times E_7$	$A_2 \times E_6$	$A_3 \times D_5$	$A_4 \times A_4$	$A_5 \times A_2 \times A_1$	$\mathbf{A_8}$	$A_7 \times A_1$	$\mathbf{D_8}$

The pattern $|v_i|^2 = \frac{k}{k+1}$ for a cell with $k+1$ points is the squared circumradius of a regular k-simplex of edge $\sqrt{2}$, which is what the nearest points span in all but the last column. Only two deletions leave a connected diagram, so E_8 has exactly two kinds of holes (Figure H.5):

- the *shallow* holes, from α_6 (mark 3): Delaunay cell the regular 8-simplex (the A_8 diagram ringed at an end), circumradius $\sqrt{8/9}$, and $|W(E_8)| \,/\, |W(A_7)| = 17280$ of them around each lattice point;

- the *deep* holes, from α_8 (mark 2): Delaunay cell the 8-dimensional cross-polytope (D_8 ringed at the end of the long arm, 16 vertices), circumradius 1, and $|W(E_8)| \,/\, |W(D_7)| = 2160$ of them.

So the Voronoi cell of E_8 has $17280 + 2160 = 19440$ vertices. Both numbers can be found in the theta series of Interlude B. The 2160 deep holes around 0 are the halves of the 2160 vectors of norm 4. The 17280 shallow holes are the thirds of the 17520 vectors of norm 8, minus the 240 that are twice a root ($17520 - 240 = 17280$).

Lecture 6 again. Part (a) of the theorem now explains both constructions of E_8. Deleting α_8 (mark 2) leaves D_8, a sublattice of index 2: this is $E_8 = D_8 \cup (D_8 + h)$. Of the three nonzero classes of D_8^*/D_8, the spinor class of h (distance $\sqrt{2}$) has been filled, and the vector class (distance 1) survives as the deep holes of E_8. Deleting α_6 (mark 3) leaves A_8, of index 3, and gives a construction we did not see in Lecture 6:

$$E_8 = A_8 \cup (A_8 + [3]) \cup (A_8 + [6]),$$

where $[3]$ is the class of ω_3 in $A_8^*/A_8 \cong \mathbb{Z}/9$. It has norm $\frac{3 \cdot 6}{9} = 2$, so the glued lattice is even, and it is unimodular since $9/3^2 = 1$.

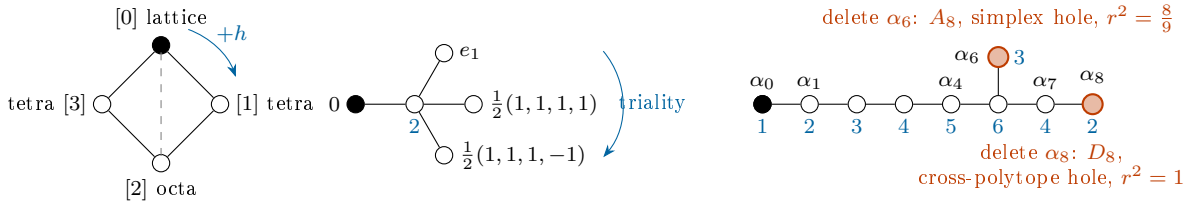
How much space is Rogers'? Per lattice point there are $17280/9 = 1920$ simplices and $2160/16 = 135$ cross-polytopes. With edge $\sqrt{2}$ their volumes are $\frac{1}{13440}$ and $\frac{256}{8!}$, so

$$1920 \cdot \frac{1}{13440} = \frac{1}{7}, \qquad 135 \cdot \frac{256}{40320} = \frac{6}{7}.$$

One seventh of $\mathbb{R}^8$ is tiled by Rogers' regular simplices, where the local density is $\sigma_8 \approx 0.2568$, and six sevenths by cross-polytopes, where it is about 0.2531. This is why E_8 comes within 1.2% of Rogers' bound (Section 2.7): its non-simplex cells, the cross-polytopes, are almost as efficient as Rogers' simplices. The deep holes give the covering radius 1, and the covering density $\operatorname{vol} B^8(1) = \pi^4/24 \approx 4.06$.

H.5 Triality

The diagram D_4 has a threefold symmetry, and so does $\tilde{D}_4$ after fixing node 0 (Figure H.5). On the lattice it acts as follows.



$\tilde{A}_3$ (FCC) $\tilde{D}_4$ $\tilde{E}_8 = T_{2,3,6}$

Figure H.5: Extended diagrams (extra node black, marks in blue). Left: the square $\tilde{A}_3$ with the four cosets of FCC in BCC; the dashed axis is the flip (inversion), the arrow the quarter turn (translation by $h = \frac{1}{2}(1,1,1)$). Middle: $\tilde{D}_4$, whose three outer nodes other than 0 are the three hole classes of D_4, permuted by triality. Right: $\tilde{E}_8$; the two shaded leaves are the only nodes whose deletion leaves a connected diagram, and they give the two kinds of holes of E_8.

Three classes of holes. $D_4^*/D_4 \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ has three nonzero classes, represented by e_1, $\frac{1}{2}(1,1,1,1)$ and $\frac{1}{2}(1,1,1,-1)$. Each has exactly 8 vectors of norm 1:

$$\pm e_i, \qquad \tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ (even number of } -), \qquad \tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ (odd number of } -),$$

and each octet is the vertex set of a 16-cell (a cross-polytope). By the previous section these are the deep holes around 0, all at distance 1, and each one's Delaunay cell is again a 16-cell. The group $W(D_4)$ acts trivially on D_4^*/D_4, and $\mathrm{Aut}(D_4) = W(F_4)$ acts on the three classes through Σ_3. That is the order count $1152 = 192 \cdot 6$ of Proposition H.5.

F_4 is D_4 together with its holes. The 24 holes around 0 form a 24-cell, the Voronoi cell of D_4. Compare with Lecture 6's description of F_4: the long roots $\pm e_i \pm e_j$ are the 24 roots of D_4, and the short roots $\pm e_i$, $\frac{1}{2}(\pm 1, \ldots, \pm 1)$ are exactly the 24 deep holes. So

$$\Delta(F_4) = \{\text{nearest neighbours of 0 in } D_4\} \cup \{\text{vertices of the Voronoi cell of 0}\},$$

two 24-cells, polar to each other (the facets of the Voronoi cell are perpendicular to the roots), the second smaller by a factor $\sqrt{2}$. The self-duality of the 24-cell $\{3,4,3\}$ is the left–right symmetry of its Coxeter diagram $\circ\!-\!\circ\overset{4}{-}\!\circ\!-\!\circ$, which as a Dynkin diagram reverses the arrow and exchanges long and short roots. And Lecture 6's remark that the F_4 root lattice is D_4^* is the statement that the holes generate it.

Representations. In Interlude A's language (Section A.9, item 8), the three outer nodes of D_4 label the three 8-dimensional representations of $\mathfrak{so}_8$: the vector representation and the two half-spin representations. Their weights are exactly the three octets above. Triality permutes them, and the $\mathfrak{so}_8$-invariant trilinear map $V \times S_+ \times S_- \to \mathbb{R}$ is octonion multiplication in disguise (Baez, *The octonions*). The subgroup of Spin(8) fixed by triality is the automorphism group G_2 of the octonions.

Folding by triality, on the roots. Interlude A listed the folding $D_4 \rightsquigarrow G_2$. Here it is on the 24 roots. Triality fixes the plane spanned by the central root α_2 and $\alpha_1 + \alpha_3 + \alpha_4$. Projecting all roots onto this plane, the 6 roots fixed by triality keep norm 2, and the other 18 fall in 6 orbits of 3, each projecting to a vector of norm $\frac{2}{3}$. The result is 12 vectors in two hexagons with squared length ratio 3: the root system G_2 of Lecture 4.

Lamination. Laminating D_4 means placing a new layer over a deep hole (Section 6.3). There are three classes to choose from, and triality makes the choices equivalent. So the laminated lattice Λ_5 is unique; it is D_5.

H.6 Accidental isomorphisms, and the E-series below E_6

The ranges B_n $(n \geq 2)$, C_n $(n \geq 3)$, D_n $(n \geq 4)$ in Theorem 5.2 hide coincidences of small diagrams. Interlude A recorded the Lie algebra isomorphisms. Here is what each says about lattices and geometry.

diagrams	groups	lattices and geometry
$A_1 = B_1 = C_1$	$\mathrm{SU}(2)/\pm 1 = \mathrm{SO}(3)$	roots $\pm\alpha$; unit quaternions rotate $\mathbb{R}^3$ by conjugation (Interlude A)
$B_2 = C_2$	$\mathrm{Sp}(4) \cong \mathrm{Spin}(5)$	B_2 and C_2 differ by a 45° rotation and a factor $\sqrt{2}$ (Lecture 6); root lattices $\mathbb{Z}^2$ and the checkerboard D_2
$D_2 = A_1 \times A_1$	$\mathrm{SO}(4) = \frac{\mathrm{SU}(2)\times\mathrm{SU}(2)}{\pm(1,1)}$	roots $e_1 \pm e_2$ orthogonal; rotations of $\mathbb{R}^4 = \mathbb{H}$ are $x \mapsto qx\bar{r}$
$D_3 = A_3$	$\mathrm{Spin}(6) \cong \mathrm{SU}(4)$	FCC in two coordinate systems; lines in $\mathbb{P}^3$ = Klein quadric
$E_5 = D_5$	$\mathrm{Spin}(10)$	del Pezzo surfaces of degree $4, 5, 6$ (below)
$E_4 = A_4$	$\mathrm{SU}(5)$	
$E_3 = A_2 \times A_1$	$\mathrm{SU}(3) \times \mathrm{SU}(2)$	

The E-series continued downward

Delete nodes one at a time from the end of the long arm of E_8 (α_1, then α_2, and so on, in our labelling). This gives $E_7 = T_{2,3,4}$, $E_6 = T_{2,3,3}$, and then $E_5 = T_{2,3,2} = D_5$, $E_4 = T_{2,3,1} = A_4$ and finally $E_3 = A_2 \times A_1$, when the branch point is reached and the T falls apart. Lecture 6 found the same chain inside E_8 by collapsing chains (Section 6.2). The determinants obey (H.1) all the way down: $\det C(E_n) = 9-n$ gives $6, 5, 4, 3, 2, 1$ for $n = 3, \ldots, 8$, and indeed $\det(A_2 \times A_1) = 3 \cdot 2$, $\det A_4 = 5$, $\det D_5 = 4$. Physicists will recognize the chain $\mathrm{SU}(5) \subset \mathrm{Spin}(10) \subset E_6$ of grand unified theories, with E_3 the $\mathrm{SU}(3) \times \mathrm{SU}(2)$ of the standard model.

The number $9 - n$ has a geometric meaning. Blow up the projective plane in r points in general position $(3 \leq r \leq 8)$. The surface X_r has $H_2(X_r, \mathbb{Z}) = \mathbb{Z}h \oplus \mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_r$ with intersection form $\mathrm{diag}(1, -1, \ldots, -1)$ (a line h, and the exceptional curves e_i), and canonical class $K = -3h + e_1 + \cdots + e_r$, with $K \cdot K = 9 - r$.

Proposition H.8. *For $3 \leq r \leq 8$ the orthogonal complement $K^\perp \subset H_2(X_r, \mathbb{Z})$, with the sign of the form reversed, is the root lattice E_r. Its roots are $e_i - e_j$ and $\pm(h - e_i - e_j - e_k)$ (and, for $r \geq 6$, more), with simple roots $e_1 - e_2, \ldots, e_{r-1} - e_r$, $h - e_1 - e_2 - e_3$. Its determinant is $K \cdot K = 9 - r$.*

Sketch. The listed simple roots are orthogonal to K, have square -2, and their intersection graph is a chain $e_1 - e_2, \ldots, e_{r-1} - e_r$ with $h - e_1 - e_2 - e_3$ attached to $e_3 - e_4$: the diagram $T_{2,3,r-3} = E_r$ (for $r = 3$ there is no $e_3 - e_4$, and $h - e_1 - e_2 - e_3$ is orthogonal to the chain: $A_2 \times A_1$). They span $K^\perp$ (exercise). For the determinant, H_2 is unimodular and K is primitive,

and for a primitive vector v of a unimodular lattice with $v\cdot v\neq 0$ the discriminant of $v^\perp$ is $|v\cdot v|$. Since $K\cdot K>0$ and the form has signature $(1,r)$, $K^\perp$ is negative definite. □

For $r\le 8$ these X_r are the *del Pezzo surfaces* of degree $K\cdot K=9-r$ ($-K$ is ample; for $r\le 6$ it embeds X_r in $\mathbb{P}^{9-r}$ as a surface of degree $9-r$, the cubic surface for $r=6$). The classes E with $E\cdot E=E\cdot K=-1$ are the lines (exceptional curves), and $W(E_r)$ permutes them transitively. For $r\ge 4$ their numbers are *also* read off the diagrams, as the index $|W(E_r)|\,/\,|W(E_{r-1})|$ (a line can be blown down, leaving X_{r-1}):

r	3	4	5	6	7	8
degree $9-r$	6	5	4	3	2	1
E_r	$A_2\times A_1$	A_4	D_5	E_6	E_7	E_8
$\lvert W(E_r)\rvert$	12	120	1920	51840	2903040	696729600
lines	6	10	16	27	56	240

For $r=6$ these are the 27 lines on a smooth cubic surface (Cayley and Salmon, 1849), whose configuration has symmetry group $W(E_6)$ of order 51840. For $r=7$ the 56 lines come in 28 pairs that map to the 28 bitangents of a plane quartic. For $r=8$ the 240 exceptional classes E correspond to the 240 roots $E+K$ of E_8. For $r=4$ the 10 lines meet in the pattern of the Petersen graph. At $r=9$, $K\cdot K=0$ and the lattice is the affine $\tilde{E}_8$, and at $r=10$ we reach E_{10} with $\det=-1$, as in Section H.1.

H.7 ADE elsewhere

The simply laced diagrams A_n, D_n, E_6, E_7, E_8 classify a surprising number of things. In each case below we say what the diagram tells us there. Interlude A (Section A.9, item 7) already mentioned McKay, du Val and Gabriel; here we look more closely.

Finite subgroups of SU(2) and the icosahedron's E_8

The finite subgroups of SU(2) are the cyclic groups and the binary dihedral, tetrahedral ($2T$, order 24), octahedral ($2O$, 48) and icosahedral ($2I$, 120) groups, the preimages of the rotation groups $G^+_{p,q,r}$ of Section H.1. McKay (1980) attached to Γ the graph with a node for each irreducible representation ρ_i and a_{ij} edges when ρ_j occurs a_{ij} times in $\mathbb{C}^2\otimes\rho_i$. The result is an extended diagram: $\tilde{A}_{n-1}$ for $\mathbb{Z}/n$, $\tilde{D}_{n+2}$ for the binary dihedral group of order $4n$, and $\tilde{E}_6$, $\tilde{E}_7$, $\tilde{E}_8$ for $2T$, $2O$, $2I$. What the diagram tells us here:

- *The marks are the dimensions of the representations.* For $2I$ the nine marks of $\tilde{E}_8$ (Figure H.5) give
$$\textstyle\sum m_i^2 = 1^2+2^2+3^2+4^2+5^2+6^2+4^2+2^2+3^2 = 120 = |2I|\,.$$
(The relation $2m_i=\sum_{j\sim i} m_j$, which says that the marks are a null vector of the affine Cartan matrix, is the dimension count $\dim(\mathbb{C}^2\otimes\rho_i)=2\dim\rho_i$.)
- *The mark-1 nodes are the one-dimensional representations*, so their number $\det C$ is the order of the abelianization $\Gamma/[\Gamma,\Gamma]$. We checked by computer that $2T$, $2O$, $2I$ have abelianizations of order $3,2,1$, the determinants of E_6, E_7, E_8.
- *The number of nodes is the number of conjugacy classes*: $7,8,9$ for $2T, 2O, 2I$.

Simple singularities

Klein (1884) showed that the ring of Γ-invariant polynomials on $\mathbb{C}^2$ is generated by three invariants with one relation. For $2I$ the relation is

$$x^2 + y^3 + z^5 = 0,$$

so $\mathbb{C}^2/2I$ is the surface in $\mathbb{C}^3$ cut out by this equation, singular at the origin. The exponents $2, 3, 5$ are again the orders of the rotation axes of the icosahedron. Du Val (1934) resolved these singularities. For $\mathbb{C}^2/\Gamma$, the minimal resolution replaces the singular point by a configuration of smooth rational curves $C_1, \dots, C_r$, each with self-intersection -2, meeting transversally.

The dual graph of the curves is the Dynkin diagram, and the intersection matrix is $-C$.

For $x^2 + y^3 + z^5$ we get eight curves in the shape $T_{2,3,5}$, and the lattice they span in H_2 of the resolution is $-E_8$. So the E_8 lattice of Lecture 6 *grows out of the icosahedron.* This also explains the coincidence of Section H.1. The binary group $\langle p, q, r\rangle$ gives a star-shaped graph with three arms of $p-1$, $q-1$, $r-1$ curves, that is, $T_{p,q,r}$. Algebraic geometry requires the intersection matrix of exceptional curves to be negative definite (Mumford, Grauert), and by Proposition H.1 this is the condition $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$.

The boundary of a small neighbourhood of the singular point is $\mathbb{S}^3/\Gamma$. Its first homology is $\Gamma/[\Gamma,\Gamma]$, and for the plumbed neighbourhood it is also the cokernel of the intersection matrix, of order $|\det C|$. (Use the exact sequence $H_2(X) \to H_2(X, \partial X) \to H_1(\partial X) \to 0$, in which the first map is the intersection matrix.) For E_8, $\det C = 1$, and $\mathbb{S}^3/2I$ is a homology sphere: the *Poincaré dodecahedral space*, obtained from a solid dodecahedron by gluing opposite faces with a twist of $\frac{\pi}{5}$. The E_8 lattice is its "filling".

Functions instead of surfaces. Arnold (1972) classified the *simple* critical points of functions: those near which a small deformation produces only finitely many inequivalent singularities (no continuous moduli). They are

$$A_k:\ x^{k+1}, \quad D_k:\ x^2y + y^{k-1}\ (k \geq 4), \quad E_6:\ x^3 + y^4, \quad E_7:\ x^3 + xy^3, \quad E_8:\ x^3 + y^5,$$

plus a nondegenerate quadratic form in the remaining variables. The subscript is the Milnor number, the number of nondegenerate critical points into which the singularity splits under a small perturbation. The lattice of vanishing cycles (for surfaces) is the negative root lattice, and the monodromy group is the Weyl group. Lecture 9's Morse lemma (Lemma 9.3) is the case A_1. The birth or death of a pair of critical points in a one-parameter family of functions is A_2: $x^3 + tx$. Thom's seven elementary catastrophes are the members of Arnold's list of codimension 1 to 4: A_2 (fold), A_3 (cusp), A_4 (swallowtail), A_5 (butterfly), $D_4^{\pm}$ (hyperbolic and elliptic umbilic) and D_5 (parabolic umbilic).

Representations of quivers and cluster algebras

A *quiver* is a directed graph, and a representation assigns a vector space to each node and a linear map to each arrow. Gabriel (1972) proved that a connected quiver has only finitely many indecomposable representations (up to isomorphism) if and only if its underlying graph is A_n, D_n, E_6, E_7 or E_8. The dimension vectors of the indecomposables are exactly the positive roots, written in the basis of simple roots; Bernstein, Gelfand and Ponomarev (1973) explained this

with reflection functors. So the E_8 quiver has 120 indecomposables. Here the diagram's *positive roots* become objects.

Fomin and Zelevinsky's cluster algebras (2003) are of finite type exactly for the finite root systems. The cluster variables correspond to the "almost positive roots" (positive roots and negative simple roots), so type A_n has $\frac{n(n+1)}{2} + n = \frac{n(n+3)}{2}$ of them, the diagonals of an $(n+3)$-gon. Its clusters correspond to the triangulations, counted by a Catalan number. The *Coxeter number* reappears in Zamolodchikov's periodicity conjecture, proved by Keller (gallery).

Conformal field theory and subfactors

Cappelli, Itzykson and Zuber (1987) classified the modular invariant partition functions of the $\mathfrak{sl}_2$ conformal field theories at level k. They are labelled by the ADE diagrams with *Coxeter number* $h = k+2$, and the diagonal terms are given by the *exponents*. So E_6, E_7, E_8 ($h = 12, 18, 30$) occur at levels 10, 16, 28. In Jones' theory of subfactors (1983), an index less than 4 must be of the form $4\cos^2\frac{\pi}{h}$. The principal graph of such a subfactor is then an ADE diagram of Coxeter number h, and by the classification of Ocneanu, Izumi, Kawahigashi and others (gallery) the graphs that occur are exactly A_n, D_{2n}, E_6 and E_8, not D_{odd} or E_7. That restriction is not visible on the diagram alone; it needs the fusion rules. The number $4\cos^2\frac{\pi}{h}$ is the largest eigenvalue of the adjacency matrix of the diagram, squared. It is also Lecture 4's $4\cos^2\theta$, whose possible values $0, 1, 2, 3$ are the values for $h = 2, 3, 4, 6$.

Niemeier lattices: the Coxeter number is a design constant

Interlude B (Section B.4) used modular forms to count vectors of even unimodular lattices, and mentioned Niemeier's 24 lattices in dimension 24. Their root systems obey a remarkable rule: *all components have the same Coxeter number h, and there are $24h$ roots.* For example A_1^{24} ($h = 2$, 48 roots), D_4^6 ($h = 6$, 144 roots), E_8^3 and $D_{16}E_8$ (both $h = 30$, 720 roots). Here is a proof from two facts already in these notes, with one input from the theory of modular forms.

Proposition H.9 (Venkov)**.** *Let $L \subset \mathbb{R}^{24}$ be an even unimodular lattice with root system $\Delta = \Delta_1 \cup \cdots \cup \Delta_k$. Then either $\Delta = \emptyset$, or Δ spans $\mathbb{R}^{24}$ and every Δ_j has Coxeter number $h = |\Delta|/24$.*

Proof. Input from modular forms. For a harmonic polynomial P of degree 2, the series

$$\Theta_{L,P} = \sum_{x \in L} P(x)\, \mathsf{q}^{x\cdot x/2}$$

is a cusp form of weight $12 + 2 = 14$ for $\mathrm{SL}_2(\mathbb{Z})$ (Hecke; see Ebeling, *Lattices and Codes*, Ch. 3), and there are no nonzero cusp forms of weight 14 (Serre, *A Course in Arithmetic*, Ch. VII). So the coefficient of q vanishes: $\sum_{\alpha\in\Delta} P(\alpha) = 0$. Take $P(x) = \langle x, y\rangle^2 - \frac{1}{24}|x|^2\,|y|^2$, which is harmonic. Then

$$\sum_{\alpha\in\Delta} \langle \alpha, y\rangle^2 = \frac{|\Delta|}{12}\,|y|^2 \qquad \text{for all } y \in \mathbb{R}^{24}.$$

Input from Lecture 6. For an irreducible root system of rank r with roots of norm 2, eutaxy (Section 6.6, item 2) gives $\sum_{\alpha\in\Delta_j}\langle\alpha, y\rangle^2 = \frac{2|\Delta_j|}{r}\,|y_j|^2$, where y_j is the projection of y to the span of Δ_j. By Interlude A (item 3), $|\Delta_j| = rh_j$, so the constant is $2h_j$. If some $y \neq 0$ is orthogonal to all roots, the left side vanishes and $|\Delta| = 0$. Otherwise take y in the span of Δ_j: $2h_j = |\Delta|/12$. □

So the Coxeter number, read off from a diagram, is the constant in the 2-design property of its roots, and a modular form forces all the constants to agree. Checking the list: $D_{16}E_8$ is allowed because $h(D_{16}) = 2\cdot 16 - 2 = 30 = h(E_8)$, and D_{24} has $h = 46$ with $1104 = 24\cdot 46$ roots.

Regular polytopes

Finally the oldest appearance. The symmetry groups of the regular polytopes are Coxeter groups with linear diagrams, and the labels are the Schläfli symbol:

diagram	$\lvert W\rvert$	polytopes	what the diagram shows
$A_3 = \{3,3\}$	24	tetrahedron	symmetric diagram: self-dual
$B_3 = \{4,3\}$	48	cube, octahedron	reversing the diagram: duality
$H_3 = \{5,3\}$	120	dodecahedron, icosahedron	label 5: no lattice
$A_4 = \{3,3,3\}$	120	5-cell	self-dual
$B_4 = \{4,3,3\}$	384	tesseract, 16-cell	16-cell = half-tesseract (D_4)
$F_4 = \{3,4,3\}$	1152	24-cell	self-dual; Voronoi cell of D_4
$H_4 = \{5,3,3\}$	14400	120-cell, 600-cell	$\det G$ barely positive (Section H.2)

In dimensions $n \geq 5$ only A_n and B_n remain, giving the simplex, the cube and the cross-polytope. The exceptional diagrams F_4 and H_4 are why dimension 4 has six regular polytopes.

H.8 Summary: what the diagram knows

feature of the diagram	what it tells us, and where
$\det C$; $\det C(T_{p,q,r}) = pqr(\frac{1}{p} + \frac{1}{q} + \frac{1}{r} - 1)$	density of the root lattice (A, Lecture 6); Platonic solids and triangle groups (§H.1); $\lvert L^*/L\rvert$, centre of the simply connected group, $\lvert\Gamma^{\mathrm{ab}}\rvert$ (§H.7); degree $9-r$ of a del Pezzo surface (§H.6)
sign of the Schläfli determinant	spherical, Euclidean or hyperbolic; the icosahedron is not made of regular tetrahedra (§H.2)
labels $m_{ij} \in \{2,3,4,6\}$	crystallographic restriction; a 5 forces a quasicrystal (§H.2, C)
extended diagram, leaves	holes and their Delaunay cells (§H.4)
marks m_i	alcove vertices ω_i/m_i; index of maximal-rank root sublattices; dimensions of the representations of $\Gamma \subset \mathrm{SU}(2)$
nodes of mark 1	classes of L^*/L (FCC: lattice, octahedral and two tetrahedral classes); one-dimensional representations (McKay)
symmetries of the diagram	lattice automorphisms beyond W (inversion of FCC, triality of D_4); folding (A, C)
symmetries of the extended diagram	$(L^*/L) \rtimes \mathrm{Aut}$: translations by holes, as in BCC
Coxeter number h	$\lvert\Delta\rvert = rh$; eutaxy constant $2h$; Niemeier lattices; CIZ level $k = h-2$; subfactor index $4\cos^2\frac{\pi}{h}$; periodicity $2(h+h')$
exponents	$\lvert W\rvert = \prod(m_j+1)$ (A); diagonal of the CIZ invariants
positive roots	indecomposable quiver representations; cluster variables
small coincidences	accidental isomorphisms: $A_3 = D_3$ is FCC = checkerboard; $E_5 = D_5$, $E_4 = A_4$

H.9 Gallery of theorems

- **Finite reflection groups** (H. S. M. Coxeter, *Discrete groups generated by reflections*, Ann. of Math. 35 (1934), 588–621). Theorem H.2: the finite groups generated by reflections are classified by the positive definite Coxeter diagrams, and the Euclidean ones by the extended diagrams. The non-crystallographic H_3, H_4 are the ones with a 5.

- **Schläfli's regular polytopes** (L. Schläfli, *Theorie der vielfachen Kontinuität*, written 1850–52, published 1901). There are six regular polytopes in $\mathbb{R}^4$ and three in each $\mathbb{R}^n$, $n \geq 5$. The criterion is the positivity of the determinant in Proposition H.3. Coxeter's *Regular Polytopes* (1948) is the standard account.

- **Maximal subgroups of maximal rank** (A. Borel and J. de Siebenthal, Comment. Math. Helv. 23 (1949), 200–221). The maximal closed connected subgroups of maximal rank of a compact simple Lie group are obtained by deleting a node of the extended diagram whose mark is prime (or a node of mark 1 from the ordinary diagram, giving a centralizer of a circle). Its lattice shadow is Theorem H.7(a): D_8 and A_8 in E_8, of index 2 and 3.

- **Voronoi and Delaunay cells of root lattices** (J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Ch. 21 (1988), and *The cell structures of certain lattices*, in *Miscellanea Mathematica*, Springer 1991; R. V. Moody and J. Patera, J. Phys. A 25 (1992), 5089–5134). Conway and Sloane determine the Voronoi and Delaunay cells of all root lattices and their duals (the relevant vectors of a root lattice are its roots). Moody and Patera classify the faces of these cells by decorated Coxeter–Dynkin diagrams, a systematic form of Theorem H.7.

- **Thinnest lattice coverings** (R. P. Bambah, *On lattice coverings by spheres*, Proc. Nat. Inst. Sci. India 20 (1954), 25–52, for $n = 3$; Delone–Ryshkov 1963 for $n = 4$; Ryshkov–Baranovskii 1975 for $n = 5$). The thinnest lattice covering of $\mathbb{R}^n$ for $n \leq 5$ is A_n^*. For $n = 3$ it is BCC, the union of FCC with all of its holes. For $n \geq 6$ the lattice covering problem is open.

- **McKay correspondence** (J. McKay, *Graphs, singularities, and finite groups*, Proc. Sympos. Pure Math. 37 (1980), 183–186; geometric version: G. Gonzalez-Sprinberg and J.-L. Verdier, Ann. Sci. Éc. Norm. Sup. (4) 16 (1983), 409–449). The McKay graph of a finite $\Gamma \subset \mathrm{SU}(2)$ is an extended ADE diagram. The geometric version matches the nontrivial irreducible representations with the exceptional curves of the minimal resolution of $\mathbb{C}^2/\Gamma$.

- **Simple surface singularities** (F. Klein, *Vorlesungen über das Ikosaeder*, 1884; P. Du Val, Proc. Cambridge Philos. Soc. 30 (1934)). The quotients $\mathbb{C}^2/\Gamma$ are hypersurfaces $x^2+y^3+z^5 = 0$ etc., and the dual graphs of their minimal resolutions are the ADE diagrams, with intersection matrix $-C$.

- **The E_8 manifold** (M. Freedman, J. Differential Geom. 17 (1982), 357–453; S. Donaldson, J. Differential Geom. 18 (1983), 279–315). There is a closed simply connected topological 4-manifold with intersection form E_8: cap the E_8 plumbing, whose boundary is the Poincaré sphere $\mathbb{S}^3/2I$, with a contractible topological manifold. It has no smooth structure (Rokhlin: the signature of a smooth spin 4-manifold is divisible by 16). Donaldson: the intersection form of a smooth closed simply connected 4-manifold that is definite must be diagonal.

- **Simple critical points** (V. I. Arnold, Funct. Anal. Appl. 6 (1972), 254–272). The simple germs of functions are exactly A_k, D_k, E_6, E_7, E_8, with Weyl groups as monodromy groups. Morse's nondegenerate critical point of Lecture 9 is A_1.

- **Gabriel's theorem** (P. Gabriel, Manuscripta Math. 6 (1972), 71–103; I. N. Bernstein, I. M. Gelfand, V. A. Ponomarev, Russian Math. Surveys 28 (1973)). A connected quiver has finite representation type if and only if its graph is ADE; indecomposables correspond to positive roots.

- **Finite type cluster algebras** (S. Fomin and A. Zelevinsky, *Cluster algebras II: Finite type classification*, Invent. Math. 154 (2003), 63–121). Cluster algebras of finite type are classified by the Cartan matrices of finite type; the cluster complex is the dual of a generalized associahedron.

- **Periodicity of Y-systems** (B. Keller, *The periodicity conjecture for pairs of Dynkin diagrams*, Ann. of Math. 177 (2013), 111–170; the case (Δ, A_1): Fomin–Zelevinsky, Ann. of Math. 158 (2003), 977–1018). For a pair (Δ, Δ') of simply laced Dynkin diagrams the Y-system $Y_{i,i',t-1}Y_{i,i',t+1} = \prod_{j\sim i}(1+Y_{j,i',t})/\prod_{j'\sim i'}(1+Y_{i,j',t}^{-1})$ is periodic, with period dividing $2(h+h')$, twice the sum of the Coxeter numbers. Conjectured by Zamolodchikov (for $\Delta' = A_1$) in the thermodynamic Bethe ansatz.

- **ADE modular invariants** (A. Cappelli, C. Itzykson, J.-B. Zuber, Comm. Math. Phys. 113 (1987), 1–26). The modular invariant partition functions of $\widehat{\mathfrak{sl}}_2$ at level k are labelled by ADE diagrams of Coxeter number $k+2$ (with D only for even k), with diagonal terms given by the exponents.

- **Subfactors of small index** (V. F. R. Jones, Invent. Math. 72 (1983), 1–25; classification announced by A. Ocneanu (1988); published proofs by M. Izumi, Publ. RIMS 27 (1991), 953–994 (no D_{odd}, E_7) and Y. Kawahigashi, J. Funct. Anal. 127 (1995), 63–107 (flatness, and the D_{2n} subfactors), with the E_6, E_8 cases completed by several authors; survey: Jones, Morrison, Snyder, Bull. Amer. Math. Soc. 51 (2014), 277–327). The index of a subfactor lies in $\{4\cos^2\frac{\pi}{n} : n \geq 3\} \cup [4, \infty]$, and below 4 the principal graphs are A_n, D_{2n}, E_6, E_8.

- **Niemeier lattices** (H.-V. Niemeier, J. Number Theory 5 (1973), 142–178; B. B. Venkov, Trudy Mat. Inst. Steklov 148 (1978), 65–76). Proposition H.9, and the classification: the 24 even unimodular lattices in $\mathbb{R}^{24}$ are determined by their root systems, which are empty (Leech) or of rank 24 with all components of the same Coxeter number.

- **Lines on del Pezzo surfaces** (A. Cayley and G. Salmon, 1849; C. Jordan, *Traité des substitutions*, 1870; P. Du Val, Proc. London Math. Soc. 35 (1933); Yu. I. Manin, *Cubic Forms*, 1972). Proposition H.8: the lines on a del Pezzo surface of degree $9-r$ are permuted by $W(E_r)$, and the 27 lines on a cubic surface have symmetry group $W(E_6)$ of order 51840.

Exercises

Exercise H.10. Compute $\det C(T_{1,q,r})$ directly and check that it is $q+r$, the determinant of A_{q+r-1}. Show that $E_{10} = T_{2,3,7}$ has determinant -1, and that its Cartan matrix has exactly one negative eigenvalue.

Exercise H.11. Using Proposition H.3, compute the Schläfli determinants of $\{4,3,3\}$, $\{3,4,3\}$, $\{4,3,4\}$ and $\{5,3,5\}$. Which are spherical, which Euclidean, which hyperbolic? Interpret $\{4,3,4\}$ geometrically.

Exercise H.12. Count the vertices, edges, 2-faces and cells of the 24-cell from its diagram $F_4 = \{3,4,3\}$, using $|W(F_4)| = 1152$ and the rule that the stabilizer of a k-face is generated by all nodes except the $(k+1)$-st. Check Euler's relation $V - E + F - C = 0$.

Exercise H.13. For the FCC lattice D_3, verify that each lattice point lies on 8 tetrahedra and 6 octahedra of the Delaunay tiling, and that each edge lies on 2 of each. Compute the solid angles at the vertices of the tetrahedron and the octahedron, and confirm $8\Omega_T + 6\Omega_O = 4\pi$.

Exercise H.14. Show that the hole ω_i of A_n has $|\omega_i|^2 = i(n+1-i)/(n+1)$ and that its nearest lattice points form a hypersimplex with $\binom{n+1}{i}$ vertices. For which n are all Delaunay cells of A_n simplices?

Exercise H.15. In D_n, compute the distances from e_1 and from $h = \frac{1}{2}(1,\dots,1)$ to the lattice, and the number of nearest lattice points. At which n does the deepest hole change type? Relate this to the construction $E_8 = D_8 \cup (D_8 + h)$.

Exercise H.16. Let $g = \omega_3 \in A_8^*$. Show that $A_8 \cup (A_8 + g) \cup (A_8 + 2g)$ is an even unimodular lattice with 240 vectors of norm 2: 72 from A_8 and 84 from each nontrivial coset. Conclude that it is E_8.

Exercise H.17. Project the 24 roots of D_4 onto the plane fixed by triality, and show that the images form the root system G_2, with 6 long roots of norm 2 and 6 short roots of norm $\frac{2}{3}$.

Exercise H.18. On the cubic surface X_6, list the classes e_i, $h - e_i - e_j$, $2h - \sum_{k \neq i} e_k$, and check that each satisfies $E \cdot E = E \cdot K = -1$. How many are there? Show that the simple roots of Proposition H.8 span $K^{\perp}$.

Exercise H.19. For the extended diagrams $\tilde{E}_6$ and $\tilde{E}_7$, find the marks, check that $\sum m_i^2$ equals 24 and 48, and that the number of nodes of mark 1 is $\det C$. Which nodes give holes of the lattices E_6 and E_7, and what are their Delaunay cells?

Further reading

- M. Hazewinkel, W. Hesselink, D. Siersma and F. D. Veldkamp, *The ubiquity of Coxeter–Dynkin diagrams (an introduction to the A–D–E problem)*, Nieuw Arch. Wisk. 25 (1977), 257–307. The classic survey of where ADE appears.
- H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973. Schläfli's criterion, Wythoff's construction, and the 600-cell.
- J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge, 1990. Coxeter's theorem, affine Weyl groups and alcoves.
- J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Ch. 4 and 21. Glue, holes and Voronoi cells of root lattices.
- P. Slodowy, *Simple Singularities and Simple Algebraic Groups*, Lecture Notes in Math. 815, Springer, 1980. The bridge between $\mathbb{C}^2/\Gamma$ and Lie theory.
- J. C. Baez, *The octonions*, Bull. Amer. Math. Soc. 39 (2002), 145–205. Triality, G_2 and the exceptional diagrams.
- I. V. Dolgachev, *Classical Algebraic Geometry: A Modern View*, Cambridge, 2012, Ch. 8–9. Del Pezzo surfaces and the root systems E_r.